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Upgrading survival models with CARE

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Abstract

Clinical risk prediction models are regularly updated as new data, often with additional covariates, become available. We propose CARE (Convex Aggregation of relative Risk Estimators) as a general approach for combining existing 'external' estimates with a new data set in a time-to-event survival analysis setting. Our method initially employs the new data to fit a flexible family of reproducing kernel estimators via penalised partial likelihood maximisation. The final relative risk estimator is then constructed as a convex combination of the kernel and external estimators, with the regularisation parameters and convex combination coefficients selected using cross-validation. We establish high-probability bounds for the L_2-error of our proposed aggregated estimator, showing that it achieves a rate of convergence that is at least as good as both the optimal kernel estimator and the best existing external model. Empirical results from simulation studies align with the theoretical results, and we illustrate the improvements our methods provide for cardiovascular disease risk modelling. Our methodology is implemented in the Python package `care-survival`.

Keywords: Model aggregation, penalised estimation, disease risk modelling, relative risk, reproducing kernels, nonparametric estimation, survival analysis.

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1 Introduction

In biomedicine, clinical prediction models for health outcomes based on covariates provide an important resource to aid medical decision-making. These models are typically updated regularly as additional potentially relevant variables are measured, usually on a separate set of individuals. A motivating example for this work is the SCORE2 model (SCORE2 Working Group and ESC Cardiovascular Risk Collaboration, 2021) for estimating 10-year fatal and non-fatal cardiovascular disease risk in Europe. The model was initially trained using data on 677,684 individuals under a Fine–Gray competing risks framework (Fine and Gray, 1999), including age, smoking status, blood pressure and cholesterol as covariates. A similar time-to-event model named SCORE2-OP was developed concurrently (SCORE2-OP Working Group and ESC Cardiovascular Risk Collaboration, 2021), specifically targeted towards individuals over 70 years old. Since the initial release of these models, several updates have been proposed, adding new predictor variables to improve risk estimation (Matsushita et al., 2023; Cosmi et al., 2024; Dienhart et al., 2024; Wong et al., 2025; Xie et al., 2025). Although the new predictors can sometimes be calculated directly from existing covariates by applying appropriate transformations, it is often the case that the model must be updated using an alternative data source (Matsushita et al., 2023; Xie et al., 2025). Other existing survival models include PREDICT for breast cancer and prostate cancer prognosis (Wishart et al., 2010, 2012, 2014; Thurtle et al., 2019a,b; Pandiaraja et al., 2024), QRISK for cardiovascular disease risk prediction (Hippisley-Cox et al., 2007, 2008, 2017) and $\text{PLCO}_{\text{M2012}}$ for lung cancer diagnosis (Tammemägi et al., 2011, 2013). A common approach for upgrading such models is to incorporate directly fitted coefficients from an existing study, here in the form of estimated hazard ratios. While validation studies can provide some reassurance, the underlying strong assumption is that the imported coefficients, which may be based on a reduced set of variables, remain valid in the presence of the expanded set of covariates in the model. For example, the PREDICT tool for breast cancer prognosis was updated twice using this approach, adding the biomarker proteins HER2 and KI67 as new covariates (see Wishart et al., 2012, 2014, respectively).

In this paper, we propose general methodology for fitting and upgrading survival analysis prediction models when new data (potentially with additional covariates) become available. Our framework encompasses the common situations where the original individual-level training data may no longer be available, for instance due to patient confidentiality, regulatory or proprietary reasons. Instead, we may only be able to query the model in the sense of obtaining predictions at an arbitrary test point of our choosing, typically through an online user interface, or in the case of a parametric model, through published fitted parameter estimates. In this setting, we have access to one or more existing pre-trained models, as well as a new independent data set consisting of the full expanded set of covariates, censored event times and censoring indicators.

Following some background material in Section 2, our first contribution, in Section 3, is to introduce a principled reproducing kernel Hilbert space (RKHS) framework that provides considerable flexibility for modelling the relative risk function. In the special case of a linear kernel, this reduces to (a regularised version of) the celebrated Cox proportional hazards model (Cox, 1972), but fully nonparametric choices are also possible. We demonstrate that such models can be fitted in a computationally efficient manner via a restricted penalised

partial likelihood approach. In order to borrow strength across all available information sources, we propose a data-driven procedure named CARE (Convex Aggregation of relative Risk Estimators), which combines the existing models and our new RKHS estimator. The final prediction function is constructed as a convex combination of the old and new prediction functions, with the convex combination weights and the RKHS regularisation parameter chosen via a joint partial likelihood cross-validation strategy.

In Section 4, we turn to the theoretical properties of our proposed CARE procedure. Our first main results consider the RKHS estimator alone. Theorem 4 provides a high-probability bound on the error of this estimator in terms of the sample size, the regularisation parameter and a quantity reflecting the spectral properties of the natural integral operator associated with the kernel, while Theorem 8 establishes minimax rate-optimality of our procedure, under some further mild conditions on the kernel spectrum. Theorem 9 then gives an oracle inequality to justify the use of cross-validation for parameter tuning. Theorem 10 in Section 4.4 presents our main performance guarantee for the CARE procedure that combines the RKHS estimator with the prediction functions from the existing models. It demonstrates that our joint cross-validation strategy performs nearly as well as the oracle convex combination of the pre-trained models and the RKHS estimators. In particular, CARE is able to adapt both to situations where the best pre-trained model is a superior predictor compared to the newly constructed RKHS estimator (for example, because the sample size of the new study is too small), and to settings where the optimal RKHS estimator is a better predictor than all of the existing estimators.

Section 5 provides practical implementation details for our CARE procedure, including fitting recommendations via first- and second-order optimisation methods. In Section 6, we present our empirical results, first verifying our theoretical results through a series of simulation studies. We then consider upgrading an existing cardiovascular disease risk model (SCORE2) using cohort data from the UK Biobank (Littlejohns et al., 2019). By incorporating a new data set with additional covariates (relating to area-specific social deprivation levels and polygenic scores) we show that CARE can improve the concordance index (Harrell et al., 1982) of the SCORE2 model by 1.52% for females and by 2.26% for males. Proofs and further technical and computational details are gathered in the appendix, along with additional empirical results.

Recent years have seen a growing interest among health researchers in developing methodologies for aggregating estimators and data sets. In the parametric setting, approaches include joint likelihood maximisation (Zhang et al., 2020), penalised constrained maximum likelihood for heterogeneous populations (Zhai and Han, 2022), empirical Bayes composite estimation (Gu et al., 2023), constrained empirical likelihood (Ni et al., 2023) and generalised method of moments (Fang et al., 2025). Semiparametric and nonparametric methods include universal aggregation (Goldenshluger, 2009), affine aggregation (Dalalyan and Salmon, 2012), efficient data fusion (Li and Luedtke, 2023; Graham et al., 2026; Hu et al., 2026), model aggregation via cross-validation (Yu et al., 2025) and pseudo-labelling (Wang, 2026). In the context of time-to-event models, Liu et al. (2014) use external data sources to improve estimation of the baseline hazard function for within-cohort Cox proportional hazards analysis of rare diseases. Huang et al. (2016) describe a method for fitting a Cox model by an empirical likelihood approach, incorporating external information in the form of published group-level survival probabilities; see also Chen et al. (2021) and Sheng et al. (2021) for extensions.

Ding et al. (2023, 2024) consider models in which auxiliary information is available in the form of fixed-time survival probabilities, stratified by subgroups; estimation then proceeds by penalised likelihood estimation or a penalised generalised method of moments. Wang et al. (2025) propose a procedure for fitting a generalised parametric Cox model by maximising a penalised version of the log-likelihood that imposes Kullback–Leibler-based shrinkage towards external pre-trained survival models. In our methodology, we opt for cross-validation of convex combinations rather than explicit penalisation in order to obtain an automatic data-driven procedure for incorporating the external information.

Related to our model upgrading task, *transfer learning* (Cai and Wei, 2021; Reeve et al., 2021) also considers settings of multiple data sources. Specifically, one typically studies source and target data sets, both of which are available and measure the same features. The primary emphasis is on adapting to the distributional shift between source and target, with the aim of improving inference regarding some aspect of the target distribution.

A separate line of work has sought to develop general theory and methodology for estimation in survival analysis models where the relative risk is modelled more flexibly than via a linear predictor. Early works include the introduction of local polynomials for partial likelihood estimation (Fan et al., 1997; Chen and Zhou, 2007); see also more recent work on high-dimensional regularised Cox models (Bradic et al., 2011; Huang et al., 2013; Yu et al., 2021). Liu et al. (2020) propose RKHS methods with Sobolev kernels for censored survival data, building on work of O'Sullivan (1988, 1993). Chung et al. (2018) incorporate nonparametric shape constraints (Samworth and Sen, 2018) into hazard ratio modelling, while Qu et al. (2016) consider kernel estimation in the functional Cox model. In a more applied vein, Montaseri et al. (2025) study heart failure via an RKHS framework (without regularisation) for an accelerated failure time model.

An alternative approach to survival modelling is landmark analysis (Van Houwelingen, 2007; van Houwelingen and Putter, 2008; Parast and Cai, 2013; Fournier et al., 2018), in which survival is modelled conditional on being event-free up to a fixed landmark time. Covariate values are typically fixed at the landmark, and survival beyond this time is analysed within the landmark cohort, often using a Cox model. This yields interpretable, dynamic predictions, but does not model the full time-varying hazard function. Since existing studies for disease risk modelling only rarely use landmark analysis, we do not address this framework in our methodology.

2 Set-up and preliminary results

We begin by providing some notation used throughout the paper, and give a description of the model for the data generating process based on a censored time-to-event formulation. We also present some further introductory material on reproducing kernel Hilbert spaces and associated norms.

2.1 Notation

For $n \in \mathbb{N}$, let $[n] := \{1, \ldots, n\}$. We write $a \wedge b$ for the minimum of two real numbers a and b, and $a \vee b$ for their maximum. For positive real-valued sequences (a_n) and (b_n), we write

$a_n \lesssim b_n$ if there exists $C > 0$ and $N \in \mathbb{N}$ with $a_n \leq C b_n$ for all $n \geq N$. Similarly, we write $a_n \asymp b_n$ if there exists $C \geq 1$ and $N \in \mathbb{N}$ with $1/C \leq a_n/b_n \leq C$ for all $n \geq N$. For $a \in \mathbb{R}$ and $d \in \mathbb{N}$, define $a^{\otimes d} := (a, \ldots, a)^\mathsf{T} \in \mathbb{R}^d$. For a real square symmetric matrix $\boldsymbol{A}$, we write $\boldsymbol{A} \succeq 0$ if $\boldsymbol{A}$ is positive semi-definite and $\boldsymbol{A} \succ 0$ if $\boldsymbol{A}$ is positive definite, with $\preceq$ and $\prec$ defined analogously. We write $\lambda_{\min}(\boldsymbol{A})$ and $\lambda_{\max}(\boldsymbol{A})$ for the minimum and maximum eigenvalues of $\boldsymbol{A}$, respectively. The operator norm is denoted by $\|\boldsymbol{A}\|_{\mathrm{op}}$, while the entrywise ℓ_∞-norm is $\|\boldsymbol{A}\|_\infty$. The $d \times d$ identity matrix is denoted $\boldsymbol{I}_d$, and we write $0_d := (0, \ldots, 0)^\mathsf{T} \in \mathbb{R}^d$ and $1_d := (1, \ldots, 1)^\mathsf{T} \in \mathbb{R}^d$. For a measurable space $(\mathcal{X}, \mathcal{A})$, define $\mathcal{B}(\mathcal{X})$ to be the set of bounded, measurable functions from $\mathcal{X}$ to $\mathbb{R}$. The supremum norm of $f \in \mathcal{B}(\mathcal{X})$ is $\|f\|_\infty := \sup_{x \in \mathcal{X}} |f(x)|$. For a totally ordered finite set $\mathcal{S}$ and a function $f : \mathcal{S} \to \mathbb{R}$, we write $\operatorname{sargmin}_{s \in \mathcal{S}} f(s) := \min\bigl\{s \in \mathcal{S} : f(s) = \min\{f(t) : t \in \mathcal{S}\}\bigr\}$. The uniform distribution on a compact interval $[a, b]$ is denoted by $\mathcal{U}[a, b]$.

2.2 Data and partial likelihood functions

Let $(\mathcal{X}, \mathcal{A})$ denote a measurable space, and let $(X, T_\mathrm{S}, T_\mathrm{C})$ be a random vector taking values in $\mathcal{X} \times (0, \infty) \times (0, \infty)$. Here, X represents a covariate, T_S a survival time and T_C a censoring time. We emphasise that X may take values in a wide range of possible covariate spaces, including $\mathbb{R}^d$ for any $d \in \mathbb{N}$ as well as non-Euclidean domains such as sequence or graph spaces. As such, we refrain from explicitly referring to the 'dimension' of the covariate. Suppose that T_S and T_C are conditionally independent given X, and define the observed time $T := T_\mathrm{S} \wedge T_\mathrm{C}$ and the censoring indicator $I := \mathbb{1}_{\{T_\mathrm{C} < T_\mathrm{S}\}}$. The *at-risk process* is given by $R(t) := \mathbb{1}_{\{T \geq t\}}$ for $t \in [0, 1]$. Define $q : \mathcal{X} \times [0, 1] \to \mathbb{R}$ by $q(x, t) := \mathbb{P}(T \geq t \mid X = x)$ and assume that $q_1 := \inf_{x \in \mathcal{X}} q(x, 1) > 0$. Let P_X denote the distribution of X. We consider a proportional hazards model with conditional hazard function

$$h(t \mid x) := -\frac{\mathrm{d}}{\mathrm{d}t} \log \mathbb{P}(T_\mathrm{S} > t \mid X = x) = \lambda_0(t) e^{f_0(x)} \tag{1}$$

for $t \in [0, 1]$ and P_X-almost all x, where $\lambda_0 : [0, 1] \to [0, \infty)$ is a measurable baseline hazard function and $f_0 \in \mathcal{B}(\mathcal{X})$ is a function representing the relative risk. For any P_X-integrable $f : \mathcal{X} \to \mathbb{R}$, define $P_X(f) := \int_{\mathcal{X}} f(x) \, \mathrm{d}P_X(x)$. To ensure the identifiability of λ_0 and f_0, we impose the constraint that $P_X(f_0) = 0$ (see O'Sullivan, 1993). Observe that

$$\begin{aligned} \Lambda := \int_0^1 \lambda_0(t) \, \mathrm{d}t &= e^{-f_0(x)} \bigl\{\log \mathbb{P}(T_\mathrm{S} > 0 \mid X = x) - \log \mathbb{P}(T_\mathrm{S} \geq 1 \mid X = x)\bigr\} \\ &\leq e^{\|f_0\|_\infty} \log(1/q_1) < \infty, \end{aligned}$$

where the second equality holds for P_X-almost all $x \in \mathcal{X}$. The observed data $(X_i, T_i, I_i)_{i \in [2n]}$ are assumed to be independent and identically distributed copies of (X, T, I). We refer to the first n triples as the *training set*, while the rest form the *validation set*. For $i \in [2n]$, write $N_i(t) := \mathbb{1}_{\{T_i \leq t, \, I_i = 0\}}$ for the event process (Fleming and Harrington, 2013) and $R_i(t) := \mathbb{1}_{\{T_i \geq t\}}$. Defining $N(t) := \sum_{i=1}^n N_i(t)/n$ and

$$S_n(f, t) := \frac{1}{n} \sum_{i=1}^n R_i(t) e^{f(X_i)}$$

for $f : \mathcal{X} \to \mathbb{R}$, the negative log-partial likelihood of this model is

$$\ell_n(f) := \int_0^1 \log S_n(f,t)\,\mathrm{d}N(t) - \frac{1}{n}\sum_{i=1}^n f(X_i)N_i(1). \tag{2}$$

This is well-defined because $S_n(f,T_i) \geq R_i(T_i)e^{f(X_i)}/n = e^{f(X_i)}/n > 0$ for $i \in [n]$. For $f \in \mathcal{B}(\mathcal{X})$ and $t \in [0,1]$, let $S(f,t) := \int_{\mathcal{X}} q(x,t)e^{f(x)}\,\mathrm{d}P_X(x)$ and define the limiting negative log-partial likelihood by

$$\ell_\star(f) := \int_0^1 \log\bigl(S(f,t)\bigr)S(f_0,t)\lambda_0(t)\,\mathrm{d}t - \int_0^1 DS(f_0,t)(f)\lambda_0(t)\,\mathrm{d}t,$$

where $DS(f_0,t)$ is a Gateaux derivative (Aliprantis and Border, 2006, Chapter 7). This is finite, since $S(f_0,t) \leq e^{\|f-f_0\|_\infty}S(f,t)$, and we interpret $0\log 0$ as zero. Both $\ell_n(f)$ and $\ell_\star(f)$ are invariant under addition of constants to f. The following lemma may be regarded as a Fisher consistency property of f_0 (Fisher, 1922).

Lemma 1 (Characterisation of f_0)**.** *We have $f_0 \in \operatorname{argmin}_{f\in\mathcal{B}(\mathcal{X})} \ell_\star(f)$. Further, f_0 is unique in the sense that if $\tilde{f} \in \operatorname{argmin}_{f\in\mathcal{B}(\mathcal{X})} \ell_\star(f)$ and $P_X(\tilde{f}) = 0$, then $\tilde{f}(x) = f_0(x)$ for P_X-almost every $x \in \mathcal{X}$.*

2.3 Reproducing kernel Hilbert spaces

A *kernel* is a function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ that is symmetric and is positive definite in the sense that for any $n \in \mathbb{N}$ and $x_1, \ldots, x_n \in \mathcal{X}$, the matrix $\boldsymbol{K} \in \mathbb{R}^{n\times n}$ with entries $\boldsymbol{K}_{ij} := k(x_i, x_j)$ is symmetric and positive semi-definite. We will assume that the measurable space $(\mathcal{X}, \mathcal{A})$ and the kernel k are sufficiently regular; see Appendix B for precise conditions and further technical details. By Mercer's theorem (Samworth and Shah, 2026, Theorem A.8.8), there exists a decreasing sequence $(\nu_r)_{r\in\mathcal{R}}$ of positive numbers with $\sum_{r\in\mathcal{R}} \nu_r < \infty$ and associated continuous functions $(e_r)_{r\in\mathcal{R}}$ in $L_2(P_X)$, where either $\mathcal{R} = [R]$ for some $R \in \mathbb{N}$ or $\mathcal{R} = \mathbb{N}$, endowed with the usual order, such that $k(x,y) = \sum_{r\in\mathcal{R}} \nu_r e_r(x)e_r(y)$ for $x, y \in \mathcal{X}$. Write $\langle\cdot,\cdot\rangle_{L_2} := \langle\cdot,\cdot\rangle_{L_2(P_X)}$. Let $(\mathcal{H}, \langle\cdot,\cdot\rangle_{\mathcal{H}})$ be the reproducing kernel Hilbert space (RKHS) associated with k, and suppose further that $(x,y) \mapsto k(x,y) - \delta$ is a positive definite function on $\mathcal{X} \times \mathcal{X}$ for some $\delta > 0$. This ensures that $\mathcal{H}$ contains the unit constant function on $\mathcal{X}$, which we denote by $1_{\mathcal{X}}$. We assume throughout that the model is correctly specified in the sense that the true relative risk function satisfies $f_0 \in \mathcal{H}$.

In preparation for the upcoming regularisation framework for RKHS estimation, we define another inner product, for $\gamma \geq 0$, by $\langle f, g\rangle_{\mathcal{H}_\gamma} := \langle f, g\rangle_{L_2} + \gamma\langle f, g\rangle_{\mathcal{H}}$. For $r \in \mathcal{R}$, write $a_r := 1/(1+\gamma/\nu_r)$ and define $K := \sup_{x\in\mathcal{X}} k(x,x)$, $H_{2,\gamma} := \sum_{r\in\mathcal{R}} a_r$ and $H_{\infty,\gamma} := \sup_{x\in\mathcal{X}} \sum_{r\in\mathcal{R}} a_r e_r(x)^2$, so that $H_{2,\gamma} \leq H_{\infty,\gamma} \leq K/\gamma$.

We give some examples of kernel functions that satisfy our assumptions in Section 4.1.1, while practical guidance on the choice of kernel is presented in Section 5.5. In Lemma 2, we bound $\|\cdot\|_\infty$ on $\mathcal{H}$ in terms of $\|\cdot\|_{\mathcal{H}}$ and $\|\cdot\|_{\mathcal{H}_\gamma}$ respectively.

Lemma 2 (Supremum norm bounds)**.** *Let $f \in \mathcal{H}$ and $\gamma > 0$. Then $\|f\|_\infty^2 \leq K\|f\|_{\mathcal{H}}^2$ and $\|f\|_\infty^2 \leq H_{\infty,\gamma}\|f\|_{\mathcal{H}_\gamma}^2$.*

3 Methodology

3.1 Kernel estimator

We begin by introducing our basic estimator, which is obtained by minimising a penalised version of the negative log-partial likelihood over the RKHS $\mathcal{H}$. For $\gamma > 0$ and $f \in \mathcal{H}$, define the penalised negative log-partial likelihood

$$\ell_{n,\gamma}(f) := \ell_n(f) + \gamma \|f\|_{\mathcal{H}}^2, \tag{3}$$

where $\ell_n(f)$ is defined in (2). We show in Proposition 3 below that there is a unique estimator

$$\hat{f}_{n,\gamma} := \operatorname*{argmin}_{f \in \mathcal{H}:\, P_n(f)=0} \ell_{n,\gamma}(f), \tag{4}$$

where $P_n(f) := \sum_{i=1}^n f(X_i)/n$. The centring constraint $P_n(f) = 0$ in our optimisation is natural in view of the identifiability condition $P_X(f_0) = 0$ and the fact that the log-likelihood is unaffected by the addition of constants. Before stating this result, we introduce some notation. Let $\mathcal{A}_n$ be a maximal subset of $[n]$ such that $\big\{(\|1_{\mathcal{X}}\|_{\mathcal{H}}^2, 1, \ldots, 1)^{\mathsf{T}}, \big(1, k(X_i, X_1), \ldots, k(X_i, X_n)\big)^{\mathsf{T}} : i \in \mathcal{A}_n\big\} \subseteq \mathbb{R}^{n+1}$ is linearly independent. For $x, y \in \mathcal{X}$, let $\tilde{k}(x, y) := k(x, y) - \bar{k}(y)$, where $\bar{k}(y) := \sum_{i=1}^n k(X_i, y)/n$. Further, for $\beta \in \mathbb{R}^{\mathcal{A}_n}$, define $f_\beta(x) := \sum_{i \in \mathcal{A}_n} \tilde{k}(x, X_i)\beta_i$, so that $P_n(f_\beta) = 0$.

Proposition 3 (Representation of $\hat{f}_{n,\gamma}$)**.** *The function $\beta \mapsto \ell_{n,\gamma}(f_\beta)$ is strongly convex on $\mathbb{R}^{\mathcal{A}_n}$. Writing $\hat{\beta}$ for its unique minimiser, we have that $\hat{f}_{n,\gamma} := f_{\hat{\beta}}$ is the unique minimiser of $f \mapsto \ell_{n,\gamma}(f)$ over $f \in \mathcal{H}$ with $P_n(f) = 0$.*

Proposition 3 shows that in order to find the minimiser in (4), it suffices to consider functions of the form f_β, where $\beta \in \mathbb{R}^{\mathcal{A}_n}$. This representation allows the feasible set of the optimisation problem to be simplified from $\mathcal{H}$ (which may be infinite-dimensional) to $\mathbb{R}^{\mathcal{A}_n}$ (which has at most n dimensions). In Section 5, we give further details on the calculation of $\|1_{\mathcal{X}}\|_{\mathcal{H}}$ and the construction of the set $\mathcal{A}_n$, and discuss practical procedures for computing $\hat{\beta}$ and $\hat{f}_{n,\gamma}$.

3.2 Parameter tuning

Partial likelihood cross-validation can be used to select the regularisation parameter γ. Let $\Gamma \subseteq (0, \infty)$ be a finite set of candidate parameters. With $\tilde{N}(t) := \sum_{i=n+1}^{2n} N_i(t)/n$ and $\tilde{S}_n(f, t) := \sum_{i=n+1}^{2n} R_i(t) e^{f(X_i)}/n$, define the negative partial log-likelihood on the validation data by

$$\tilde{\ell}_n(f) := \int_0^1 \log \tilde{S}_n(f, t)\, \mathrm{d}\tilde{N}(t) - \frac{1}{n} \sum_{i=n+1}^{2n} f(X_i) N_i(1),$$

where $f : \mathcal{X} \to \mathbb{R}$. For each $\gamma \in \Gamma$, fit $\hat{f}_{n,\gamma}$ on the training data as in (4). Finally, set

$$\hat{\gamma} := \operatorname*{sargmin}_{\gamma \in \Gamma} \tilde{\ell}_n\big(\hat{f}_{n,\gamma}\big). \tag{5}$$

3.3 Model aggregation with CARE

We now provide a framework for combining the kernel estimators $(\hat{f}_{n,\gamma} : \gamma \in \Gamma)$ with a collection of 'external' estimators $(\hat{f}_m : m \in \mathcal{M})$, where $\mathcal{M} = [|\mathcal{M}|]$ is a non-empty finite set and $\hat{f}_m$ takes values in $\mathcal{B}(\mathcal{X})$ for each $m \in \mathcal{M}$. In applications, we do not need the external estimators $\hat{f}_m$ to use precisely the same set of covariates as the kernel estimators $\hat{f}_{n,\gamma}$. Indeed, the only requirement is that the validation likelihood $\tilde{\ell}_n$ can be evaluated at each $\hat{f}_m$. Specifically, if the validation data set contains a covariate not used by an external estimator (that is, a new covariate is being added), it can simply be ignored when evaluating the validation likelihood. Conversely, if the external estimator uses an extra covariate not present in the validation data (that is, an existing covariate is being removed), then one should marginalise over (an estimate of) the distribution of the missing covariate given the observed covariates. This may be challenging if the original data set is no longer available; however, in most practical settings, one seeks to add new covariates to existing models rather than remove them.

To ensure the validity of our cross-validation methodology, we assume that $(\hat{f}_m : m \in \mathcal{M})$ are computed on a data set that is independent of the training and validation data $(X_i, T_i, I_i)_{i\in[2n]}$. However, we do not need each $\hat{f}_m$ to itself be a function of independent and identically distributed data points; nor do we require that $\hat{f}_m$ be statistically independent of $\hat{f}_{m'}$, where $m \neq m'$. We further suppose that $\max_{m\in\mathcal{M}} \|\hat{f}_m\|_\infty \leq M$ almost surely, where M is a random variable taking values in $(0, \infty)$. For $m \in \mathcal{M}$, define $\tilde{f}_m := \hat{f}_m - P_n(\hat{f}_m)1_{\mathcal{X}}$. Let $\Theta \subseteq \Delta_{\mathcal{M}} := \{\theta \in [0,1]^{\mathcal{M}} : \sum_{m\in\mathcal{M}} \theta_m \leq 1\}$ be a finite set and consider the convex combination estimators $(\check{f}_{n,\gamma,\theta} : \gamma \in \Gamma, \theta \in \Theta)$, where we define

$$\check{f}_{n,\gamma,\theta}(x) := \left(1 - \sum_{m\in\mathcal{M}} \theta_m\right)\hat{f}_{n,\gamma}(x) + \sum_{m\in\mathcal{M}} \theta_m \tilde{f}_m(x)$$

for $x \in \mathcal{X}$. We construct $\check{\gamma}$ and $\check{\theta}$ via partial likelihood cross-validation, setting

$$(\check{\gamma}, \check{\theta}) := \operatorname*{sargmin}_{(\gamma,\theta)\in\Gamma\times\Theta} \tilde{\ell}_n\big(\check{f}_{n,\gamma,\theta}\big), \tag{6}$$

using the lexicographic order on Θ. In general, $\check{\gamma}$ defined in (6) may not take the same value as $\hat{\gamma}$ given in (5). In Section 5.4, we give some practical recommendations for solving the optimisation problem (6). We define the CARE estimator as $\check{f}_{n,\check{\gamma},\check{\theta}}$.

4 Theoretical analysis

In this section, we establish high-probability bounds for the deviations of our proposed estimators from the true relative risk function f_0. We are primarily interested in approximation under the $L_2(P_X)$-norm; that is, in controlling $\|f - f_0\|_{L_2}$. This choice of metric can be motivated by treating f as an estimator of the log-relative hazard, since for $f \in L_2(P_X)$ and $t \in [0,1]$, and with $h(t \mid x)$ as in (1),

$$\|f - f_0\|_{L_2}^2 = \mathbb{E}\left\{\left(\log\frac{h(t \mid X)}{\lambda_0(t)} - f(X)\right)^2\right\}.$$

4.1 Kernel estimator

Our first main theoretical result concerns the proximity of $\hat{f}_{n,\gamma}$ to f_0, where $\gamma > 0$ is a fixed regularisation parameter.

Theorem 4 (Rate of convergence)**.** *There exists $C_0 > 0$, depending only on $\|f_0\|_{\mathcal{H}}$, Λ, q_1, $\|1_{\mathcal{X}}\|_{\mathcal{H}}$ and K, such that if $\gamma H_{\infty,\gamma} \leq 1/C_0$ and $1 \leq s \leq \sqrt{n}/(C_0 H_{\infty,\gamma})$, then with probability at least $1 - e^{-s^2}$,*

$$\left\|\hat{f}_{n,\gamma} - f_0\right\|_{L_2} \leq \left\|\hat{f}_{n,\gamma} - f_0\right\|_{\mathcal{H}_\gamma} \leq C_0\left(\frac{\sqrt{H_{2,\gamma}} + s}{\sqrt{n}} + \sqrt{\gamma}\right). \tag{7}$$

The right-hand side of (7) consists of two terms, corresponding to a bias–variance trade-off. Specifically, the 'variance' term $H_{2,\gamma}/n$ is a decreasing function of both the sample size n and the regularisation parameter γ, while the 'bias' term is $\sqrt{\gamma}$. As such, choosing a small value of $\gamma > 0$ allows for less bias at the expense of potentially increasing the variance. The rate of convergence is driven by the kernel complexity quantity $H_{2,\gamma}$, corresponding to the effective dimension of the problem (Mendelson, 2003; Caponnetto and De Vito, 2007; Koltchinskii, 2011). The conditions under which (7) holds depend on the quantity $H_{\infty,\gamma}$, corresponding to a uniform upper bound on the leverage scores (Rudi et al., 2015, Definition 2) associated with the γ-regularised kernel. If $\mathcal{X}$ is compact, the assumption $\gamma H_{\infty,\gamma} \leq 1/C_0$ is satisfied for sufficiently small $\gamma > 0$ since $\gamma H_{\infty,\gamma} \to 0$ as $\gamma \searrow 0$.

The best bias–variance trade-off in (7), in terms of the order in n, is obtained by taking $\gamma = \gamma^\star$, where $\gamma^\star \in \operatorname{argmin}_{\gamma>0}(H_{2,\gamma}/n + \gamma) \to 0$ as $n \to \infty$. Moreover, we show that this rate is minimax optimal in Theorem 8 below. For this choice of $\gamma^\star$, and when $H_{2,\gamma^\star} \asymp H_{\infty,\gamma^\star}$, we have $\sqrt{n}/H_{\infty,\gamma^\star} \asymp 1/\sqrt{\gamma^\star H_{2,\gamma^\star}}$, so for every fixed $s \geq 1$, the condition $s \leq \sqrt{n}/(C_0 H_{\infty,\gamma^\star})$ is satisfied for sufficiently large n. If $\gamma^a H_{2,\gamma} \asymp 1$ as $\gamma \searrow 0$ for some $a \in [0, 1)$, as is common in applications (see Section 4.1.1 below), then we obtain $\gamma^\star \asymp n^{-1/(1+a)}$, yielding an overall convergence rate for $\hat{f}_{n,\gamma^\star}$ of $n^{-1/(2+2a)}$.

When covariates take values in $\mathbb{R}^d$ for some $d \in \mathbb{N}$, the dimension d affects the rate of convergence only through the quantities $H_{2,\gamma}$ and $H_{\infty,\gamma}$; our framework therefore handles covariates with diverging dimension in a fully non-asymptotic fashion.

4.1.1 Examples of kernel functions

While $\gamma H_{\infty,\gamma} \to 0$ as $\gamma \searrow 0$ when $\mathcal{X}$ is compact, sharper upper bounds on $H_{\infty,\gamma}$ (and, therefore, on $H_{2,\gamma}$) are sometimes available when a particular kernel is specified. In this section, we present some examples of kernels that satisfy our assumptions and, where possible, derive bounds on the corresponding quantities $H_{\infty,\gamma}$ and $\|1_{\mathcal{X}}\|_{\mathcal{H}}$. See Appendix E for details.

First, in Example 5, we demonstrate that for a multivariate polynomial kernel on a bounded domain, $H_{\infty,\gamma}$ is bounded as $\gamma \to 0$, provided that the polynomial degree p and the distribution of X are fixed. In Example 6, we consider shifted first- and second-order Sobolev kernels under a uniform measure. For the first-order Sobolev kernel, we show that $H_{\infty,\gamma}$ increases at a rate no faster than $1/\sqrt{\gamma}$ as $\gamma \searrow 0$. Lastly, in Example 7, we present a shifted Gaussian kernel.

Example 5 (Polynomial kernel). *Let $d \in \mathbb{N}$ and $\mathcal{X} \subseteq \mathbb{R}^d$, and take $p \in \mathbb{N}$ and $a \geq 0$. For $x, y \in \mathcal{X}$, define the polynomial kernel of degree p by*

$$k_{p,a}^{\mathrm{poly}}(x, y) := \big(x^{\mathsf{T}} y + a\big)^p.$$

Assume that $\mathcal{X}$ is a bounded, non-empty, Borel measurable subset of $\mathbb{R}^d$ containing 0_d and let P_X be a Borel distribution on $\mathcal{X}$. Then $\mathcal{R}$ is a finite set with $|\mathcal{R}| \leq \binom{d+p}{p}$. Moreover, there exists $C(p, d, P_X) > 0$, depending only on p, d and P_X, such that for all $a \geq 0$ and $\gamma > 0$, we have $\max_{r\in\mathcal{R}} \nu_r \leq C(p, d, P_X)$, $\max_{r\in\mathcal{R}} \|e_r\|_\infty \leq C(p, d, P_X)$ and $H_{2,\gamma} \leq H_{\infty,\gamma} \leq C(p, d, P_X)$. Further, $1_{\mathcal{X}} \in \mathcal{H}$ if and only if $a > 0$; in that case, $\|1_{\mathcal{X}}\|_{\mathcal{H}}^2 = 1/a^p$.

Example 6 (Shifted Sobolev kernels). *Let $\mathcal{X} = [0, 1]$ and take $a \geq 0$. For $x, y \in \mathcal{X}$, the first- and second-order shifted Sobolev kernels are defined respectively by*

$$k_{1,a}^{\mathrm{Sob}}(x, y) := a + (x \wedge y), \qquad k_{2,a}^{\mathrm{Sob}}(x, y) := a + \int_0^{x\wedge y} (x - z)(y - z)\,\mathrm{d}z.$$

Let P_X be Lebesgue measure on $\mathcal{X} := [0, 1]$ and take $\gamma > 0$. Then, for $k_{1,a}^{\mathrm{Sob}}$, we have $\mathcal{R} = \mathbb{N}$, $\nu_r \leq 1/\{(r - 1)^2\pi^2\}$ for each $r \geq 2$, $\sup_{r\in\mathcal{R}} \|e_r\|_\infty \leq \sqrt{2}$ and $H_{2,\gamma} \leq H_{\infty,\gamma} \leq 2 + 1/\sqrt{\gamma}$. Let $\mathcal{H}_1$, $\mathcal{H}_2$ denote the RKHSs associated respectively with $k_{1,a}^{\mathrm{Sob}}$ and $k_{2,a}^{\mathrm{Sob}}$. Then $1_{\mathcal{X}} \in \mathcal{H}_1$ if and only if $a > 0$; in that case, $\|1_{\mathcal{X}}\|_{\mathcal{H}_1}^2 = 1/a$. Likewise, $1_{\mathcal{X}} \in \mathcal{H}_2$ if and only if $a > 0$, and again in that case, $\|1_{\mathcal{X}}\|_{\mathcal{H}_2}^2 = 1/a$.

Example 7 (Shifted Gaussian kernel). *Let $d \in \mathbb{N}$ and $\mathcal{X} \subseteq \mathbb{R}^d$, and take $a \geq 0$ and $\Sigma \in \mathbb{R}^{d\times d}$ with $\Sigma \succ 0$. For $x, y \in \mathcal{X}$, the d-dimensional shifted Gaussian kernel is given by*

$$k_{\Sigma,a}^{\mathrm{Gauss}}(x, y) := a + \exp\big(-(x - y)^{\mathsf{T}}\Sigma^{-1}(x - y)\big).$$

If $\mathcal{X}$ is unbounded, then $1_{\mathcal{X}} \in \mathcal{H}$ if and only if $a > 0$; in that case, $\|1_{\mathcal{X}}\|_{\mathcal{H}}^2 = 1/a$.

By Theorem 4, we deduce that for the polynomial kernel, $\gamma^\star \asymp 1/n$, giving a parametric rate of convergence of $1/\sqrt{n}$. For the shifted first-order Sobolev kernel, $\gamma^\star \asymp n^{-2/3}$ and the corresponding high-probability bound from Theorem 4 is $\|\hat{f}_{n,\gamma} - f_0\|_{L_2} \lesssim n^{-1/3}$. More generally, if the eigenfunctions $(e_r)_{r\in\mathcal{R}}$ are uniformly bounded, in the sense that $\sup_{x\in\mathcal{X}} \sup_{r\in\mathcal{R}} |e_r(x)| \leq C$, then both $H_{2,\gamma}$ and $H_{\infty,\gamma}$ are controlled by the decay rate of the eigenvalues $(\nu_r)_{r\in\mathcal{R}}$. For example, if $\mathcal{R} = \mathbb{N}$ and $\nu_r \leq 1/r^b$ for some $b > 1$, then $H_{\infty,\gamma} \leq C \int_0^\infty 1/(1 + \gamma t^b)\,\mathrm{d}t = C\pi/\big\{b\gamma^{1/b}\sin(\pi/b)\big\}$. Thus, $\gamma^\star \asymp n^{-b/(1+b)}$ and the high-probability bound from Theorem 4 is $\|\hat{f}_{n,\gamma} - f_0\|_{L_2} \lesssim n^{-b/(2+2b)}$.

4.2 Minimax lower bound

In this section we present a minimax lower bound for relative risk function estimation in an RKHS, establishing rate-optimality of our Theorem 4. In order to demonstrate that the fundamental difficulty of this problem is driven by estimation in the RKHS, we consider a subset of the model described in Section 2.2 in which no censoring is present, the baseline hazard function is constant, and the distribution of the covariates is fixed. More precisely,

suppose that $T_C = 1$ almost surely, so that $I = 0$ and $T = T_S$, and assume that (1) holds with $\lambda_0(t) = 1$ for all $t \in [0,1]$. Since the baseline hazard function λ_0 is fixed, we may drop the identifiability condition that $P_X(f_0) = 0$. Assume without further loss of generality that $K := \sup_{x \in \mathcal{X}} k(x,x) = 1$. Recalling that either $\mathcal{R}$ is finite or $\mathcal{R} = \mathbb{N}$, we suppose in the latter case that for all $r \in \mathcal{R}$, we have (i) $\sum_{r' \in \mathcal{R}, r' > r} \nu_{r'} \leq C_K r \nu_r$; and (ii) $\nu_{r+1} \geq \nu_r / C_K$, for some $C_K > 0$. These conditions are satisfied by all commonly used kernels, including first- and second-order Sobolev kernels, as well as kernels with exponentially decaying eigenvalues; see Yang et al. (2017, Section 2) and Ma et al. (2023, Section 3). Let $\mathcal{H}_1 := \{f \in \mathcal{H} : \|f\|_{\mathcal{H}} \leq 1\}$ and write $\mathcal{F}_n$ for the set of measurable functions from $\big(\mathcal{X} \times [0,1]\big)^n$ to $\mathcal{H}_1$.

Theorem 8 (Minimax lower bound)**.** *There exists $C_0 > 0$ depending only on ν_1, R (if $\mathcal{R} = [R]$) and C_K (if $\mathcal{R} = \mathbb{N}$) such that*

$$\inf_{\hat{f}_n \in \mathcal{F}_n} \sup_{f_0 \in \mathcal{H}_1} \mathbb{E}\big(\|\hat{f}_n - f_0\|_{L_2}^2\big)^{1/2} \geq \frac{1}{C_0} \inf_{\gamma > 0} \left(\sqrt{\frac{H_{2,\gamma}}{n}} + \sqrt{\gamma} \right).$$

The lower bound in Theorem 8 matches the leading bias and variance terms in the upper bound obtained in Theorem 4, thereby establishing rate-optimality of our procedure.

4.3 Parameter tuning

To state our main result on the RKHS estimator with the cross-validated choice of γ, define $\gamma^+ := \max \Gamma$ and $\gamma^- := \min \Gamma$, and suppose that $\Gamma \subseteq [n^{-\xi}, n^{\xi}]$ for some $\xi \geq 1$.

Theorem 9 (Parameter tuning)**.** *Let $\hat{\gamma}$ be as in (5). There exists $C_0 > 0$ depending only on $\|f_0\|_{\mathcal{H}}$, Λ, q_1, $\|1_{\mathcal{X}}\|_{\mathcal{H}}$, K and ξ such that if $\gamma^+ H_{\infty,\gamma^+} \leq 1/C_0$, $H_{\infty,\gamma^-} \sqrt{(\log n)/n} \leq 1/C_0$ and $1 \leq s \leq \sqrt{n}/(C_0 H_{\infty,\gamma^-})$, then with probability at least $1 - e^{-s^2}$,*

$$\big\|\hat{f}_{n,\hat{\gamma}} - f_0\big\|_{L_2} \leq C_0 \left(\min_{\gamma \in \Gamma} \big\|\hat{f}_{n,\gamma} - f_0\big\|_{L_2} + \frac{s + \sqrt{\log n}}{\sqrt{n}} \right).$$

The cross-validated choice $\hat{\gamma}$ therefore achieves the same convergence rate as the oracle choice of γ, up to a constant factor and an additive term that is at most $\sqrt{(\log n)/n}$. Applying Theorem 4, and under the conditions specified in Theorem 9, we deduce that with probability at least $1 - e^{-s^2}$,

$$\big\|\hat{f}_{n,\hat{\gamma}} - f_0\big\|_{L_2} \leq C_0 \left(\sqrt{\frac{H_{2,\gamma^\star}}{n}} + \sqrt{\gamma^\star} + \frac{s + \sqrt{\log n}}{\sqrt{n}} \right).$$

Whenever $\gamma^a H_{2,\gamma} \asymp 1$ as $\gamma \searrow 0$ for some $a \in [0,1)$, the optimal rate according to Theorem 4 and Theorem 8 is $n^{-1/(2+2a)}$. If $a > 0$, such as with the first-order Sobolev kernel in Example 6 where $a = 1/2$, then $n^{1/(2+2a)} \sqrt{(\log n)/n} \to 0$ as $n \to \infty$, so the logarithmic term appearing in Theorem 9 is asymptotically negligible. In such a setting, we obtain the same rate of convergence when using $\hat{\gamma}$ as when using the oracle choice $\gamma^\star$. When $a = 0$, such as for the polynomial kernel of Example 5, the upper bound on the convergence rate with cross-validation is worse by a factor of at most $\sqrt{\log n}$ than with the optimal tuning parameter $\gamma^\star$.

Our cross-validation scheme for γ can be extended to include other parameters, such as the choice of kernel, as well as hyperparameters including bandwidths, polynomial degrees and shape parameters. While these may each induce a different RKHS, Theorem 9 is stated with respect to the $L_2(P_X)$-norm rather than any kernel-dependent metrics and can therefore be applied to arbitrary collections of estimators in $L_2(P_X)$. See Appendix C for more details.

4.4 Model aggregation with CARE

Our final main theoretical result gives an upper bound on the rate of convergence of the CARE estimator $\check{f}_{n,\check{\gamma},\check{\theta}}$.

Theorem 10 (CARE procedure)**.** *Let $(\check{\gamma}, \check{\theta})$ be the cross-validated choices as defined in (6). There exists $C_0 > 0$ depending only on $\|f_0\|_{\mathcal{H}}$, Λ, q_1, $\|1_{\mathcal{X}}\|_{\mathcal{H}}$, M, K and ξ such that if $\gamma^+ H_{\infty,\gamma^+} \leq 1/C_0$, $H_{\infty,\gamma^-}\sqrt{(\log n)/n} \leq 1/C_0$ and $1 \leq s \leq \sqrt{n}/(C_0 H_{\infty,\gamma^-})$, then with probability at least $1 - e^{-s^2}$,*

$$\big\|\check{f}_{n,\check{\gamma},\check{\theta}} - f_0\big\|_{L_2} \leq C_0\left(\min_{(\gamma,\theta)\in\Gamma\times\Theta}\big\|\check{f}_{n,\gamma,\theta} - f_0\big\|_{L_2} + \frac{s + \sqrt{|\mathcal{M}| + \log n}}{\sqrt{n}}\right).$$

The cross-validated CARE estimator $\check{f}_{n,\check{\gamma},\check{\theta}}$ therefore attains a convergence rate under $L_2(P_X)$ comparable to that of the oracle choices of $\gamma \in \Gamma$ and $\theta \in \Theta$. In particular, $\check{\gamma}$ may depend on properties of the external estimators $\tilde{f}_m$.

Denote the index of the best external estimator by $m^\star := \operatorname{sargmin}_{m\in\mathcal{M}} \|\tilde{f}_m - f_0\|_{L_2}$. Then under the conditions of Theorem 10, by combining this result with Theorem 4, we have with probability at least $1 - e^{-s^2}$ that

$$\big\|\check{f}_{n,\check{\gamma},\check{\theta}} - f_0\big\|_{L_2} \leq C_0\left(\left\{\sqrt{\frac{H_{2,\gamma^\star}}{n}} + \sqrt{\gamma^\star}\right\} \wedge \big\|\tilde{f}_{m^\star} - f_0\big\|_{L_2} + \frac{s + \sqrt{|\mathcal{M}| + \log n}}{\sqrt{n}}\right). \tag{8}$$

Therefore, $\check{f}_{n,\check{\gamma},\check{\theta}}$ attains a rate of convergence that is at least as good as both (i) the optimal kernel estimator $\hat{f}_{n,\gamma^\star}$, and (ii) the best empirically centred external estimator $\tilde{f}_{m^\star}$. In this sense, CARE is able to automatically adapt to the predictive values of the external models $\tilde{f}_m$ relative to those of the kernel estimators $\hat{f}_{n,\gamma}$. In particular, the performance of the CARE procedure is not significantly degraded when some of the external estimators may be poor predictors, for example, if they are misspecified, or if they were originally fitted to a small data set. When the kernel estimator is well-specified (a relatively mild assumption in practice when using a nonparametric infinite-dimensional RKHS), the CARE procedure may be thought of as a double regularisation, applying shrinkage towards the origin (via γ) and also towards the external estimators (via θ).

Our method can accommodate certain forms of covariate shift with no modification; this is important when the external estimators may have been trained on samples from populations that differ from that of the data $(X_i, T_i, I_i)_{i\in[2n]}$. To this end, suppose that $\|\tilde{f}_{\tilde{m}} - f_0\|_{\tilde{L}_2} \leq \delta$ for some $\tilde{m} \in \mathcal{M}$, where $\tilde{L}_2 := L_2(\tilde{P}_X)$ and $\tilde{P}_X$ is a distribution on $\mathcal{X}$ with Radon–Nikodym derivative $\mathrm{d}P_X/\mathrm{d}\tilde{P}_X \leq C^2$ almost surely for some non-random $C \in (0, \infty)$.

Then $\|\tilde{f}_{\tilde{m}} - f_0\|_{L_2} \leq C\delta$ and the high-probability bound given in (8) holds with $\|\tilde{f}_{m^\star} - f_0\|_{L_2}$ replaced by $C\delta$.

In practice, an estimate of the survival probability $\mathbb{P}(T_{\mathrm{S}} > t \mid X = x)$ is often required, where t is a fixed time and x is a specified covariate value. Given an estimator f of the relative risk function (which may be obtained using the methods in Section 4.4), we propose the plug-in estimator

$$\hat{P}(t \mid x, f) := \hat{P}(t \mid f)^{\exp\{f(x)\}}, \tag{9}$$

where $\hat{P}(t \mid f) := \exp\bigl(-\int_0^t \{1/S_n(f,s)\}\,\mathrm{d}N(s)\bigr)$.

5 Implementation details

In this section, we present some further details on the practical implementation of our CARE methodology. Specifically, we describe procedures for calculating $\|1_{\mathcal{X}}\|_{\mathcal{H}}$, computing the set $\mathcal{A}_n$, solving the optimisation problem (4) and conducting cross-validation as detailed in (5) and (6). We also provide guidance on the choice of kernel. In Appendix A.1, we provide an alternative 'feature map' formulation for superior computational efficiency when using certain kernels, while in Appendix A.2, we demonstrate how Nyström approximation can be used to attain better computational performance with general kernels.

5.1 Calculating $\|1_{\mathcal{X}}\|_{\mathcal{H}}$

In Proposition 11 below, we give the exact general form of $\|1_{\mathcal{X}}\|_{\mathcal{H}}$ based on properties of the kernel. For $n \in \mathbb{N}$ and $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ with $\boldsymbol{A} \succeq 0$, define

$$\kappa(\boldsymbol{A}) := \lim_{\delta \searrow 0} \frac{1}{1_n^{\mathsf{T}}(\boldsymbol{A} + \delta \boldsymbol{I}_n)^{-1} 1_n} = \begin{cases} 0 & \text{if there is } v \in \mathbb{R}^n \text{ with } \boldsymbol{A}v = 0, 1_n^{\mathsf{T}} v = 1, \\ \frac{1}{1_n^{\mathsf{T}} \boldsymbol{A}^{+} 1_n} & \text{otherwise,} \end{cases}$$

where $\boldsymbol{A}^{+}$ is the Moore–Penrose pseudo-inverse of $\boldsymbol{A}$ (e.g., Friedberg et al., 2003, Section 6.7).

Proposition 11 (Characterisation of $\|1_{\mathcal{X}}\|_{\mathcal{H}}$)**.** *For $n \in \mathbb{N}$ and $x_1, \ldots, x_n \in \mathcal{X}$, define the matrix $\boldsymbol{K}(x_1, \ldots, x_n) \in \mathbb{R}^{n\times n}$ by $\boldsymbol{K}(x_1, \ldots, x_n)_{ij} := k(x_i, x_j)$ for $i, j \in [n]$. Then $1_{\mathcal{X}} \in \mathcal{H}$ if and only if $\kappa_k := \inf_{n\in\mathbb{N}} \inf_{x_1,\ldots,x_n \in \mathcal{X}} \kappa\bigl(\boldsymbol{K}(x_1, \ldots, x_n)\bigr) > 0$, in which case $\|1_{\mathcal{X}}\|_{\mathcal{H}}^2 = 1/\kappa_k$.*

See Examples 5, 6 and 7 for explicit calculations of $\|1_{\mathcal{X}}\|_{\mathcal{H}}$ using Proposition 11.

5.2 Constructing the set $\mathcal{A}_n$

Let $\boldsymbol{K} \in \mathbb{R}^{n\times n}$ be given by $\boldsymbol{K}_{ij} := k(X_i, X_j)$, and define the block matrix

$$\tilde{\boldsymbol{K}} := \begin{pmatrix} \|1_{\mathcal{X}}\|_{\mathcal{H}}^2 & 1_n^{\mathsf{T}} \\ 1_n & \boldsymbol{K} \end{pmatrix} \in \mathbb{R}^{(n+1)\times(n+1)}. \tag{10}$$

Let $\boldsymbol{R} \in \mathbb{R}^{(n+1)\times(n+1)}$ be the reduced row echelon form of $\tilde{\boldsymbol{K}}$, obtained by applying Gaussian elimination (e.g. Friedberg et al., 2003, Theorem 3.14) to $\tilde{\boldsymbol{K}}$, and define $\mathcal{A}_n := \bigl\{j \in [n] : \exists i \in [n+1] : \bigl(\boldsymbol{R}_{i,1}, \ldots, \boldsymbol{R}_{i,j+1}\bigr) = (0, \ldots, 0, 1)\bigr\}$, that is, the one-decremented indices of the columns in $\boldsymbol{R}$ that contain a leading one. Then by Friedberg et al. (2003, Theorem 3.16), $\bigl\{\bigl(\tilde{\boldsymbol{K}}_{i+1,1}, \ldots, \tilde{\boldsymbol{K}}_{i+1,n+1}\bigr) : i \in \{0\} \cup \mathcal{A}_n\bigr\}$ is a basis for $\operatorname{span}\bigl\{\bigl(\tilde{\boldsymbol{K}}_{i,1}, \ldots, \tilde{\boldsymbol{K}}_{i,n+1}\bigr) : i \in [n+1]\bigr\}$, and hence $\mathcal{A}_n$ satisfies the desired properties.

5.3 Computing the estimator $\hat{f}_{n,\gamma}$

Proposition 3 shows that for $\gamma > 0$, the kernel estimator $\hat{f}_{n,\gamma}$ is the solution to a strongly convex optimisation problem. Here we give the explicit form of this problem and discuss iterative methods for its solution. We use notation from Section 3.1 throughout and, for $x, y \in \mathcal{X}$, we define $\hat{k}(x,y) := k(x,y) - \bar{k}(x) - \bar{k}(y) + \bar{k}(x)\bar{k}(y)\|1_{\mathcal{X}}\|^2_{\mathcal{H}}$. First, for $\beta \in \mathbb{R}^{\mathcal{A}_n}$, we have $P_n(f_\beta) = 0$, so the objective function in (4) can be written as

$$\ell_{n,\gamma}(f_\beta) = \frac{1}{n}\sum_{i=1}^{n} \log S_n(f_\beta, T_i) N_i(1) - \frac{1}{n}\sum_{i=1}^{n} f_\beta(X_i) N_i(1) + \gamma \sum_{i\in\mathcal{A}_n}\sum_{j\in\mathcal{A}_n} \hat{k}(X_i, X_j)\beta_i\beta_j.$$

For $j \in \mathcal{A}_n$, define $\partial_j \ell_{n,\gamma}(f_\beta) := \partial \ell_{n,\gamma}(f_\beta)/\partial\beta_j$. The gradient of the objective function is

$$\partial_j \ell_{n,\gamma}(f_\beta) = \frac{1}{n}\sum_{i=1}^{n} \frac{DS_n(f_\beta, T_i)\big(\tilde{k}(\cdot, X_j)\big)}{S_n(f_\beta, T_i)} N_i(1) - \frac{1}{n}\sum_{i=1}^{n} \tilde{k}(X_i, X_j) N_i(1) + 2\gamma \sum_{i\in\mathcal{A}_n} \hat{k}(X_i, X_j)\beta_i.$$

Since this optimisation problem is strongly convex by Proposition 3, a variety of methods can be used for approximate computation of the optimal solution $\hat{\beta}$. One first-order approach is to run steepest gradient descent with the Armijo backtracking step size rule; this is guaranteed to converge as the number of iterations diverges to infinity (Lange, 2013, Section 12.6). However, computational speed-ups can be achieved by incorporating second-order quasi-Newton methods such as the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm (Broyden, 1970; Fletcher, 1970; Goldfarb, 1970; Shanno, 1970); this is how we implemented our procedure.

5.4 Cross-validation of γ and θ

To compute $\hat{\gamma}$ using (5), we suggest taking Γ to be a finite set of geometrically spaced values and applying a grid search procedure, computing $\ell_n(\hat{f}_{n,\gamma})$ for each $\gamma \in \Gamma$. Our implementation starts with $\gamma = \max\Gamma$, initialising the optimisation algorithm at zero. The remaining values of γ may be evaluated in decreasing order, initialising the algorithm each time with a warm start at the previous optimal solution $\hat{\beta}_{n,\gamma}$.

The computational burden of solving (6) for the CARE estimator is higher than that of (5), requiring a joint optimisation over $(\gamma, \theta) \in \Gamma \times \Theta$. However, for each $\gamma \in \Gamma$, the expensive step of computing $\hat{f}_{n,\gamma}$ only needs to be performed once. In practice we therefore recommend taking Θ to be an approximately evenly spaced set of points in $\Delta_{\mathcal{M}}$ and using the following procedure. For each $\gamma \in \Gamma$, compute $\hat{f}_{n,\gamma}$ as before, then search over $\theta \in \Theta$ to find $\check{\theta}(\gamma) := \operatorname{sargmin}_{\theta\in\Theta} \tilde{\ell}_n(\check{f}_{n,\gamma,\theta})$. It remains to search over $\gamma \in \Gamma$ to find $\check{\gamma} := \operatorname{sargmin}_{\gamma\in\Gamma} \tilde{\ell}_n(\check{f}_{n,\gamma,\check{\theta}(\gamma)})$; finally set $\check{\theta} := \check{\theta}(\check{\gamma})$.

5.5 Choice of kernel

The choice of kernel in applications depends on the nature of the covariates under consideration. For continuous features, the most flexible kernel functions are those that induce large, fully nonparametric Hilbert spaces. For example, the RKHS generated by the first-order Sobolev kernel (Example 6) contains all absolutely continuous functions $f : [0,1] \to \mathbb{R}$ with weak

derivative $f' \in L_2([0,1])$ (Samworth and Shah, 2026, Example 6.16). Somewhat smaller is the RKHS arising from the second-order Sobolev kernel, which only includes functions possessing absolutely continuous first derivative f' and weak second derivative $f'' \in L_2([0,1])$ (Samworth and Shah, 2026, Example 6.17). The RKHS generated by a Gaussian kernel on $\mathcal{X} \subseteq \mathbb{R}^d$ (Example 7) is dense in the space of continuous functions on $\mathcal{X}$ equipped with the topology induced by uniform convergence on compact sets (Micchelli et al., 2006, Theorem 17). For such nonparametric kernels, we typically have $H_{2,\gamma} \to \infty$ as $\gamma \searrow 0$ (Example 6); appropriate regularisation is therefore crucial to prevent overfitting (see Theorem 4).

At the other end of the spectrum, polynomial kernels with fixed degree induce finite-dimensional Hilbert spaces and $H_{\infty,\gamma}$ is bounded as $\gamma \searrow 0$ (Example 5). While this property limits the approximation power of the resulting kernel estimators, they are also less prone to overfitting. Further, the existence of a finite-dimensional feature map permits significant computational improvements in low-dimensional settings; see Appendix A.1.

When working with a diverse collection of covariates, we construct multivariate kernels by combining lower-dimensional kernel functions as follows. If k_j is a kernel on $\mathcal{X}_j$ for $j \in [d]$, then with $\mathcal{X} := \prod_{j=1}^d \mathcal{X}_j$, the function $k_{\mathrm{sum}} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ defined by

$$k_{\mathrm{sum}}(x, y) := \sum_{j=1}^{d} k_j(x_j, y_j)$$

is a kernel (Samworth and Shah, 2026, Proposition 6.28). This formulation is especially appropriate for an additive model structure of the form $f_0(x) = \sum_{j=1}^d f_j(x_j)$.

6 Empirical studies

We investigate the empirical performance of our procedures through experiments with simulated and real data sets. Our methodology is implemented in the Python package `care-survival` (Underwood et al., 2025).

6.1 Simulated data study

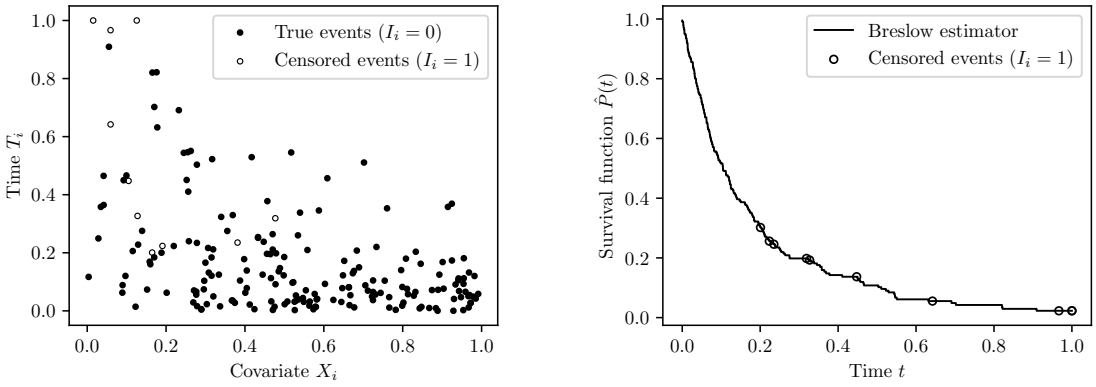
To simulate data from the model specified in (1), we construct $T_{\mathrm{S}} := \Lambda^{-1}\big(-e^{-f_0(X)} \log U\big)$ where $U \sim \mathcal{U}[0,1]$ and $\Lambda(t) := \int_0^t \lambda_0(s)\,\mathrm{d}s$. Since estimation of the baseline hazard is not the focus of this paper, we restrict to relatively simple forms for λ_0, for which the function Λ^{-1} can be computed in closed form. We define the Breslow estimator (Breslow, 1972; see also Huang and Strawderman, 2006) of the survival function $t \mapsto \mathbb{P}(T_{\mathrm{S}} > t)$ by

$$\hat{P}(t) := \exp\biggl(-\int_0^t \frac{n}{\sum_{i=1}^n R_i(s)}\,\mathrm{d}N(s)\biggr). \tag{11}$$

6.1.1 Univariate regressors

We begin by considering a setting where X is univariate, not because this is realistic for modern prediction models in biomedicine, but because it allows us visualise our methodology by

plotting the survival times and the fitted relative risk function. Let $P_X = \mathcal{U}[0,1]$ and $\lambda_0(t) = 6$ for $t \in [0,1]$, so that $\Lambda(t) = 6t$ and $\Lambda^{-1}(t) = t/6$. We take $f_0(x) := 2\sin(2x) - 2\sin^2(1)$ for $x \in [0,1]$, so that $P_X(f_0) = 0$, and sample $T_C \sim \mathcal{U}[1/5, 2] \wedge 1$ independent of X. Figure 1 illustrates a sample of size $n = 200$ from this distribution, along with the Breslow estimator (11). We use the shifted first-order Sobolev kernel $k^{\mathrm{Sob}}_{1,a}$ defined in Example 6 with $a = 1$. By the remark in Section 5.5, we have $f_0 \in \mathcal{H}$. Since f_0 is not a polynomial, using a shifted polynomial kernel as in Example 5 would result in misspecification bias. We take Γ to consist of 50 geometrically increasing values from 10^{-5} to 10.

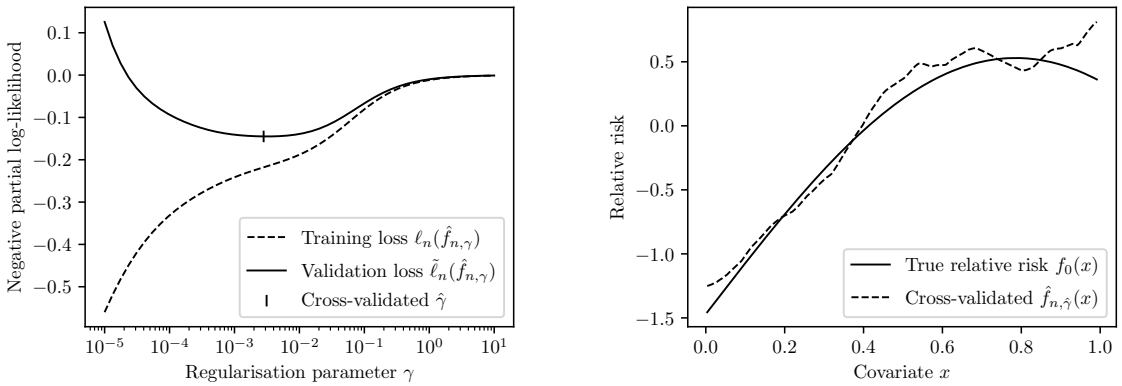


(a) Survival time against covariate value.

(b) Breslow survival estimator.

Figure 1: In panel (a), we plot a sample of size $n = 200$ from the specified data distribution. In panel (b), we display the associated Breslow estimator of the survival function.

Figure 2(a) illustrates our proposed cross-validation scheme for choosing the regularisation parameter γ based on a sample of size $n = 200$. The negative partial log-likelihood $\ell_n(\hat{f}_{n,\gamma})$ on the training data is an increasing function of γ, reflecting overfitting for small values of γ. However, the corresponding negative partial log-likelihood $\tilde{\ell}_n(\hat{f}_{n,\gamma})$ on the validation data attains a global minimum at $\gamma = \hat{\gamma}$. In Figure 2(b), the resulting cross-validated estimator $\hat{f}_{n,\hat{\gamma}}$ is seen to approximate the true relative risk function f_0 reasonably well when $n = 200$.



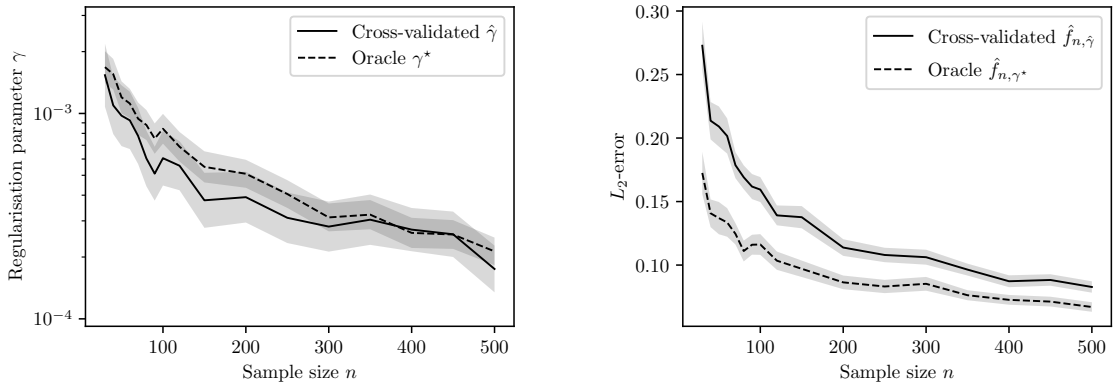
(a) Partial likelihood cross-validation curves.

(b) Fitted estimator and true risk function.

Figure 2: In panel (a), we show how cross-validation is conducted by minimising the negative partial log-likelihood on an independent sample of size $n = 200$. In panel (b), we illustrate how the resulting estimator approximates the true relative risk function.

Next, we investigate the behaviour of the cross-validated RKHS estimator as we vary the sample size n. For computational efficiency reasons, we use Nyström approximation with subsampling number $m \coloneqq n \wedge 100$ (see Appendix A.2 for details). In all of the subsequent figures, results are averaged over $n_{\text{rep}} = 200$ repetitions. More precisely, for each value presented, the mean $\hat{\mu}$ and standard deviation $\hat{\sigma}$ are calculated across n_{rep} repetitions; we then plot the point estimates $\hat{\mu}$ in black, and display $[\hat{\mu} \pm 2\hat{\sigma}/\sqrt{n_{\text{rep}}}]$ as grey shaded regions.

In Figure 3(a), we illustrate the dependence of the cross-validated and oracle regularisation parameters on the sample size. The oracle choice $\gamma^\star$ is calculated by directly minimising $\gamma \mapsto \|\hat{f}_{n,\gamma} - f_0\|_{L_2}$, which we approximate by Monte Carlo integration using an independent sample of 500 data points. As the sample size increases, $\gamma^\star$ decreases, reflecting the bias–variance trade-off suggested by Theorems 4 and 8. Further, the cross-validated choice $\hat{\gamma}$ provides a good approximation to $\gamma^\star$, even for small sample sizes. In Figure 3(b), we see that the L_2-error $\|\hat{f}_{n,\hat{\gamma}} - f_0\|_{L_2}$ of the cross-validated estimator decreases as the sample size increases, and it is competitive with the oracle estimator (which requires knowledge of f_0).

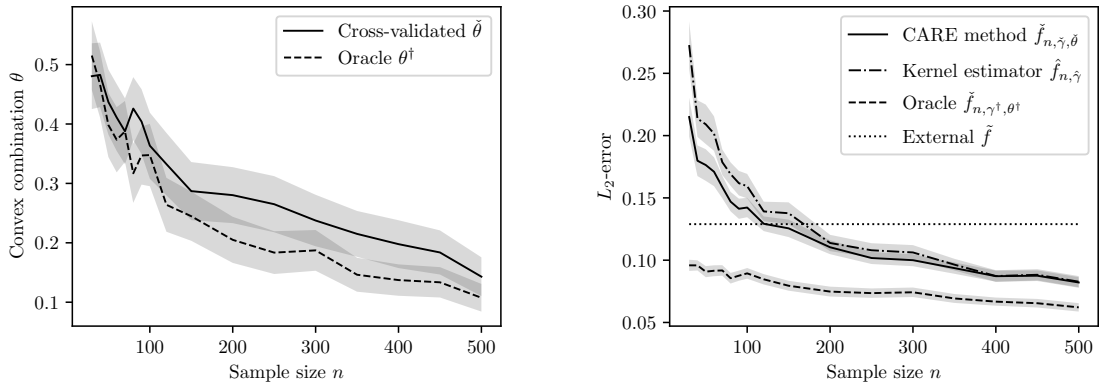


(a) Cross-validated and oracle choices of γ.

(b) Cross-validated and oracle L_2-error.

Figure 3: In panel (a), we plot the cross-validated and oracle choices of regularisation parameter, as a function of the sample size. In panel (b), we plot the corresponding L_2-errors of the fitted prediction functions.

We now assess the performance of the aggregated CARE estimator $\check{f}_{n,\check{\gamma},\check{\theta}}$. We define a fixed 'external' estimator $\tilde{f}$ by perturbing the function f_0, setting $\tilde{f}(x) := 2\sin(3x/2) - (8/3)\sin^2(3/4)$. The convex combination parameter θ is selected from the set $\Theta := \{(k-1)/20 : k \in [21]\}$. In Figure 4(a), we see that the oracle choice $\theta^\dagger$, where $(\gamma^\dagger, \theta^\dagger)$ minimises $(\gamma, \theta) \mapsto \|\check{f}_{n,\gamma,\theta} - f_0\|_{L_2}$, decreases as the sample size increases. For small sample sizes, $\tilde{f}$ provides valuable information due to the large variance of the kernel estimator, and hence $\theta^\dagger$ is relatively large. However, with more new data, the kernel estimators $\hat{f}_{n,\gamma}$ approximate f_0 well and the external model $\tilde{f}$ becomes less informative, yielding a value of $\theta^\dagger$ that is closer to zero. The cross-validation scheme produces estimates $\check{\theta}$ that are reasonably close to the oracle choices $\theta^\dagger$. From Figure 4(b), we also observe that CARE consistently outperforms the cross-validated kernel estimator that ignores the external estimator, although in this smooth, one-dimensional problem the effect is smaller at larger sample sizes. When the sample size is sufficiently large, the CARE estimator is also seen to be superior to the external estimator $\tilde{f}$.



(a) Cross-validation and oracle choices of θ.

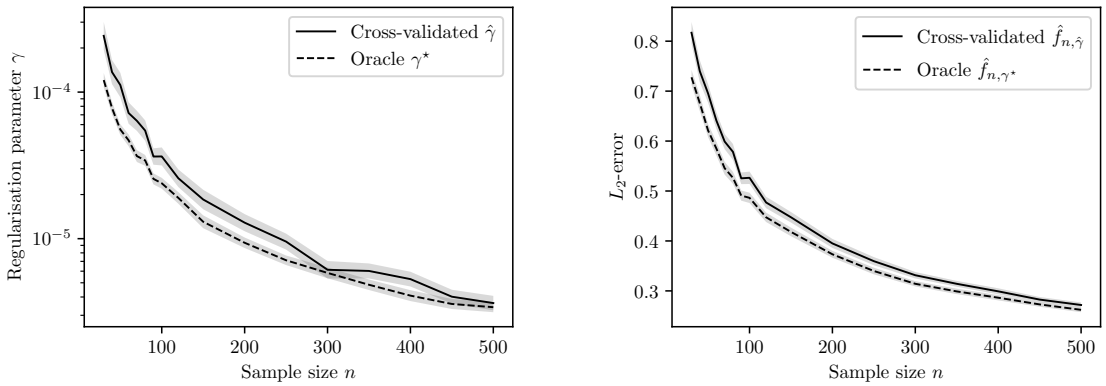
(b) Cross-validated and oracle performance.

Figure 4: In panel (a), we plot the cross-validated and oracle choices of the convex combination parameter as a function of the sample size. In panel (b), we compare the corresponding CARE, kernel, oracle and external estimators.

6.1.2 Multivariate regressors

We now turn to a setting with multiple covariates. With $d = 10$, let $P_X = \mathcal{U}[0,1]^d$ and $\lambda_0(t) = 6$ as before. We consider the sparsely supported relative risk function f_0 given by $f_0(x) = \sum_{j=1}^{5}\{2\sin(2x_j) - 2\sin^2(1)\}$ for $x = (x_1, \ldots, x_d)^\mathsf{T} \in [0,1]^d$, which depends only on the first five components of x. We draw $T_\mathrm{C} \sim \mathcal{U}[1/5, 2] \wedge 1$ independently of X, and employ an additive shifted first-order Sobolev kernel k defined by $k(x,y) := a + \sum_{j=1}^{d}(x_j \wedge y_j)$ with $a = 1$. We take Γ to consist of 50 geometrically increasing values from 10^{-7} to 0.1.

In Figure 5(a), we illustrate the dependence of the cross-validated and oracle regularisation parameters on the sample size. As before, $\gamma^\star$ decreases as the sample size increases. The cross-validated choice $\hat{\gamma}$ provides a reasonable approximation to $\gamma^\star$, although it appears to exhibit some upward bias. In Figure 5(b), the L_2-error of the cross-validated estimator decreases as the sample size increases, and is competitive with the oracle estimator.

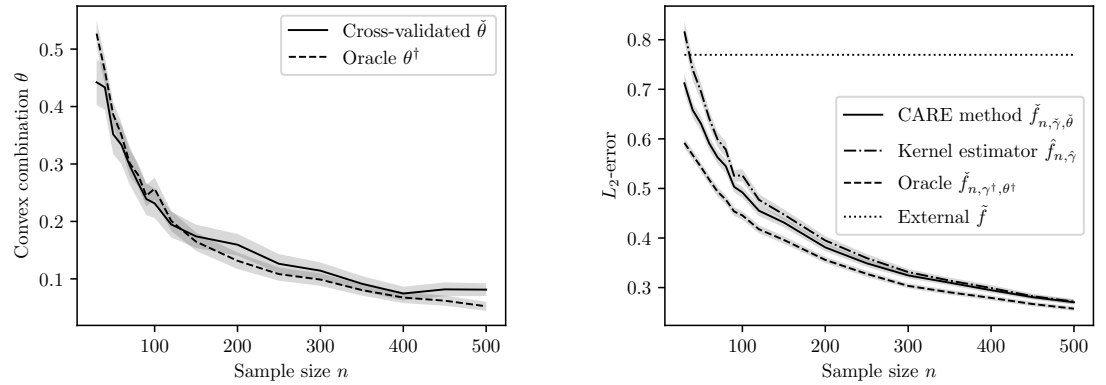


(a) Cross-validated and oracle choices of γ.

(b) Cross-validated and oracle performance.

Figure 5: In panel (a), we plot the cross-validated and oracle choices of regularisation parameter, as a function of the sample size. In panel (b), we plot the corresponding L_2-errors of the fitted prediction functions.

Finally, we investigate the CARE estimator. For the 'external' estimator $\tilde{f}$, we retain the additive structure, defining $\tilde{f}(x) := \sum_{j=1}^{4}(\sin 2 - \cos 2 - 1)(6x_j - 3)$. That is, for the first four entries of x, we take the optimal linear approximation to f_0 under the $L_2(P_X)$-norm. In Figure 6(a), we see that again the oracle choice $\theta^\dagger$ decreases as the sample size increases. For small sample sizes, the linear approximation $\tilde{f}$ provides useful information and hence $\theta^\dagger$ is relatively large. With more data, the kernel estimators $\hat{f}_{n,\gamma}$ approximate the non-linear function f_0 well and reliance on the external model $\tilde{f}$ diminishes, yielding $\theta^\dagger$ closer to zero. The cross-validated estimates $\check{\theta}$ are reasonably close to the oracle choices $\theta^\dagger$. In Figure 6(b), we also observe that CARE consistently outperforms the cross-validated kernel estimator that ignores the external estimator. For most sample sizes, the CARE estimator is also superior to the external estimator $\tilde{f}$. Some additional simulation results are given in Appendix D.1.



(a) Cross-validation and oracle choices of θ.

(b) Cross-validated and oracle performance.

Figure 6: In panel (a), we plot the cross-validated and oracle choices of the convex combination parameter as a function of the sample size. In panel (b), we compare the corresponding CARE, kernel, oracle and external estimators.

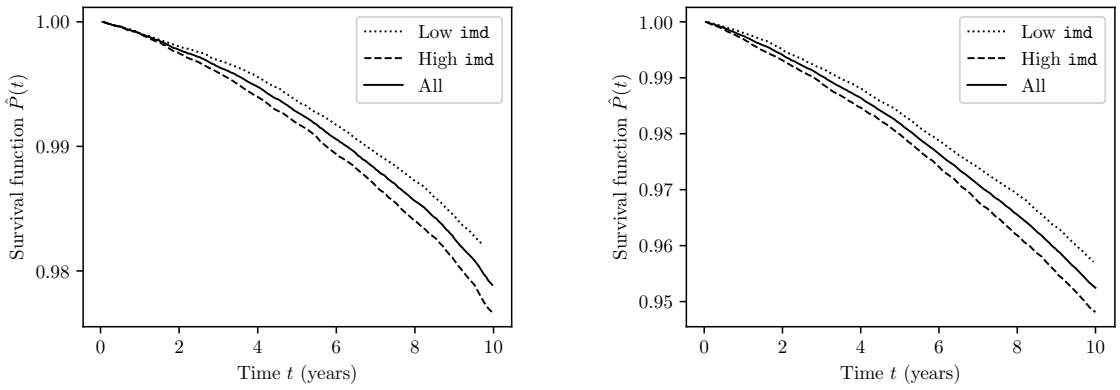
6.2 Cardiovascular disease risk prediction

We consider the problem of estimating 10-year fatal and non-fatal cardiovascular disease risk in England, using cohort data taken from the UK Biobank (UKB, Littlejohns et al., 2019). As an external estimator, we employ a version of the SCORE2 model (SCORE2 Working Group and ESC Cardiovascular Risk Collaboration, 2021), trained on a data set of size[1] 173,438 from the Emerging Risk Factor Collaboration (Emerging Risk Factors Collaboration, 2007) that is independent of the UK Biobank data. This estimator was fitted using a Fine–Gray model (Fine and Gray, 1999) and, as such, targets the subdistribution hazard function rather than the standard hazard function given in (1). However, our framework makes no assumptions on the consistency properties (or lack thereof) of the external estimators; it suffices that SCORE2 contains predictive information about the cardiovascular risk. Under appropriate assumptions, our main methodology and theory could be adapted for the Fine–Gray setting, but we prefer the relative risk formulation for two main reasons. First, it is simpler, avoiding complex modelling assumptions and unusual risk sets. Second, the primary motivation of our methodology is to estimate the impact of a new covariate, treatment or test on the incidence of an event of interest. For such explanatory tasks, it is generally preferable to target the standard hazard function (1) rather than the subdistribution hazard function (Andersen et al., 2012).

SCORE2 uses the following covariates for prediction: `age` (at baseline survey), `smoking` (binary-valued smoking status), `sbp` (systolic blood pressure), `tchol` (total cholesterol) and `hdl` (high density lipoprotein cholesterol), along with the two-way interactions of `age` with `smoking`, `sbp`, `tchol` and `hdl`. We propose to upgrade the SCORE2 model using the CARE methodology defined in Section 3.3 by including three new covariates, also available in the UK Biobank data, which are known to be associated with cardiovascular disease risk: `imd` (multiple social deprivation index), `pgs000018` (polygenic score associated with coronary artery disease; see Inouye et al., 2018) and `pgs000039` (polygenic score associated with ischaemic stroke; see Abraham et al., 2019). To be precise, we set $|\mathcal{M}| = 1$ and the external estimator is taken to be the fitted SCORE2 relative risk function $\hat{f}_1$. Kernel estimators $\hat{f}_{n,\gamma}$ are fitted to the UK Biobank data, using an expanded set of covariates, and the CARE methodology combines these two sources of information. Polygenic scores were calculated using data and software (`pgsc_calc`) from the Polygenic Score Catalog (Lambert et al., 2021, 2024). Approximately 15% of the individuals in the study had missing covariates, and these people were removed from the data set prior to analysis. Estimators for female and male participants were constructed separately, as for the original SCORE2 model. We used a quadratic polynomial kernel, allowing for non-linear effects while taking advantage of the computationally efficient feature map representation; see Appendix A.1. We also tried using a first-order additive Sobolev kernel and Nyström approximation but found this did not improve the performance; see Appendix D.2.

In Figure 7, we plot Breslow estimators (11) of the survival functions, stratified by sex and `imd`, with `imd` scores at most the median labelled 'low' and scores above the median labelled 'high'. Higher levels of area-specific deprivation are associated with an increased incidence of cardiovascular disease and a lower survival probability (Conrad et al., 2024).

[1]Large sample sizes are necessary for effective models since diagnoses of cardiovascular disease are relatively rare (less than 5% of the population in any 10-year period, as indicated by Figure 7).



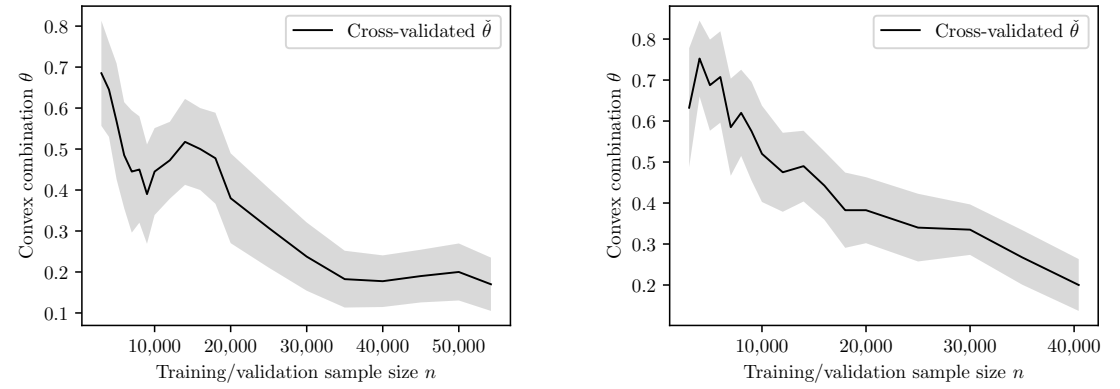
(a) Breslow survival estimators for females.

(b) Breslow survival estimators for males.

Figure 7: We display the Breslow estimators of the survival function. Panel (a) uses data from the female participants and panel (b) is based on the male participants.

6.2.1 Model 1: upgrading SCORE2 for UKB risk stratification

We update the SCORE2 model by incorporating data from UK Biobank. To fit the cross-validated CARE estimator, we first set aside $n_{\text{test}} := \lfloor N/3 \rfloor$ data points as a test set, where N is the total number of UK Biobank data points available ($N = 162{,}682$ for females and $N = 121{,}333$ for males). For a range of values n satisfying $1 \leq n \leq n_{\text{test}}$, we then draw n training points and n validation points at random from the remaining data, without replacement. The sampling of training and validation data is repeated 20 times with different randomisation seeds and the results are averaged; uncertainty estimates are calculated as for the simulated data in Section 6.1. In Figure 8 we see that, for both the female and male participants, the cross-validated choice of the convex combination parameter tends to decrease as n increases.



(a) Cross-validated choices of θ for females.

(b) Cross-validated choices of θ for males.

Figure 8: We plot the cross-validated choices of the convex combination parameter θ in Model 1 as a function of the training and validation sample sizes.

Since the true relative risk function f_0 is unknown, we cannot compute the L_2-error of the estimators; instead, we measure performance of a relative risk estimate f using the concordance index (Harrell et al., 1982) and the integrated Brier score (Graf et al., 1999; Li et al., 2021). The concordance index is defined on the test data $(X_i, T_i, I_i)_{i \in [n_{\text{test}}]}$ as

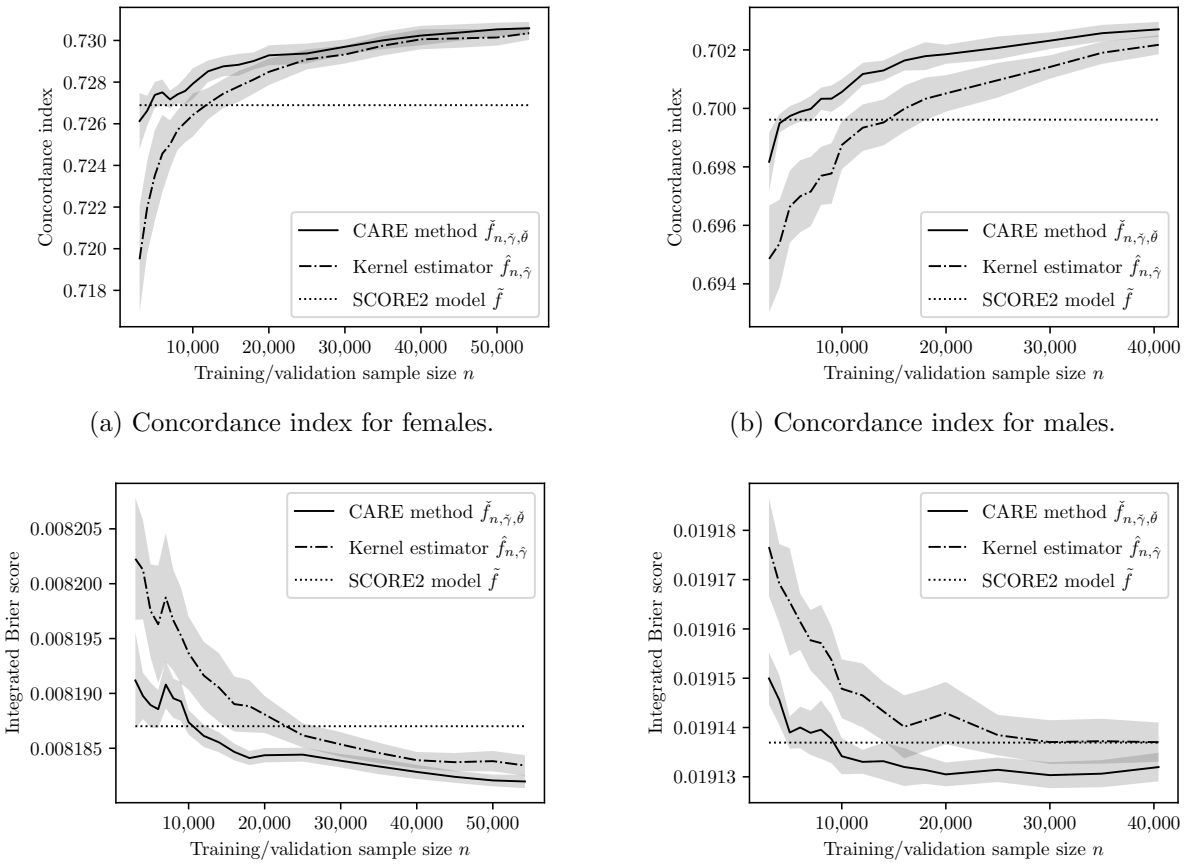
$$\text{Concordance}(f) := \frac{\sum_{i=1}^{n_{\text{test}}} \sum_{j=1}^{n_{\text{test}}} \mathbb{1}_{\{f(X_i) < f(X_j)\}} \mathbb{1}_{\{T_i > T_j\}} (1 - I_j)}{\sum_{i=1}^{n_{\text{test}}} \sum_{j=1}^{n_{\text{test}}} \mathbb{1}_{\{T_i > T_j\}} (1 - I_j)}.$$

The integrated Brier score is $\int_0^1 \text{Brier}(f, t) \, \mathrm{d}w(t)$, where w is a measure on $[0, 1]$ and

$$\text{Brier}(f, t) := \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \left(\frac{\hat{P}(t \mid X_i, f)^2 \mathbb{1}\{T_i \leq t\}(1 - I_i)}{\hat{P}_{\text{C}}(T_i)} + \frac{\{1 - \hat{P}(t \mid X_i, f)\}^2 \mathbb{1}\{T_i > t\}}{\hat{P}_{\text{C}}(t)} \right),$$

with $\hat{P}(t \mid x, f)$ and $\hat{P}(t \mid f)$ as in (9), and $\hat{P}_{\text{C}}(t) := \exp\bigl(-\sum_{i=1}^{n_{\text{test}}} \mathbb{1}\{T_i \leq t\} I_i / \sum_{j=1}^{n_{\text{test}}} \mathbb{1}\{T_j \geq T_i\}\bigr)$. For our experiments, we take w to be the discrete uniform measure on $\bigl\{i/24 : i \in \{0, \ldots, 24\}\bigr\}$, after scaling the time variable to $[0, 1]$.

Figure 9 illustrates the dependence of the concordance index and integrated Brier score on n. The CARE estimator $\check{f}_{n,\check{\gamma},\check{\theta}}$ consistently improves upon the external estimator from the SCORE2 model as soon as $n \geq 10{,}000$, indicating that only a relatively small additional sample size is needed for the benefits of aggregation to become visible. Likewise, CARE tends to perform better than the kernel estimator $\hat{f}_{n,\hat{\gamma}}$ (which ignores the SCORE2 data), particularly for smaller values of n. When using all of the available data ($n = n_{\text{test}}$), the concordance index improved by 0.51% for females and by 0.44% for males, while the integrated Brier score improved by 0.06% for females and by 0.03% for males. We emphasise that these integrated Brier scores are computed using estimated survival probabilities that are recalibrated for the UKB data (9). As such, the observed improvements from CARE are probably due to the non-linear features providing a superior predictive signal, and not because of any potential miscalibration in SCORE2.



(a) Concordance index for females.

(b) Concordance index for males.

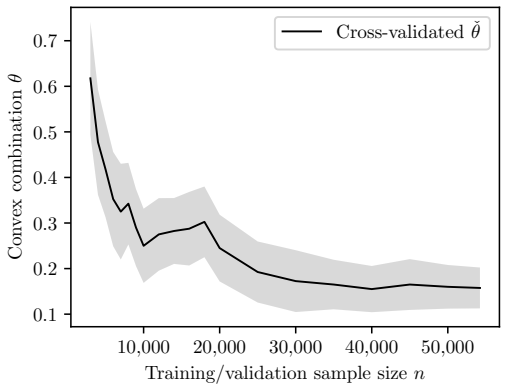
(c) Integrated Brier score for females.

(d) Integrated Brier score for males.

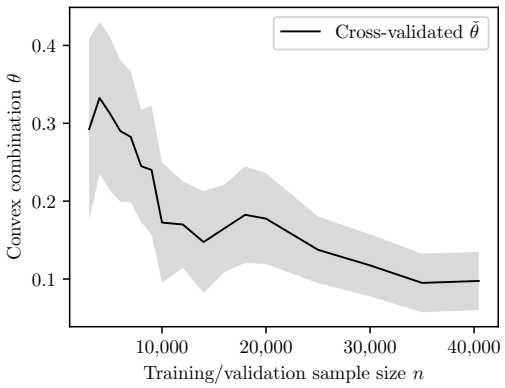
Figure 9: We plot the concordance index and integrated Brier score in Model 1 against the training and validation sample size. CARE, kernel and external SCORE2 estimators are shown. For concordance index, higher is better; for integrated Brier score, lower is better.

6.2.2 Model 2: adding features to SCORE2 for improved risk stratification

For our second model, we use the `imd` area-level deprivation feature and the polygenic scores `pgs000018` and `pgs000039` from UKB as additional covariates, in combination with SCORE2. From Figure 10, we see a similar pattern as in Figure 8, with the weight assigned to the external estimator decreasing with n. Figure 11 indicates the benefits of incorporating the deprivation index and polygenic scores into the model, using our cross-validated estimator. When using all of the available data, the concordance index improved by 1.52% for females and by 2.26% for males, while the integrated Brier score improved by 0.19% for females and by 0.44% for males. We conclude that the new covariates `imd`, `pgs000018` and `pgs000039` accounted for approximately 66.6% of the total concordance improvement for females and 80.4% of the improvement for males, with the rest due to the use of a quadratic kernel. Further analysis in Appendix D.2 shows that the polygenic score covariates `pgs000018` and `pgs000039` contributed approximately 77.9% of these improvements for females and 86.4% for males, with the remainder attributable to the deprivation index variable `imd`.

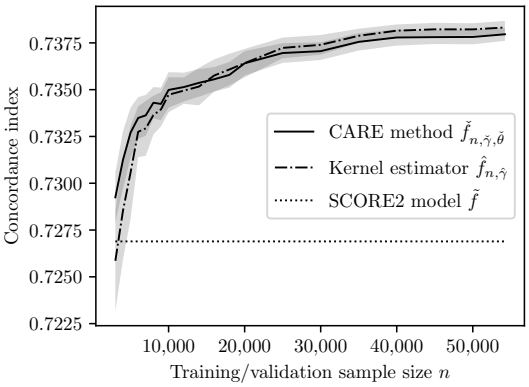


(a) Cross-validated choices of θ for females.

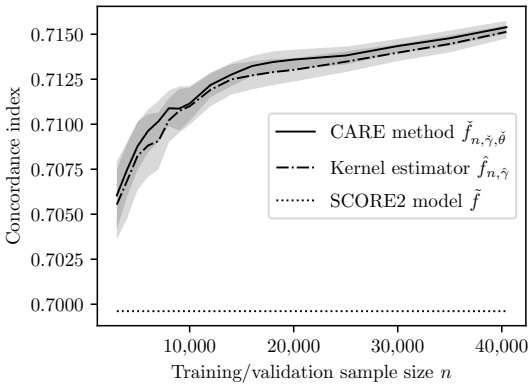


(b) Cross-validated choices of θ for males.

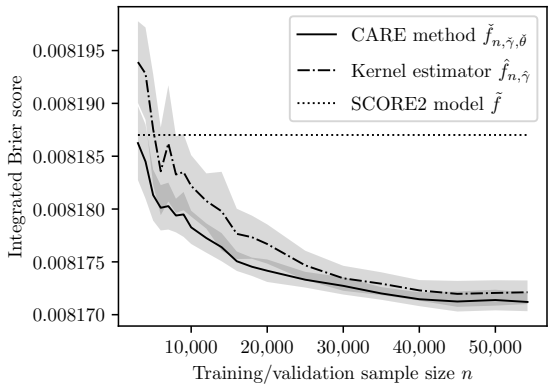
Figure 10: We plot the cross-validated choices of the convex combination parameter θ in Model 2 as a function of the training and validation sample sizes.



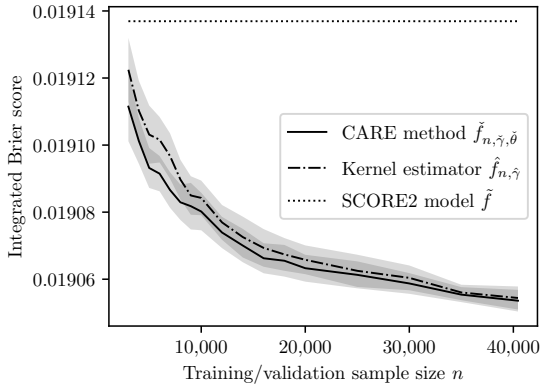
(a) Concordance index for females.



(b) Concordance index for males.



(c) Integrated Brier score for females.



(d) Integrated Brier score for males.

Figure 11: We plot the concordance index and integrated Brier score in Model 2 against the training and validation sample size, for the CARE, kernel and external SCORE2 estimators.

These models highlight an important robustness property of the CARE methodology: when the existing SCORE2 model can be improved, it is successfully ignored by our convex aggregation procedure in favour of the superior kernel estimator that uses the new covariates. See Appendix D.2 for further empirical results; in summary, we show that CARE can be used to upgrade a fitted survival model for the task of in-cohort prediction, and can incorporate new features to further improve performance.

7 Acknowledgements and funding

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A Further computational considerations

We discuss two techniques for improving the computational properties of our procedure.

A.1 Feature map formulation of the RKHS estimator

Certain kernel functions, for example polynomial kernels, admit finite-dimensional feature maps, and the aim of this subsection is to show how this can be exploited for substantial computational gains. In fact, even when the feature map is infinite-dimensional, the scheme we outline below can be used to compute an approximate solution to (4). Suppose that there exist $q \in \mathbb{N}$ and a measurable function $\phi = (\phi_1, \ldots, \phi_{q+1}) : \mathcal{X} \to \mathbb{R}^{q+1}$ with linearly independent coordinate functions satisfying $k(x, y) = \phi(x)^\mathsf{T}\phi(y)$ and $\phi_{q+1}(x) = c$ for all $x \in \mathcal{X}$ and some $c \in \mathbb{R} \setminus \{0\}$. Define $\tilde{\phi}(\cdot) := \big(\phi_j(\cdot) - P_n(\phi_j) : j \in [q]\big)$ and, for $\alpha \in \mathbb{R}^q$, let $f_\alpha(\cdot) := \alpha^\mathsf{T}\tilde{\phi}(\cdot)$. Noting that $P_n(\hat{f}_{n,\gamma}) = 0$, the representer theorem (Proposition 3) asserts the existence of $\hat{\alpha}$ taking values in $\mathbb{R}^q$ with

$$\hat{f}_{n,\gamma}(\cdot) = \hat{\alpha}^\mathsf{T}\tilde{\phi}(\cdot) = f_{\hat{\alpha}}(\cdot).$$

For $j \in [q+1]$ and $x \in \mathcal{X}$, by the reproducing property, we have

$$\phi_j(x) = \langle \phi_j, k(\cdot, x)\rangle_\mathcal{H} = \langle \phi_j, \phi(x)^\mathsf{T}\phi\rangle_\mathcal{H} = \sum_{r=1}^{q+1} \phi(x)_r \langle \phi_j, \phi_r\rangle_\mathcal{H},$$

so $\langle \phi_j, \phi_r\rangle_\mathcal{H} = \mathbb{1}_{\{j=r\}}$ for each $j, r \in [q+1]$ by linear independence. Thus, for every $\alpha = (\alpha_1, \ldots, \alpha_q)^\mathsf{T} \in \mathbb{R}^q$,

$$\begin{aligned}\|f_\alpha\|_\mathcal{H}^2 &= \sum_{j=1}^q \sum_{r=1}^q \alpha_j \alpha_r \langle \tilde{\phi}_j, \tilde{\phi}_r\rangle_\mathcal{H} = \sum_{j=1}^q \sum_{r=1}^q \alpha_j \alpha_r \left\langle \phi_j - P_n(\phi_j)\frac{\phi_{q+1}}{c}, \phi_r - P_n(\phi_r)\frac{\phi_{q+1}}{c} \right\rangle_\mathcal{H} \\ &= \sum_{j=1}^q \alpha_j^2 + \left(\frac{1}{c}\sum_{j=1}^q \alpha_j P_n(\phi_j)\right)^2.\end{aligned}$$

This allows us to calculate the penalised negative partial log-likelihood (3) for any function of the form f_α, and hence to solve the optimisation problem in (4) over $\alpha \in \mathbb{R}^q$. When the sample size is much larger than the dimension of the features, this problem can be significantly easier to solve than the more general version obtained through Proposition 3.

A.2 Nyström approximation

When a nonparametric kernel is used, there is no finite-dimensional feature map available. However, when the number of training data points is large, the computational complexity of calculating $\hat{f}_{n,\gamma}$ can be reduced using Nyström approximation (Williams and Seeger, 2000). This method proceeds by writing $\hat{f}_{n,\gamma} \approx \sum_{i \in \mathcal{S}_n} \tilde{k}(x, X_i)\beta_i$, where $\mathcal{S}_n$ is a randomly selected subset of $[n]$ of cardinality $m < n$. This increases the bias of the procedure by shrinking the space of admissible functions. However, when $m \ll n$, this approximation can greatly reduce

the computational cost of solving (4), as the decision variable in the resulting optimisation problem is now only m-dimensional. See Appendix D for empirical results on using Nyström approximation to upgrade the SCORE2 model. We refer the reader to Bach (2013) and Alaoui and Mahoney (2015) for some theoretical guarantees on Nyström approximation for RKHS methods; typically, it suffices to take m on the order of the effective dimension $H_{2,\gamma}$ of the γ-regularised kernel.

B Kernel assumptions

We provide some technical conditions and results regarding the kernel function k which are required for our main theoretical results. Let $\mathcal{X}$ be a locally compact metric space, equipped with its Borel σ-algebra $\mathcal{A}$, and assume that every open set in $\mathcal{X}$ is σ-compact, i.e. a countable union of compact sets. This ensures that P_X is a Radon probability measure (Folland, 1999, Theorem 7.8). We will further assume that k is continuous, that $P_X(U) > 0$ for every non-empty open set $U \subseteq \mathcal{X}$, and that $x \mapsto k(x,x)$ is P_X-integrable on $\mathcal{X}$. By Mercer's theorem (Samworth and Shah, 2026, Theorem A.8.8), T_k admits a decreasing sequence $(\nu_r)_{r\in\mathcal{R}}$ of positive eigenvalues with $\sum_{r\in\mathcal{R}} \nu_r < \infty$ and associated continuous orthonormal eigenfunctions $(e_r)_{r\in\mathcal{R}}$ in $L_2(P_X)$, with $T_k(e_r) = \nu_r e_r$ for $r \in \mathcal{R}$, where either $\mathcal{R} = [R]$ for some $R \in \mathbb{N}$ or $\mathcal{R} = \mathbb{N}$, endowed with the usual order. We then have the spectral decomposition $k(x,y) = \sum_{r\in\mathcal{R}} \nu_r e_r(x) e_r(y)$ for $x, y \in \mathcal{X}$, where the convergence is absolute and uniform on compact subsets of $\mathcal{X} \times \mathcal{X}$. Writing $\ell_2(\mathcal{R}) := \big\{(\alpha_r)_{r\in\mathcal{R}} : \sum_{r\in\mathcal{R}} \alpha_r^2 < \infty\big\}$, let

$$\mathcal{H} := \left\{ f_\alpha := \sum_{r\in\mathcal{R}} \alpha_r \sqrt{\nu_r} e_r : \alpha \in \ell_2(\mathcal{R}) \right\},$$

and for $\alpha := (\alpha_r)_{r\in\mathcal{R}} \in \ell_2(\mathcal{R})$ and $\beta := (\beta_r)_{r\in\mathcal{R}} \in \ell_2(\mathcal{R})$, define the inner product $\langle f_\alpha, f_\beta\rangle_{\mathcal{H}} := \sum_{r\in\mathcal{R}} \alpha_r\beta_r$. Then by Samworth and Shah (2026, Corollary A.8.9), the pair $(\mathcal{H}, \langle\cdot,\cdot\rangle_{\mathcal{H}})$ forms a Hilbert space with orthonormal basis $(\sqrt{\nu_r} e_r)_{r\in\mathcal{R}}$, and is in fact the unique reproducing kernel Hilbert space (RKHS) associated with k. Further, every $f \in \mathcal{H}$ is continuous (Samworth and Shah, 2026, Theorem A.8.8 and Corollary A.8.9) and satisfies $f \in L_2(P_X)$, so $f = \sum_{r\in\mathcal{R}} \langle f, e_r\rangle_{L_2} e_r$ where $\langle\cdot,\cdot\rangle_{L_2} := \langle\cdot,\cdot\rangle_{L_2(P_X)}$. From this it follows that $\langle f, g\rangle_{\mathcal{H}} = \sum_{r\in\mathcal{R}} \frac{1}{\nu_r}\langle f, e_r\rangle_{L_2}\langle g, e_r\rangle_{L_2}$, so $\langle f, e_r\rangle_{\mathcal{H}} = \frac{1}{\nu_r}\langle f, e_r\rangle_{L_2}$ for each $r \in \mathcal{R}$, and $\|f\|_{\mathcal{H}}^2 = \sum_{r\in\mathcal{R}} \frac{1}{\nu_r}\langle f, e_r\rangle_{L_2}^2$. We deduce that $\langle f, e_r\rangle_{\mathcal{H}_\gamma} = \frac{1}{a_r}\langle f, e_r\rangle_{L_2}$ and $\|f\|_{\mathcal{H}_\gamma}^2 = \sum_{r\in\mathcal{R}} \frac{1}{a_r}\langle f, e_r\rangle_{L_2}^2$. Further, by Paulsen and Raghupathi (2016, Theorem 3.11), $\mathcal{H}$ contains the unit constant function on $\mathcal{X}$. Note that $H_{\infty,\gamma}$ does not depend on the choice of $L_2(P_X)$-orthonormal eigenfunctions $(e_r)_{r\in\mathcal{R}}$ due to the uniqueness of the eigenspace associated with ν_r, for each $r \in \mathcal{R}$. Observe that $\gamma \sum_{r\in\mathcal{R}} a_r e_r(x)^2 \searrow 0$ as $\gamma \searrow 0$ for each $x \in \mathcal{X}$, by the dominated convergence theorem. Since $\gamma a_r \le \nu_r$ and $\sum_{r\in\mathcal{R}} \nu_r e_r(x)^2 = k(x,x)$, with the convergence uniform on compact sets, we deduce that $\sup_{x\in\mathcal{X}'} \gamma \sum_{r\in\mathcal{R}} a_r e_r(x)^2 \searrow 0$ as $\gamma \searrow 0$ whenever $\mathcal{X}' \subseteq \mathcal{X}$ is compact. In particular, $\gamma H_{\infty,\gamma} \searrow 0$ as $\gamma \searrow 0$ if $\mathcal{X}$ is compact.

C Automatic kernel selection

Suppose that $\mathcal{K}$ is a finite set of kernels such that each $k \in \mathcal{K}$ induces an RKHS $\mathcal{H}_k$ satisfying the conditions of Section 2.3. For each kernel $k \in \mathcal{K}$, define

$$\hat{f}_{n,\gamma,k} := \operatorname*{argmin}_{f \in \mathcal{H}_k :\, P_n(f)=0} \ell_{n,\gamma}(f)$$

and take K_k, $H_{2,\gamma,k}$ and $H_{\infty,\gamma,k}$ as in Section 2.3. Set $K_{\mathcal{K}} := \max_{k\in\mathcal{K}} K_k$ and $H_{\infty,\gamma,\mathcal{K}} := \max_{k\in\mathcal{K}} H_{\infty,\gamma,k}$, and let

$$\big(\hat{\gamma}_{\mathcal{K}}, \hat{k}\big) := \operatorname*{sargmin}_{\gamma\in\Gamma,\, k\in\mathcal{K}} \tilde{\ell}_n\big(\hat{f}_{n,\gamma,k}\big).$$

Recall the notation $\gamma^+ := \max\Gamma$ and $\gamma^- := \min\Gamma$ from Section 4.3.

Proposition 12 (Kernel selection)**.** *There exists $C_0 > 0$ depending only on $\max_{k\in\mathcal{K}} \|f_0\|_{\mathcal{H}_k}$, Λ, q_1, $\max_{k\in\mathcal{K}} \|1_{\mathcal{X}}\|_{\mathcal{H}_k}$, $K_{\mathcal{K}}$ and ξ such that if $\gamma^+ H_{\infty,\gamma^+,\mathcal{K}} \le 1/C_0$, $H_{\infty,\gamma^-,\mathcal{K}}\sqrt{\{\log(n|\mathcal{K}|)\}/n} \le 1/C_0$ and $1 \le s \le \sqrt{n}/(C_0 H_{\infty,\gamma^-,\mathcal{K}})$, then with probability at least $1 - e^{-s^2}$,*

$$\big\|\hat{f}_{n,\hat{\gamma}_{\mathcal{K}},\hat{k}} - f_0\big\|_{L_2} \le C_0\left(\min_{\gamma\in\Gamma,\, k\in\mathcal{K}} \big\|\hat{f}_{n,\gamma,k} - f_0\big\|_{L_2} + \frac{s + \sqrt{\log|\mathcal{K}| + \log n}}{\sqrt{n}}\right).$$

This oracle inequality permits adaptation across Sobolev scales or automatic kernel selection (Blanchard et al., 2008; Goldenshluger and Lepski, 2011); we omit details for brevity.

D Additional empirical results

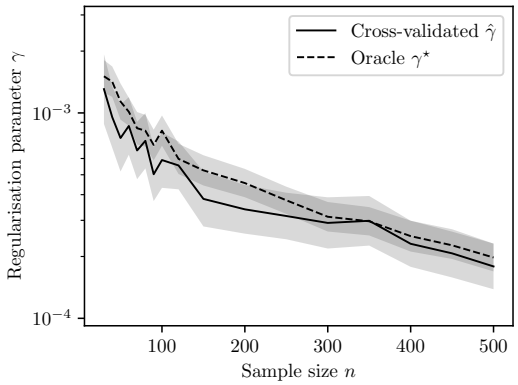
We present additional results relating to our simulation studies and SCORE2 model analysis.

D.1 Simulated data study

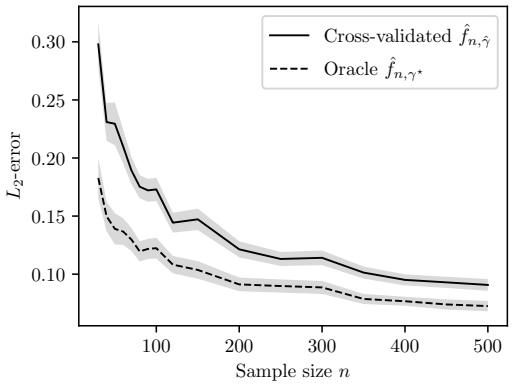
We consider some further censoring distributions to better understand the behaviour our kernel method and the CARE procedure. The set-up is identical to the univariate and multivariate situations of Sections 6.1.1 and 6.1.2 respectively, modifying only the law of the censoring variable T_{C}. For 'heavy censoring,' we take $T_{\mathrm{C}} \sim \mathcal{U}[0,1]$ and for 'no censoring,' we consider $T_{\mathrm{C}} = 1$.

D.1.1 Univariate regressors, heavy censoring

We present results for univariate regressors with heavy censoring in Figures 12 and 13.

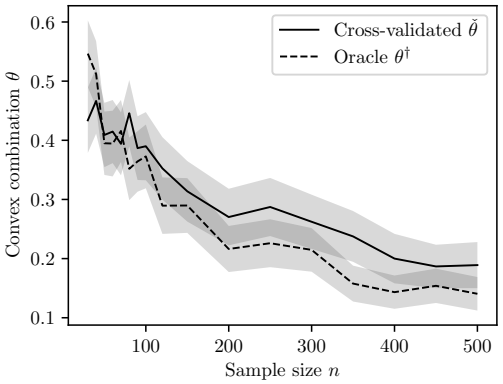


(a) Cross-validated and oracle choices of γ.

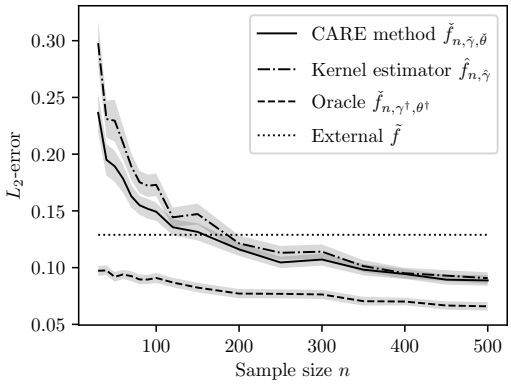


(b) Cross-validated and oracle L_2-error.

Figure 12: Panel (a): regularisation parameter. Panel (b): L_2-error.



(a) Cross-validation and oracle choices of θ.

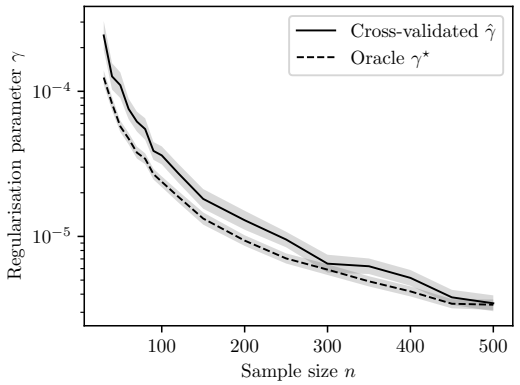


(b) Cross-validated and oracle performance.

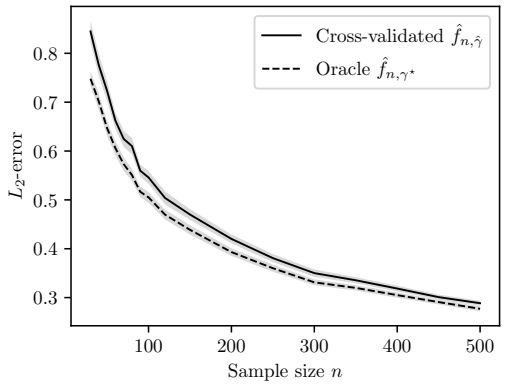
Figure 13: Panel (a): convex combination parameter. Panel (b): comparison of estimators.

D.1.2 Multivariate regressors, heavy censoring

We present results for multivariate regressors with heavy censoring in Figures 14 and 15.

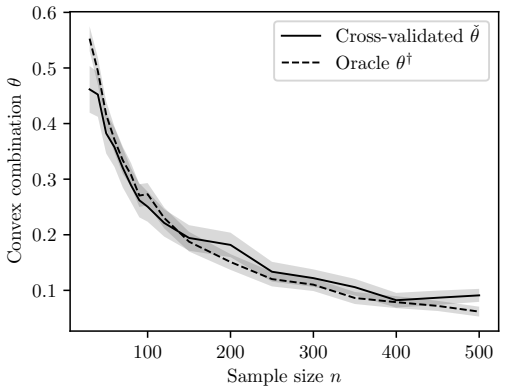


(a) Cross-validated and oracle choices of γ.

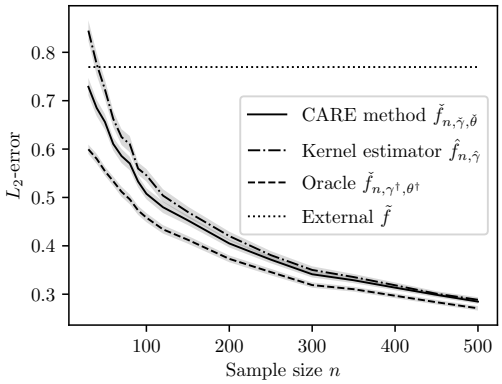


(b) Cross-validated and oracle L_2-error.

Figure 14: Panel (a): regularisation parameter. Panel (b): L_2-error.



(a) Cross-validation and oracle choices of θ.

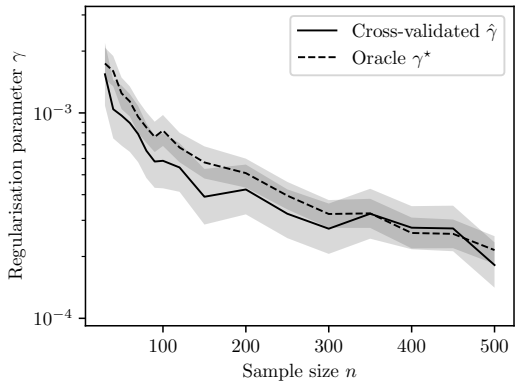


(b) Cross-validated and oracle performance.

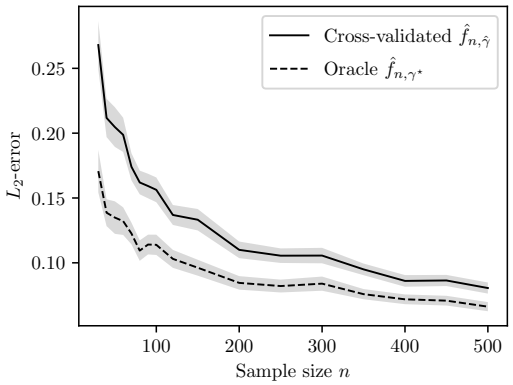
Figure 15: Panel (a): convex combination parameter. Panel (b): comparison of estimators.

D.1.3 Univariate regressors, no censoring

We present results for univariate regressors with no censoring in Figures 16 and 17.

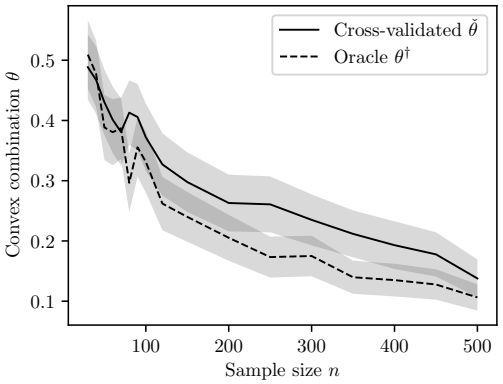


(a) Cross-validated and oracle choices of γ.

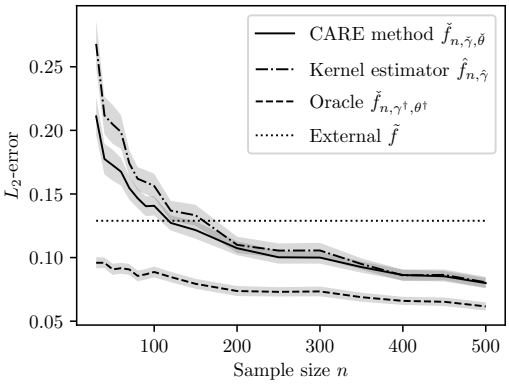


(b) Cross-validated and oracle L_2-error.

Figure 16: Panel (a): regularisation parameter. Panel (b): L_2-error.



(a) Cross-validation and oracle choices of θ.

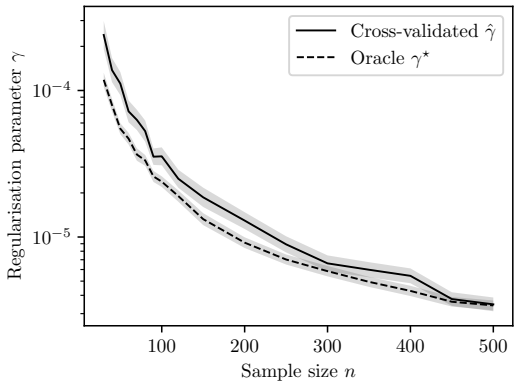


(b) Cross-validated and oracle performance.

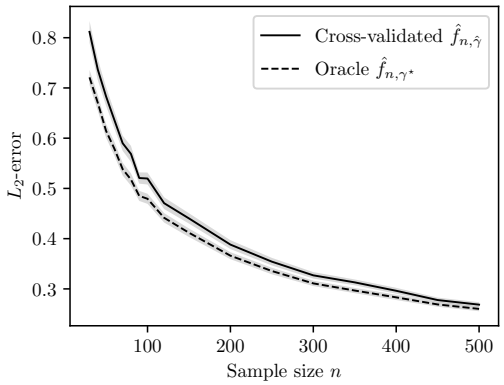
Figure 17: Panel (a): convex combination parameter. Panel (b): comparison of estimators.

D.1.4 Multivariate regressors, no censoring

We present results for multivariate regressors with no censoring in Figures 18 and 19.

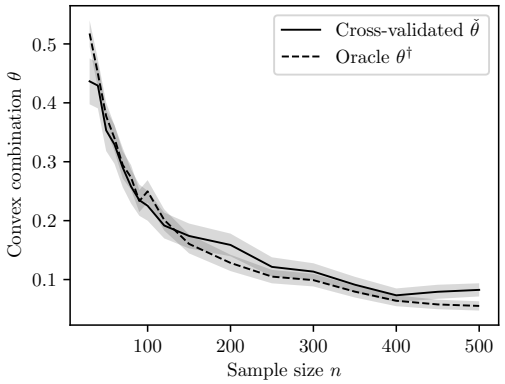


(a) Cross-validated and oracle choices of γ.

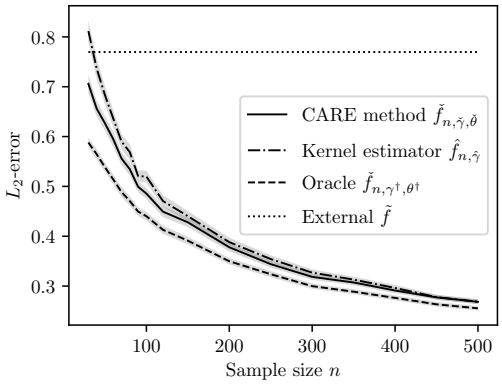


(b) Cross-validated and oracle L_2-error.

Figure 18: Panel (a): regularisation parameter. Panel (b): L_2-error.



(a) Cross-validation and oracle choices of θ.



(b) Cross-validated and oracle performance.

Figure 19: Panel (a): convex combination parameter. Panel (b): comparison of estimators.

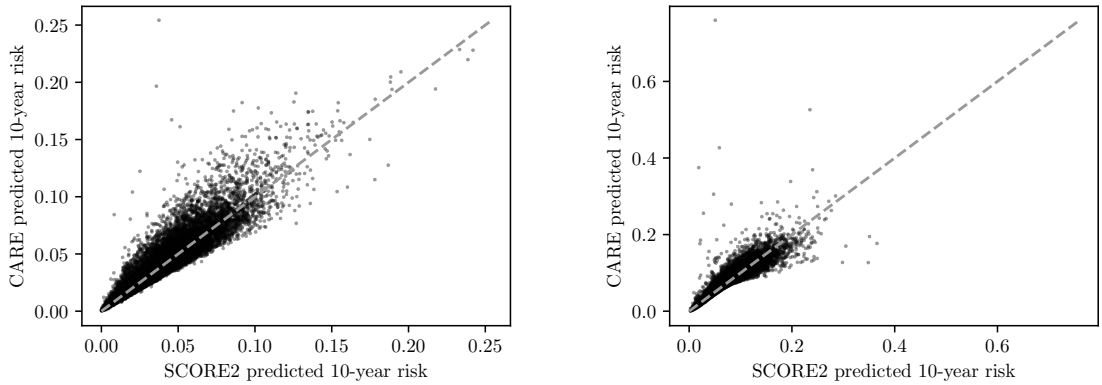
By comparing with the results in Section 6.1, we conclude that the relative performance of our methodology is broadly similar across different censoring levels, both for univariate and multivariate regressors. The 'no censoring' case generally leading to slightly better results in terms of L_2-error and integrated Brier score.

D.2 Cardiovascular disease risk prediction

We present additional results on upgrading the SCORE2 cardiovascular disease risk model.

D.2.1 Model 1

In Figure 20, we plot the predicted 10-year risks in Model 1 against the corresponding predicted risks for the recalibrated SCORE2 model. We see that most predicted risks increase slightly under CARE as opposed to SCORE2, perhaps particularly among low-risk males.



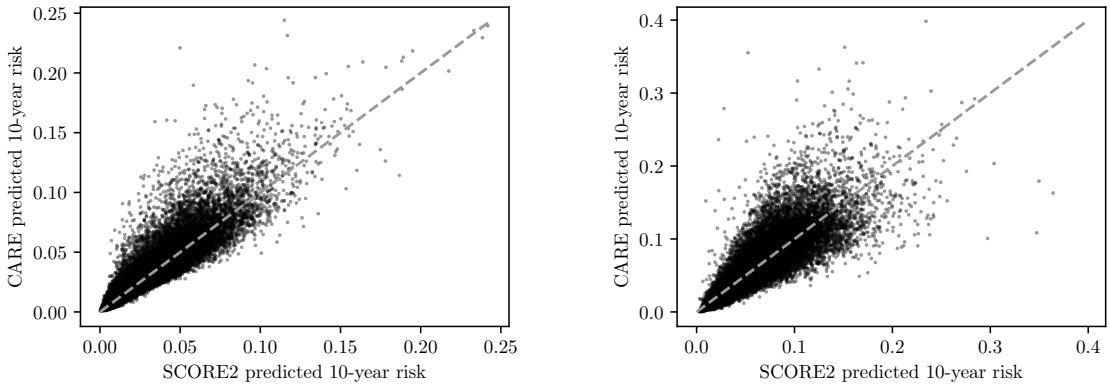
(a) Predicted risk for females.

(b) Predicted risk for males.

Figure 20: Risk predictions for Model 1 compared to SCORE2.

D.2.2 Model 2

In Figure 21, we plot the predicted 10-year risks in Model 2 against the corresponding predicted risks for the recalibrated SCORE2 model. We see more deviation between the SCORE2 and CARE predicted risks than for Model 1, due to the extra predictive power offered by the inclusion of area-level deprivation and polygenic score covariates.



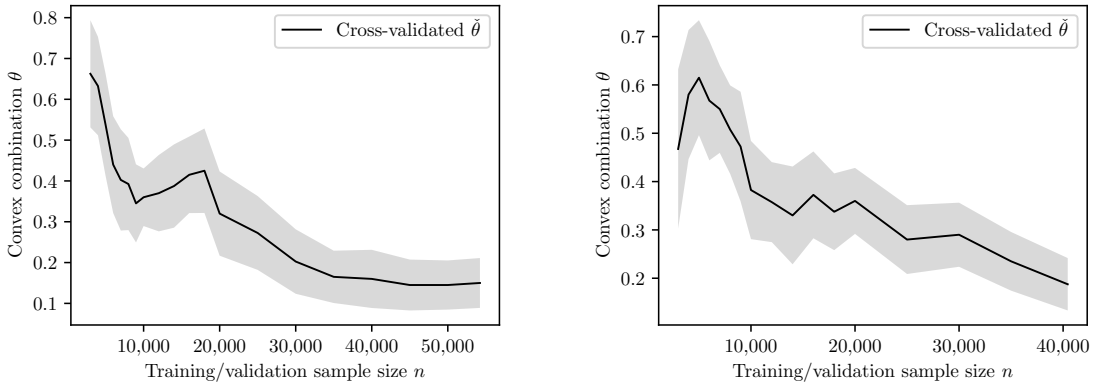
(a) Predicted risk for females.

(b) Predicted risk for males.

Figure 21: Risk predictions for Model 2 compared to SCORE2.

D.2.3 Model 3: SCORE2 and UKB with deprivation index

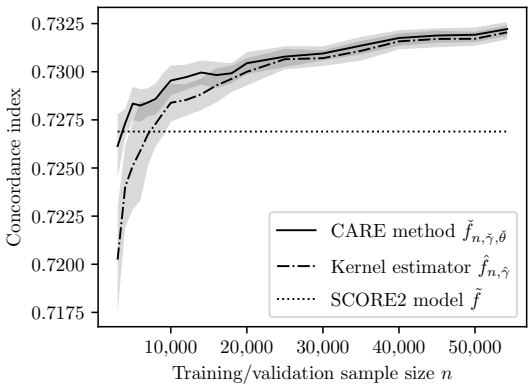
In Model 3, we consider upgrading SCORE2 using CARE, adding just the deprivation index variable `imd`. The cross-validated convex combination parameter is shown in Figure 22, while Figure 23 illustrates the performance of this model as measured by concordance index and integrated Brier score. A scatter plot of the predicted risks against recalibrated SCORE2 is shown in Figure 24. When using all of the available data, the concordance index improved by 0.73% for females and by 0.69% for males, while the integrated Brier score improved by 0.07% for females and by 0.15% for males. We conclude that when moving from Model 1 to Model 2, the polygenic score covariates `pgs000018` and `pgs000039` accounted for approximately 77.9% of the total concordance improvement (upon to the UKB-adjusted Model 1) for females and 86.4% of the improvement for males, with the remainder of the improvement attributable to inclusion of the deprivation index variable `imd`.



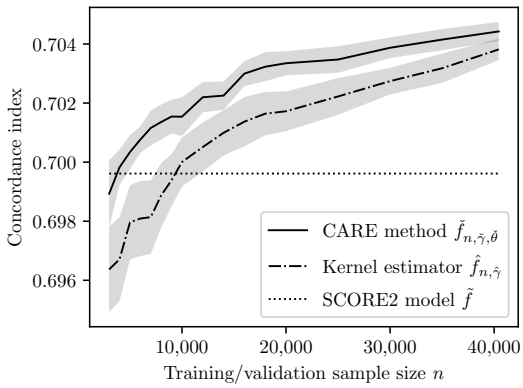
(a) Cross-validated choices of θ for females.

(b) Cross-validated choices of θ for males.

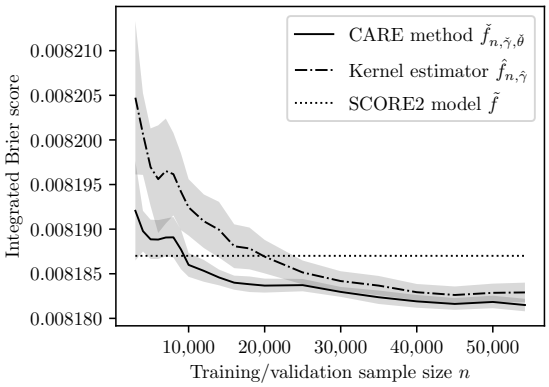
Figure 22: Convex combination parameter choices in Model 3.



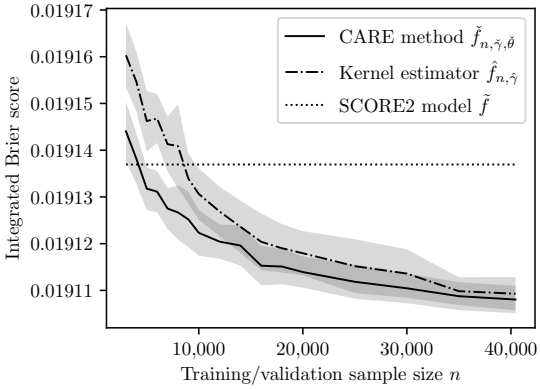
(a) Concordance index for females.



(b) Concordance index for males.

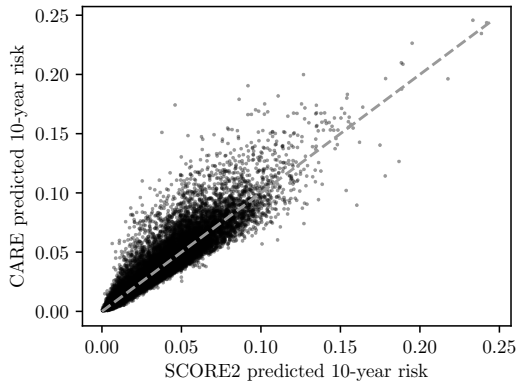


(c) Integrated Brier score for females.

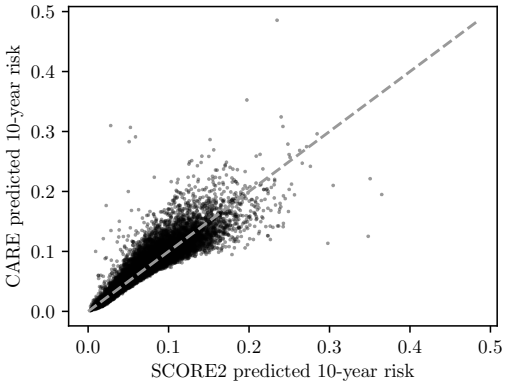


(d) Integrated Brier score for males.

Figure 23: Comparison of estimators in Model 3.



(a) Predicted risk for females.

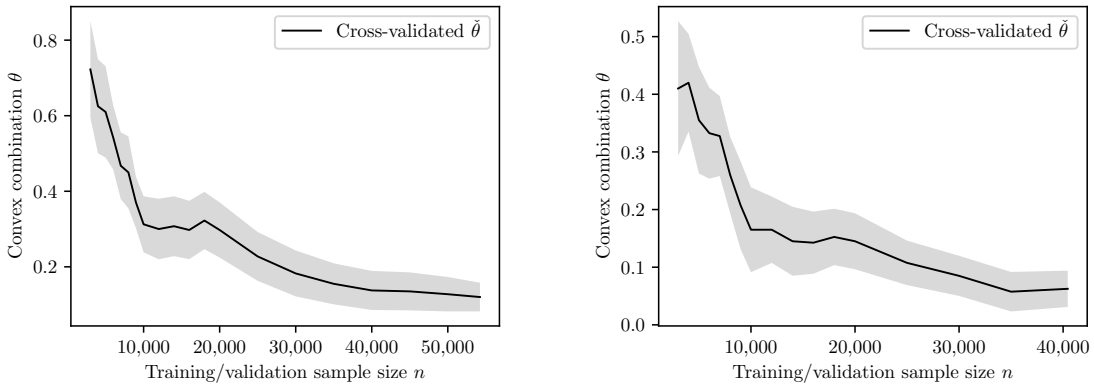


(b) Predicted risk for males.

Figure 24: Risk predictions for Model 3 compared to SCORE2.

D.2.4 Model 4: SCORE2 and UKB with deprivation index and polygenic scores, linear kernel

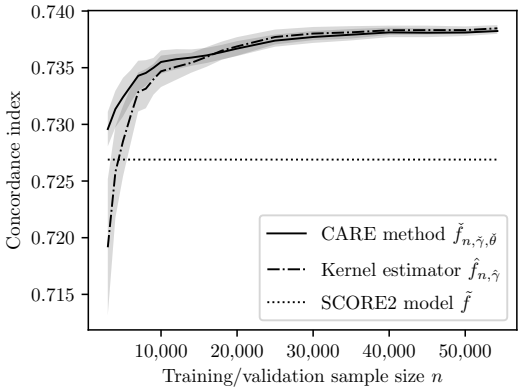
In Model 4, we add the deprivation index variable `imd` and the polygenic scores `pgs000018` and `pgs000039` from UKB to SCORE2, this time using a linear kernel with no regularisation (corresponding to a standard Cox model). The cross-validated convex combination parameter is shown in Figure 25, while Figure 26 illustrates the performance of this model as measured by concordance index and integrated Brier score. A scatter plot of the predicted risks against recalibrated SCORE2 is shown in Figure 27. When using all of the available data, the concordance index improved by 1.56% for females and by 2.00% for males, while the integrated Brier score improved by 0.17% for females and by 0.41% for males.



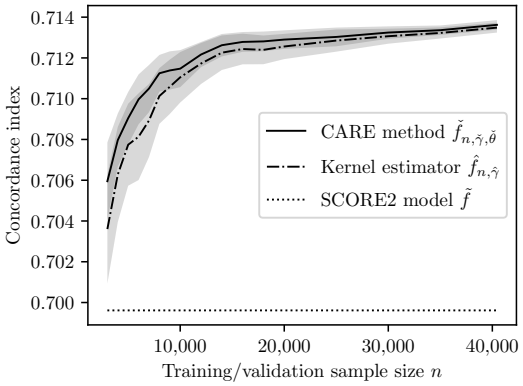
(a) Cross-validated choices of θ for females.

(b) Cross-validated choices of θ for males.

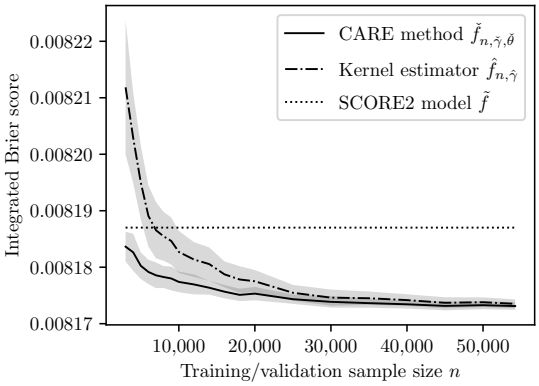
Figure 25: Convex combination parameter choices in Model 4.



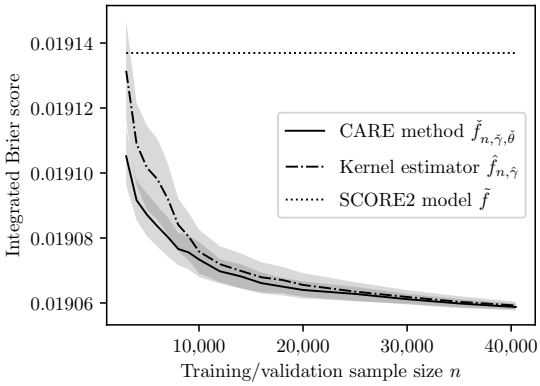
(a) Concordance index for females.



(b) Concordance index for males.

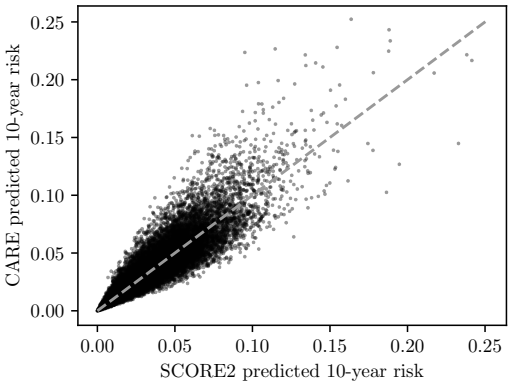


(c) Integrated Brier score for females.

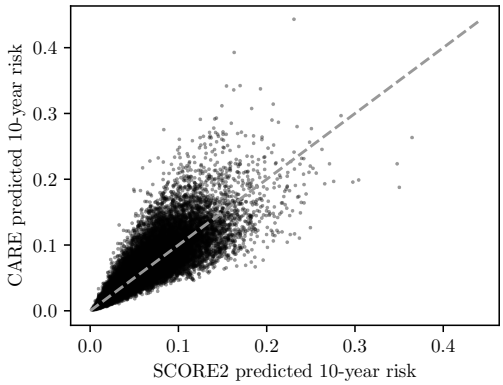


(d) Integrated Brier score for males.

Figure 26: Comparison of estimators in Model 4.



(a) Predicted risk for females.

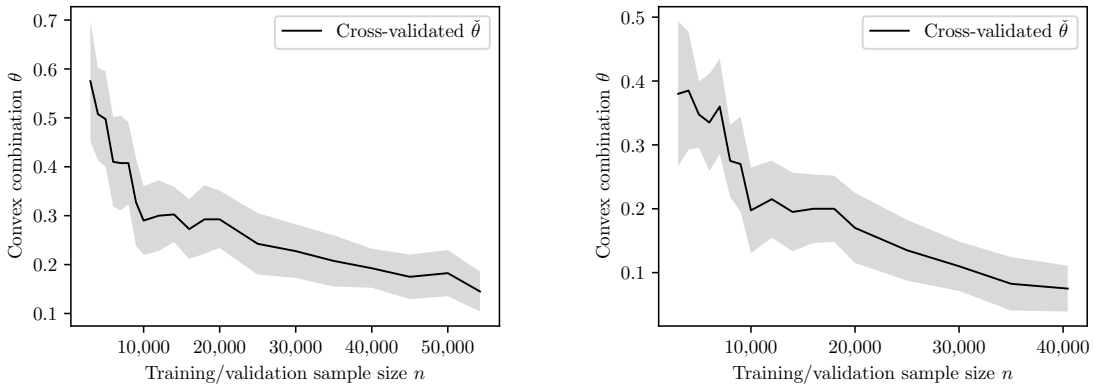


(b) Predicted risk for males.

Figure 27: Risk predictions for Model 4 compared to SCORE2.

D.2.5 Model 5: SCORE2 and UKB with deprivation index and polygenic scores, Sobolev kernel

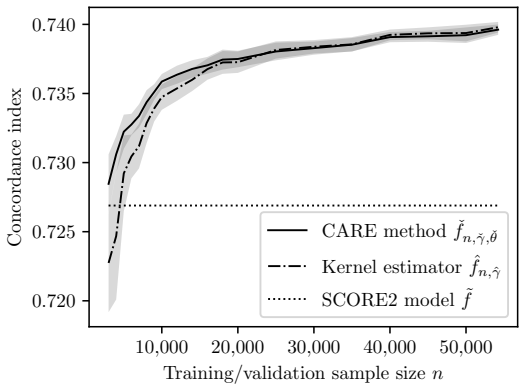
In Model 5, we add the deprivation index variable `imd` and the polygenic scores `pgs000018` and `pgs000039` from UKB to SCORE2, this time using a shifted first order Sobolev kernel, computing the RKHS estimators using a Nyström subsampling size of 100. The cross-validated convex combination parameter is shown in Figure 28, while Figure 29 illustrates the performance of this model as measured by concordance index and integrated Brier score. A scatter plot of the predicted risks against recalibrated SCORE2 is shown in Figure 30. When using all of the available data, the concordance index improved by 1.75% for females and by 2.00% for males, while the integrated Brier score improved by 0.17% for females and by 0.32% for males.



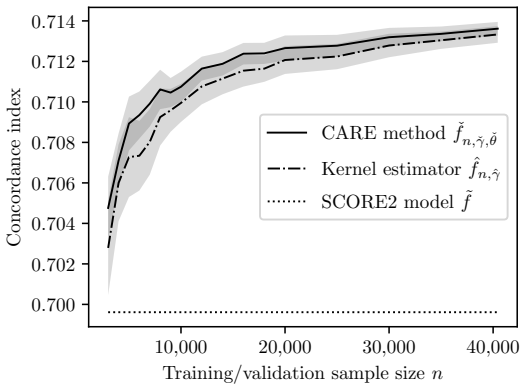
(a) Cross-validated choices of θ for females.

(b) Cross-validated choices of θ for males.

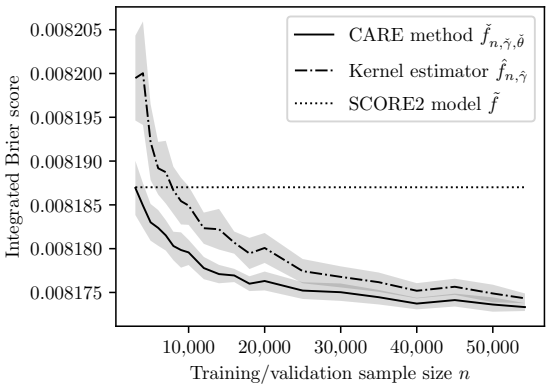
Figure 28: Convex combination parameter choices in Model 5.



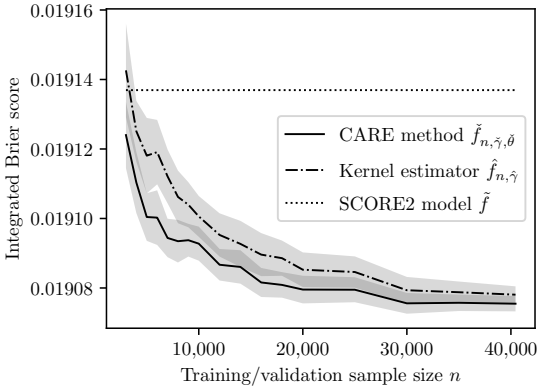
(a) Concordance index for females.



(b) Concordance index for males.

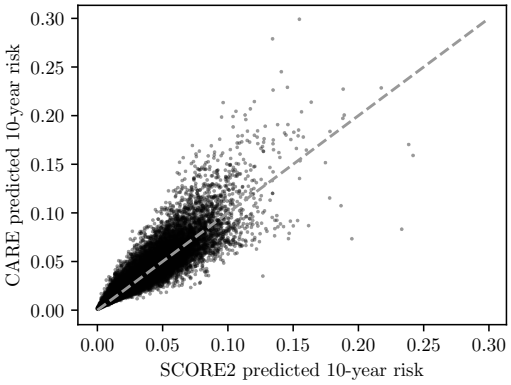


(c) Integrated Brier score for females.

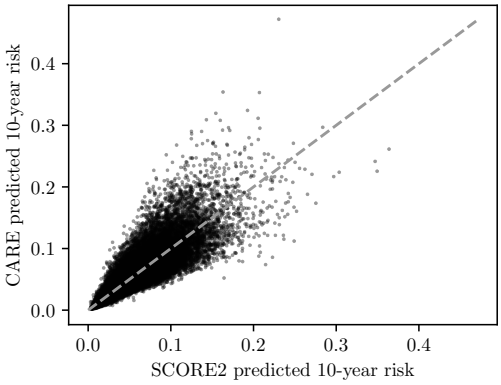


(d) Integrated Brier score for males.

Figure 29: Comparison of estimators in Model 5.



(a) Predicted risk for females.



(b) Predicted risk for males.

Figure 30: Risk predictions for Model 5 compared to SCORE2.

E Proofs and technical details

In this section, we present full proofs for all of our main results, along with auxiliary lemmas and technical details. Several of our proofs rely on the decoupling and symmetrisation inequalities for U-processes given in Appendix E.6, which may be of independent interest.

E.1 Proofs for Section 2

We begin with some properties of the partial likelihood functions introduced in Section 2.2.

E.1.1 Gateaux derivatives of ℓ_n and $\ell_\star$

Firstly, Lemmas 13 and 14 characterise the Gateaux derivatives of the negative log-partial likelihood ℓ_n. Next, Lemmas 15 and 16 yield the Gateaux derivatives of the limiting negative log-partial likelihood $\ell_\star$.

Lemma 13 (Gateaux derivatives of S_n)**.** *Let $f, f_1, \ldots, f_m : \mathcal{X} \to \mathbb{R}$. Then for $m \in \mathbb{N}_0$, the m-th order Gateaux derivative of $S_n(f,t)$ in the direction $(f_1, \ldots, f_m)$ is given by*

$$D^m S_n(f,t)(f_1, \ldots, f_m) = \frac{1}{n} \sum_{i=1}^{n} R_i(t) e^{f(X_i)} \prod_{j=1}^{m} f_j(X_i).$$

Proof of Lemma 13. We proceed by induction on m; the result holds for $m = 0$ by definition of $S_n(f,t)$. Now for $m \in \mathbb{N}$ and $f, f_1, \ldots, f_m : \mathcal{X} \to \mathbb{R}$, suppose that

$$D^{m-1} S_n(f,t)\,(f_1, \ldots, f_{m-1}) = \frac{1}{n} \sum_{i=1}^{n} R_i(t) e^{f(X_i)} \prod_{j=1}^{m-1} f_j(X_i).$$

By the definition of the Gateaux derivative and the induction hypothesis, for every $\varepsilon \in \mathbb{R} \setminus \{0\}$ and $t \in [0,1]$, we have

$$\begin{aligned}
&\frac{1}{\varepsilon}\big(D^{m-1} S_n(f + \varepsilon f_m, t)(f_1, \ldots, f_{m-1}) - D^{m-1} S_n(f,t)(f_1, \ldots, f_{m-1})\big) \\
&\quad = \frac{1}{n} \sum_{i=1}^{n} R_i(t) \frac{1}{\varepsilon}\big(e^{f(X_i) + \varepsilon f_m(X_i)} - e^{f(X_i)}\big) \prod_{j=1}^{m-1} f_j(X_i) \to \frac{1}{n} \sum_{i=1}^{n} R_i(t) e^{f(X_i)} \prod_{j=1}^{m} f_j(X_i)
\end{aligned}$$

as $\varepsilon \to 0$, as required. □

Lemma 14 (Gateaux derivatives of ℓ_n)**.** *Let $f, f_1, f_2 : \mathcal{X} \to \mathbb{R}$. Then the first-order Gateaux derivative of $\ell_n(f)$ in the direction f_1 exists and is given by*

$$D\ell_n(f)(f_1) = \int_0^1 \frac{D S_n(f,t)(f_1)}{S_n(f,t)} \, \mathrm{d}N(t) - \frac{1}{n} \sum_{i=1}^{n} f_1(X_i) N_i(1).$$

This is well-defined as $S_n(f, T_i) > 0$ for $i \in [n]$. The second-order Gateaux derivative of $\ell_n(f)$ in the direction (f_1, f_2) also exists, and is given by

$$D^2\ell_n(f)(f_1, f_2) = \int_0^1 \left(\frac{D^2 S_n(f,t)(f_1, f_2)}{S_n(f,t)} - \frac{D S_n(f,t)(f_1)}{S_n(f,t)} \frac{D S_n(f,t)(f_2)}{S_n(f,t)} \right) \mathrm{d}N(t).$$

Further, $D^2\ell_n(f)(f_1, f_1) \geq 0$, so ℓ_n is a convex function. Finally, for $f, f_1, f_2, f_3 : \mathcal{X} \to \mathbb{R}$, the third-order Gateaux derivative $D^3\ell_n(f)(f_1, f_2, f_3)$ exists.

Proof of Lemma 14. By definition of the negative log-partial likelihood, for any $f : \mathcal{X} \to \mathbb{R}$,

$$\begin{aligned}\ell_n(f) &= \int_0^1 \log S_n(f,t)\,\mathrm{d}N(t) - \frac{1}{n}\sum_{i=1}^n f(X_i)N_i(1) \\ &= \frac{1}{n}\sum_{i=1}^n (1-I_i)\log S_n(f,T_i) - \frac{1}{n}\sum_{i=1}^n f(X_i)N_i(1).\end{aligned}$$

For $\varepsilon \in \mathbb{R}\setminus\{0\}$, it follows from the chain rule that

$$\begin{aligned}&\frac{1}{\varepsilon}\big(\ell_n(f+\varepsilon f_1) - \ell_n(f)\big) \\ &\qquad= \frac{1}{\varepsilon n}\sum_{i=1}^n (1-I_i)\big(\log S_n(f+\varepsilon f_1, T_i) - \log S_n(f,T_i)\big) - \frac{1}{n}\sum_{i=1}^n f_1(X_i)N_i(1) \\ &\qquad\to \frac{1}{n}\sum_{i=1}^n (1-I_i)\frac{DS_n(f,T_i)(f_1)}{S_n(f,T_i)} - \frac{1}{n}\sum_{i=1}^n f_1(X_i)N_i(1) \\ &\qquad= \int_0^1 \frac{DS_n(f,t)(f_1)}{S_n(f,t)}\,\mathrm{d}N(t) - \frac{1}{n}\sum_{i=1}^n f_1(X_i)N_i(1)\end{aligned}$$

as $\varepsilon \to 0$. Similarly, by the quotient rule, for $\varepsilon \in \mathbb{R}\setminus\{0\}$,

$$\begin{aligned}&\frac{1}{\varepsilon}\big(D\ell_n(f+\varepsilon f_2)(f_1) - D\ell_n(f)(f_1)\big) \\ &\qquad= \frac{1}{\varepsilon n}\sum_{i=1}^n (1-I_i)\left(\frac{DS_n(f+\varepsilon f_2, T_i)(f_1)}{S_n(f+\varepsilon f_2, T_i)} - \frac{DS_n(f,T_i)(f_1)}{S_n(f,T_i)}\right) \\ &\qquad\to \frac{1}{n}\sum_{i=1}^n (1-I_i)\left(\frac{D^2S_n(f,T_i)(f_1,f_2)}{S_n(f,T_i)} - \frac{DS_n(f,T_i)(f_1)}{S_n(f,T_i)}\frac{DS_n(f,T_i)(f_2)}{S_n(f,T_i)}\right) \\ &\qquad= \int_0^1 \left(\frac{D^2S_n(f,t)(f_1,f_2)}{S_n(f,t)} - \frac{DS_n(f,t)(f_1)}{S_n(f,t)}\frac{DS_n(f,t)(f_2)}{S_n(f,t)}\right)\mathrm{d}N(t)\end{aligned}$$

as $\varepsilon \to 0$, as required. To show convexity of ℓ_n, note that

$$\begin{aligned}0 &\leq \int_0^1 \frac{1}{n}\sum_{i=1}^n \frac{R_i(t)e^{f(X_i)}}{S_n(f,t)}\left(f_1(X_i) - \frac{DS_n(f,t)(f_1)}{S_n(f,t)}\right)^2 \mathrm{d}N(t) \\ &= \int_0^1 \frac{1}{S_n(f,t)}\left(D^2S_n(f,t)(f_1,f_1) - 2\frac{DS_n(f,t)(f_1)^2}{S_n(f,t)} + \frac{DS_n(f,t)(f_1)^2}{S_n(f,t)}\right)\mathrm{d}N(t) \\ &= \int_0^1 \left(\frac{D^2S_n(f,t)(f_1,f_1)}{S_n(f,t)} - \frac{DS_n(f,t)(f_1)^2}{S_n(f,t)^2}\right)\mathrm{d}N(t) = D^2\ell_n(f)(f_1,f_1).\end{aligned}$$

Finally, we show that the third-order Gateaux derivative exists. For $\varepsilon \in \mathbb{R} \setminus \{0\}$, by the dominated convergence theorem, the chain rule, the product rule and the quotient rule,

$$
\begin{aligned}
&\frac{1}{\varepsilon}\Big\{D^2\ell_n(f+\varepsilon f_3)(f_1,f_2) - D^2\ell_n(f)(f_1,f_2)\Big\} \\
&= \int_0^1 \frac{1}{\varepsilon}\left(\frac{D^2S_n(f+\varepsilon f_3,t)(f_1,f_2)}{S_n(f+\varepsilon f_3,t)} - \frac{D^2S_n(f,t)(f_1,f_2)}{S_n(f,t)}\right)\mathrm{d}N(t) \\
&\quad - \int_0^1 \frac{1}{\varepsilon}\left(\frac{DS_n(f+\varepsilon f_3,t)(f_1)}{S_n(f+\varepsilon f_3,t)}\frac{DS_n(f+\varepsilon f_3,t)(f_2)}{S_n(f+\varepsilon f_3,t)} - \frac{DS_n(f,t)(f_1)}{S_n(f,t)}\frac{DS_n(f,t)(f_2)}{S_n(f,t)}\right)\mathrm{d}N(t) \\
&\overset{\epsilon\to 0}{\to} \int_0^1 \left(\frac{D^3S_n(f,t)(f_1,f_2,f_3)}{S_n(f,t)} - \frac{D^2S_n(f,t)(f_1,f_2)DS_n(f,t)(f_3)}{S_n(f,t)^2}\right)\mathrm{d}N(t) \\
&\quad - \int_0^1 \frac{DS_n(f,t)(f_1)}{S_n(f,t)}\left(\frac{D^2S_n(f,t)(f_2,f_3)}{S_n(f,t)} - \frac{DS_n(f,t)(f_2)DS_n(f,t)(f_3)}{S_n(f,t)^2}\right)\mathrm{d}N(t) \\
&\quad - \int_0^1 \frac{DS_n(f,t)(f_2)}{S_n(f,t)}\left(\frac{D^2S_n(f,t)(f_1,f_3)}{S_n(f,t)} - \frac{DS_n(f,t)(f_1)DS_n(f,t)(f_3)}{S_n(f,t)^2}\right)\mathrm{d}N(t) \\
&= \int_0^1 \left(\frac{D^3S_n(f,t)(f_1,f_2,f_3)}{S_n(f,t)} - \frac{D^2S_n(f,t)(f_1,f_2)DS_n(f,t)(f_3)}{S_n(f,t)^2}\right. \\
&\qquad - \frac{D^2S_n(f,t)(f_1,f_3)DS_n(f,t)(f_2)}{S_n(f,t)^2} - \frac{D^2S_n(f,t)(f_2,f_3)DS_n(f,t)(f_1)}{S_n(f,t)^2} \\
&\qquad \left. + 2\frac{DS_n(f,t)(f_1)DS_n(f,t)(f_2)DS_n(f,t)(f_3)}{S_n(f,t)^3}\right)\mathrm{d}N(t).
\end{aligned}
\tag{12}
$$

□

Lemma 15 (Gateaux derivatives of S)**.** *Take $m \in \mathbb{N}_0$ and let $f, f_1, \ldots, f_m \in \mathcal{B}(\mathcal{X})$. Then*

$$
\mathbb{E}\big(D^mS_n(f,t)(f_1,\ldots,f_m)\big) = \int_{\mathcal{X}} q(x,t)e^{f(x)}\prod_{j=1}^m f_j(x)\,\mathrm{d}P_X(x) = D^mS(f,t)\,(f_1,\ldots,f_m)\,.
$$

Proof of Lemma 15. We proceed by induction on m. For $m = 0$, we have

$$
\mathbb{E}\big(S_n(f,t)\big) = \mathbb{E}\big(R(t)e^{f(X)}\big) = \int_{\mathcal{X}} q(x,t)e^{f(x)}\,\mathrm{d}P_X(x) = S(f,t).
$$

For $m \in \mathbb{N}$ and $f, f_1, \ldots, f_m \in \mathcal{B}(\mathcal{X})$, suppose that

$$
\begin{aligned}
\mathbb{E}\big(D^{m-1}S_n(f,t)(f_1,\ldots,f_{m-1})\big) &= \int_{\mathcal{X}} q(x,t)e^{f(x)}\prod_{j=1}^{m-1} f_j(x)\,\mathrm{d}P_X(x) \\
&= D^{m-1}S(f,t)(f_1,\ldots,f_{m-1}).
\end{aligned}
$$

Then by Lemma 13,

$$
\mathbb{E}\big(D^mS_n(f,t)(f_1,\ldots,f_m)\big) = \mathbb{E}\left\{R(t)e^{f(X)}\prod_{j=1}^m f_j(X)\right\} = \int_0^1 q(x,t)e^{f(x)}\prod_{j=1}^m f_j(x)\,\mathrm{d}P_X(x).
$$

It remains to find the Gateaux derivative $D^m S(f,t)(f_1,\dots,f_m)$. By the induction hypothesis and the dominated convergence theorem, for $\varepsilon \in \mathbb{R}\setminus\{0\}$,

$$\begin{aligned}&\frac{1}{\varepsilon}\big(D^{m-1}S(f+\varepsilon f_m,t)(f_1,\dots,f_{m-1}) - D^{m-1}S(f,t)(f_1,\dots,f_{m-1})\big)\\ &\quad= \frac{1}{\varepsilon}\bigg(\int_{\mathcal{X}} q(x,t)e^{f(x)}\big(e^{\varepsilon f_m(x)}-1\big)\prod_{j=1}^{m-1} f_j(x)\,\mathrm{d}P_X(x)\bigg) \to \int_{\mathcal{X}} q(x,t)e^{f(x)}\prod_{j=1}^{m} f_j(x)\,\mathrm{d}P_X(x)\end{aligned}$$

as $\varepsilon\to 0$, as required. □

Lemma 16 (Gateaux derivatives of $\ell_\star$)**.** *Let $f, f_1, f_2 \in \mathcal{B}(\mathcal{X})$. Then the first-order Gateaux derivative of $\ell_\star(f)$ in the direction of f_1 exists and is given by*

$$D\ell_\star(f)(f_1) = \int_0^1 DS(f,t)(f_1)\frac{S(f_0,t)}{S(f,t)}\lambda_0(t)\,\mathrm{d}t - \int_0^1 DS(f_0,t)(f_1)\lambda_0(t)\,\mathrm{d}t.$$

Similarly, the second-order Gateaux derivative of $\ell_\star(f)$ in the direction of (f_1,f_2) is

$$D^2\ell_\star(f)(f_1,f_2) = \int_0^1\bigg(\frac{D^2S(f,t)(f_1,f_2)}{S(f,t)} - \frac{DS(f,t)(f_1)}{S(f,t)}\frac{DS(f,t)(f_2)}{S(f,t)}\bigg)S(f_0,t)\lambda_0(t)\,\mathrm{d}t.$$

In each display we interpret $0/0$ to be zero.

Proof of Lemma 16. By Lemma 15,

$$\ell_\star(f) = \int_0^1 \log\big(S(f,t)\big)S(f_0,t)\lambda_0(t)\,\mathrm{d}t - \int_0^1\int_{\mathcal{X}} q(x,t)e^{f_0(x)}f(x)\,\mathrm{d}P_X(x)\lambda_0(t)\,\mathrm{d}t.$$

Note that $\int_0^1\lambda_0(t)\,\mathrm{d}t = \Lambda < \infty$. Hence by the dominated convergence theorem, since $f, f_0, f_1 \in \mathcal{B}(\mathcal{X})$, and as $S(f_0,t)\le e^{\|\tilde f - f_0\|_\infty}S(\tilde f,t)$ for all $\tilde f\in\mathcal{B}(\mathcal{X})$, we have

$$\begin{aligned}\frac{1}{\varepsilon}\big(\ell_\star(f+\varepsilon f_1)-\ell_\star(f)\big) &= \frac{1}{\varepsilon}\int_0^1\big(\log S(f+\varepsilon f_1,t)-\log S(f,t)\big)S(f_0,t)\lambda_0(t)\,\mathrm{d}t\\ &\qquad - \int_0^1\int_{\mathcal{X}} q(x,t)e^{f_0(x)}f_1(x)\,\mathrm{d}P_X(x)\lambda_0(t)\,\mathrm{d}t\\ &\to \int_0^1\frac{DS(f,t)(f_1)}{S(f,t)}S(f_0,t)\lambda_0(t)\,\mathrm{d}t - \int_0^1 DS(f_0,t)(f_1)\lambda_0(t)\,\mathrm{d}t\end{aligned}$$

as $\varepsilon\to 0$. Similarly, for the second derivative, again by the dominated convergence theorem,

$$\begin{aligned}&\frac{1}{\varepsilon}\big(D\ell_\star(f+\varepsilon f_2)(f_1) - D\ell_\star(f)(f_1)\big)\\ &\qquad= \frac{1}{\varepsilon}\int_0^1\bigg(\frac{DS(f+\varepsilon f_2,t)(f_1)}{S(f+\varepsilon f_2,t)} - \frac{DS(f,t)(f_1)}{S(f,t)}\bigg)S(f_0,t)\lambda_0(t)\,\mathrm{d}t\\ &\qquad\to \int_0^1\bigg(\frac{D^2S(f,t)(f_1,f_2)}{S(f,t)} - \frac{DS(f,t)(f_1)}{S(f,t)}\frac{DS(f,t)(f_2)}{S(f,t)}\bigg)S(f_0,t)\lambda_0(t)\,\mathrm{d}t\end{aligned}$$

as $\varepsilon\to 0$, as required. □

E.1.2 Proofs of Lemmas 1 and 2

The proof of Lemma 1 relies on the following restricted strong convexity result for $\ell_\star$.

Lemma 17 (Restricted strong convexity of $\ell_\star$)**.** *If $f, f_1 \in \mathcal{B}(\mathcal{X})$, then*

$$e^{-\|f-f_0\|_\infty} e^{-\|f\|_\infty} \Lambda q_1 \|f_1 - P_X(f_1)\|_{L_2}^2 \le D^2\ell_\star(f)(f_1,f_1) \le e^{\|f-f_0\|_\infty} e^{\|f\|_\infty} \Lambda \|f_1\|_{L_2}^2.$$

Proof of Lemma 17. For the upper bound, observe by Lemma 15 that

$$D^2 S(f,t)(f_1,f_1) = \int_{\mathcal{X}} q(x,t) e^{f(x)} f_1(x)^2 \,\mathrm{d}P_X(x) \le e^{\|f\|_\infty} \|f_1\|_{L_2}^2,$$

so by Lemma 16,

$$D^2\ell_\star(f)(f_1,f_1) \le \int_0^1 \frac{D^2 S(f,t)(f_1,f_1)}{S(f,t)} S(f_0,t)\lambda_0(t)\,\mathrm{d}t \le e^{\|f-f_0\|_\infty} e^{\|f\|_\infty} \Lambda \|f_1\|_{L_2}^2.$$

Since $\ell_\star(f) = \ell_\star(f + c1_{\mathcal{X}})$ for all $c \in \mathbb{R}$, we have

$$D^2\ell_\star(f)(f_1,f_1) = D^2\ell_\star(f)(f_2,f_2),$$

where $f_2 := f_1 - P_X(f_1)1_{\mathcal{X}}$. For the lower bound, by Lemma 16 and Lemma 15 again,

$$\begin{aligned} D^2\ell_\star(f)(f_2,f_2) &= \int_0^1 \left(\frac{D^2 S(f,t)(f_2,f_2)}{S(f,t)} - \frac{DS(f,t)(f_2)^2}{S(f,t)^2} \right) S(f_0,t)\lambda_0(t)\,\mathrm{d}t \\ &= \int_0^1 \int_{\mathcal{X}} \left(f_2(x) - \frac{DS(f,t)(f_2)}{S(f,t)} \right)^2 e^{f(x)} q(x,t)\,\mathrm{d}P_X(x)\, \frac{S(f_0,t)}{S(f,t)} \lambda_0(t)\,\mathrm{d}t. \end{aligned}$$

Since $P_X(f_2) = 0$, we have $\int_{\mathcal{X}} \big(f_2(x) - c\big)^2 \,\mathrm{d}P_X(x) \ge \|f_2\|_{L_2}^2$ for every $c \in \mathbb{R}$, so because $S(f_0,t) \ge e^{-\|f-f_0\|_\infty} S(f,t)$, we have

$$D^2\ell_\star(f)(f_2,f_2) \ge e^{-\|f-f_0\|_\infty} e^{-\|f\|_\infty} \Lambda q_1 \|f_2\|_{L_2}^2. \qquad \square$$

Proof of Lemma 1. By Lemma 16, $D\ell_\star(f_0) = 0$. Let $f \in \mathcal{B}(\mathcal{X})$. By Taylor's theorem and Lemma 17, as $P_X(f_0) = 0$, there exists $\check{f}$ on the line segment between f_0 and f with

$$\ell_\star(f) - \ell_\star(f_0) = D^2\ell(\check{f})(f-f_0)^{\otimes 2} \ge e^{-\|\check{f}-f_0\|_\infty} e^{-\|\check{f}\|_\infty} \Lambda q_1 \|f - P_X(f) - f_0\|_{L_2}^2 \ge 0, \tag{13}$$

so $f_0 \in \operatorname{argmin}_{f \in \mathcal{B}(\mathcal{X})} \ell_\star(f)$. Moreover, if $\tilde{f} \in \operatorname{argmin}_{f \in \mathcal{B}(\mathcal{X})} \ell_\star(f)$ with $P_X(\tilde{f}) = 0$, then from (13), we have $\|\tilde{f} - f_0\|_{L_2} = 0$, so $\tilde{f}(x) = f_0(x)$ holds for P_X-almost every $x \in \mathcal{X}$. $\square$

Proof of Lemma 2. By Cauchy–Schwarz, for $x \in \mathcal{X}$,

$$\begin{aligned} f(x)^2 &= \left(\sum_{r \in \mathcal{R}} \langle f, e_r \rangle_{L_2} e_r(x) \right)^2 = \left(\sum_{r \in \mathcal{R}} \frac{1}{\sqrt{\nu_r}} \langle f, e_r \rangle_{L_2} \cdot \sqrt{\nu_r} e_r(x) \right)^2 \\ &\le \|f\|_{\mathcal{H}}^2 \sum_{r \in \mathcal{R}} \nu_r e_r(x)^2 = \|f\|_{\mathcal{H}}^2 k(x,x) \le K \|f\|_{\mathcal{H}}^2. \end{aligned}$$

Similarly,

$$f(x)^2 = \left(\sum_{r \in \mathcal{R}} \frac{1}{\sqrt{a_r}} \langle f, e_r \rangle_{L_2} \cdot \sqrt{a_r} e_r(x) \right)^2 \le \|f\|_{\mathcal{H}_\gamma}^2 \sum_{r \in \mathcal{R}} a_r e_r(x)^2 \le H_{\infty,\gamma} \|f\|_{\mathcal{H}_\gamma}^2. \qquad \square$$

E.2 Proofs for Section 3

We prove our result on representation of the kernel estimator $\hat{f}_{n,\gamma}$.

Proof of Proposition 3. We treat $(X_i, T_i, I_i)_{i\in[n]}$ as fixed for now; measurability issues are addressed later. For $\gamma > 0$ and $f \in \mathcal{H}$, define $\check{\ell}_{n,\gamma}(f) := \ell_n(f) + \gamma\|f - P_n(f)1_{\mathcal{X}}\|_{\mathcal{H}}^2 + \gamma P_n(f)^2$. Since $f \mapsto \ell_n(f)$ is invariant to addition of constants to f, the optimisation problem (4) is equivalent to solving

$$\hat{f}_{n,\gamma} := \operatorname*{argmin}_{f\in\mathcal{H}} \check{\ell}_{n,\gamma}(f). \tag{14}$$

Define the subspace $\mathcal{H}_{\mathbf{X}} := \operatorname{span}\{1_{\mathcal{X}}, k(\cdot, X_1), \ldots, k(\cdot, X_n)\}$ of $\mathcal{H}$, which is finite-dimensional and hence closed. Further define $\pi_{\mathbf{X}} : \mathcal{H} \to \mathcal{H}_{\mathbf{X}}$ to be the projection operator onto $\mathcal{H}_{\mathbf{X}}$ under $\langle\cdot,\cdot\rangle_{\mathcal{H}}$, and for $f \in \mathcal{H}$ write $f_{\mathbf{X}} := \pi_{\mathbf{X}}(f)$. By the reproducing property and orthogonality, $f(X_i) - f_{\mathbf{X}}(X_i) = \langle f - f_{\mathbf{X}}, k(\cdot, X_i)\rangle_{\mathcal{H}} = 0$ for each $i \in [n]$. Thus, $\ell_n(f_{\mathbf{X}}) = \ell_n(f)$ and $P_n(f_{\mathbf{X}}) = P_n(f)$. Hence, since $\pi_{\mathbf{X}}(1_{\mathcal{X}}) = 1_{\mathcal{X}}$, we have $\pi_{\mathbf{X}}\big(f - P_n(f)1_{\mathcal{X}}\big) = f_{\mathbf{X}} - P_n(f_{\mathbf{X}})1_{\mathcal{X}}$. It follows that

$$\|f - P_n(f)1_{\mathcal{X}}\|_{\mathcal{H}}^2 = \|f_{\mathbf{X}} - P_n(f_{\mathbf{X}})1_{\mathcal{X}}\|_{\mathcal{H}}^2 + \|f - f_{\mathbf{X}}\|_{\mathcal{H}}^2,$$

so $\check{\ell}_{n,\gamma}(f_{\mathbf{X}}) \le \check{\ell}_{n,\gamma}(f)$, with equality if and only if $f \in \mathcal{H}_{\mathbf{X}}$. But every $f \in \mathcal{H}_{\mathbf{X}}$ is of the form $f(x) = \beta_0 + \sum_{i=1}^n k(x, X_i)\beta_i$, where $\beta_i \in \mathbb{R}$ for $i \in \{0\} \cup [n]$. Since $\ell_n(f) = \ell_n(f + a1_{\mathcal{X}})$ and $f + a1_{\mathcal{X}} - P_n(f + a1_{\mathcal{X}}) = f - P_n(f)$ for all $a \in \mathbb{R}$, we deduce that any minimiser f of (14) must satisfy $P_n(f) = 0$ and $f \in \mathcal{H}_{\mathbf{X}}$.

Define $\tilde{\mathcal{H}}_{\mathbf{X}} := \big\{f \in \mathcal{H} : f(x) = \beta_0 + \sum_{i\in\mathcal{A}_n} k(x, X_i)\beta_i, P_n(f) = 0, \beta \in \mathbb{R}^{\{0\}\cup\mathcal{A}_n}\big\}$. Our final reduction is to show that it suffices to consider functions $f \in \tilde{\mathcal{H}}_{\mathbf{X}}$. Indeed, take $\delta \in \mathbb{R}^{\{0\}\cup[n]}$ such that the function $f \in \mathcal{H}_{\mathbf{X}}$ defined by $f(x) := \delta_0 + \sum_{i=1}^n k(x, X_i)\delta_i$ satisfies $P_n(f) = 0$. Then with $\tilde{\boldsymbol{K}}$ defined as in (10), by the reproducing property,

$$\begin{aligned}\|f - P_n(f)\|_{\mathcal{H}}^2 = \|f\|_{\mathcal{H}}^2 &= \Big\langle \delta_0 1_{\mathcal{X}} + \sum_{i=1}^n k(\cdot, X_i)\delta_i,\ \delta_0 1_{\mathcal{X}} + \sum_{i=1}^n k(\cdot, X_i)\delta_i \Big\rangle_{\mathcal{H}} \\ &= \delta_0^2\|1_{\mathcal{X}}\|_{\mathcal{H}}^2 + 2\delta_0\sum_{i=1}^n \delta_i + \sum_{i=1}^n\sum_{j=1}^n \delta_i\delta_j k(X_i, X_j) = \delta^{\mathsf{T}}\tilde{\boldsymbol{K}}\delta.\end{aligned}$$

Now, by definition of $\mathcal{A}_n$, there exist $\beta_i \in \mathbb{R}$ for $i \in \{0\} \cup \mathcal{A}_n$ such that

$$\begin{aligned}\beta_0\|1_{\mathcal{X}}\|_{\mathcal{H}}^2 + \sum_{i\in\mathcal{A}_n}\beta_i &= \delta_0\|1_{\mathcal{X}}\|_{\mathcal{H}}^2 + \sum_{i=1}^n \delta_i, \\ \beta_0 + \sum_{j\in\mathcal{A}_n} k\big(X_i, X_j\big)\beta_j &= \delta_0 + \sum_{j=1}^n k(X_i, X_j)\delta_j \qquad \text{for all } i \in [n].\end{aligned}$$

Thus, with $\tilde{f} \in \tilde{\mathcal{H}}_{\mathbf{X}}$ of the form $\tilde{f}(\cdot) := \beta_0 + \sum_{i\in\mathcal{A}_n} k(\cdot, X_i)\beta_i$, we have $\tilde{f}(X_i) = f(X_i)$ for all

$i \in [n]$ and so $\ell_n(\tilde{f}) = \ell_n(f)$. Further, $P_n(\tilde{f}) = P_n(f) = 0$, so

$$\begin{aligned}
\big\|\tilde{f} - P_n(\tilde{f})\big\|_{\mathcal{H}}^2 = \|\tilde{f}\|_{\mathcal{H}}^2 &= \beta_0^2 \|1_{\mathcal{X}}\|_{\mathcal{H}}^2 + 2\beta_0 \sum_{i \in \mathcal{A}_n} \beta_i + \sum_{i \in \mathcal{A}_n} \sum_{j \in \mathcal{A}_n} \beta_i \beta_j k(X_i, X_j) \\
&= \beta_0 \bigg(\beta_0 \|1_{\mathcal{X}}\|_{\mathcal{H}}^2 + \sum_{i \in \mathcal{A}_n} \beta_i \bigg) + \sum_{i \in \mathcal{A}_n} \beta_i \bigg(\beta_0 + \sum_{j \in \mathcal{A}_n} k(X_i, X_j) \beta_j \bigg) \\
&= \beta_0 \bigg(\delta_0 \|1_{\mathcal{X}}\|_{\mathcal{H}}^2 + \sum_{i=1}^n \delta_i \bigg) + \sum_{i \in \mathcal{A}_n} \beta_i \bigg(\delta_0 + \sum_{j=1}^n k(X_i, X_j) \delta_j \bigg) \\
&= \delta_0 \bigg(\beta_0 \|1_{\mathcal{X}}\|_{\mathcal{H}}^2 + \sum_{i \in \mathcal{A}_n} \beta_i \bigg) + \sum_{i=1}^n \delta_i \bigg(\beta_0 + \sum_{j \in \mathcal{A}_n} k(X_i, X_j) \beta_j \bigg) \\
&= \delta_0 \bigg(\delta_0 \|1_{\mathcal{X}}\|_{\mathcal{H}}^2 + \sum_{i=1}^n \delta_i \bigg) + \sum_{i=1}^n \delta_i \bigg(\delta_0 + \sum_{j=1}^n k(X_i, X_j) \delta_j \bigg) \\
&= \delta^{\mathsf{T}} \tilde{\boldsymbol{K}} \delta = \|f - P_n(f)\|_{\mathcal{H}}^2. \qquad (15)
\end{aligned}$$

Thus, $\check{\ell}_{n,\gamma}(\tilde{f}) = \check{\ell}_{n,\gamma}(f)$, so we may indeed restrict the feasible set in (14) to $f \in \tilde{\mathcal{H}}_{\mathbf{X}}$. Since $P_n(f) = 0$ for such a function f, we may write it as $f(\cdot) = \sum_{i \in \mathcal{A}_n} \tilde{k}(\cdot, X_i)\beta_i$.

Next, we show that the objective in (14), and hence also in (4), is strongly convex as a function of $\beta \in \mathbb{R}^{\mathcal{A}_n}$. Indeed, if $\boldsymbol{A}$ is a $d \times d$ positive semi-definite matrix whose first k columns $\boldsymbol{B}_k \in \mathbb{R}^{d \times k}$ are linearly independent, then $\boldsymbol{A}_k \succ 0$, where $\boldsymbol{A}_k$ is the leading $k \times k$ principal submatrix of $\boldsymbol{A}$. To see this, suppose that $\boldsymbol{A}_k u = 0$ for some $u = (u_1, \ldots, u_k)^{\mathsf{T}} \in \mathbb{R}^k$, and define $v := (u_1, \ldots, u_k, 0, \ldots, 0)^{\mathsf{T}} \in \mathbb{R}^d$. Then $v^{\mathsf{T}} \boldsymbol{A} v = u^{\mathsf{T}} \boldsymbol{A}_k u = 0$, so $0 = \boldsymbol{A} v = \boldsymbol{B}_k u$ and hence $u = 0$. Thus, with $\boldsymbol{K}_{\mathcal{A}_n} \in \mathbb{R}^{\mathcal{A}_n \times \mathcal{A}_n}$ defined by $(\boldsymbol{K}_{\mathcal{A}_n})_{ij} := \boldsymbol{K}_{ij}$ for $i, j \in \mathcal{A}_n$ and setting

$$\tilde{\boldsymbol{K}}_{\mathcal{A}_n} := \begin{pmatrix} \|1_{\mathcal{X}}\|_{\mathcal{H}}^2 & 1_{\mathcal{A}_n}^{\mathsf{T}} \\ 1_{\mathcal{A}_n} & \boldsymbol{K}_{\mathcal{A}_n} \end{pmatrix} \in \mathbb{R}^{(\{0\} \cup \mathcal{A}_n) \times (\{0\} \cup \mathcal{A}_n)},$$

we have $\tilde{\boldsymbol{K}}_{\mathcal{A}_n} \succ 0$. Therefore, by (15),

$$\beta \equiv (\beta_i)_{i \in \{0\} \cup \mathcal{A}_n} \mapsto \bigg\| \beta_0 + \sum_{i \in \mathcal{A}_n} k(\cdot, X_i)\beta_i \bigg\|_{\mathcal{H}}^2 = \beta^{\mathsf{T}} \tilde{\boldsymbol{K}}_{\mathcal{A}_n} \beta \qquad (16)$$

is a strongly convex quadratic function on $\mathbb{R}^{\{0\} \cup \mathcal{A}_n}$. Now, the linear map from $\mathbb{R}^{\mathcal{A}_n}$ to $\mathbb{R}^{\{0\} \cup \mathcal{A}_n}$ defined by $(\beta_i)_{i \in \mathcal{A}_n} \mapsto (\beta_i)_{i \in \{0\} \cup \mathcal{A}_n}$ with $\beta_0 := -\sum_{i \in \mathcal{A}_n} \bar{k}(X_i)\beta_i$ has (full) rank $|\mathcal{A}_n|$. Hence by (16), the function $(\beta_i)_{i \in \mathcal{A}_n} \mapsto \big\| \sum_{i \in \mathcal{A}_n} \tilde{k}(\cdot, X_i)\beta_i \big\|_{\mathcal{H}}^2$ is a strongly convex quadratic on $\mathbb{R}^{\mathcal{A}_n}$. Moreover, the negative log-partial likelihood ℓ_n is convex by Lemma 14. Since $\gamma > 0$,

$$(\beta_i)_{i \in \mathcal{A}_n} \mapsto \ell_n \bigg(\sum_{i \in \mathcal{A}_n} \tilde{k}(\cdot, X_i)\beta_i \bigg) + \gamma \bigg\| \sum_{i \in \mathcal{A}_n} \tilde{k}(\cdot, X_i)\beta_i \bigg\|_{\mathcal{H}}^2$$

is strongly convex, so has a unique minimiser $(\hat{\beta}_i)_{i \in \mathcal{A}_n}$. Measurability of $\hat{\beta}$ follows since it is the limit of a sequence of iterates produced by gradient descent started from the origin with step sizes chosen by backtracking line search (Lange, 2013, Proposition 12.6.1).

For the final part, let $f_1, f_2 \in \operatorname{argmin}_{f\in\mathcal{H}} \ell_{n,\gamma}(f)$ with $P_n(f_1) = P_n(f_2) = 0$. Define $g : [0,1] \to \mathbb{R}$ by $g(a) := \ell_{n,\gamma}\big((1-a)f_1 + af_2\big)$, which is convex by Lemma 14. Since $g(0) = g(1)$, we have $g(a) = g(0)$ for all $a \in [0,1]$. Thus, it follows that $0 = g''(a) = D^2\ell_n\big(af_1 + (1-a)f_2\big)(f_2 - f_1)^{\otimes 2} + \gamma\|f_1 - f_2\|_{\mathcal{H}}^2$, and hence $f_1 = f_2$ by Lemma 14 again. □

E.3 Proofs for Section 4

We now present the proofs of our main theoretical results. First, we define a projected estimator $\tilde{f}_{n,\gamma}$ and derive the basic inequality (17).

E.3.1 Projection and the basic inequality

For a fixed $b \geq 1$ to be determined, let

$$\tilde{f}_{n,\gamma} := f_0 - P_n(f_0)1_{\mathcal{X}} + \big(\hat{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}}\big)\left\{1 \wedge \frac{1}{b\sqrt{H_{\infty,\gamma}}\big\|\hat{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}}\big\|_{\mathcal{H}_\gamma}}\right\}.$$

Then $\big\|\tilde{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}}\big\|_{\mathcal{H}_\gamma}^2 \leq 1/(b^2 H_{\infty,\gamma})$ and hence $\big\|\tilde{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}}\big\|_\infty \leq 1/b$ by Lemma 2. Now $\tilde{f}_{n,\gamma}$ lies on the line segment between $f_0 - P_n(f_0)1_{\mathcal{X}}$ and $\hat{f}_{n,\gamma}$, so $P_n(\tilde{f}_{n,\gamma}) = 0$ by Proposition 3. Since $\ell_{n,\gamma}$ is convex by Lemma 14, by definition of $\hat{f}_{n,\gamma}$,

$$\ell_{n,\gamma}\big(\tilde{f}_{n,\gamma}\big) \leq \ell_{n,\gamma}\big(f_0 - P_n(f_0)1_{\mathcal{X}}\big) \vee \ell_{n,\gamma}\big(\hat{f}_{n,\gamma}\big) \leq \ell_{n,\gamma}\big(f_0 - P_n(f_0)1_{\mathcal{X}}\big).$$

Since ℓ_n is invariant under addition of constants, we deduce that

$$0 \geq \ell_n\big(\tilde{f}_{n,\gamma}\big) - \ell_n(f_0) + \gamma\big\|\tilde{f}_{n,\gamma}\big\|_{\mathcal{H}}^2 - \gamma\big\|f_0 - P_n(f_0)1_{\mathcal{X}}\big\|_{\mathcal{H}}^2.$$

For $f_1, f_2 \in \mathcal{H}$, we have $\|f_1\|_{\mathcal{H}}^2 - \|f_2\|_{\mathcal{H}}^2 = \|f_1 - f_2\|_{\mathcal{H}}^2 + 2\langle f_1 - f_2, f_2\rangle_{\mathcal{H}}$. By Taylor's theorem, there is a function $\check{f}_{n,\gamma}$ on the segment between $f_0 - P_n(f_0)1_{\mathcal{X}}$ and $\tilde{f}_{n,\gamma}$ such that

$$\begin{aligned} 0 \geq{}& D\ell_n(f_0)\big(\tilde{f}_{n,\gamma} - f_0\big) + \frac{1}{2}D^2\ell_n(f_0)\big(\tilde{f}_{n,\gamma} - f_0\big)^{\otimes 2} + \frac{1}{6}D^3\ell_n\big(\check{f}_{n,\gamma}\big)\big(\tilde{f}_{n,\gamma} - f_0\big)^{\otimes 3} \\ &+ \gamma\big\|\tilde{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}}\big\|_{\mathcal{H}}^2 + 2\gamma\big\langle\tilde{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}}, f_0 - P_n(f_0)1_{\mathcal{X}}\big\rangle_{\mathcal{H}}. \end{aligned} \tag{17}$$

E.3.2 Convergence of S_n, DS_n and D^2S_n

Lemmas 18 and 19 are auxiliary results which are applied repeatedly in subsequent proofs. In Lemmas 20, 21 and 22, we show convergence of S_n, DS_n and D^2S_n respectively.

Lemma 18 (A Cauchy–Schwarz inequality for integrals)**.** *Let $h : [0,1] \to \mathbb{R}$ be a function of bounded variation. Given a countable index set $\mathcal{R}$, suppose that $g_r : [0,1] \to \mathbb{R}$ for $r \in \mathcal{R}$ satisfy $\int_0^1 |g_r(t)|\,|\mathrm{d}h(t)| < \infty$. Then*

$$\sum_{r\in\mathcal{R}}\left(\int_0^1 g_r(t)\,\mathrm{d}h(t)\right)^2 \leq \left(\int_0^1 |\mathrm{d}h(t)|\right)^2 \sup_{t\in[0,1]}\sum_{r\in\mathcal{R}} g_r^2(t).$$

Proof of Lemma 18. By the Cauchy–Schwarz inequality and Fubini's theorem,

$$\sum_{r\in\mathcal{R}}\left(\int_0^1 g_r(t)\,\mathrm{d}h(t)\right)^2 \le \sum_{r\in\mathcal{R}}\int_0^1 g_r^2(t)\,|\mathrm{d}h(t)|\cdot\int_0^1|\mathrm{d}h(t)|$$
$$\le \left(\int_0^1|\mathrm{d}h(t)|\right)^2 \sup_{t\in[0,1]}\sum_{r\in\mathcal{R}} g_r^2(t). \qquad \square$$

Lemma 19 (Operator norm bounds)**.** *Let $G:\mathcal{H}\to\mathbb{R}$ be a linear operator and $\gamma\ge 0$. Define $\|G\|_{\mathcal{H},\mathrm{op}} := \sup_{f\in\mathcal{H}:\|f\|_{\mathcal{H}}=1}|G(f)|$ and $\|G\|_{\mathcal{H}_\gamma,\mathrm{op}} := \sup_{f\in\mathcal{H}:\|f\|_{\mathcal{H}_\gamma}=1}|G(f)|$. Then*

$$\|G\|_{\mathcal{H},\mathrm{op}}^2 \le \sum_{r\in\mathcal{R}}\nu_r G(e_r)^2, \qquad \|G\|_{\mathcal{H}_\gamma,\mathrm{op}}^2 \le \sum_{r\in\mathcal{R}} a_r G(e_r)^2.$$

Proof of Lemma 19. By linearity of G and Cauchy–Schwarz, for any $f\in\mathcal{H}$,

$$G(f)^2 = \left(\sum_{r\in\mathcal{R}}\langle f,e_r\rangle_{L_2}G(e_r)\right)^2 = \left(\sum_{r\in\mathcal{R}}\frac{1}{\sqrt{\nu_r}}\langle f,e_r\rangle_{L_2}\cdot\sqrt{\nu_r}G(e_r)\right)^2 \le \|f\|_{\mathcal{H}}^2\sum_{r\in\mathcal{R}}\nu_r G(e_r)^2.$$

Similarly,

$$G(f)^2 = \left(\sum_{r\in\mathcal{R}}\frac{1}{\sqrt{a_r}}\langle f,e_r\rangle_{L_2}\cdot\sqrt{a_r}G(e_r)\right)^2 \le \|f\|_{\mathcal{H}_\gamma}^2\sum_{r\in\mathcal{R}} a_r G(e_r)^2. \qquad \square$$

Lemma 20 (Convergence of S_n)**.** *There exists a universal constant $C>0$ such that for all $s\ge 1$ and $f\in\mathcal{B}(\mathcal{X})$, with probability at least $1-e^{-s^2}$,*

$$\sup_{t\in[0,1]}\left|S_n(f,t)-S(f,t)\right| \le \frac{Cse^{\|f\|_\infty}}{\sqrt{n}}.$$

Proof of Lemma 20. For $i\in[n]$ and $t\in[0,1]$, define

$$\xi_i(t) := R_i(t)e^{f(X_i)} - \int_{\mathcal{X}} q(x,t)e^{f(x)}\,\mathrm{d}P_X(x),$$

noting that $|\xi_i(t)|\le 2e^{\|f\|_\infty}$. It follows that with $\mathcal{I} := \{(i,j): i,j\in[n], i<j\}$,

$$\sup_{t\in[0,1]}\big(S_n(f,t)-S(f,t)\big)^2 = \sup_{t\in[0,1]}\frac{1}{n^2}\sum_{i=1}^n\sum_{j=1}^n\xi_i(t)\xi_j(t)$$
$$\le \sup_{t\in[0,1]}\frac{1}{n^2}\sum_{i=1}^n\xi_i(t)^2 + \sup_{t\in[0,1]}\frac{2}{n^2}\sum_{(i,j)\in\mathcal{I}}\xi_i(t)\xi_j(t)$$
$$\le \frac{4e^{2\|f\|_\infty}}{n} + \sup_{t\in[0,1]}\frac{2}{n(n-1)}\left|\sum_{(i,j)\in\mathcal{I}}\xi_i(t)\xi_j(t)\right|. \tag{18}$$

The second term in the above display is a degenerate U-process (de la Peña and Giné, 1999, Chapter 5) as $\mathbb{E}\{\xi_i(t)\xi_j(t)\mid X_i,T_i,I_i\} = \xi_i(t)\,\mathbb{E}\{\xi_j(t)\} = 0$ for $(i,j)\in\mathcal{I}$ and $t\in[0,1]$. We

now let $\mathcal{Q}$ denote the set of finitely-supported probability measures on $(\mathcal{X} \times [0,1] \times \{0,1\})^2$, and for $Q \in \mathcal{Q}$, seek to bound the $L_2(Q)$-covering number (see Definition 46) of the class

$$\mathcal{G} = \Big\{ \big((X_i, T_i, I_i), (X_j, T_j, I_j)\big) \mapsto \xi_i(t)\xi_j(t) : t \in [0,1] \Big\}.$$

For $0 \le t < t' \le 1$ and $i, j \in [n]$,

$$|\xi_i(t)||\xi_j(t) - \xi_j(t')| \le 2e^{2\|f\|_\infty} \big\{ \mathbb{1}_{\{t \le T_j < t'\}} + \mathbb{P}(t \le T < t') \big\}.$$

As such, we deduce that

$$\begin{aligned} |\xi_i(t)\xi_j(t) - \xi_i(t')\xi_j(t')| &\le |\xi_i(t)||\xi_j(t) - \xi_j(t')| + |\xi_j(t')||\xi_i(t) - \xi_i(t')| \\ &\le 2e^{2\|f\|_\infty} \big\{ \mathbb{1}_{\{t \le T_i < t'\}} + \mathbb{1}_{\{t \le T_j < t'\}} + 2\mathbb{P}(t \le T < t') \big\}. \end{aligned}$$

It follows that

$$\begin{aligned} &\mathbb{E}_Q \Big[\big(\xi_i(t)\xi_j(t) - \xi_i(t')\xi_j(t') \big)^2 \Big] \\ &\quad \le 12e^{4\|f\|_\infty} \big\{ \mathbb{P}_Q(t \le T_i < t') + \mathbb{P}_Q(t \le T_j < t') + 4\mathbb{P}(t \le T < t')^2 \big\}. \end{aligned}$$

Given $\varepsilon \in (0,1)$, define $\mathcal{T}_1 := \{t_1, \dots, t_{\lceil 1/\varepsilon^2 \rceil}\}$, where

$$t_m := \inf\{ t \in [0,1] : \mathbb{P}_Q(T_i \le t) \ge m\varepsilon^2 \}$$

for $m \in [\lceil 1/\varepsilon^2 \rceil - 1]$ and $t_{\lceil 1/\varepsilon^2 \rceil} := 1$, so that given any $t \in [0,1]$, there exists $m \in [\lceil 1/\varepsilon^2 \rceil]$ such that $\mathbb{P}_Q(t \le T_i < t_m) \le \varepsilon^2$. Let $\mathcal{T}_2$ and $\mathcal{T}_3$ satisfy the same properties but for T_j under Q and T under P_T respectively. Then the set $\mathcal{T} = \mathcal{T}_1 \cup \mathcal{T}_2 \cup \mathcal{T}_3$ is a $\sqrt{72}e^{2\|f\|_\infty}\varepsilon$-cover of $\mathcal{G}$ under $L_2(Q)$, of cardinality at most $6/\varepsilon^2$. Defining $\varepsilon' := 3\varepsilon/\sqrt{2} \in (0, 3/\sqrt{2})$ and $M := 4e^{2\|f\|_\infty}$,

$$N\big(\varepsilon' M, \mathcal{G}, \|\cdot\|_{Q,2}\big) \le \frac{27}{\varepsilon'^2}.$$

Hence, since $\log(1+x) \le 1 + \log x$ for $x \ge 1$,

$$J := \sup_{Q \in \mathcal{Q}} \int_0^1 \log\big(1 + N\big(\varepsilon M, \mathcal{G}, \|\cdot\|_{Q,2}\big)\big)\, \mathrm{d}\varepsilon \le 1 + \int_0^1 \log(27/\varepsilon^2)\, \mathrm{d}\varepsilon = 1 + \log(27) + 2 \le 7.$$

It follows by Lemma 49 that there exists a universal constant $C_1 > 0$ such that for all $s \ge 1$,

$$\mathbb{P}\left(\sup_{t \in [0,1]} \frac{2}{n(n-1)} \left| \sum_{(i,j) \in \mathcal{I}} \xi_i(t)\xi_j(t) \right| > \frac{C_1 s e^{2\|f\|_\infty}}{n} \right) \le e^{-s}.$$

By combining this with the bound (18) on the diagonal elements, there exists a universal constant $C_2 > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,

$$\sup_{t \in [0,1]} \big(S_n(f,t) - S(f,t) \big)^2 \le \frac{C_2 s^2 e^{2\|f\|_\infty}}{n}. \qquad \square$$

Lemma 21 (Convergence of DS_n)**.** *Let $f \in \mathcal{B}(\mathcal{X})$. There exists a universal constant $C > 0$ such that for all $s \geq 1$, with probability at least $1 - e^{-s^2}$,*

$$\sup_{t\in[0,1]} \left\| DS_n(f,t) - DS(f,t) \right\|_{\mathcal{H}_\gamma,\mathrm{op}} \leq C e^{\|f\|_\infty} \left(\frac{\sqrt{H_{2,\gamma}} + s}{\sqrt{n}} + \frac{s^2 \sqrt{H_{\infty,\gamma}}}{n} \right).$$

Proof of Lemma 21. Let $\varepsilon_1, \ldots, \varepsilon_n$ be independent Rademacher variables that are independent of the data. Noting that

$$\sup_{f_1 \in \mathcal{H}:\, \|f_1\|_{\mathcal{H}_\gamma} \leq 1} \sup_{t\in[0,1]} \left| DS_n(f,t)(f_1) - DS(f,t)(f_1) \right|$$

is measurable by arguments similar to those in the proof of Lemma 29, by Lemmas 13 and 15, a symmetrisation inequality (van der Vaart and Wellner, 1996, Lemma 2.3.1), a contraction principle (van der Vaart and Wellner, 1996, Proposition A.1.10), Jensen's inequality and Cauchy–Schwarz, we have

$$\begin{aligned}
&\mathbb{E}\left(\sup_{f_1 \in \mathcal{H}:\, \|f_1\|_{\mathcal{H}_\gamma} \leq 1} \sup_{t\in[0,1]} \left| DS_n(f,t)(f_1) - DS(f,t)(f_1) \right| \right) \\
&\quad \leq 2\,\mathbb{E}\left(\sup_{f_1 \in \mathcal{H}:\, \|f_1\|_{\mathcal{H}_\gamma} \leq 1} \sup_{t\in[0,1]} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i R_i(t) e^{f(X_i)} f_1(X_i) \right| \right) \\
&\quad \leq 2e^{\|f\|_\infty} \mathbb{E}\left(\sup_{f_1 \in \mathcal{H}:\, \|f_1\|_{\mathcal{H}_\gamma} \leq 1} \sup_{t\in[0,1]} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i R_i(t) f_1(X_i) \right| \right) \\
&\quad \leq 2e^{\|f\|_\infty} \mathbb{E}\left(\sup_{(\alpha_r)_{r\in\mathcal{R}}:\, \sum_{r\in\mathcal{R}} \alpha_r^2/a_r \leq 1} \sup_{t\in[0,1]} \frac{1}{n} \sum_{i=1}^n \varepsilon_i R_i(t) \sum_{r\in\mathcal{R}} \alpha_r e_r(X_i) \right) \\
&\quad \leq 2e^{\|f\|_\infty} \left[\mathbb{E}\left\{ \sup_{(\alpha_r)_{r\in\mathcal{R}}:\, \sum_{r\in\mathcal{R}} \alpha_r^2/a_r \leq 1} \sup_{t\in[0,1]} \left(\sum_{r\in\mathcal{R}} \alpha_r \frac{1}{n} \sum_{i=1}^n \varepsilon_i R_i(t) e_r(X_i) \right)^2 \right\} \right]^{1/2} \\
&\quad \leq 2e^{\|f\|_\infty} \left[\mathbb{E}\left\{ \sup_{(\alpha_r)_{r\in\mathcal{R}}:\, \sum_{r\in\mathcal{R}} \alpha_r^2/a_r \leq 1} \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} a_r \left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i R_i(t) e_r(X_i) \right)^2 \sum_{r'\in\mathcal{R}} \frac{\alpha_{r'}^2}{a_{r'}} \right\} \right]^{1/2} \\
&\quad = 2e^{\|f\|_\infty} \left[\mathbb{E}\left\{ \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} a_r \left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i R_i(t) e_r(X_i) \right)^2 \right\} \right]^{1/2}. \qquad (19)
\end{aligned}$$

For $i \in [n]$, $r \in \mathcal{R}$ and $t \in [0,1]$, define $\xi_{ir}(t) := \varepsilon_i R_i(t) e_r(X_i)$, so that

$$\sup_{t\in[0,1]} \left| \sum_{r\in\mathcal{R}} a_r \xi_{ir}(t) \xi_{jr}(t) \right| \leq \left| \sum_{r\in\mathcal{R}} a_r e_r(X_i) e_r(X_j) \right| =: F(X_i, X_j).$$

Thus, with $\mathcal{I} := \{(i,j) : i,j \in [n], i < j\}$, we have

$$\begin{aligned}
&\sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} a_r \left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i R_i(t) e_r(X_i) \right)^2 \\
&\qquad = \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n a_r \xi_{ir}(t) \xi_{jr}(t) \\
&\qquad \le \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} \frac{1}{n^2} \sum_{i=1}^n a_r \xi_{ir}(t)^2 + \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} \frac{2}{n^2} \sum_{(i,j)\in\mathcal{I}} a_r \xi_{ir}(t) \xi_{jr}(t) \\
&\qquad \le \frac{1}{n^2} \sum_{i=1}^n F(X_i, X_i) + \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} \frac{2}{n^2} \sum_{(i,j)\in\mathcal{I}} a_r \xi_{ir}(t) \xi_{jr}(t). \qquad (20)
\end{aligned}$$

The first term in (20) satisfies

$$\mathbb{E}\left(\frac{1}{n^2} \sum_{i=1}^n F(X_1, X_1) \right) = \frac{1}{n} \mathbb{E}\big\{ F(X_1, X_1) \big\} = \frac{1}{n} \sum_{r\in\mathcal{R}} a_r \mathbb{E}\big\{ e_r(X_1)^2 \big\} = \frac{1}{n} \sum_{r\in\mathcal{R}} a_r = \frac{H_{2,\gamma}}{n}. \quad (21)$$

The second term in (20) is a degenerate second-order U-process (de la Peña and Giné, 1999, Chapter 5), since by the dominated convergence theorem, for each $(i,j) \in \mathcal{I}$ and $t \in [0,1]$,

$$\mathbb{E}\left(\sum_{r\in\mathcal{R}} a_r \xi_{ir}(t) \xi_{jr}(t) \;\middle|\; X_i, T_i, I_i, \varepsilon_i \right) = \sum_{r\in\mathcal{R}} a_r \xi_{ir}(t) \mathbb{E}\big\{ \xi_{jr}(t) \big\} = 0.$$

We now let $\mathcal{Q}$ denote the set of finitely-supported probability measures on $(\mathcal{X} \times [0,1] \times \{0,1\})^2$, and for $Q \in \mathcal{Q}$, seek to bound the $L_2(Q)$-covering number of the function class

$$\mathcal{G} = \left\{ \big((X_i, T_i, I_i, \varepsilon_i), (X_j, T_j, I_j, \varepsilon_j) \big) \mapsto \sum_{r\in\mathcal{R}} a_r \xi_{ir}(t) \xi_{jr}(t) : t \in [0,1] \right\}.$$

For $0 \le t \le t' \le 1$ and $i, j \in [n]$, observe that

$$\begin{aligned}
&\left\| \sum_{r\in\mathcal{R}} a_r \big\{ \xi_{ir}(t') \xi_{jr}(t') - \xi_{ir}(t) \xi_{jr}(t) \big\} \right\|_{Q,2}^2 \\
&\quad = \left\| \sum_{r\in\mathcal{R}} a_r \xi_{ir}(t') \big\{ \xi_{jr}(t') - \xi_{jr}(t) \big\} + \sum_{r\in\mathcal{R}} a_r \xi_{jr}(t) \big\{ \xi_{ir}(t') - \xi_{ir}(t) \big\} \right\|_{Q,2}^2 \\
&\quad = \left\| \Big(R_i(t') \big\{ R_j(t) - R_j(t') \big\} + R_j(t) \big\{ R_i(t) - R_i(t') \big\} \Big) \left| \sum_{r\in\mathcal{R}} a_r e_r(X_j) e_r(X_i) \right| \right\|_{Q,2}^2 \\
&\quad \le \left\| \big(\mathbb{1}_{\{t \le T_i < t'\}} + \mathbb{1}_{\{t \le T_j < t'\}} \big) F(X_i, X_j) \right\|_{Q,2}^2 \\
&\quad \le 2\, \mathbb{E}_Q \Big\{ \big(\mathbb{1}_{\{t \le T_i < t'\}} + \mathbb{1}_{\{t \le T_j < t'\}} \big) F(X_i, X_j)^2 \Big\} =: \mu(t, t').
\end{aligned}$$

Noting that $\mu(0,1) = 4\|F\|_{Q,2}^2$ and $t' \mapsto \mu(t,t')$ is increasing and left continuous for each $t \in [0,1]$, given $\varepsilon \in (0,1)$, define $\mathcal{T} := \{t_1, \ldots, t_{\lceil 1/\varepsilon^2 \rceil}\}$, where $t_m := \inf\{t \in [0,1] : \mu(0,t) \ge$

$4m\|F\|_{Q,2}^2\varepsilon^2\}$ for $m \in [\lceil 1/\varepsilon^2\rceil - 1]$ and $t_{\lceil 1/\varepsilon^2\rceil} := 1$. Moreover, $t \mapsto \mu(t,t')$ is decreasing and left continuous for each $t' \in [0,1]$; therefore, given any $t \in [0,1]$, there exists $m \in [\lceil 1/\varepsilon^2\rceil]$ such that $t_m \le t$ and $\mu(t_m, t) \le 4\|F\|_{Q,2}^2\varepsilon^2$. Thus $\mathcal{T}$ is a $2\|F\|_{Q,2}\varepsilon$-cover of $\mathcal{G}$ under $L_2(Q)$, of cardinality at most $2/\varepsilon^2$. Defining $\varepsilon' := 2\varepsilon \in (0,2)$, we deduce that

$$N\big(\varepsilon'\|F\|_{Q,2}, \mathcal{G}, \|\cdot\|_{Q,2}\big) \le \frac{8}{\varepsilon'^2}.$$

Hence, since $\log(1+x) \le 1 + \log x$ for $x \ge 1$,

$$\begin{aligned} J := \sup_{Q\in\mathcal{Q}} \int_0^1 \log\big\{1 + N\big(\varepsilon\|F\|_{Q,2}, \mathcal{G}, \|\cdot\|_{Q,2}\big)\big\}\,\mathrm{d}\varepsilon &\le 1 + \int_0^1 \log(8/\varepsilon^2)\,\mathrm{d}\varepsilon \\ &= 1 + \log 8 + 2 \le 6. \end{aligned}$$

Further, by independence and orthonormality of the eigenfunctions $(e_r)_{r\in\mathcal{R}}$, we have

$$\begin{aligned} \|F\|_{P_X,2}^2 &= \mathbb{E}\bigg\{\bigg(\sum_{r\in\mathcal{R}} a_r e_r(X_1)e_r(X_2)\bigg)^2\bigg\} \\ &= \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'}\Big(\mathbb{E}\big\{e_r(X_1)e_{r'}(X_1)\big\}\Big)^2 \le \sum_{r\in\mathcal{R}} a_r^2 \le \bigg(\sum_{r\in\mathcal{R}} a_r\bigg)^2 = H_{2,\gamma}^2. \end{aligned}$$

Applying Chen and Kato (2020, Corollary 5.6), there exists a universal constant $C > 0$ such that

$$\mathbb{E}\bigg\{\sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} \frac{2}{n^2} \sum_{(i,j)\in\mathcal{I}} a_r \xi_{ir}(t)\xi_{jr}(t)\bigg\} \le \frac{CJ\|F\|_{P_X,2}}{n} \le \frac{6CH_{2,\gamma}}{n}, \tag{22}$$

By (19), (20), (21) and (22), and by increasing the universal constant $C > 0$ if necessary, we deduce that

$$\mathbb{E}\bigg(\sup_{t\in[0,1]} \big\|DS_n(f,t) - DS(f,t)\big\|_{\mathcal{H}_\gamma,\mathrm{op}}\bigg) \le \frac{Ce^{\|f\|_\infty}\sqrt{H_{2,\gamma}}}{\sqrt{n}}.$$

It remains to control the stochastic fluctuations of this supremum above its expected value. To this end, by Lemma 2, we have

$$\begin{aligned} \sigma^2 &:= \sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\le 1} \sup_{t\in[0,1]} \sum_{i=1}^n \mathbb{E}\bigg\{\bigg(\frac{1}{n}R_i(t)e^{f(X_i)}f_1(X_i)\bigg)^2\bigg\} \\ &\le \sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\le 1} \frac{e^{2\|f\|_\infty}\|f_1\|_{L_2}^2}{n} \le \frac{e^{2\|f\|_\infty}}{n}, \\ M &:= \sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\le 1} \sup_{t\in[0,1]} \bigg|\frac{1}{n}R_i(t)e^{f(X_i)}f_1(X_i)\bigg| \\ &\le \sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\le 1} \frac{e^{\|f\|_\infty}\|f_1\|_\infty}{n} \le \frac{e^{\|f\|_\infty}\sqrt{H_{\infty,\gamma}}}{n}. \end{aligned}$$

By Bousquet's inequality (Boucheron et al., 2013, Theorem 12.5), we conclude that with probability at least $1 - e^{-s^2}$,

$$\sup_{t\in[0,1]} \left\| DS_n(f,t) - DS(f,t) \right\|_{\mathcal{H}_\gamma,\mathrm{op}} \le Ce^{\|f\|_\infty} \left(\frac{\sqrt{H_{2,\gamma}} + s}{\sqrt{n}} + \frac{s^2\sqrt{H_{\infty,\gamma}}}{n} \right),$$

as required. □

Lemma 22 (Convergence of D^2S_n)**.** *Let $f \in \mathcal{B}(\mathcal{X})$. There is a universal constant $C > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,*

$$\sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \left(D^2S_n(f,t)(e_r,e_{r'}) - D^2S(f,t)(e_r,e_{r'}) \right)^2 \le \frac{Cs^2 e^{2\|f\|_\infty} H_{\infty,\gamma}^2}{n}.$$

Proof of Lemma 22. For $i \in [n]$ and $r, r' \in \mathcal{R}$, define

$$\xi_{irr'}(t) := R_i(t) e^{f(X_i)} e_r(X_i) e_{r'}(X_i) - \int_{\mathcal{X}} q(x,t) e^{f(x)} e_r(x) e_{r'}(x) \, \mathrm{d}P_X(x).$$

For every $t \in [0,1]$ and $i \in [n]$,

$$\begin{aligned}
\sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \xi_{irr'}(t)^2 &\le 2 \sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} R_i(t) e^{2f(X_i)} e_r(X_i)^2 e_{r'}(X_i)^2 \\
&\qquad + 2 \sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \left(\int_{\mathcal{X}} q(x,t) e^{f(x)} e_r(x) e_{r'}(x) \, \mathrm{d}P_X(x) \right)^2 \\
&\le 4e^{2\|f\|_\infty} H_{\infty,\gamma}^2 =: M. \qquad (23)
\end{aligned}$$

Therefore with $\mathcal{I} := \{(i,j) : i,j \in [n], i < j\}$, by Lemmas 13 and 15,

$$\begin{aligned}
&\sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \left(D^2S_n(f,t)(e_r,e_{r'}) - D^2S(f,t)(e_r,e_{r'}) \right)^2 \\
&= \sup_{t\in[0,1]} \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \xi_{irr'}(t) \xi_{jrr'}(t) \\
&\le \sup_{t\in[0,1]} \frac{1}{n^2} \sum_{i=1}^n \sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \xi_{irr'}(t)^2 + \sup_{t\in[0,1]} \frac{2}{n^2} \sum_{(i,j)\in\mathcal{I}} \sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \xi_{irr'}(t) \xi_{jrr'}(t) \\
&\le \frac{4e^{2\|f\|_\infty} H_{\infty,\gamma}^2}{n} + \sup_{t\in[0,1]} \left| \frac{2}{n(n-1)} \sum_{(i,j)\in\mathcal{I}} \sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \xi_{irr'}(t) \xi_{jrr'}(t) \right|. \qquad (24)
\end{aligned}$$

The second term above is a degenerate second-order U-process, since by (23), the Cauchy–Schwarz inequality and the dominated convergence theorem, $(i,j) \in \mathcal{I}$ and $t \in [0,1]$,

$$\mathbb{E}\left(\sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \xi_{irr'}(t) \xi_{jrr'}(t) \;\middle|\; X_i, T_i, I_i \right) = \sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \xi_{irr'}(t) \, \mathbb{E}\{\xi_{jrr'}(t)\} = 0.$$

We now let $\mathcal{Q}$ denote the set of finitely-supported probability measures on $(\mathcal{X}\times[0,1]\times\{0,1\})^2$, and for $Q\in\mathcal{Q}$, seek to bound the $L_2(Q)$-covering number of the function class

$$\mathcal{G} := \Big\{ \big((X_i,T_i,I_i),(X_j,T_j,I_j)\big) \mapsto \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \xi_{irr'}(t)\xi_{jrr'}(t) : t\in[0,1] \Big\}.$$

For $0\le t<t'\le 1$ and $i,j\in[n]$, by the Cauchy–Schwarz inequality,

$$\begin{aligned}
&\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} |\xi_{irr'}(t)||\xi_{jrr'}(t)-\xi_{jrr'}(t')| \\
&\quad\le e^{2\|f\|_\infty} \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \bigg(|e_r(X_i)e_{r'}(X_i)| + \int_{\mathcal{X}} |e_r(x)e_{r'}(x)| \,\mathrm{d}P_X(x) \bigg) \\
&\qquad\times \bigg(|e_r(X_j)e_{r'}(X_j)||R_j(t)-R_j(t')| + \int_{\mathcal{X}} |e_r(x)e_{r'}(x)||q(x,t)-q(x,t')| \,\mathrm{d}P_X(x) \bigg) \\
&\quad\le 2e^{2\|f\|_\infty} H_{\infty,\gamma}^2 \big\{ \mathbb{1}_{\{t\le T_j<t'\}} + \mathbb{P}(t\le T<t') \big\}.
\end{aligned}$$

As such, we deduce that

$$\begin{aligned}
&\bigg| \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \xi_{irr'}(t)\xi_{jrr'}(t) - \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \xi_{irr'}(t')\xi_{jrr'}(t') \bigg| \\
&\qquad\le \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} |\xi_{irr'}(t)||\xi_{jrr'}(t)-\xi_{jrr'}(t')| + \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} |\xi_{jrr'}(t')||\xi_{irr'}(t)-\xi_{irr'}(t')| \\
&\qquad\le 2e^{2\|f\|_\infty} H_{\infty,\gamma}^2 \big\{ \mathbb{1}_{\{t\le T_i<t'\}} + \mathbb{1}_{\{t\le T_j<t'\}} + 2\mathbb{P}(t\le T<t') \big\}.
\end{aligned}$$

It follows that

$$\begin{aligned}
&\mathbb{E}_Q \bigg\{ \bigg(\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \big\{ \xi_{irr'}(t)\xi_{jrr'}(t) - \xi_{irr'}(t')\xi_{jrr'}(t') \big\} \bigg)^2 \bigg\} \\
&\qquad\le 12 e^{4\|f\|_\infty} H_{\infty,\gamma}^4 \big\{ \mathbb{P}_Q(t\le T_i<t') + \mathbb{P}_Q(t\le T_j<t') + 4\mathbb{P}(t\le T<t')^2 \big\}.
\end{aligned}$$

For $\varepsilon\in(0,1)$ and $m\in\big[\lceil 1/\varepsilon^2\rceil - 1\big]$, define

$$t_m := \inf\big\{ t\in[0,1] : \mathbb{P}_Q(T_i\le t)\ge m\varepsilon^2 \big\}$$

and set $t_{\lceil 1/\varepsilon^2\rceil} := 1$ and $\mathcal{T}_1 := \big\{t_1,\dots,t_{\lceil 1/\varepsilon^2\rceil}\big\}$. Thus, given any $t\in[0,1]$, there exists $m\in\big[\lceil 1/\varepsilon^2\rceil\big]$ such that $\mathbb{P}_Q(t\le T_i<t_m)\le\varepsilon^2$. Let $\mathcal{T}_2$ and $\mathcal{T}_3$ satisfy the same properties but for T_j under Q and T under P_T respectively. Then the set $\mathcal{T} := \mathcal{T}_1\cup\mathcal{T}_2\cup\mathcal{T}_3$ is a $\sqrt{72}e^{2\|f\|_\infty}H_{\infty,\gamma}^2\varepsilon$-cover of $\mathcal{G}$ under $L_2(Q)$, of cardinality at most $6/\varepsilon^2$. Defining $\varepsilon' := 3\varepsilon/\sqrt{2}\in(0,3/\sqrt{2})$ and with M as in (23), we deduce that

$$N\big(\varepsilon' M,\mathcal{G},\|\cdot\|_{Q,2}\big) \le \frac{27}{\varepsilon'^2}.$$

Hence

$$J := \sup_{Q\in\mathcal{Q}} \int_0^1 \log N\big(1+\varepsilon M,\mathcal{G},\|\cdot\|_{Q,2}\big)\,\mathrm{d}\varepsilon \le 1 + \int_0^1 \log(27/\varepsilon^2)\,\mathrm{d}\varepsilon = 1+\log(27)+2 \le 7.$$

By Lemma 49, there exists a universal constant $C_1 > 0$ such that for all $s \geq 1$,

$$\mathbb{P}\left(\sup_{t\in[0,1]} \left| \frac{2}{n(n-1)} \sum_{(i,j)\in\mathcal{I}} \sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \xi_{irr'}(t)\xi_{jrr'}(t) \right| \geq \frac{C_1 s^2 e^{2\|f\|_\infty} H_{\infty,\gamma}^2}{n} \right) \leq e^{-s^2}.$$

By combining this with the bound (24) on the diagonal elements, there exists a universal constant $C_2 > 0$ such that for all $s \geq 1$, with probability at least $1 - e^{-s^2}$,

$$\sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} \sum_{r'\in\mathcal{R}} a_r a_{r'} \big(D^2 S_n(f,t)(e_r, e_{r'}) - D^2 S(f,t)(e_r, e_{r'}) \big)^2 \leq \frac{C_2 s^2 e^{2\|f\|_\infty} H_{\infty,\gamma}^2}{n}. \qquad \square$$

E.3.3 Bounding first-order terms from the basic inequality

By the Cauchy–Schwarz inequality, the inner product term from (17) can be bounded using

$$\big| \big\langle \tilde{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}}, f_0 - P_n(f_0)1_{\mathcal{X}} \big\rangle_{\mathcal{H}} \big| \leq \big\| f_0 - P_n(f_0)1_{\mathcal{X}} \big\|_{\mathcal{H}} \big\| \tilde{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}} \big\|_{\mathcal{H}}.$$

For the term in (17) involving $D\ell_n$, the definition of the operator norm gives

$$\big| D\ell_n(f_0)\big(\tilde{f}_{n,\gamma} - f_0\big) \big| \leq \|D\ell_n(f_0)\|_{\mathcal{H}_\gamma,\mathrm{op}} \big\| \tilde{f}_{n,\gamma} - f_0 \big\|_{\mathcal{H}_\gamma}.$$

We now present a high-probability bound for $\|D\ell_n(f_0)\|_{\mathcal{H}_\gamma,\mathrm{op}}$.

Proposition 23 (Convergence of $D\ell_n$)**.** *There exists a universal constant $C > 0$ such that for all $s \geq 1$, with probability at least $1 - e^{-s^2}$,*

$$\|D\ell_n(f_0)\|_{\mathcal{H}_\gamma,\mathrm{op}} \leq \frac{C e^{2\|f_0\|_\infty}}{q_1^2} \left(\frac{\sqrt{H_{2,\gamma}} + s}{\sqrt{n}} + s^2 \frac{\sqrt{H_{\infty,\gamma}}}{n} \right),$$

Proof of Proposition 23. For any $f_1 \in \mathcal{H}$, we have by Lemma 14 that

$$\begin{aligned}
D\ell_n(f_0)(f_1) &= \int_0^1 \frac{DS_n(f_0,t)(f_1)}{S_n(f_0,t)} \,\mathrm{d}N(t) - \frac{1}{n}\sum_{i=1}^n f_1(X_i)N_i(1) \\
&= \int_0^1 \frac{DS_n(f_0,t)(f_1) - DS(f_0,t)(f_1)}{S_n(f_0,t)} \,\mathrm{d}N(t) && (25) \\
&\quad + \int_0^1 \frac{DS(f_0,t)(f_1)\{S(f_0,t) - S_n(f_0,t)\}}{S(f_0,t)S_n(f_0,t)} \,\mathrm{d}N(t) && (26) \\
&\quad + \int_0^1 \frac{DS(f_0,t)(f_1)}{S(f_0,t)} \big(\mathrm{d}N(t) - S(f_0,t)\lambda_0(t)\,\mathrm{d}t \big) && (27) \\
&\quad + \int_0^1 DS(f_0,t)(f_1)\lambda_0(t)\,\mathrm{d}t - \frac{1}{n}\sum_{i=1}^n f_1(X_i)N_i(1). && (28)
\end{aligned}$$

We begin by working on the event $\mathcal{A}_n := \{\inf_{t\in[0,1]} S_n(f_0,t) \geq q_1/2\}$, so that $\mathbb{P}(\mathcal{A}_n) \geq 1 - \exp\big(1 - nq_1^2 e^{-2\|f_0\|_\infty}/C\big)$ for a universal constant $C > 0$, by Lemma 20, as $S(f_0,t) =$

$\mathbb{E}\{q(X_1,t)e^{f_0(X_1)}\} \geq q_1 e^{\mathbb{E}\{f_0(X_1)\}} = q_1$ by Jensen's inequality. By Lemma 21, we may bound (25) as

$$\begin{aligned}
&\sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\left|\int_0^1 \frac{DS_n(f_0,t)(f_1) - DS(f_0,t)(f_1)}{S_n(f_0,t)}\,\mathrm{d}N(t)\mathbb{1}_{\mathcal{A}_n}\right| \\
&\qquad\leq \frac{2}{q_1}\sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\sup_{t\in[0,1]}\left|DS_n(f_0,t)(f_1) - DS(f_0,t)(f_1)\right|\mathbb{1}_{\mathcal{A}_n} \\
&\qquad\leq \frac{Ce^{\|f_0\|_\infty}}{q_1}\left(\frac{\sqrt{H_{2,\gamma}}+s}{\sqrt{n}} + \frac{s^2\sqrt{H_{\infty,\gamma}}}{n}\right),
\end{aligned}$$

with probability at least $1-e^{-s^2}$, after increasing the universal constant $C>0$ if necessary. For all $f_1\in\mathcal{H}$ and $t\in[0,1]$, by Lemma 15 and Jensen's inequality,

$$\begin{aligned}
DS(f_0,t)(f_1)^2 &= \left(\int_{\mathcal{X}} q(x,t)e^{f_0(x)}f_1(x)\,\mathrm{d}P_X(x)\right)^2 \\
&\leq \int_{\mathcal{X}} q(x,t)^2 e^{2f_0(x)}f_1(x)^2\,\mathrm{d}P_X(x) \leq e^{2\|f_0\|_\infty}\|f_1\|_{L_2}^2. \qquad (29)
\end{aligned}$$

Therefore, by Lemma 20, we may bound (26) as

$$\begin{aligned}
&\sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\left|\int_0^1 \frac{DS(f_0,t)(f_1)\{S(f_0,t)-S_n(f_0,t)\}}{S(f_0,t)S_n(f_0,t)}\,\mathrm{d}N(t)\right|\mathbb{1}_{\mathcal{A}_n} \\
&\qquad\leq \frac{2e^{\|f_0\|_\infty}}{q_1^2}\sup_{t\in[0,1]}\left|S_n(f_0,t)-S(f_0,t)\right|\mathbb{1}_{\mathcal{A}_n} \leq \frac{Cse^{2\|f_0\|_\infty}}{\sqrt{n}q_1^2},
\end{aligned}$$

with probability at least $1-e^{-s^2}$, again after increasing the universal constant $C>0$ if necessary. We now control the term in (27). By Lemma 19,

$$\sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\left(\int_0^1 \frac{DS(f_0,t)(f_1)}{S(f_0,t)}\big(\mathrm{d}N(t) - S(f_0,t)\lambda_0(t)\mathrm{d}t\big)\right)^2 \leq \sum_{r\in\mathcal{R}} a_r\left(\frac{1}{n}\sum_{i=1}^n \xi_{ir}\right)^2,$$

where

$$\xi_{ir} := \int_0^1 \frac{DS(f_0,t)(e_r)}{S(f_0,t)}\big(\mathrm{d}N_i(t) - S(f_0,t)\lambda_0(t)\,\mathrm{d}t\big). \qquad (30)$$

Note that ξ_{ir} is (X_i,T_i,I_i)-measurable, and $\mathbb{E}(\xi_{ir})=0$ because the integrand in (30) is non-random and bounded, and $1\geq\mathbb{E}\{N_i(t)\} = \int_0^t \lambda_0(s)S(f_0,s)\,\mathrm{d}s$ for all $t\in[0,1]$ by Fleming and Harrington (2013, Theorem 1.3.1). Further, by Lemma 18 and (29),

$$\begin{aligned}
\sum_{r\in\mathcal{R}} a_r\xi_{ir}^2 &\leq \sup_{t\in[0,1]}\sum_{r\in\mathcal{R}} a_r\frac{DS(f_0,t)(e_r)^2}{S(f_0,t)^2}\left(N_i(1) + \int_0^1 S(f_0,t)\lambda_0(t)\,\mathrm{d}t\right)^2 \\
&\leq \frac{4}{q_1^2}\sup_{t\in[0,1]}\sum_{r\in\mathcal{R}} a_r DS(f_0,t)(e_r)^2 \leq \frac{4e^{2\|f_0\|_\infty}H_{2,\gamma}}{q_1^2}. \qquad (31)
\end{aligned}$$

Therefore, by Cauchy–Schwarz, with $\mathcal{I} := \{(i,j) : i,j \in [n], i < j\}$, for $n \geq 2$,

$$\sum_{r\in\mathcal{R}} a_r \left(\frac{1}{n}\sum_{i=1}^n \xi_{ir}\right)^2 \leq \frac{1}{n^2}\sum_{i=1}^n\sum_{r\in\mathcal{R}} a_r\xi_{ir}^2 + \frac{2}{n^2}\sum_{(i,j)\in\mathcal{I}}\sum_{r\in\mathcal{R}} a_r\xi_{ir}\xi_{jr}$$
$$\leq \frac{4e^{2\|f_0\|_\infty}H_{2,\gamma}}{nq_1^2} + \left|\frac{2}{n(n-1)}\sum_{(i,j)\in\mathcal{I}}\sum_{r\in\mathcal{R}} a_r\xi_{ir}\xi_{jr}\right|.$$

In the case $n = 1$, the second term in the bound is not needed. Otherwise, since ξ_{ir} and ξ_{jr} are independent when $i < j$, this term is a degenerate second-order U-statistic so by Cauchy–Schwarz, (31) and Lemma 49, for all $s \geq 1$,

$$\left|\frac{2}{n(n-1)}\sum_{(i,j)\in\mathcal{I}}\sum_{r\in\mathcal{R}} a_r\xi_{ir}\xi_{jr}\right| \leq \frac{Cs^2e^{2\|f_0\|_\infty}H_{2,\gamma}}{nq_1^2}$$

with probability at least $1 - e^{-s^2}$, after increasing the universal constant $C > 0$ if necessary. As such, for (27),

$$\sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\left(\int_0^1 \frac{DS(f_0,t)(f_1)}{S(f_0,t)}\big(\mathrm{d}N(t) - S(f_0,t)\lambda_0(t)\mathrm{d}t\big)\right)^2 \leq \frac{Cs^2e^{2\|f_0\|_\infty}H_{2,\gamma}}{nq_1^2}$$

with probability at least $1 - e^{-s^2}$ for all $s \geq 1$, once more increasing $C > 0$ if necessary.

For (28), by Fleming and Harrington (2013, Theorem 1.3.1), symmetrisation (van der Vaart and Wellner, 1996, Lemma 2.3.1) and a contraction principle (van der Vaart and Wellner, 1996, Proposition A.1.10), with independent Rademacher variables $\varepsilon_1, \ldots, \varepsilon_n$ that are independent of the data, and by the proof of Lemma 29,

$$\mathbb{E}\left\{\sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\left|\frac{1}{n}\sum_{i=1}^n f_1(X_i)N_i(1) - \int_0^1 DS(f_0,t)(f_1)\lambda_0(t)\,\mathrm{d}t\right|\right\}$$
$$\leq 2\,\mathbb{E}\left\{\sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\frac{1}{n}\sum_{i=1}^n \varepsilon_i f_1(X_i)\right\} \leq 2\sqrt{\frac{H_{2,\gamma}}{n}}.$$

By Lemma 2, we have

$$\sigma^2 := \sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\sum_{i=1}^n \mathbb{E}\left\{\left(\frac{1}{n}f_1(X_i)N_i(1)\right)^2\right\} \leq \sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\frac{\|f_1\|_{L_2}^2}{n} \leq \frac{1}{n},$$
$$M := \sup_{x\in\mathcal{X}}\sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\left|\frac{1}{n}f_1(x)N_i(1)\right| \leq \sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\frac{\|f_1\|_\infty}{n} \leq \frac{\sqrt{H_{\infty,\gamma}}}{n}.$$

By Boucheron et al. (2013, Theorem 12.5), by increasing the universal constant $C > 0$ if necessary, with probability at least $1 - e^{-s^2}$,

$$\sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\left|\frac{1}{n}\sum_{i=1}^n f_1(X_i)N_i(1) - \int_0^1 DS(f_0,t)(f_1)\lambda_0(t)\,\mathrm{d}t\right| \leq C\left(\frac{\sqrt{H_{2,\gamma}}+s}{\sqrt{n}} + \frac{s^2\sqrt{H_{\infty,\gamma}}}{n}\right).$$

We deduce that

$$\|D\ell_n(f_0)\|_{\mathcal{H}_\gamma,\mathrm{op}} \leq \frac{Ce^{2\|f_0\|_\infty}}{q_1^2}\left(\frac{\sqrt{H_{2,\gamma}}+s}{\sqrt{n}} + s^2\frac{\sqrt{H_{\infty,\gamma}}}{n}\right), \tag{32}$$

with probability at least $1-4e^{-s^2}-\mathbb{P}(\mathcal{A}_n) \geq 1-4e^{-s^2}-\exp\bigl(-nq_1^2e^{-2\|f_0\|_\infty}/C\bigr)$, for all $s \geq 1$. By increasing the universal constant $C>0$ again, we conclude that (32) holds with probability at least $1-e^{-s^2}$ whenever $1 \leq s^2 \leq nq_1^2e^{-2\|f_0\|_\infty}/C$. Now suppose that $s^2 > nq_1^2e^{-2\|f_0\|_\infty}/C$. Since

$$\bigl|DS_n(f_0,t)(f_1)\bigr| = \left|\frac{1}{n}\sum_{i=1}^n R_i(t)e^{f_0(X_i)}f_1(X_i)\right| \leq \|f_1\|_\infty S_n(f_0,t),$$

we have by Lemma 14 and Lemma 2 that

$$\begin{aligned}\|D\ell_n(f_0)\|_{\mathcal{H}_\gamma,\mathrm{op}} &\leq \sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\left\{\sup_{t\in[0,1]}\left|\frac{DS_n(f_0,t)(f_1)}{S_n(f_0,t)}\right| + \|f_1\|_\infty\right\}\\ &\leq 2\sup_{f_1\in\mathcal{H}:\,\|f_1\|_{\mathcal{H}_\gamma}\leq 1}\|f_1\|_\infty \leq 2\sqrt{H_{\infty,\gamma}}.\end{aligned}$$

By increasing C again, with $s^2 > nq_1^2e^{-2\|f_0\|_\infty}/C$, we have

$$\frac{C^2e^{2\|f_0\|_\infty}}{q_1^2}\left(\frac{\sqrt{H_{2,\gamma}}+s}{\sqrt{n}} + s^2\frac{\sqrt{H_{\infty,\gamma}}}{n}\right) \geq C\sqrt{H_{\infty,\gamma}},$$

and so (32) holds with probability one, concluding the proof. □

E.3.4 Bounding second-order terms from the basic inequality

We establish a lower bound for the term in (17) involving $D^2\ell_n$. Observe that

$$D^2\ell_n(f_0)\bigl(\tilde{f}_{n,\gamma}-f_0\bigr)^{\otimes 2} \geq D^2\ell_\star(f_0)\bigl(\tilde{f}_{n,\gamma}-f_0\bigr)^{\otimes 2} - \bigl\|D^2\ell_n(f_0)-D^2\ell_\star(f_0)\bigr\|_{\mathcal{H}_\gamma,\mathrm{op}}\bigl\|\tilde{f}_{n,\gamma}-f_0\bigr\|_{\mathcal{H}_\gamma}^2.$$

Lemma 17 already provides a lower bound based on restricted strong convexity for $D^2\ell_\star(f_0)$; we now show convergence of $D^2\ell_n(f_0)$ to $D^2\ell_\star(f_0)$ in Proposition 24.

Proposition 24 (Convergence of $D^2\ell_n$)**.** *There exists a universal constant $C>0$ such that for all $s \geq 1$, with probability at least $1-e^{-s^2}$,*

$$\bigl\|D^2\ell_n(f_0)-D^2\ell_\star(f_0)\bigr\|_{\mathcal{H}_\gamma,\mathrm{op}} \leq \frac{Cse^{2\|f_0\|_\infty}}{q_1^2}\sqrt{\frac{H_{\infty,\gamma}^2}{n}}.$$

Proof of Proposition 24. Since the second-order Gateaux derivative is a bilinear operator, we first state a general result allowing us to bound the bilinear $\|\cdot\|_{\mathcal{H}_\gamma,\mathrm{op}}$-operator norm in terms

of actions on the eigenfunctions. To this end, let $G : \mathcal{H} \times \mathcal{H} \to \mathbb{R}$ be bilinear. We have by the Cauchy–Schwarz inequality that

$$\begin{aligned}
\|G\|_{\mathcal{H}_\gamma,\mathrm{op}}^2 &:= \sup_{f,g\in\mathcal{H}:\|f\|_{\mathcal{H}_\gamma}=1,\|g\|_{\mathcal{H}_\gamma}=1} G(f,g)^2 \\
&= \sup_{f,g\in\mathcal{H}:\|f\|_{\mathcal{H}_\gamma}=1,\|g\|_{\mathcal{H}_\gamma}=1} \left(\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} \langle f, e_r\rangle_{L_2}\langle g, e_{r'}\rangle_{L_2} G(e_r, e_{r'})\right)^2 \\
&\le \sup_{f,g\in\mathcal{H}:\|f\|_{\mathcal{H}_\gamma}=1,\|g\|_{\mathcal{H}_\gamma}=1} \left(\sum_{r\in\mathcal{R}} \frac{\langle f, e_r\rangle_{L_2}^2}{a_r}\right)\left(\sum_{r'\in\mathcal{R}} \frac{\langle g, e_{r'}\rangle_{L_2}^2}{a_{r'}}\right) \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} G(e_r, e_{r'})^2 \\
&= \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} G(e_r, e_{r'})^2. \qquad (33)
\end{aligned}$$

We see from this that

$$\left\|D^2\ell_n(f_0) - D^2\ell_\star(f_0)\right\|_{\mathcal{H}_\gamma,\mathrm{op}}^2 \le \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'}\left(D^2\ell_n(f_0)(e_r, e_{r'}) - D^2\ell_\star(f_0)(e_r, e_{r'})\right)^2. \qquad (34)$$

Now, for any $r, r' \in \mathcal{R}$, by Lemmas 14 and 16,

$$\begin{aligned}
&D^2\ell_n(f_0)(e_r, e_{r'}) - D^2\ell_\star(f_0)(e_r, e_{r'}) \\
&= \int_0^1 \left(\frac{D^2 S_n(f_0,t)(e_r,e_{r'})}{S_n(f_0,t)} - \frac{D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)}\right) \mathrm{d}N(t) \qquad (35) \\
&\quad - \int_0^1 \left(\frac{DS_n(f_0,t)(e_r)}{S_n(f_0,t)}\frac{DS_n(f_0,t)(e_{r'})}{S_n(f_0,t)} - \frac{DS(f_0,t)(e_r)}{S(f_0,t)}\frac{DS(f_0,t)(e_{r'})}{S(f_0,t)}\right) \mathrm{d}N(t) \qquad (36) \\
&\quad + \int_0^1 \left(\frac{D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)} - \frac{DS(f_0,t)(e_r)}{S(f_0,t)}\frac{DS(f_0,t)(e_{r'})}{S(f_0,t)}\right)\left(\mathrm{d}N(t) - S(f_0,t)\lambda_0(t)\,\mathrm{d}t\right). \qquad (37)
\end{aligned}$$

For (35), observe that

$$\begin{aligned}
&\frac{D^2 S_n(f_0,t)(e_r,e_{r'})}{S_n(f_0,t)} - \frac{D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)} \\
&\quad = \frac{D^2 S_n(f_0,t)(e_r,e_{r'})\left(S(f_0,t) - S_n(f_0,t)\right)}{S(f_0,t)S_n(f_0,t)} + \frac{D^2 S_n(f_0,t)(e_r,e_{r'}) - D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)}.
\end{aligned}$$

Now, by the Cauchy–Schwarz inequality, for $t \in [0,1]$,

$$\begin{aligned}
&\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} D^2 S_n(f_0,t)(e_r,e_{r'})^2 \\
&\quad = \frac{1}{n^2}\sum_{i=1}^n\sum_{j=1}^n R_i(t)R_j(t)e^{f_0(X_i)}e^{f_0(X_j)}\left(\sum_{r\in\mathcal{R}} a_r e_r(X_i)e_r(X_j)\right)^2 \\
&\quad \le \frac{1}{n^2}\sum_{i=1}^n\sum_{j=1}^n R_i(t)R_j(t)e^{f_0(X_i)}e^{f_0(X_j)}\left(\sum_{r\in\mathcal{R}} a_r e_r(X_i)^2\right)\left(\sum_{r\in\mathcal{R}} a_r e_r(X_j)^2\right) \\
&\quad \le H_{\infty,\gamma}^2 S_n(f_0,t)^2. \qquad (38)
\end{aligned}$$

For all $t \in [0,1]$, by Jensen's inequality and as $\int_{\mathcal{X}} f_0(x)\,\mathrm{d}P_X(x) = 0$,

$$S(f_0,t) = \int_{\mathcal{X}} q(x,t)e^{f_0(x)}\,\mathrm{d}P_X(x) \geq q_1 \int_{\mathcal{X}} e^{f_0(x)}\,\mathrm{d}P_X(x) \geq q_1 e^{\int_{\mathcal{X}} f_0(x)\,\mathrm{d}P_X(x)} = q_1. \tag{39}$$

Hence, since $\int_0^1 \mathrm{d}N(t) = N(1) \leq 1$, it follows from Lemma 18, (38) and (39) that

$$\begin{aligned}
&\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left\{ \int_0^1 \left(\frac{D^2 S_n(f_0,t)(e_r,e_{r'})}{S_n(f_0,t)} - \frac{D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)} \right) \mathrm{d}N(t) \right\}^2 \\
&\quad \leq \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} 2a_r a_{r'} \left(\int_0^1 \frac{D^2 S_n(f_0,t)(e_r,e_{r'})\big(S(f_0,t) - S_n(f_0,t)\big)}{S(f_0,t)S_n(f_0,t)}\,\mathrm{d}N(t) \right)^2 \\
&\qquad + \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} 2a_r a_{r'} \left(\int_0^1 \frac{D^2 S_n(f_0,t)(e_r,e_{r'}) - D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)}\,\mathrm{d}N(t) \right)^2 \\
&\quad \leq \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} 2a_r a_{r'} \left(\frac{D^2 S_n(f_0,t)(e_r,e_{r'})\big(S(f_0,t) - S_n(f_0,t)\big)}{S(f_0,t)S_n(f_0,t)} \right)^2 \\
&\qquad + \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} 2a_r a_{r'} \left(\frac{D^2 S_n(f_0,t)(e_r,e_{r'}) - D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)} \right)^2 \\
&\quad \leq \frac{2H_{\infty,\gamma}^2}{q_1^2} \sup_{t\in[0,1]} \big(S_n(f_0,t) - S(f_0,t)\big)^2 \qquad (40) \\
&\qquad + \frac{2}{q_1^2} \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \big(D^2 S_n(f_0,t)(e_r,e_{r'}) - D^2 S(f_0,t)(e_r,e_{r'})\big)^2. \qquad (41)
\end{aligned}$$

By Lemma 20 applied to (40) and Lemma 22 applied to (41), there exists a universal constant $C_2 > 0$ such that for all $s \geq 1$, with probability at least $1 - e^{-s^2}$,

$$\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left\{ \int_0^1 \left(\frac{D^2 S_n(f_0,t)(e_r,e_{r'})}{S_n(f_0,t)} - \frac{D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)} \right) \mathrm{d}N(t) \right\}^2 \leq \frac{C_2 s^2 e^{2\|f_0\|_\infty} H_{\infty,\gamma}^2}{n q_1^2}. \tag{42}$$

This concludes the analysis for (35), so we now turn to (36). For $t \in [0,1]$ and $r, r' \in \mathcal{R}$,

$$\begin{aligned}
&\frac{DS_n(f_0,t)(e_r)}{S_n(f_0,t)} \frac{DS_n(f_0,t)(e_{r'})}{S_n(f_0,t)} - \frac{DS(f_0,t)(e_r)}{S(f_0,t)} \frac{DS(f_0,t)(e_{r'})}{S(f_0,t)} \\
&\qquad = \frac{DS_n(f_0,t)(e_r) DS_n(f_0,t)(e_{r'}) \big(S(f_0,t) - S_n(f_0,t)\big)\big(S(f_0,t) + S_n(f_0,t)\big)}{S(f_0,t)^2 S_n(f_0,t)^2} \\
&\qquad\qquad + \frac{DS_n(f_0,t)(e_r)\big(DS_n(f_0,t)(e_{r'}) - DS(f_0,t)(e_{r'})\big)}{S(f_0,t)^2} \\
&\qquad\qquad + \frac{DS(f_0,t)(e_{r'})\big(DS_n(f_0,t)(e_r) - DS(f_0,t)(e_r)\big)}{S(f_0,t)^2}.
\end{aligned}$$

Therefore by Lemma 18, since $\int_0^1 \mathrm{d}N(t) \le 1$,

$$\begin{aligned}
&\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left\{ \int_0^1 \left(\frac{DS_n(f_0,t)(e_r)}{S_n(f_0,t)} \frac{DS_n(f_0,t)(e_{r'})}{S_n(f_0,t)} - \frac{DS(f_0,t)(e_r)}{S(f_0,t)} \frac{DS(f_0,t)(e_{r'})}{S(f_0,t)} \right) \mathrm{d}N(t) \right\}^2 \\
&\quad \le \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left(\frac{DS_n(f_0,t)(e_r)}{S_n(f_0,t)} \frac{DS_n(f_0,t)(e_{r'})}{S_n(f_0,t)} - \frac{DS(f_0,t)(e_r)}{S(f_0,t)} \frac{DS(f_0,t)(e_{r'})}{S(f_0,t)} \right)^2 \\
&\quad \le 3 \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left(\frac{DS_n(f_0,t)(e_r) DS_n(f_0,t)(e_{r'})}{S(f_0,t)^2 S_n(f_0,t)^2} \right)^2 \\
&\qquad\qquad \times \big(S(f_0,t) - S_n(f_0,t)\big)^2 \big(S(f_0,t) + S_n(f_0,t)\big)^2 \\
&\qquad + 3 \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left(\frac{DS_n(f_0,t)(e_r) \big(DS_n(f_0,t)(e_{r'}) - DS(f_0,t)(e_{r'})\big)}{S(f_0,t)^2} \right)^2 \\
&\qquad + 3 \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left(\frac{DS(f_0,t)(e_{r'}) \big(DS_n(f_0,t)(e_r) - DS(f_0,t)(e_r)\big)}{S(f_0,t)^2} \right)^2 \\
&\quad \le 6 \sup_{t\in[0,1]} \frac{S(f_0,t)^2 + S_n(f_0,t)^2}{S(f_0,t)^4} \big(S_n(f_0,t) - S(f_0,t)\big)^2 \left(\sum_{r\in\mathcal{R}} a_r \frac{DS_n(f_0,t)(e_r)^2}{S_n(f_0,t)^2} \right)^2 \\
&\qquad + 6 \sup_{t\in[0,1]} \frac{1}{S(f_0,t)^4} \sum_{r\in\mathcal{R}} a_r DS_n(f_0,t)(e_r)^2 \sum_{r'\in\mathcal{R}} a_{r'} \big(DS_n(f_0,t)(e_{r'}) - DS(f_0,t)(e_{r'})\big)^2.
\end{aligned}$$

By Lemma 15, the dominated convergence theorem and the Cauchy–Schwarz inequality, for each $t \in [0,1]$,

$$\begin{aligned}
\sum_{r\in\mathcal{R}} a_r DS(f_0,t)(e_r)^2 &= \int_{\mathcal{X}}\int_{\mathcal{X}} q(x,t)q(x',t) e^{f_0(x)} e^{f_0(x')} \sum_{r\in\mathcal{R}} a_r e_r(x) e_r(x') \,\mathrm{d}P_X(x)\,\mathrm{d}P_X(x') \\
&\le \int_{\mathcal{X}}\int_{\mathcal{X}} q(x,t)q(x',t) e^{f_0(x)} e^{f_0(x')} \left\{ \sum_{r\in\mathcal{R}} a_r e_r(x)^2 \right\}^{1/2} \left\{ \sum_{r\in\mathcal{R}} a_r e_r(x')^2 \right\}^{1/2} \mathrm{d}P_X(x)\,\mathrm{d}P_X(x') \\
&\le H_{\infty,\gamma} S(f_0,t)^2. \qquad (43)
\end{aligned}$$

Moreover, by Lemma 13 and the Cauchy–Schwarz inequality,

$$\begin{aligned}
\sum_{r\in\mathcal{R}} a_r DS_n(f_0,t)(e_r)^2 &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n R_i(t) R_j(t) e^{f_0(X_i)} e^{f_0(X_j)} \sum_{r\in\mathcal{R}} a_r e_r(X_i) e_r(X_j) \\
&\le \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n R_i(t) R_j(t) e^{f_0(X_i)} e^{f_0(X_j)} \left\{ \sum_{r\in\mathcal{R}} a_r e_r(X_i)^2 \right\}^{1/2} \left\{ \sum_{r\in\mathcal{R}} a_r e_r(X_j)^2 \right\}^{1/2} \\
&\le H_{\infty,\gamma} S_n(f_0,t)^2. \qquad (44)
\end{aligned}$$

Applying (39), (44) and (43), and noting that $|S(f_0,t)| \le e^{\|f_0\|_\infty}$ and $|S_n(f_0,t)| \le e^{\|f_0\|_\infty}$,

$$\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left\{ \int_0^1 \left(\frac{DS_n(f_0,t)(e_r)}{S_n(f_0,t)} \frac{DS_n(f_0,t)(e_{r'})}{S_n(f_0,t)} - \frac{DS(f_0,t)(e_r)}{S(f_0,t)} \frac{DS(f_0,t)(e_{r'})}{S(f_0,t)} \right) \mathrm{d}N(t) \right\}^2$$
$$\le \frac{12 e^{2\|f_0\|_\infty} H_{\infty,\gamma}^2}{q_1^4} \sup_{t\in[0,1]} \big(S_n(f_0,t) - S(f_0,t)\big)^2$$
$$+ \frac{6 e^{2\|f_0\|_\infty} H_{\infty,\gamma}}{q_1^4} \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}} a_r \big(DS_n(f_0,t)(e_r) - DS(f_0,t)(e_r)\big)^2.$$

By Lemma 20 and Lemma 21, there is a universal constant $C_3 > 0$ such that for all $s \ge 1$,

$$\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left\{ \int_0^1 \left(\frac{DS_n(f_0,t)(e_r)}{S_n(f_0,t)} \frac{DS_n(f_0,t)(e_{r'})}{S_n(f_0,t)} - \frac{DS(f_0,t)(e_r)}{S(f_0,t)} \frac{DS(f_0,t)(e_{r'})}{S(f_0,t)} \right) \mathrm{d}N(t) \right\}^2$$
$$\le \frac{C_3 s^2 e^{4\|f_0\|_\infty} H_{\infty,\gamma}^2}{n q_1^4} \tag{45}$$

with probability at least $1 - e^{-s^2}$. We finally consider (37). For $i \in [n]$ and $r, r' \in \mathcal{R}$, define

$$\xi'_{irr'} := \int_0^1 \left(\frac{D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)} - \frac{DS(f_0,t)(e_r)}{S(f_0,t)} \frac{DS(f_0,t)(e_{r'})}{S(f_0,t)} \right) \big(\mathrm{d}N_i(t) - S(f_0,t)\lambda_0(t)\,\mathrm{d}t\big). \tag{46}$$

By Lemma 15 and Cauchy–Schwarz, for $t \in [0,1]$,

$$\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} D^2 S(f_0,t)(e_r,e_{r'})^2$$
$$= \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left(\int_{\mathcal{X}} q(x,t) e^{f_0(x)} e_r(x) e_{r'}(x)\,\mathrm{d}P_X(x) \right)^2$$
$$= \int_{\mathcal{X}}\int_{\mathcal{X}} q(x,t) q(x',t) e^{f_0(x)} e^{f_0(x')} \sum_{r\in\mathcal{R}} a_r e_r(x) e_r(x') \sum_{r'\in\mathcal{R}} a_{r'} e_{r'}(x) e_{r'}(x')\,\mathrm{d}P_X(x)\,\mathrm{d}P_X(x')$$
$$\le H_{\infty,\gamma}^2 \left(\int_{\mathcal{X}} q(x,t) e^{f_0(x)}\,\mathrm{d}P_X(x) \right)^2 = H_{\infty,\gamma}^2 S(f_0,t)^2.$$

Note that $\xi'_{irr'}$ is (X_i,T_i,I_i)-measurable, and $\mathbb{E}(\xi'_{irr'}) = 0$ because the integrand in (46) is non-random and bounded, and $\mathbb{E}\{N(t)\} = \int_0^t \lambda_0(s) S(f_0,s)\,\mathrm{d}s \le 1$ for all $t \in [0,1]$. Thus, by Lemma 18 and (43), since $|S(f_0,t)| \le e^{\|f_0\|_\infty}$,

$$\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \xi'^2_{irr'} \le \left(N_i(1) + \int_0^1 S(f_0,t)\lambda_0(t)\,\mathrm{d}t \right)^2$$
$$\times \sup_{t\in[0,1]} \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left(\frac{D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)} - \frac{DS(f_0,t)(e_r)}{S(f_0,t)} \frac{DS(f_0,t)(e_{r'})}{S(f_0,t)} \right)^2$$
$$\le 16 H_{\infty,\gamma}^2. \tag{47}$$

Therefore by Cauchy–Schwarz, with $\mathcal{I} := \{(i,j) : i,j \in [n], i < j\}$,

$$\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'}\left(\frac{1}{n}\sum_{i=1}^{n}\xi'_{irr'}\right)^2 \le \frac{1}{n^2}\sum_{i=1}^{n}\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'}\xi'^2_{irr'} + \frac{2}{n^2}\sum_{(i,j)\in\mathcal{I}}\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'}\xi'_{irr'}\xi'_{jrr'}$$
$$\le \frac{16H^2_{\infty,\gamma}}{n} + \left|\frac{2}{n(n-1)}\sum_{(i,j)\in\mathcal{I}}\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'}\xi'_{irr'}\xi'_{jrr'}\right|.$$

The last term above is a degenerate second-order U-statistic so by Cauchy–Schwarz, (47) and Lemma 49, there exists a universal constant $C_4 > 0$ such that for all $s \ge 1$,

$$\left|\frac{2}{n(n-1)}\sum_{(i,j)\in\mathcal{I}}\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'}\xi'_{irr'}\xi'_{jrr'}\right| \le \frac{C_4 s^2 H^2_{\infty,\gamma}}{n}$$

with probability at least $1 - e^{-s^2}$. Thus, there is a universal constant $C_5 > 0$ such that

$$\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'}\left\{\int_0^1\left(\frac{D^2S(f_0,t)(e_r,e_{r'})}{S(f_0,t)} - \frac{DS(f_0,t)(e_r)}{S(f_0,t)}\frac{DS(f_0,t)(e_{r'})}{S(f_0,t)}\right)\right.$$
$$\left.\times\left(\mathrm{d}N(t) - S(f_0,t)\lambda_0(t)\,\mathrm{d}t\right)\right\}^2$$
$$\le \frac{C_5 s^2 H^2_{\infty,\gamma}}{n} \tag{48}$$

with probability at least $1 - e^{-s^2}$. Combining (34), (35), (36), (37), (42), (45) and (48), it follows that there exists a universal constant $C > 0$ such that

$$\|D^2\ell_n(f_0) - D^2\ell_\star(f_0)\|^2_{\mathcal{H}_\gamma,\mathrm{op}} \le \frac{C^2 s^2 H^2_{\infty,\gamma} e^{4\|f_0\|_\infty}}{n q_1^4}$$

with probability at least $1 - e^{-s^2}$. □

E.3.5 Bounding third-order terms from the basic inequality

We bound the term in (17) involving $D^3\ell_n$. For $f, f_1, f_2 \in \mathcal{B}(\mathcal{X})$, define

$$V_n(f)(f_1,f_2) := \int_0^1 \frac{D^2S_n(f,t)(f_1,f_2)}{S_n(f,t)}\,\mathrm{d}N(t),$$
$$V(f)(f_1,f_2) := \int_0^1 \frac{D^2S(f,t)(f_1,f_2)}{S(f,t)}S(f_0,t)\lambda_0(t)\,\mathrm{d}t.$$

In Lemma 25 we provide an upper bound for $D^3\ell_n$ in terms of V_n. Lemma 26 then establishes convergence of V_n to V, while Lemma 27 presents an upper bound for V.

Lemma 25 (Convergence of $D^3\ell_n$)**.** *Let $f, f_1 \in \mathcal{B}(\mathcal{X})$. Then*

$$\left|D^3\ell_n(f)(f_1)^{\otimes 3}\right| \le 6\|f_1\|_\infty e^{2\|f-f_0\|_\infty} V_n(f_0)(f_1,f_1).$$

Proof of Lemma 25. By Lemma 13 and Cauchy–Schwarz,

$$|D^3 S_n(f,t)(f_1,f_1,f_1)| \le \frac{1}{n}\sum_{i=1}^n \left|R_i(t) f_1(X_i)^3 e^{f(X_i)}\right| \le \|f_1\|_\infty D^2 S_n(f,t)(f_1,f_1),$$

$$|DS_n(f,t)(f_1)| \le \frac{1}{n}\sum_{i=1}^n \left|R_i(t) f_1(X_i) e^{f(X_i)}\right| \le \|f_1\|_\infty S_n(f,t),$$

$$\begin{aligned} DS_n(f,t)(f_1)^2 &= \left(\frac{1}{n}\sum_{i=1}^n R_i(t) f_1(X_i) e^{f(X_i)}\right)^2 \\ &\le \left(\frac{1}{n}\sum_{i=1}^n R_i(t) e^{f(X_i)}\right)\left(\frac{1}{n}\sum_{i=1}^n R_i(t) f_1(X_i)^2 e^{f(X_i)}\right) \\ &= S_n(f,t) D^2 S_n(f,t)(f_1,f_1). \end{aligned}$$

Further,

$$S_n(f,t) = \frac{1}{n}\sum_{i=1}^n R_i(t) e^{f_0(X_i)} e^{f(X_i)-f_0(X_i)} \ge e^{-\|f-f_0\|_\infty} S_n(f_0,t)$$

and

$$D^2 S_n(f,t)(f_1,f_1) = \frac{1}{n}\sum_{i=1}^n R_i(t) f_1(X_i)^2 e^{f_0(X_i)} e^{f(X_i)-f_0(X_i)} \le e^{\|f-f_0\|_\infty} D^2 S_n(f_0,t)(f_1,f_1).$$

Applying these bounds to (12) and using Lemma 14, we obtain

$$\begin{aligned} \left|D^3 \ell_n(f)(f_1,f_1,f_1)\right| &\le 6\|f_1\|_\infty \int_0^1 \frac{D^2 S_n(f,t)(f_1,f_1)}{S_n(f,t)}\,\mathrm{d}N(t) \\ &\le 6\|f_1\|_\infty e^{2\|f-f_0\|_\infty} \int_0^1 \frac{D^2 S_n(f_0,t)(f_1,f_1)}{S_n(f_0,t)}\,\mathrm{d}N(t). \end{aligned}$$

□

Lemma 26 (Convergence of V_n)**.** *Consider $V_n(f_0)$ and $V(f_0)$ as bilinear functionals from $\mathcal{H}\times\mathcal{H}$ to $\mathbb{R}$. There is a universal constant $C>0$ such that for all $s\ge 1$, with probability at least $1-e^{-s^2}$,*

$$\|V_n(f_0) - V(f_0)\|_{\mathcal{H}_\gamma,\mathrm{op}} \le \frac{C s e^{2\|f_0\|_\infty}}{q_1^2}\sqrt{\frac{H_{\infty,\gamma}^2}{n}}.$$

Proof of Lemma 26. By (33),

$$\begin{aligned} \|V_n(f_0) - V(f_0)\|_{\mathcal{H}_\gamma,\mathrm{op}}^2 &\le \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \big(V_n(f_0)(e_r,e_{r'}) - V(f_0)(e_r,e_{r'})\big)^2 \\ &= \sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left(\int_0^1 \frac{D^2 S_n(f_0,t)(e_r,e_{r'})}{S_n(f_0,t)}\,\mathrm{d}N(t) - \int_0^1 D^2 S(f_0,t)(e_r,e_{r'})\lambda_0(t)\,\mathrm{d}t\right)^2 \\ &\le 2\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left\{\int_0^1 \left(\frac{D^2 S_n(f_0,t)(e_r,e_{r'})}{S_n(f_0,t)} - \frac{D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)}\right)\mathrm{d}N(t)\right\}^2 \qquad (49) \\ &\quad + 2\sum_{r\in\mathcal{R}}\sum_{r'\in\mathcal{R}} a_r a_{r'} \left\{\int_0^1 \frac{D^2 S(f_0,t)(e_r,e_{r'})}{S(f_0,t)}\big(\mathrm{d}N(t) - S(f_0,t)\lambda_0(t)\,\mathrm{d}t\big)\right\}^2. \qquad (50) \end{aligned}$$

A probabilistic bound for (49) is given in (42), while (50) is controlled by a slightly simpler argument than that used to control (37). We therefore recover the same probabilistic bound as that given in Proposition 24. □

Lemma 27 (Upper bound for V)**.** *For $f_1 \in L_2(P_X)$, we have*

$$V(f_0)(f_1, f_1) \le e^{\|f_0\|_\infty} \Lambda \|f_1\|_{L_2}^2.$$

Proof of Lemma 27. We have

$$\begin{aligned} V(f_0)(f_1, f_1) &= \int_0^1 \int_{\mathcal{X}} q(x,t) e^{f_0(x)} f_1(x)^2 \, \mathrm{d}P_X(x) \lambda_0(t) \, \mathrm{d}t \\ &\le e^{\|f_0\|_\infty} \int_0^1 \lambda_0(t) \, \mathrm{d}t \int_{\mathcal{X}} f_1^2 \, \mathrm{d}P_X = e^{\|f_0\|_\infty} \Lambda \|f_1\|_{L_2}^2. \end{aligned}$$

□

E.3.6 Convergence of P_n

In Lemma 28 we bound the difference between $P_n(f_0)$ and $P_X(f_0)$, while in Lemma 29 we give a bound for the supremum of the empirical process $P_n - P_X$ over $\mathcal{H}_\gamma$-balls in $\mathcal{H}$.

Lemma 28 (Convergence of P_n)**.** *For all $s \ge 1$, we have $|P_n(f_0)| \le 2s\|f_0\|_\infty / \sqrt{n}$ with probability at least $1 - e^{-s^2}$.*

Proof of Lemma 28. This follows by Hoeffding's inequality, noting that $P_X(f_0) = 0$. □

Lemma 29 (Uniform convergence of P_n on $\mathcal{H}_\gamma$-balls)**.** *Let $M > 0$. Then for all $s \ge 1$,*

$$\mathbb{P}\left(\sup_{f \in \mathcal{H} : \|f\|_{\mathcal{H}_\gamma} \le M} \left| P_n(f) - P_X(f) \right| > 4Ms \sqrt{\frac{H_{\infty,\gamma}}{n}} \right) \le e^{-s^2}.$$

Proof of Lemma 29. Recall that $f \in \mathcal{H}$ with $\|f\|_{\mathcal{H}_\gamma} \le M$ if and only if $f(\cdot) = \sum_{r \in \mathcal{R}} \alpha_r e_r(\cdot)$ with $\sum_{r \in \mathcal{R}} \alpha_r^2 / a_r \le M^2$, and define the set

$$\mathcal{H}_{\gamma,M,\mathbb{Q}} := \left\{ f : \mathcal{X} \to \mathbb{R} : f(\cdot) = \sum_{r \in \mathcal{R}} \tilde{\alpha}_r e_r(\cdot), (\tilde{\alpha}_r)_{r \in \mathcal{R}} \in \mathbb{Q}^{\mathcal{R}}, \sum_{r \in \mathcal{R}} \frac{\tilde{\alpha}_r^2}{a_r} \le M^2 \right\}.$$

Suppose that $f(\cdot) = \sum_{r \in \mathcal{R}} \alpha_r e_r(\cdot) \in \mathcal{H}_{\gamma,M}$ and let $N \in \mathbb{N}$. For each $r \in \mathcal{R}$, let $\tilde{\alpha}_{r,N} \in \mathbb{Q}$ satisfy $|\tilde{\alpha}_{r,N}| \le \alpha_r$ and $|\tilde{\alpha}_{r,N} - \alpha_r| \le \nu_r / N$, and define $\tilde{f}_N(\cdot) := \sum_{r \in \mathcal{R}} \tilde{\alpha}_{r,N} e_r(\cdot) \in \mathcal{H}_{\gamma,M,\mathbb{Q}}$. Then for each $x \in \mathcal{X}$, by Cauchy–Schwarz and Mercer's theorem,

$$\begin{aligned} \left| \tilde{f}_N(x) - f(x) \right| &\le \sum_{r \in \mathcal{R}} \left| \tilde{\alpha}_{r,N} - \alpha_r \right| |e_r(x)| \le \frac{1}{N} \sum_{r \in \mathcal{R}} \nu_r |e_r(x)| \\ &\le \frac{1}{N} \left(\sum_{r \in \mathcal{R}} \nu_r \right)^{1/2} \left(\sum_{r \in \mathcal{R}} \nu_r e_r(x)^2 \right)^{1/2} \le \frac{\sqrt{K}}{N} \left(\sum_{r \in \mathcal{R}} \nu_r \right)^{1/2} \to 0 \end{aligned}$$

as $N \to \infty$. Thus $\mathcal{H}_{\gamma,M,\mathbb{Q}}$ is countable and dense in $\mathcal{H}_{\gamma,M}$ under the topology of pointwise convergence; it follows that $\mathcal{H}_{\gamma,M}$ is pointwise measurable in the terminology of van der

Vaart and Wellner (1996, Definition 2.3.3). Let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher variables (taking values ± 1 with equal probability) that are independent of the data. By a symmetrisation inequality (van der Vaart and Wellner, 1996, Lemma 2.3.1), Jensen's inequality, Cauchy–Schwarz, since $\varepsilon_1, \dots, \varepsilon_n$ are conditionally independent given the data, and as $\varepsilon_i^2 = 1$ and $\mathbb{E}(\varepsilon_i) = 0$ for each $i \in [n]$,

$$\begin{aligned}
\mathbb{E}\biggl(\sup_{f \in \mathcal{H}_{\gamma,M}} \bigl|P_n(f) - P_X(f)\bigr|\biggr) &\leq 2\,\mathbb{E}\biggl(\sup_{f \in \mathcal{H}_{\gamma,M}} \biggl|\frac{1}{n}\sum_{i=1}^n \varepsilon_i f(X_i)\biggr|\biggr) \\
&= 2\,\mathbb{E}\biggl(\sup_{(\alpha_r)_{r\in\mathcal{R}}:\, \sum_{r\in\mathcal{R}} \alpha_r^2/a_r \leq M^2} \frac{1}{n}\sum_{i=1}^n \varepsilon_i \sum_{r\in\mathcal{R}} \alpha_r e_r(X_i)\biggr) \\
&\leq 2\biggl[\mathbb{E}\biggl\{\sup_{(\alpha_r)_{r\in\mathcal{R}}:\, \sum_{r\in\mathcal{R}} \alpha_r^2/a_r \leq M^2} \biggl(\sum_{r\in\mathcal{R}} \alpha_r \frac{1}{n}\sum_{i=1}^n \varepsilon_i e_r(X_i)\biggr)^2\biggr\}\biggr]^{1/2} \\
&\leq 2\biggl[\mathbb{E}\biggl\{\sup_{(\alpha_r)_{r\in\mathcal{R}}:\, \sum_{r\in\mathcal{R}} \alpha_r^2/a_r \leq M^2} \sum_{r\in\mathcal{R}} a_r \biggl(\frac{1}{n}\sum_{i=1}^n \varepsilon_i e_r(X_i)\biggr)^2 \sum_{r'\in\mathcal{R}} \frac{\alpha_{r'}^2}{a_{r'}}\biggr\}\biggr]^{1/2} \\
&\leq 2M\biggl\{\sum_{r\in\mathcal{R}} \frac{a_r}{n^2} \sum_{i=1}^n \sum_{j=1}^n \mathbb{E}\bigl(\varepsilon_i \varepsilon_j e_r(X_i) e_r(X_j)\bigr)\biggr\}^{1/2} \\
&= 2M\biggl\{\frac{1}{n^2}\sum_{i=1}^n \mathbb{E}\biggl(\sum_{r\in\mathcal{R}} a_r e_r(X_i)^2\biggr)\biggr\}^{1/2} \leq 2M\sqrt{\frac{H_{2,\gamma}}{n}} \leq 2M\sqrt{\frac{H_{\infty,\gamma}}{n}}.
\end{aligned}$$

For each $f \in \mathcal{H}_{\gamma,M}$, Lemma 2 gives $\|f\|_\infty \leq \sqrt{H_{\infty,\gamma}}\|f\|_{\mathcal{H}_\gamma} \leq M\sqrt{H_{\infty,\gamma}}$, so by the bounded differences inequality (Boucheron et al., 2013, Theorem 6.2), for $s \geq 1$,

$$\mathbb{P}\biggl(\sup_{f \in \mathcal{H}_{\gamma,M}} \bigl|P_n(f) - P_X(f)\bigr| > 4Ms\sqrt{\frac{H_{\infty,\gamma}}{n}}\biggr) \leq e^{-s^2}. \qquad \square$$

E.3.7 Proof of Theorem 4

We prove our convergence result for $\hat{f}_{n,\gamma}$, bounding each term in the basic inequality (17).

Proof of Theorem 4. We write $C_1^0, C_2^0, \dots$ for positive quantities depending only on $\|f_0\|_{\mathcal{H}}$, Λ, q_1, $\|1_{\mathcal{X}}\|_{\mathcal{H}}$ and K, noting that this allows us to absorb increasing functions of $\|f_0\|_\infty$ by Lemma 2. We begin by deriving a lower bound for $H_{\infty,\gamma}$. If $\gamma \geq \nu_1$, then

$$H_{\infty,\gamma} = \sup_{x\in\mathcal{X}} \sum_{r\in\mathcal{R}} \frac{e_r(x)^2}{1+\gamma/\nu_r} \geq \frac{1}{2\gamma}\sup_{x\in\mathcal{X}} \sum_{r\in\mathcal{R}} \nu_r e_r(x)^2 = \frac{1}{2\gamma}\sup_{x\in\mathcal{X}} k(x,x) = \frac{K}{2\gamma},$$

while if $\gamma < \nu_1$, then

$$H_{\infty,\gamma} > \frac{1}{2}\sup_{x\in\mathcal{X}} e_1(x)^2 \geq \frac{\|e_1\|_{L_2}^2}{2} = \frac{1}{2}.$$

Therefore, $2H_{\infty,\gamma} \geq 1 \wedge (K/\gamma)$, so $1 \leq 2H_{\infty,\gamma} + 2\gamma H_{\infty,\gamma}/K$. Taking $(C_0)^2 \geq 4/K$, we have for $\gamma H_{\infty,\gamma} \leq 1/(C_0)^2$ that $1 \leq 2H_{\infty,\gamma} + 2/\{(C_0)^2 K\} \leq 2H_{\infty,\gamma} + 1/2$. Thus, $H_{\infty,\gamma} \geq 1/4$.

We now use the previous results to control each term in (17). By Proposition 23, there exists $C_1^0 > 0$ such that for all $s \geq 1$, with probability at least $1 - e^{-s^2}$,

$$\left|D\ell_n(f_0)(\tilde{f}_{n,\gamma} - f_0)\right| \leq C_1^0 \left(\frac{\sqrt{H_{2,\gamma}} + s}{\sqrt{n}} + s^2 \frac{\sqrt{H_{\infty,\gamma}}}{n} \right) \left\| \tilde{f}_{n,\gamma} - f_0 \right\|_{\mathcal{H}_\gamma}.$$

By Cauchy–Schwarz and Lemma 28, for each $s \geq 1$, with probability at least $1 - e^{-s^2}$,

$$\begin{aligned}
&\left|\gamma \left\langle \tilde{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}}, f_0 - P_n(f_0)1_{\mathcal{X}} \right\rangle_{\mathcal{H}}\right| \\
&\quad \leq \gamma \|f_0 - P_n(f_0)1_{\mathcal{X}}\|_{\mathcal{H}} \left\| \tilde{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}} \right\|_{\mathcal{H}} \\
&\quad \leq \gamma \left(\|f_0\|_{\mathcal{H}} + \|f_0\|_\infty \|1_{\mathcal{X}}\|_{\mathcal{H}} \right) \left(\left\| \tilde{f}_{n,\gamma} - f_0 \right\|_{\mathcal{H}} + |P_n(f_0)| \|1_{\mathcal{X}}\|_{\mathcal{H}} \right) \\
&\quad \leq \gamma \left(\|f_0\|_{\mathcal{H}} + \|f_0\|_\infty \|1_{\mathcal{X}}\|_{\mathcal{H}} \right) \left\{ \left\| \tilde{f}_{n,\gamma} - f_0 \right\|_{\mathcal{H}} + \frac{2s\|f_0\|_\infty \|1_{\mathcal{X}}\|_{\mathcal{H}}}{\sqrt{n}} \right\}.
\end{aligned}$$

Since $P_X(f_0) = 0$, by Lemma 17 and Proposition 24, there exists $C_2^0 > 0$ such that for each $s \geq 1$, with probability at least $1 - e^{-s^2}$,

$$\begin{aligned}
D^2\ell_n(f_0)\left(\tilde{f}_{n,\gamma} - f_0\right)^{\otimes 2} &\geq D^2\ell_\star(f_0)\left(\tilde{f}_{n,\gamma} - f_0\right)^{\otimes 2} - \left\| D^2\ell_n(f_0) - D^2\ell_\star(f_0) \right\|_{\mathcal{H}_\gamma, \mathrm{op}} \left\| \tilde{f}_{n,\gamma} - f_0 \right\|_{\mathcal{H}_\gamma}^2 \\
&\geq \frac{1}{C_2^0} \left\| \tilde{f}_{n,\gamma} - P_X\left(\tilde{f}_{n,\gamma}\right)1_{\mathcal{X}} - f_0 \right\|_{L_2}^2 - C_2^0 s \sqrt{\frac{H_{\infty,\gamma}^2}{n}} \left\| \tilde{f}_{n,\gamma} - f_0 \right\|_{\mathcal{H}_\gamma}^2.
\end{aligned}$$

Note that $P_n(\tilde{f}_{n,\gamma}) = 0$ and $P_X(f_0) = 0$, and recall that $\left\| \tilde{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}} \right\|_{\mathcal{H}_\gamma} \leq M$, where $M^2 := 1/(b^2 H_{\infty,\gamma}) \leq 1/H_{\infty,\gamma}$. Therefore, with $\mathcal{H}_{\gamma,M} := \{f \in \mathcal{H} : \|f\|_{\mathcal{H}_\gamma} \leq M\}$, by Lemmas 28 and 29, there exists $C_3^0 > 0$ such that for $s \geq 1$, with probability at least $1 - e^{-s^2}$,

$$\begin{aligned}
\left|P_X\left(\tilde{f}_{n,\gamma}\right)\right| &= \left|(P_X - P_n)\left(\tilde{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}}\right) - P_n(f_0)\right| \\
&\leq \left|P_n(f_0)\right| + \sup_{f \in \mathcal{H}_{\gamma,M}} \left|P_n(f) - P_X(f)\right| \leq \frac{2s\|f_0\|_\infty}{\sqrt{n}} + CMs\sqrt{\frac{H_{\infty,\gamma}}{n}} \leq \frac{C_3^0 s}{\sqrt{n}}.
\end{aligned}$$

Thus for $s \geq 1$, with probability at least $1 - e^{-s^2}$,

$$\begin{aligned}
\left\| \tilde{f}_{n,\gamma} - f_0 \right\|_{L_2}^2 &= \left\| \tilde{f}_{n,\gamma} - P_X\left(\tilde{f}_{n,\gamma}\right)1_{\mathcal{X}} - f_0 \right\|_{L_2}^2 + P_X\left(\tilde{f}_{n,\gamma}\right)^2 \\
&\qquad + 2P_X\left(\tilde{f}_{n,\gamma}\right)\left\langle \tilde{f}_{n,\gamma} - P_X\left(\tilde{f}_{n,\gamma}\right)1_{\mathcal{X}} - f_0, 1_{\mathcal{X}} \right\rangle_{L_2} \\
&= \left\| \tilde{f}_{n,\gamma} - P_X\left(\tilde{f}_{n,\gamma}\right)1_{\mathcal{X}} - f_0 \right\|_{L_2}^2 + P_X\left(\tilde{f}_{n,\gamma}\right)^2 \\
&\leq \left\| \tilde{f}_{n,\gamma} - P_X\left(\tilde{f}_{n,\gamma}\right)1_{\mathcal{X}} - f_0 \right\|_{L_2}^2 + \frac{(C_3^0)^2 s^2}{n}.
\end{aligned}$$

Therefore, there is $C_4^0 > 0$ such that for $1 \leq s \leq \sqrt{n}/(C_4^0 H_{\infty,\gamma})$, with probability at least $1 - e^{-s^2}$,

$$D^2\ell_n(f_0)\left(\tilde{f}_{n,\gamma} - f_0\right)^{\otimes 2} \geq \frac{1}{C_4^0} \left\| \tilde{f}_{n,\gamma} - f_0 \right\|_{L_2}^2 - \frac{\gamma}{4} \left\| \tilde{f}_{n,\gamma} - f_0 \right\|_{\mathcal{H}}^2 - \frac{C_4^0 s^2}{n}.$$

For $f_1, f_2 \in \mathcal{H}$ we have $\|f_1+f_2\|_{\mathcal{H}}^2 \geq \|f_1\|_{\mathcal{H}}^2/2 - \|f_2\|_{\mathcal{H}}^2$ by the triangle inequality, so by Lemma 28, with probability at least $1-e^{-s^2}$,

$$\begin{aligned}\gamma\big\|\tilde{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}}\big\|_{\mathcal{H}}^2 &\geq \frac{\gamma}{2}\big\|\tilde{f}_{n,\gamma} - f_0\big\|_{\mathcal{H}}^2 - \gamma P_n(f_0)^2\|1_{\mathcal{X}}\|_{\mathcal{H}}^2 \\ &\geq \frac{\gamma}{2}\big\|\tilde{f}_{n,\gamma} - f_0\big\|_{\mathcal{H}}^2 - \frac{4\gamma s^2\|f_0\|_\infty^2\|1_{\mathcal{X}}\|_{\mathcal{H}}^2}{n}.\end{aligned}$$

By Lemmas 25, 26 and 27, there exists a universal constant $C>0$ such that for all $f, f_1 \in \mathcal{H}$ and $s \geq 1$, with probability at least $1-e^{-s^2}$,

$$\begin{aligned}\big|D^3\ell_n(f)(f_1)^{\otimes 3}\big| &\leq 6\|f_1\|_\infty e^{2\|f-f_0\|_\infty}\big\{V(f_0)(f_1,f_1) + \|V_n(f_0)-V(f_0)\|_{\mathcal{H}_\gamma,\mathrm{op}}\|f_1\|_{\mathcal{H}_\gamma}^2\big\} \qquad (51)\\ &\leq 6e^{\|f_0\|_\infty}e^{2\|f-f_0\|_\infty}\Lambda\|f_1\|_\infty\|f_1\|_{L_2}^2 + \frac{Cse^{2\|f_0\|_\infty}e^{2\|f-f_0\|_\infty}}{q_1^2}\sqrt{\frac{H_{\infty,\gamma}^2}{n}}\|f_1\|_\infty\|f_1\|_{\mathcal{H}_\gamma}^2.\end{aligned}$$

As ℓ_n is invariant under addition of constants, we have

$$\big|D^3\ell_n\big(\check{f}_{n,\gamma}\big)\big(\tilde{f}_{n,\gamma}-f_0\big)^{\otimes 3}\big| = \big|D^3\ell_n\big(\check{f}_{n,\gamma}+P_n(f_0)1_{\mathcal{X}}\big)\big(\tilde{f}_{n,\gamma}-f_0+P_n(f_0)1_{\mathcal{X}}\big)^{\otimes 3}\big|.$$

Since $\check{f}_{n,\gamma}$ is on the segment between $f_0 - P_n(f_0)1_{\mathcal{X}}$ and $\tilde{f}_{n,\gamma}$, we have $\|\check{f}_{n,\gamma} - f_0 + P_n(f_0)1_{\mathcal{X}}\|_\infty \leq 1/b$. Thus, by (51), and Lemma 28, there exists $C_5^0>0$ such that for $1 \leq s \leq \sqrt{n}/(C_5^0 H_{\infty,\gamma})$, with probability at least $1-e^{-s^2}$,

$$\begin{aligned}\big|D^3\ell_n\big(\check{f}_{n,\gamma}\big)\big(\tilde{f}_{n,\gamma}-f_0\big)^{\otimes 3}\big| &\leq \frac{C_5^0}{b}\big\|\tilde{f}_{n,\gamma}-f_0+P_n(f_0)1_{\mathcal{X}}\big\|_{L_2}^2 + \frac{C_5^0}{b}\gamma\big\|\tilde{f}_{n,\gamma}-f_0+P_n(f_0)1_{\mathcal{X}}\big\|_{\mathcal{H}}^2 \\ &\leq \frac{2C_5^0}{b}\big\|\tilde{f}_{n,\gamma}-f_0\big\|_{\mathcal{H}_\gamma}^2 + \frac{2C_5^0}{b}\big(1+\|1_{\mathcal{X}}\|_{\mathcal{H}}^2\gamma\big)P_n(f_0)^2 \\ &\leq \frac{2C_5^0}{b}\big\|\tilde{f}_{n,\gamma}-f_0\big\|_{\mathcal{H}_\gamma}^2 + \frac{16C_5^0 s^2\|f_0\|_\infty^2}{bn}\big(1+\|1_{\mathcal{X}}\|_{\mathcal{H}}^2\gamma\big).\end{aligned}$$

Therefore, by substituting these bounds into (17), there exists $C_6^0>0$ such that for $\gamma \leq 1/C_6^0$ and $1 \leq s \leq \sqrt{n}/(C_6^0 H_{\infty,\gamma})$, with probability at least $1-e^{-s^2}$, as $b \geq 1$,

$$\begin{aligned}&\frac{1}{2}D^2\ell_n(f_0)\big(\tilde{f}_{n,\gamma}-f_0\big)^{\otimes 2} + \gamma\big\|\tilde{f}_{n,\gamma}-f_0+P_n(f_0)1_{\mathcal{X}}\big\|_{\mathcal{H}}^2 \\ &\quad\leq \big|D\ell_n(f_0)\big(\tilde{f}_{n,\gamma}-f_0\big)\big| + \frac{1}{6}\big|D^3\ell_n\big(\check{f}_{n,\gamma}\big)\big(\tilde{f}_{n,\gamma}-f_0\big)^{\otimes 3}\big| \\ &\qquad + 2\gamma\big|\big\langle\tilde{f}_{n,\gamma}-f_0+P_n(f_0)1_{\mathcal{X}}, f_0-P_n(f_0)1_{\mathcal{X}}\big\rangle_{\mathcal{H}}\big|, \\ &\frac{1}{C_6^0}\big\|\tilde{f}_{n,\gamma}-f_0\big\|_{L_2}^2 + \frac{3\gamma}{8}\big\|\tilde{f}_{n,\gamma}-f_0\big\|_{\mathcal{H}}^2 \\ &\quad\leq C_6^0\left(\frac{\sqrt{H_{2,\gamma}}+s}{\sqrt{n}} + s^2\frac{\sqrt{H_{\infty,\gamma}}}{n}\right)\big\|\tilde{f}_{n,\gamma}-f_0\big\|_{\mathcal{H}_\gamma} \\ &\qquad + \frac{C_6^0}{b}\big\|\tilde{f}_{n,\gamma}-f_0\big\|_{\mathcal{H}_\gamma}^2 + C_6^0\gamma\big\|\tilde{f}_{n,\gamma}-f_0\big\|_{\mathcal{H}} + C_6^0\left(\frac{s}{\sqrt{n}}+\gamma\right)^2.\end{aligned}$$

By taking $C_6^0 \geq 8/3$ and setting $b := 2(C_6^0)^2$, with probability at least $1-e^{-s^2}$,

$$\frac{1}{2C_6^0}\big\|\tilde{f}_{n,\gamma}-f_0\big\|_{\mathcal{H}_\gamma}^2 \leq C_6^0\left(\frac{\sqrt{H_{2,\gamma}}+s}{\sqrt{n}} + s^2\frac{\sqrt{H_{\infty,\gamma}}}{n} + \sqrt{\gamma}\right)\big\|\tilde{f}_{n,\gamma}-f_0\big\|_{\mathcal{H}_\gamma} + C_6^0\left(\frac{s}{\sqrt{n}}+\gamma\right)^2.$$

Now if $x^2 \le ax + c$ for some $x, a, c \ge 0$, then $x \le a + \sqrt{c}$. It follows that there exists $C_7^0 > 0$ such that for $\gamma \le 1/C_7^0$ and $1 \le s \le \sqrt{n}/(C_7^0 H_{\infty,\gamma})$, with probability at least $1 - e^{-s^2}$,

$$\left\|\tilde{f}_{n,\gamma} - f_0\right\|_{\mathcal{H}_\gamma} \le C_7^0 \left(\frac{\sqrt{H_{2,\gamma}} + s}{\sqrt{n}} + s^2 \frac{\sqrt{H_{\infty,\gamma}}}{n} + \sqrt{\gamma} \right). \tag{52}$$

Next, we show that $\hat{f}_{n,\gamma} = \tilde{f}_{n,\gamma}$ with high probability. By definition of $\tilde{f}_{n,\gamma}$, we have $\hat{f}_{n,\gamma} = \tilde{f}_{n,\gamma}$ whenever $b\sqrt{H_{\infty,\gamma}} \left\|\tilde{f}_{n,\gamma} - f_0 + P_n(f_0) 1_{\mathcal{X}}\right\|_{\mathcal{H}_\gamma} < 1$. By definition of b, this holds if both $4(C_6^0)^2 \sqrt{H_{\infty,\gamma}} \left\|\tilde{f}_{n,\gamma} - f_0\right\|_{\mathcal{H}_\gamma} < 1$ and $4(C_6^0)^2 \sqrt{H_{\infty,\gamma}} |P_n(f_0)| \|1_{\mathcal{X}}\|_{\mathcal{H}_\gamma} < 1$. The first of these holds by (52) with probability at least $1 - e^{-s^2}$ if

$$4(C_6^0)^2 C_7^0 \left(\sqrt{\frac{H_{2,\gamma} H_{\infty,\gamma}}{n}} + s\sqrt{\frac{H_{\infty,\gamma}}{n}} + s^2 \frac{H_{\infty,\gamma}}{n} + \sqrt{\gamma H_{\infty,\gamma}} \right) < 1, \tag{53}$$

while, by Lemma 28, the second holds with probability at least $1 - e^{-s^2}$ if

$$8(C_6^0)^2 s \|f_0\|_\infty \|1_{\mathcal{X}}\|_{\mathcal{H}} \sqrt{\frac{H_{\infty,\gamma}}{n}} < 1. \tag{54}$$

For sufficiently large C_8^0, (53) and (54) both hold simultaneously when $\gamma H_{\infty,\gamma} \le 1/C_8^0$ and $1 \le s \le \sqrt{n}/(C_8^0 H_{\infty,\gamma})$.

Finally, we remove the $H_{\infty,\gamma}$ term from the right-hand side of (52). Under these conditions above, taking $C_8^0 \ge 2$, we have that $s^2\sqrt{H_{\infty,\gamma}}/n \le s/\left(2\sqrt{nH_{\infty,\gamma}}\right) \le s/\sqrt{n}$. Thus, with probability at least $1 - e^{-s^2}$,

$$\left\|\hat{f}_{n,\gamma} - f_0\right\|_{\mathcal{H}_\gamma} \le C_9^0 \left(\frac{\sqrt{H_{2,\gamma}} + s}{\sqrt{n}} + \sqrt{\gamma} \right),$$

whenever $\gamma H_{\infty,\gamma} \le 1/C_9^0$ and $1 \le s \le \sqrt{n}/(C_9^0 H_{\infty,\gamma})$, as required. □

E.3.8 Details for Examples 5, 6 and 7

We prove our results on $H_{\infty,\gamma}$ and $\|1_{\mathcal{X}}\|_{\mathcal{H}}$ for polynomial, Sobolev and Gaussian kernels respectively.

Details for Example 5. Let $\{q_r(\cdot)\}_{r\in[R]}$ be an orthonormal basis under $L_2(P_X)$ for the space $\mathrm{poly}_p(\mathcal{X})$ of d-variate polynomials on $\mathcal{X}$ of degree at most p, where $R \le \binom{d+p}{p}$. For $\alpha = (\alpha_1, \ldots, \alpha_d)^\mathsf{T} \in \mathbb{N}_0^d$ with $|\alpha| := \sum_{j=1}^d \alpha_j \le p$, and for $r \in [R]$, let $b_{\alpha,r} \in \mathbb{R}$ be such that $\prod_{j=1}^d x_j^{\alpha_j} = \sum_{r=1}^R b_{\alpha,r} q_r(x)$ for all $x \in \mathcal{X}$. Let $\boldsymbol{B} = (B_{rr'})_{r,r'\in[R]} \in \mathbb{R}^{R\times R}$ be the matrix with entries

$$B_{rr'} := \sum_{s=0}^p \binom{p}{s} a^{p-s} \sum_{\alpha:|\alpha|=s} \binom{s}{\alpha_1, \ldots, \alpha_d} b_{\alpha,r} b_{\alpha,r'},$$

for $r, r' \in [R]$. Then $\boldsymbol{B}$ is a non-negative linear combination of positive semi-definite rank-one matrices, so is positive semi-definite. It therefore has an eigendecomposition $\boldsymbol{B} = \boldsymbol{U}^\mathsf{T}\boldsymbol{D}\boldsymbol{U}$

where $\boldsymbol{U} \in \mathbb{R}^{R\times R}$ is orthogonal and $\boldsymbol{D} \in \mathbb{R}^{R\times R}$ is diagonal with non-negative diagonal entries given by $(\nu_r)_{r\in[R]}$. Define $e : \mathcal{X} \to \mathbb{R}^d$ by $e(x) := \boldsymbol{U}q(x)$, where $q(x) := \big(q_1(x), \dots, q_R(x)\big)^\mathsf{T}$. By the binomial and multinomial theorems,

$$\begin{aligned} k_{p,a}^{\mathrm{poly}}(x,y) &= \sum_{s=0}^{p} \binom{p}{s} a^{p-s} \sum_{\alpha:|\alpha|=s} \binom{s}{\alpha_1,\dots,\alpha_d} \Big(\prod_{j=1}^{d} x_j^{\alpha_j}\Big)\Big(\prod_{j=1}^{d} y_j^{\alpha_j}\Big) \\ &= \sum_{s=0}^{p} \binom{p}{s} a^{p-s} \sum_{\alpha:|\alpha|=s} \binom{s}{\alpha_1,\dots,\alpha_d} \sum_{r=1}^{R} b_{\alpha,r} q_r(x) \sum_{r'=1}^{R} b_{\alpha,r'} q_{r'}(y) \\ &= \sum_{r=1}^{R}\sum_{r'=1}^{R} q_r(x) B_{rr'} q_{r'}(y) = q(x)^\mathsf{T} \boldsymbol{B} q(y) = e(x)^\mathsf{T} \boldsymbol{D} e(y) = \sum_{r=1}^{R} \nu_r e_r(x) e_r(y). \end{aligned}$$

Further, $\{e_r(\cdot)\}_{r\in[R]}$ forms an $L_2(P_X)$-orthonormal basis of $\mathrm{poly}_p(\mathcal{X})$ as

$$\int_{\mathcal{X}} e(x)e(x)^\mathsf{T} \,\mathrm{d}P_X(x) = \int_{\mathcal{X}} \boldsymbol{U}q(x)q(x)^\mathsf{T}\boldsymbol{U}^\mathsf{T} \,\mathrm{d}P_X(x) = \boldsymbol{I}_R.$$

Note that

$$\max_{r\in[R]} \sup_{x\in\mathcal{X}} |e_r(x)| \le \sup_{x\in\mathcal{X}} \|e(x)\|_2 = \sup_{x\in\mathcal{X}} \|\boldsymbol{U}q(x)\|_2 = \sup_{x\in\mathcal{X}} \|q(x)\|_2,$$

which depends only on p, d and P_X, and is finite because each $q_r(\cdot)$ is a polynomial and $\mathcal{X}$ is bounded. Define $\mathcal{R} := \{r \in [R] : \nu_r > 0\}$, and endow it with the ordering induced by the reverse order of $(\nu_r)_{r\in\mathcal{R}}$, so that the representation $k_{p,a}^{\mathrm{poly}}(x,y) = \sum_{r\in\mathcal{R}} \nu_r e_r(x) e_r(y)$ of the kernel $k_{p,a}^{\mathrm{poly}}$ aligns with that in Mercer's theorem (Samworth and Shah, 2026, Theorem A.8.8). Since $a_r = 1/(1+\gamma/\nu_r) \le 1$ for $r \in \mathcal{R}$, there exists $C(p,d,P_X) > 0$ such that

$$H_{\infty,\gamma} \le R \max_{r\in[R]} \sup_{x\in\mathcal{X}} e_r(x)^2 \le C(p,d,P_X).$$

Finally, we compute $\|1_{\mathcal{X}}\|_{\mathcal{H}}$. By the binomial theorem,

$$k_{p,a}^{\mathrm{poly}}(x,y) = \sum_{r=0}^{p} \binom{p}{r} (x^\mathsf{T}y)^r a^{p-r}.$$

Since $(x,y) \mapsto x^\mathsf{T}y$ is a kernel, and since kernels are preserved by non-negative scaling, sums and products (Samworth and Shah, 2026, Proposition 6.4*(a)* and *(c)*), we conclude that $k_{p,a}^{\mathrm{poly}}(\cdot,\cdot) - a^p$ is a kernel on $\mathcal{X}$. By considering $x = y = 0_d$, we see that $k_{p,a}^{\mathrm{poly}}(\cdot,\cdot) - a^p - \delta$ is not a kernel if $\delta > 0$. We conclude by Lemma 40 that $1_{\mathcal{X}} \in \mathcal{H}$ if and only if $a > 0$, in which case $\|1_{\mathcal{X}}\|_{\mathcal{H}}^2 = 1/a^p$. □

Details for Example 6. We argue similarly to Samworth and Shah (2026, Example 6.22). Suppose that $e_\nu : [0,1] \to \mathbb{R}$ is a unit $L_2(P_X)$-norm eigenfunction of $\mathcal{T}_{k_{1,a}^{\mathrm{Sob}}}$ with eigenvalue $\nu > 0$. For $x \in [0,1]$,

$$\begin{aligned} \nu e_\nu(x) = \mathcal{T}_{k_{1,a}^{\mathrm{Sob}}}(e_\nu)(x) &= \int_{\mathcal{X}} k_{1,a}^{\mathrm{Sob}}(x,y) e_\nu(y) \,\mathrm{d}P_X(y) = \int_0^1 \big(a + (x \wedge y)\big) e_\nu(y) \,\mathrm{d}y \\ &= a\int_0^1 e_\nu(y)\,\mathrm{d}y + \int_0^x y e_\nu(y)\,\mathrm{d}y + x\int_x^1 e_\nu(y)\,\mathrm{d}y. \end{aligned} \tag{55}$$

Differentiating twice with respect to x, we obtain that for all $x \in [0,1]$,

$$\int_x^1 e_\nu(y)\,\mathrm{d}y = \nu e_\nu'(x), \tag{56}$$

$$-e_\nu(x) = \nu e_\nu''(x). \tag{57}$$

By (57), there exist $A_\nu, B_\nu \in \mathbb{R}$ with $e_\nu(x) = A_\nu \sin(x/\sqrt{\nu}) + B_\nu \cos(x/\sqrt{\nu})$ for all $x \in [0,1]$. Substituting into (56) and setting $x = 1$, we have

$$A_\nu \cos(1/\sqrt{\nu}) = B_\nu \sin(1/\sqrt{\nu}). \tag{58}$$

Returning to (55), with $x = 0$ we obtain $a\{A_\nu(1 - \cos(1/\sqrt{\nu})) + B_\nu \sin(1/\sqrt{\nu})\} = \sqrt{\nu} B_\nu$. Combining this with (58), we obtain

$$aA_\nu = \sqrt{\nu} B_\nu. \tag{59}$$

Now

$$\begin{aligned} 4 &= 4\int_0^1 e_\nu(x)^2\,\mathrm{d}x \\ &= A_\nu^2\{2 - \sqrt{\nu}\sin(2/\sqrt{\nu})\} + \sqrt{\nu}A_\nu B_\nu\{2 - 2\cos(2/\sqrt{\nu})\} + B_\nu^2\{2 + \sqrt{\nu}\sin(2/\sqrt{\nu})\}. \end{aligned}$$

Substituting (59) into (58), we see that $\tan(1/\sqrt{\nu}) = \sqrt{\nu}/a$, interpreted as $\tan(1/\sqrt{\nu}) = \infty$ if $a = 0$. Thus, by the half-angle identities, $\sin(2/\sqrt{\nu}) = 2\sqrt{\nu}a/(a^2 + \nu)$ and $\cos(2/\sqrt{\nu}) = (a^2 - \nu)/(a^2 + \nu)$. Thus, since e_ν is defined only up to sign, we may take

$$A_\nu = \sqrt{\frac{2\nu}{a^2 + \nu + a\nu}} \quad \text{and} \quad B_\nu = \sqrt{\frac{2a^2}{a^2 + \nu + a\nu}}.$$

Further, for every $r \in \mathbb{N}$, there exists $\nu_r \in \big[(r - 1/2)^{-2}\pi^{-2}, (r-1)^{-2}\pi^{-2}\big)$ such that $\tan(1/\sqrt{\nu_r}) = \sqrt{\nu_r}/a$, and these are the only positive values of ν to solve this equation. Hence, the eigenvalue–eigenfunction pairs for $\mathcal{T}_{k_{1,a}^{\mathrm{Sob}}}$ are, for $r \in \mathbb{N}$,

$$\begin{aligned} \nu_r &\in \left[\frac{1}{(r - 1/2)^2\pi^2}, \frac{1}{(r-1)^2\pi^2}\right), \\ e_r(\cdot) &= \sqrt{\frac{2\nu_r}{a^2 + \nu_r + a\nu_r}} \sin\left(\frac{\cdot}{\sqrt{\nu_r}}\right) + \sqrt{\frac{2a^2}{a^2 + \nu_r + a\nu_r}} \cos\left(\frac{\cdot}{\sqrt{\nu_r}}\right). \end{aligned}$$

By Cauchy–Schwarz, $\|e_r\|_\infty^2 \le 2(\nu_r + a^2)/(a^2 + \nu_r + a\nu_r) \le 2$. For $r \ge 2$, we have $a_r = 1/(1 + \gamma/\nu_r) \le 1/\{1 + \gamma(r-1)^2\pi^2\}$, while $a_1 \le 1$. Therefore,

$$H_{\infty,\gamma} \le \sum_{r=1}^\infty \frac{2}{1 + \gamma/\nu_r} \le 2 + \sum_{r=1}^\infty \frac{2}{1 + \gamma r^2\pi^2} \le 2 + \int_0^\infty \frac{2}{1 + \gamma x^2\pi^2}\,\mathrm{d}x = 2 + \frac{1}{\sqrt{\gamma}}.$$

Finally, we compute $\|1_\mathcal{X}\|_\mathcal{H}$. The functions $k_{1,a}^{\mathrm{Sob}}$ and $k_{2,a}^{\mathrm{Sob}}$ are kernels by Samworth and Shah (2026, Examples 6.8 and 6.9 and Proposition 6.4(a)). By considering $x = y = 0$, we see that $k_{1,a}^{\mathrm{Sob}} - a - \delta$ and $k_{2,a}^{\mathrm{Sob}} - a - \delta$ are not kernels if $\delta > 0$. The conclusions now follow by Lemma 40. □

Details for Example 7. Define $\tilde{k}^{\mathrm{Gauss}}_{\Sigma,a} : \mathcal{X} \times \mathcal{X} \to [0,\infty)$ by

$$\tilde{k}^{\mathrm{Gauss}}_{\Sigma,a}(x,y) = a + \exp\bigl(-(\tilde{x}-\tilde{y})^{\mathsf{T}}(\tilde{x}-\tilde{y})\bigr),$$

and set $\tilde{x} := \Sigma^{1/2}x$ and $\tilde{y} := \Sigma^{1/2}y$. Since the constant map $(x,y) \mapsto a$ is a kernel and the standard Gaussian kernel is a kernel (Samworth and Shah, 2026, Example 6.7), and as addition preserves kernels (Samworth and Shah, 2026, Proposition 6.4), we deduce that $k^{\mathrm{Gauss}}_{\Sigma,a}$ is a kernel on $\mathcal{X}$.

Now suppose that $a = 0$. Let $n \in \mathbb{N}$, let $\delta \in \bigl(0, 1/(2n^2)\bigr)$ and, recalling that $\mathcal{X}$ is unbounded, choose $x_1, \ldots, x_n \in \mathcal{X}$ so that for each $i, j \in [n]$ with $i \neq j$, we have $\|x_i - x_j\|_2^2 \geq \lambda_{\max}(\Sigma)\sqrt{\log(1/\delta)}$. Then $(x_i - x_j)^{\mathsf{T}}\Sigma^{-1}(x_i - x_j) \geq \lambda_{\min}(\Sigma^{-1})\|x_i - x_j\|_2^2 \geq \log(1/\delta)$, so $k^{\mathrm{Gauss}}_{\Sigma,0}(x_i, x_j) \leq \delta$. Note also that $k^{\mathrm{Gauss}}_{\Sigma,0}(x_i, x_i) = 1$ for each $i \in [n]$. Now define $\boldsymbol{K}_n := \boldsymbol{K}(x_1, \ldots, x_n)$ in the notation of Proposition 11, so that $\|\boldsymbol{I}_n - \boldsymbol{K}_n\|_{\mathrm{op}} \leq n\|\boldsymbol{I}_n - \boldsymbol{K}_n\|_\infty \leq n\delta < 1$. Hence $\boldsymbol{K}_n$ is invertible, and since $\boldsymbol{I}_n - \boldsymbol{K}_n^{-1} = (\boldsymbol{I}_n - \boldsymbol{K}_n^{-1})(\boldsymbol{I}_n - \boldsymbol{K}_n) - (\boldsymbol{I}_n - \boldsymbol{K}_n)$, we have

$$\bigl\|\boldsymbol{I}_n - \boldsymbol{K}_n^{-1}\bigr\|_{\mathrm{op}} \leq \frac{\bigl\|\boldsymbol{I}_n - \boldsymbol{K}_n\bigr\|_{\mathrm{op}}}{1 - \bigl\|\boldsymbol{I}_n - \boldsymbol{K}_n\bigr\|_{\mathrm{op}}} \leq \frac{n\delta}{1 - n\delta} \leq 2n\delta.$$

It follows that

$$\bigl|1_n^{\mathsf{T}}1_n - 1_n^{\mathsf{T}}\boldsymbol{K}_n^{-1}1_n\bigr| \leq 2n\delta\|1_n\|_2^2 = 2n^2\delta < 1,$$

so that

$$\kappa(\boldsymbol{K}_n) = \frac{1}{1_n^{\mathsf{T}}\boldsymbol{K}_n^{-1}1_n} \leq \frac{1}{1_n^{\mathsf{T}}1_n - 1} = \frac{1}{n-1} \to 0$$

as $n \to \infty$. We deduce from Proposition 11 that $\kappa_{k^{\mathrm{Gauss}}_{\Sigma,0}} = 0$ and so $1_{\mathcal{X}} \notin \mathcal{H}$. Moreover, by Lemma 40, if $a > 0$ then $1_{\mathcal{X}} \in \mathcal{H}$ and $\|1_{\mathcal{X}}\|_{\mathcal{H}}^2 = 1/a$. □

E.3.9 Proof of the minimax lower bound

Proof of Theorem 8. Since $\lambda_0(t) = 1$ for all $t \in [0,1]$ and no censoring is present, the conditional hazard function (1) at $f_0 \in \mathcal{H}$ is

$$h(t \mid x) = -\frac{\mathrm{d}}{\mathrm{d}t}\log \mathbb{P}(T > t \mid X = x) = e^{f_0(x)}.$$

It follows that $T \mid X \sim \mathrm{Exp}\bigl(e^{f_0(X)}\bigr)$ and so the log-likelihood ratio between $f_0, g_0 \in \mathcal{H}$ for a single observation (X,T), is

$$\hat{\Lambda}(f_0, g_0; X, T) = f_0(X) - g_0(X) + T\bigl(e^{g_0(X)} - e^{f_0(X)}\bigr).$$

Writing $P_{f_0}^{\otimes n}$ for the joint distributions of n independent copies of (X,T) when $X \sim P_X$ and $T \mid X \sim \mathrm{Exp}\bigl(e^{f_0(X)}\bigr)$ and defining $P_{g_0}^{\otimes n}$ similarly, it follows that the Kullback–Leibler

divergence from $P_{g_0}^{\otimes n}$ to $P_{f_0}^{\otimes n}$ satisfies

$$\begin{aligned}\mathrm{KL}\big(P_{f_0}^{\otimes n} \,\|\, P_{g_0}^{\otimes n}\big) &= \sum_{i=1}^{n} \mathbb{E}_{P_{f_0}^{\otimes n}}\big\{\hat{\Lambda}(f_0, g_0; X_i, T_i)\big\} \\ &= n\,\mathbb{E}_{P_{f_0}}\big\{f_0(X) - g_0(X) + T\big(e^{g_0(X)} - e^{f_0(X)}\big)\big\} \\ &= n\,\mathbb{E}_{P_X}\big\{e^{g_0(X) - f_0(X)} - 1 + f_0(X) - g_0(X)\big\}.\end{aligned}$$

Since $f \in \mathcal{H}_1$, we have $\|f\|_\infty^2 \leq K$ by Lemma 2. Since $e^x - 1 - x \leq 2x^2$ for $x \in [-2, 2]$, we have for $f_0, g_0 \in \mathcal{H}_1$ that

$$\mathrm{KL}\big(P_{f_0}^{\otimes n} \,\|\, P_{g_0}^{\otimes n}\big) \leq 2n\|f_0 - g_0\|_{L_2}^2.$$

We now proceed by cases, depending on whether (i) $\mathcal{R} = [R]$ for some $R \in \mathbb{N}$; or (ii) $\mathcal{R} = \mathbb{N}$. In case (i), consider the distributions induced by $f_0 \in \{0, \sqrt{\gamma} e_1\}$, where $\gamma \in (0, 1]$ is to be chosen later and $e_1 \in \mathcal{H}$ is an eigenfunction with $\|e_1\|_{L_2} = 1$ associated with the largest eigenvalue ν_1 of the kernel integral operator T_k. We have $\sqrt{\gamma} e_1 \in \mathcal{H}_1$ whenever $\gamma \leq 1/\|e_1\|_{\mathcal{H}}^2 = \nu_1$. Applying Le Cam's two-point lemma (e.g. Samworth and Shah, 2026, Lemma 8.4) and Pinsker's inequality (Samworth and Shah, 2026, Exercise 8.2*(b)*) then yields

$$\begin{aligned}\inf_{\hat{f}_n \in \mathcal{F}_n} \sup_{f_0 \in \mathcal{H}_1} \mathbb{E}\big(\|\hat{f}_n - f_0\|_{L_2}^2\big) &\geq \frac{\gamma \|e_1\|_{L_2}^2}{2}\Big\{1 - \mathrm{TV}\big(P_0^{\otimes n}, P_{\sqrt{\gamma} e_1}^{\otimes n}\big)\Big\} \\ &\geq \frac{\gamma}{2}\left\{1 - \frac{\mathrm{KL}\big(P_0^{\otimes n} \,\|\, P_{\sqrt{\gamma} e_1}^{\otimes n}\big)^{1/2}}{2^{1/2}}\right\} \geq \frac{\gamma}{2}\big(1 - \sqrt{n\gamma}\big).\end{aligned}$$

Setting $\gamma = \gamma^\star := \nu_1 \wedge \{1/(4n)\}$ gives, for all $n \in \mathbb{N}$,

$$\inf_{\hat{f}_n \in \mathcal{F}_n} \sup_{f_0 \in \mathcal{H}_1} \mathbb{E}\big(\|\hat{f}_n - f_0\|_{L_2}^2\big) \geq \frac{\gamma^\star}{4} = \frac{\nu_1}{4} \wedge \frac{1}{16n} \geq \frac{1}{C_1 n}$$

where $C_1 > 0$ depends only on ν_1. As $\gamma \searrow 0$ we have $H_{2,\gamma} \nearrow R$, so $\inf_{\gamma > 0}\big\{\gamma + H_{2,\gamma}/n\big\} \leq R/n$. Therefore, there exists $C_2 > 0$ depending only on R and ν_1 such that

$$\inf_{\hat{f}_n \in \mathcal{F}_n} \sup_{f_0 \in \mathcal{H}_1} \mathbb{E}\big(\|\hat{f}_n - f_0\|_{L_2}^2\big) \geq \frac{1}{C_2} \inf_{\gamma > 0}\left\{\gamma + \frac{H_{2,\gamma}}{n}\right\},$$

concluding the analysis of the setting where $\mathcal{R} = [R]$.

We now consider the case where $\mathcal{R} = \mathbb{N}$, following ideas of Yang et al. (2017). For $\gamma \in (0, 1]$, define $\tilde{\mathcal{H}}_\gamma := \{f \in \mathcal{H} : \|f\|_{\mathcal{H}_\gamma}^2 \leq \gamma\}$, so that $\tilde{\mathcal{H}}_\gamma \subseteq \tilde{\mathcal{H}}_1 \subseteq \mathcal{H}_1$. For $d \in \mathcal{R}$ and $b > 0$ to be chosen later, let $\beta := bZ$, where Z is a standard Gaussian random vector in $\mathbb{R}^d$. Taking $\gamma a_d / b^2 \geq 2d$, by a chi-squared upper tail bound (Laurent and Massart, 2000, Lemma 1), we have

$$\mathbb{P}\left(\sum_{r=1}^{d} \frac{\beta_r^2}{a_r} > \gamma\right) \leq \mathbb{P}\left(\|Z\|_2^2 > \frac{\gamma a_d}{b^2}\right) \leq \mathbb{P}\big(\|Z\|_2^2 > 2d\big) \leq e^{-d/16}.$$

For $M \in \mathbb{N}$, let $\beta^{(1)}, \dots, \beta^{(M)}$ be independent copies of β and define the random function $f_\beta := \sum_{r=1}^d \beta_r e_r$. Since $\|f_\beta\|_{\mathcal{H}_\gamma}^2 = \sum_{r=1}^d \beta_r^2/a_r$, we have by a union bound that

$$\mathbb{P}\biggl(\bigcap_{m=1}^M \bigl\{f_{\beta^{(m)}} \in \tilde{\mathcal{H}}_\gamma\bigr\}\biggr) \geq 1 - Me^{-d/16}.$$

For distinct $m, m' \in [M]$ and independent standard Gaussian vectors Z, Z' in $\mathbb{R}^d$, and assuming that $\gamma/(8b^2) \leq d/2$, we have

$$\begin{aligned}\mathbb{P}\biggl(\bigl\|f_{\beta^{(m)}} - f_{\beta^{(m')}}\bigr\|_{L_2}^2 \leq \frac{\gamma}{4}\biggr) &= \mathbb{P}\biggl(\sum_{r=1}^d \bigl(\beta_r^{(m)} - \beta_r^{(m')}\bigr)^2 \leq \frac{\gamma}{4}\biggr) = \mathbb{P}\biggl(b^2\|Z - Z'\|_2^2 \leq \frac{\gamma}{4}\biggr) \\ &= \mathbb{P}\biggl(\|Z\|_2^2 \leq \frac{\gamma}{8b^2}\biggr) \leq \mathbb{P}\biggl(\|Z\|_2^2 \leq \frac{d}{2}\biggr) \leq e^{-d/16},\end{aligned}$$

by a chi-squared lower tail bound (Laurent and Massart, 2000, Lemma 1). Thus, by another union bound, as $\binom{M}{2} + M \leq M^2$, the collection $\bigl\{f_{\beta^{(m)}} : m \in [M]\bigr\}$ forms a $\sqrt{\gamma/4}$-packing of $\tilde{\mathcal{H}}_\gamma$ under $\|\cdot\|_{L_2}$ with probability at least $1 - M^2e^{-d/16}$. Taking $\log M \leq d/64$ makes this probability positive, and so a packing $\{f_1, \dots, f_{M^\star}\}$ of cardinality $M^\star := \lfloor e^{d/64} \rfloor$ exists whenever both $\gamma a_d/b^2 \geq 2d$ and $\gamma/(8b^2) \leq d/2$. We may satisfy these inequalities by choosing $b^2 := \gamma/(4d)$ and when $a_d \geq 1/2$; that is, $\gamma \leq \nu_d$. Note that $f \in \tilde{\mathcal{H}}_\gamma$ implies $\|f\|_{L_2}^2 \leq \gamma$. Therefore, by Fano's lemma (e.g., Samworth and Shah, 2026, Corollary 8.12), taking $d \geq 64 \log 2$, and as $\log(2 - 1/x)/\log x \leq \log(3/2)/\log 2 \leq 3/4$ for $x \geq 2$,

$$\begin{aligned}\inf_{\hat{f}_n \in \mathcal{F}_n} \sup_{f_0 \in \mathcal{H}_1} \mathbb{E}\bigl(\|\hat{f}_n - f_0\|_{L_2}^2\bigr) &\geq \frac{\gamma}{4}\biggl(1 - \frac{\sum_{m=1}^{M^\star} \mathrm{KL}\bigl(P_0^{\otimes n} \,\|\, P_{f_m}^{\otimes n}\bigr)/M^\star + \log(2 - 1/M^\star)}{\log M^\star}\biggr) \\ &\geq \frac{\gamma}{4}\biggl(\frac{1}{4} - \frac{2n\sum_{m=1}^{M^\star} \|f_m\|_{L_2}^2/M^\star}{d/128}\biggr) = \frac{\gamma}{16}\biggl\{1 - \frac{1024n\gamma}{d}\biggr\},\end{aligned}$$

where we used the fact that if $d \geq 64 \log 2$ then $M^\star \geq 2 \vee \lfloor e^{d/64} \rfloor$ so $M^\star \geq e^{d/128}$. Thus, if $n\gamma/d \leq 1/2048$, $\gamma \leq \nu_d$ and $d \geq 64 \log 2$, then

$$\inf_{\hat{f}_n \in \mathcal{F}_n} \sup_{f_0 \in \mathcal{H}_1} \mathbb{E}\bigl(\|\hat{f}_n - f_0\|_{L_2}^2\bigr) \geq \frac{\gamma}{32}.$$

Now, by condition (i) on the kernel, for $\gamma \geq \nu_d$, we have

$$\begin{aligned}H_{2,\gamma} = \sum_{r \in \mathcal{R}} \frac{1}{1 + \gamma/\nu_r} &\leq \sum_{r \in \mathcal{R}, r \leq d} \frac{1}{1 + \nu_d/\nu_r} + \sum_{r \in \mathcal{R}, r > d} \frac{1}{1 + \nu_d/\nu_r} \\ &\leq d + \frac{1}{\nu_d} \sum_{r \in \mathcal{R}, r > d} \nu_r \leq (C_K + 1)d.\end{aligned}$$

Taking $\gamma := \nu_d$ and $C_3 := 2048(C_K + 1)$, and writing $\bar{d} := \lceil 64 \log 2 \rceil = 45$, we deduce that

$$\inf_{\hat{f}_n \in \mathcal{F}_n} \sup_{f_0 \in \mathcal{H}_1} \mathbb{E}\bigl(\|\hat{f}_n - f_0\|_{L_2}^2\bigr) \geq \frac{1}{32} \max\biggl\{\nu_d : d \in \mathcal{R},\, \nu_d \leq \frac{H_{2,\nu_d}}{C_3 n} \text{ and } d \geq \bar{d}\biggr\}.$$

Taking $n > H_{2,\nu_{\bar{d}}}/(C_3\nu_{\bar{d}}) =: n_0$ ensures that $\nu_d > H_{2,\nu_d}/(C_3 n)$ for all $d \le \bar{d}$. Therefore, for $n > n_0$,

$$\inf_{\hat{f}_n \in \mathcal{F}_n} \sup_{f_0 \in \mathcal{H}_1} \mathbb{E}\big(\|\hat{f}_n - f_0\|_{L_2}^2\big) \ge \frac{1}{32} \max\left\{\nu_d : d \in \mathcal{R},\, \nu_d \le \frac{H_{2,\nu_d}}{C_3 n}\right\}.$$

This maximum is attained at $d^\star := \min\big\{d \in \mathcal{R} : \nu_d \le H_{2,\nu_d}/(C_3 n)\big\}$, since $\mathcal{R} = \mathbb{N}$ and $d \mapsto \nu_d$ is decreasing with $\lim_{d\to\infty} \nu_d = 0$ and $d \mapsto H_{2,\nu_d}$ is increasing with $\lim_{d\to\infty} H_{2,\nu_d} = \infty$. Further, we have $d^\star \ge \bar{d}$. By minimality of $d^\star$ and condition (ii) on the kernel, $\nu_{d^\star-1} > H_{2,\nu_{d^\star-1}}/(C_3 n)$ and so $\nu_{d^\star-1} + H_{2,\nu_{d^\star-1}}/(C_3 n) < 2\nu_{d^\star-1} \le 2C_K \nu_{d^\star}$. Therefore, for $n > n_0$,

$$\inf_{\hat{f}_n \in \mathcal{F}_n} \sup_{f_0 \in \mathcal{H}_1} \mathbb{E}\big(\|\hat{f}_n - f_0\|_{L_2}^2\big) \ge \frac{\nu_{d^\star}}{32} \ge \frac{1}{64 C_K}\left(\nu_{d^\star-1} + \frac{H_{2,\nu_{d^\star-1}}}{C_3 n}\right).$$

Since the minimax risk is decreasing in n, we have for all $n \in \mathbb{N}$ that

$$\inf_{\hat{f}_n \in \mathcal{F}_n} \sup_{f_0 \in \mathcal{H}_1} \mathbb{E}\big(\|\hat{f}_n - f_0\|_{L_2}^2\big) \ge \frac{1}{C_4} \inf_{\gamma > 0}\left(\gamma + \frac{H_{2,\gamma}}{n}\right),$$

where $C_4 > 0$ depends only on C_K and n_0. Finally, by condition (ii) on the kernel again,

$$n_0 = \frac{H_{2,\nu_{\bar{d}}}}{C_3 \nu_{\bar{d}}} \le \frac{(C_K+1)\bar{d}}{C_3 \nu_{\bar{d}}} \le \frac{(C_K+1)\bar{d} C_K^{\bar{d}-1}}{C_3 \nu_1} = \frac{45(C_K+1)C_K^{44}}{C_3 \nu_1},$$

and depends only on C_K and ν_1, as required. □

E.3.10 Uniform convergence of S_n and DS_n

In Lemmas 30 and 31 we present uniform convergence results for $S_n(f,t)$ and $DS_n(f,t)$ respectively, uniformly over f lying on line segments. These results may be used for bounding terms appearing in mean-value forms of Taylor's theorem; the previous Lemmas 20 and 21, while similar, are unsuited for this task.

Lemma 30 (Uniform convergence of S_n)**.** *Let $f_1, f_2 \in \mathcal{B}(\mathcal{X})$ and for $a \in [0,1]$ define $\tilde{f}_a := a f_1 + (1-a) f_2$. There exists a universal constant $C > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,*

$$\sup_{a\in[0,1]} \sup_{t\in[0,1]} \big|S_n(\tilde{f}_a, t) - S(\tilde{f}_a, t)\big| \le \frac{C s e^{2(\|f_1\|_\infty \vee \|f_2\|_\infty)}}{\sqrt{n}}.$$

Proof of Lemma 30. For $i \in [n]$, $a \in [0,1]$ and $t \in [0,1]$, define

$$\xi_{iat} := R_i(t) e^{\tilde{f}_a(X_i)} - \int_{\mathcal{X}} q(x,t) e^{\tilde{f}_a(x)} \, \mathrm{d}P_X(x),$$

noting that $|\xi_{iat}| \leq 2e^{\|f_1\|_\infty \vee \|f_2\|_\infty}$. It follows that with $\mathcal{I} := \{(i,j) : i,j \in [n], i < j\}$,

$$\begin{aligned}
\sup_{a\in[0,1]} \sup_{t\in[0,1]} \big(S_n(\tilde{f}_a, t) - S(\tilde{f}_a, t)\big)^2 &= \sup_{a\in[0,1]} \sup_{t\in[0,1]} \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \xi_{iat}\xi_{jat} \\
&\leq \sup_{a\in[0,1]} \sup_{t\in[0,1]} \frac{1}{n^2} \sum_{i=1}^n \xi_{iat}^2 + \sup_{a\in[0,1]} \sup_{t\in[0,1]} \frac{2}{n^2} \sum_{(i,j)\in\mathcal{I}} \xi_{iat}\xi_{jat} \\
&\leq \frac{4e^{2(\|f_1\|_\infty \vee \|f_2\|_\infty)}}{n} + \sup_{a\in[0,1]} \sup_{t\in[0,1]} \frac{2}{n(n-1)} \bigg| \sum_{(i,j)\in\mathcal{I}} \xi_{iat}\xi_{jat} \bigg|. \qquad (60)
\end{aligned}$$

The second term in the above display is a degenerate U-process (de la Peña and Giné, 1999, Chapter 5) as $\mathbb{E}\big\{\xi_{iat}\xi_{jat} \mid X_i, T_i, I_i\big\} = \xi_{iat}\,\mathbb{E}(\xi_{jat}) = 0$ for $(i,j) \in \mathcal{I}$, $a \in [0,1]$ and $t \in [0,1]$. We now let $\mathcal{Q}$ denote the set of finitely-supported probability measures on $(\mathcal{X} \times [0,1] \times \{0,1\})^2$, and for $Q \in \mathcal{Q}$, seek to bound the $L_2(Q)$-covering number of the function class

$$\mathcal{G} = \bigg\{ \big((X_i, T_i, I_i), (X_j, T_j, I_j)\big) \mapsto \xi_{iat}\xi_{jat} : a \in [0,1], t \in [0,1] \bigg\}.$$

For $0 \leq a < a' \leq 1$, $0 \leq t < t' \leq 1$ and $i \in [n]$,

$$\begin{aligned}
|\xi_{iat} - \xi_{ia't't}| &\leq \big|R_i(t) e^{\tilde{f}_a(X_i)} - R_i(t') e^{\tilde{f}_{a'}(X_i)}\big| + \int_{\mathcal{X}} \Big| q(x,t) e^{\tilde{f}_a(x)} \,\mathrm{d}P_X(x) - q(x,t') e^{\tilde{f}_{a'}(x)} \Big| \,\mathrm{d}P_X(x) \\
&\leq e^{\|f_1\|_\infty \vee \|f_2\|_\infty} \big(\mathbb{1}_{\{t \leq T_i < t'\}} + \mathbb{P}(t \leq T < t') \big) + 4(a' - a) e^{2(\|f_1\|_\infty \vee \|f_2\|_\infty)},
\end{aligned}$$

so

$$\begin{aligned}
|\xi_{iat}\xi_{jat} - \xi_{ia't't}\xi_{ja't't}| &\leq |\xi_{iat}||\xi_{jat} - \xi_{ja't't}| + |\xi_{jat'}||\xi_{iat} - \xi_{ia't't}| \\
&\leq 2e^{2(\|f_1\|_\infty \vee \|f_2\|_\infty)} \big(\mathbb{1}_{\{t \leq T_i < t'\}} + \mathbb{1}_{\{t \leq T_j < t'\}} + 2\mathbb{P}(t \leq T < t') \big) + 16(a' - a) e^{3(\|f_1\|_\infty \vee \|f_2\|_\infty)} \\
&\leq 16 e^{3(\|f_1\|_\infty \vee \|f_2\|_\infty)} \big(\mathbb{1}_{\{t \leq T_i < t'\}} + \mathbb{1}_{\{t \leq T_j < t'\}} + \mathbb{P}(t \leq T < t') + a' - a \big).
\end{aligned}$$

It follows that there exists a universal constant $C_1 \geq 2$ such that

$$\begin{aligned}
&\mathbb{E}_Q \Big\{ \big(\xi_{iat}\xi_{jat} - \xi_{ia't't}\xi_{ja't't} \big)^2 \Big\} \\
&\quad \leq C_1^2 e^{6(\|f_1\|_\infty \vee \|f_2\|_\infty)} \big\{ \mathbb{P}_Q(t \leq T_i < t') + \mathbb{P}_Q(t \leq T_j < t') + \mathbb{P}(t \leq T < t')^2 + (a' - a)^2 \big\}.
\end{aligned}$$

Given $\varepsilon \in (0,1)$, define $\mathcal{T}_1 := \{t_1, \ldots, t_{\lceil 1/\varepsilon^2 \rceil}\}$, where

$$t_m := \inf\{t \in [0,1] : \mathbb{P}_Q(T_i \leq t) \geq m\varepsilon^2\}$$

for $m \in [\lceil 1/\varepsilon^2 \rceil - 1]$ and $t_{\lceil 1/\varepsilon^2 \rceil} := 1$, so that given any $t \in [0,1]$, there exists $m \in [\lceil 1/\varepsilon^2 \rceil]$ such that $\mathbb{P}_Q(t \leq T_i < t_m) \leq \varepsilon^2$. Let $\mathcal{T}_2$ and $\mathcal{T}_3$ satisfy the same properties but for T_j under Q and T under P_T respectively, and define $\mathcal{T} := \mathcal{T}_1 \cup \mathcal{T}_2 \cup \mathcal{T}_3$. Likewise, let $\mathcal{A} := \{0, \varepsilon, 2\varepsilon, \ldots, (\lceil 1/\varepsilon \rceil - 1)\varepsilon, 1\}$, so that given any $a \in [0,1]$, there exists $a' \in \mathcal{A}$ such that $|a' - a| \leq \varepsilon$. Then the set $\mathcal{T} \times \mathcal{A}$ induces a $\big(2C_1 e^{3(\|f_1\|_\infty \vee \|f_2\|_\infty)}\varepsilon\big)$-cover of $\mathcal{G}$ under $L_2(Q)$, of

cardinality at most $3\lceil 1/\varepsilon^2\rceil \times \lceil 1/\varepsilon\rceil \le 12/\varepsilon^3$. Defining $M := 4e^{3(\|f_1\|_\infty \vee \|f_2\|_\infty)}$, there therefore exists a universal constant $C_2 > 0$ such that for all $\varepsilon \in (0,1)$,

$$N\big(\varepsilon M, \mathcal{G}, \|\cdot\|_{Q,2}\big) \le \frac{C_2}{\varepsilon^3}.$$

As $\log(1+x) \le 1 + \log x$ for $x \ge 1$, and $\int_0^1 \log(1/x)\,\mathrm{d}x = 1$, for a universal constant $C_3 > 0$,

$$J := \sup_{Q\in\mathcal{Q}} \int_0^1 \log\big(1 + N\big(\varepsilon M, \mathcal{G}, \|\cdot\|_{Q,2}\big)\big)\,\mathrm{d}\varepsilon \le 1 + \int_0^1 \log\big(C_2/\varepsilon^3\big)\,\mathrm{d}\varepsilon \le C_3.$$

It follows by Lemma 49 that there exists a universal constant $C_4 > 0$ such that for all $s \ge 1$,

$$\mathbb{P}\bigg(\sup_{a\in[0,1]}\sup_{t\in[0,1]} \frac{2}{n(n-1)}\bigg|\sum_{(i,j)\in\mathcal{I}} \xi_i(t)\xi_j(t)\bigg| > \frac{C_4 s^2 e^{3(\|f_1\|_\infty \vee \|f_2\|_\infty)}}{n}\bigg) \le e^{-s^2}.$$

We deduce from (60) that there exists a universal constant $C_5 > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,

$$\sup_{a\in[0,1]}\sup_{t\in[0,1]} \big(S_n(f,t) - S(f,t)\big)^2 \le \frac{C_5 s^2 e^{3(\|f_1\|_\infty \vee \|f_2\|_\infty)}}{n},$$

as required. □

Lemma 31 (Uniform convergence of DS_n)**.** *Let $f, f_1, f_2 \in \mathcal{B}(\mathcal{X})$ and for $a \in [0,1]$ define $\tilde{f}_a := af_1 + (1-a)f_2$. Then there is a universal constant $C > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,*

$$\sup_{a\in[0,1]}\sup_{t\in[0,1]} \big|DS_n(\tilde{f}_a, t)(f) - DS(\tilde{f}_a, t)(f)\big| \le Ce^{2(\|f_1\|_\infty \vee \|f_2\|_\infty)}\bigg(\frac{s\|f\|_{L_2}}{\sqrt{n}} + \frac{s^2\|f\|_\infty}{n}\bigg).$$

Proof of Lemma 31. Define the class of functions

$$\mathcal{G} := \Big\{(X_i, T_i, I_i) \mapsto R_i(t)e^{\tilde{f}_a(X_i)}f(X_i) : a \in [0,1], t \in [0,1]\Big\},$$

with associated envelope function $F(X,T,I) := e^{f_1(X)\vee f_2(X)}|f(X)|$. By Lemmas 13 and 15,

$$\begin{aligned}
&\sup_{a\in[0,1]}\sup_{t\in[0,1]} \big|DS_n(\tilde{f}_a, t)(f) - DS(\tilde{f}_a, t)(f)\big| \\
&= \sup_{a\in[0,1]}\sup_{t\in[0,1]} \bigg|\frac{1}{n}\sum_{i=1}^n R_i(t)e^{\tilde{f}_a(X_i)}f(X_i) - \int_{\mathcal{X}} q(x,t)e^{\tilde{f}_a(x)}f(x)\,\mathrm{d}P_X(x)\bigg| \\
&= \sup_{g\in\mathcal{G}}\bigg|\frac{1}{n}\sum_{i=1}^n \Big\{g(X_i,T_i,I_i) - \mathbb{E}\big(g(X_i,T_i,I_i)\big)\Big\}\bigg|.
\end{aligned}$$

Define $\sigma^2 := \mathbb{E}\big(F(X,T,I)^2\big) \le e^{2(\|f_1\|_\infty \vee \|f_2\|_\infty)}\|f\|_{L_2}^2$ and $M := \|F\|_\infty \le e^{\|f_1\|_\infty \vee \|f_2\|_\infty}\|f\|_\infty$. Let $\mathcal{Q}$ be the set of discrete measures on $\mathcal{X} \times [0,1] \times \{0,1\}$ and for $Q \in \mathcal{Q}$ write $\|\cdot\|_{Q,2}$ for the $L_2(Q)$-norm. Define the uniform entropy integral

$$J := \sup_{Q \in \mathcal{Q}} \int_0^1 \sqrt{1 + \log N\big(\varepsilon \|F\|_{Q,2}, \mathcal{G}, \|\cdot\|_{Q,2}\big)}\, \mathrm{d}\varepsilon.$$

By Chernozhukov et al. (2014, Theorem 5.2), there is a universal constant $C_1 > 0$ such that

$$\begin{aligned}\mathbb{E}\bigg[\sup_{g \in \mathcal{G}} \bigg| \frac{1}{n} \sum_{i=1}^n \Big\{ g(X_i,T_i,I_i) - \mathbb{E}\big(g(X_i,T_i,I_i)\big) \Big\} \bigg| \bigg] &\le \frac{C_1 J \sigma}{\sqrt{n}} + \frac{C_1 J^2 M}{n} \\ &\le \frac{C_1 J e^{\|f_1\|_\infty \vee \|f_2\|_\infty}\|f\|_{L_2}}{\sqrt{n}} + \frac{C_1 J^2 e^{\|f_1\|_\infty \vee \|f_2\|_\infty}\|f\|_\infty}{n}. \end{aligned} \tag{61}$$

To compute the entropy integral, note that for any $i \in [n]$, $0 \le a < a' \le 1$ and $0 \le t < t' \le 1$,

$$\begin{aligned}&\big| R_i(t') e^{\tilde{f}_{a'}(X_i)} f(X_i) - R_i(t) e^{\tilde{f}_a(X_i)} f(X_i) \big| \\ &\quad \le e^{\tilde{f}_{a'}(X_i)} |f(X_i)| \big| R_i(t') - R_i(t) \big| + R_i(t) |f(X_i)| \big| e^{\tilde{f}_{a'}(X_i)} - e^{\tilde{f}_a(X_i)} \big| \\ &\quad \le e^{\|f_1\|_\infty \vee \|f_2\|_\infty} |f(X_i)| \mathbb{1}_{\{t \le T_i < t'\}} + e^{\|f_1\|_\infty \vee \|f_2\|_\infty} |f(X_i)| \big| \tilde{f}_{a'}(X_i) - \tilde{f}_a(X_i) \big| \\ &\quad \le 2 e^{2(\|f_1\|_\infty \vee \|f_2\|_\infty)} |f(X_i)| \big(\mathbb{1}_{\{t \le T_i < t'\}} + a' - a \big).\end{aligned}$$

Therefore for each $Q \in \mathcal{Q}$,

$$\begin{aligned}&\mathbb{E}_Q \Big\{ \big(R_i(t') e^{\tilde{f}_{a'}(X_i)} f(X_i) - R_i(t) e^{\tilde{f}_a(X_i)} f(X_i) \big)^2 \Big\} \\ &\quad \le 8 e^{4(\|f_1\|_\infty \vee \|f_2\|_\infty)} \Big\{ \mathbb{E}_Q\big(f(X_i)^2 \mathbb{1}_{\{t \le T_i < t'\}} \big) + (a' - a)^2 \|f\|_{Q,2}^2 \Big\}.\end{aligned}$$

Given $\varepsilon \in (0,1)$, define $\mathcal{T} := \{t_1, \ldots, t_{\lceil 1/\varepsilon^2 \rceil}\}$, where

$$t_m := \inf\Big\{ t \in [0,1] : \mathbb{E}_Q\big(f(X_i)^2 \mathbb{1}_{\{T_i \le t\}} \big) \ge m\varepsilon^2 \|f\|_{Q,2}^2 \Big\}$$

for $m \in [\lceil 1/\varepsilon^2 \rceil - 1]$ and $t_{\lceil 1/\varepsilon^2 \rceil} := 1$, so that given any $t \in [0,1]$, there exists $m \in [\lceil 1/\varepsilon^2 \rceil]$ such that $\mathbb{E}_Q\big(f(X_i)^2 \mathbb{1}_{\{t \le T_i < t_m\}} \big) \le \varepsilon^2 \|f\|_{Q,2}^2$. Likewise, let $\mathcal{A} := \{0, \varepsilon, 2\varepsilon, \ldots, (\lceil 1/\varepsilon \rceil - 1)\varepsilon, 1\}$, so that given any $a \in [0,1]$, there exists $a' \in \mathcal{A}$ such that $|a' - a| \le \varepsilon/2$. Then $\mathcal{A} \times \mathcal{T}$ induces a $\big(4e^{2(\|f_1\|_\infty \vee \|f_1\|_\infty)} \varepsilon \|f\|_{Q,2}\big)$-cover of $\mathcal{G}$ under $L_2(Q)$, of cardinality at most $\lceil 1/\varepsilon^2 \rceil \times \lceil 1/\varepsilon \rceil \le 4/\varepsilon^3$. Since $\|F\|_{Q,2} \ge e^{-(\|f_1\|_\infty \vee \|f_2\|_\infty)} \|f\|_{Q,2}$, we have for $\varepsilon \in (0,1)$ that

$$N\Big(\varepsilon \|F\|_{Q,2}, \mathcal{G}, \|\cdot\|_{Q,2} \Big) \le N\Big(\varepsilon e^{-(\|f_1\|_\infty \vee \|f_2\|_\infty)} \|f\|_{Q,2}, \mathcal{G}, \|\cdot\|_{Q,2} \Big) \le \frac{256 e^{9(\|f_1\|_\infty \vee \|f_2\|_\infty)}}{\varepsilon^3}.$$

Since $\int_0^1 \sqrt{\log(1/\varepsilon)}\, \mathrm{d}\varepsilon = \sqrt{\pi}/2$, it follows that

$$\begin{aligned}J &\le \int_0^1 \sqrt{1 + \log\big(256 e^{9(\|f_1\|_\infty \vee \|f_2\|_\infty)} / \varepsilon^3 \big)}\, \mathrm{d}\varepsilon \\ &\le \sqrt{1 + \log 256} + 3\sqrt{\|f_1\|_\infty \vee \|f_2\|_\infty} + \sqrt{3\pi}/2 \\ &\le 5 + 3\sqrt{\|f_1\|_\infty \vee \|f_2\|_\infty} \le 6 e^{(\|f_1\|_\infty \vee \|f_2\|_\infty)/2}.\end{aligned}$$

Therefore, by (61), there exists a universal constant $C_2 > 0$ with

$$\mathbb{E}\Big(\sup_{a\in[0,1]}\sup_{t\in[0,1]}\big|DS_n(\tilde{f}_a,t)(f) - DS(\tilde{f}_a,t)(f)\big|\Big) \le C_2 e^{2(\|f_1\|_\infty \vee \|f_2\|_\infty)}\Big(\frac{\|f\|_{L_2}}{\sqrt{n}} + \frac{\|f\|_\infty}{n}\Big). \quad (62)$$

We now apply a Bernstein-type inequality for empirical processes (Boucheron et al., 2013, Chapter 12) to control the tail behaviour of the supremum. Noting that the result is trivial if $\|f\|_\infty = 0$, assume that $\|f\|_\infty > 0$. For $i \in [n]$, $a \in [0,1]$ and $t \in [0,1]$, write

$$X_{iat} := \frac{1}{2e^{(\|f_1\|_\infty \vee \|f_2\|_\infty)}\|f\|_\infty}\Big(R_i(t)e^{\tilde{f}_a(X_i)}f(X_i) - \int_{\mathcal{X}} q(x,t)e^{\tilde{f}_a(x)}f(x)\,\mathrm{d}P_X(x)\Big),$$

so that $(X_{iat})_{a\in[0,1],t\in[0,1]}$ are independent and identically distributed, with $\mathbb{E}(X_{iat}) = 0$ and $|X_{iat}| \le 1$. Further,

$$\sup_{a\in[0,1]}\sup_{t\in[0,1]}\big|DS_n(\tilde{f}_a,t)(f) - DS(\tilde{f}_a,t)(f)\big| = \frac{2e^{(\|f_1\|_\infty \vee \|f_2\|_\infty)}\|f\|_\infty}{n}\sup_{a\in[0,1]}\sup_{t\in[0,1]}\Big|\sum_{i=1}^n X_{iat}\Big|,$$

and also

$$\mathbb{E}\Big(\sup_{a\in[0,1]}\sup_{t\in[0,1]}\sum_{i=1}^n X_{iat}^2\Big) \le \frac{1}{4\|f\|_\infty^2}\Big(2\sum_{i=1}^n \mathbb{E}\big(f(X_i)^2\big) + 2n\int_{\mathcal{X}} f(x)^2\,\mathrm{d}P_X(x)\Big) = \frac{n\|f\|_{L_2}^2}{\|f\|_\infty^2}.$$

Therefore by Boucheron et al. (2013, Theorem 12.2), for all $s' \ge 0$,

$$\mathbb{P}\Big(\sup_{a\in[0,1]}\sup_{t\in[0,1]}\Big|\sum_{i=1}^n X_{iat}\Big| \ge \mathbb{E}\Big\{\sup_{a\in[0,1]}\sup_{t\in[0,1]}\Big|\sum_{i=1}^n X_{iat}\Big|\Big\} + s'\Big) \le 2\exp\Big(-\frac{s'^2}{8n\|f\|_{L_2}^2/\|f\|_\infty^2 + 2s'}\Big).$$

In combination with (62), setting $s' := 3s\sqrt{n}\|f\|_{L_2}/\|f\|_\infty + 2s^2$ with $s \ge 1$, we deduce that there exists a universal constant $C > 0$ such that with probability at least $1 - e^{-s^2}$,

$$\sup_{a\in[0,1]}\sup_{t\in[0,1]}\big|DS_n(\tilde{f}_a,t)(f) - DS(\tilde{f}_a,t)(f)\big| \le Ce^{2(\|f_1\|_\infty \vee \|f_2\|_\infty)}\Big(\frac{s\|f\|_{L_2}}{\sqrt{n}} + \frac{s^2\|f\|_\infty}{n}\Big). \qquad \square$$

E.3.11 Local convergence of ℓ_n

We now apply the uniform bounds given in Lemmas 30 and 31 to deduce a local convergence result for $D\ell_n$ in Lemma 32. Lemma 33 demonstrates that ℓ_n and $\ell_\star$ are Lipschitz with respect to the supremum norm. Finally, Lemma 34 provides a bound on $\ell_n(f)$ which is local in the sense that the error reduces when f is close to f_0 under the $L_2(P_X)$-norm. This property is essential when used in combination with the $L_2(P_X)$-strong convexity result for $\ell_\star$ from Lemma 17, and allows us to derive our main theorems on cross-validation.

Lemma 32 (Local convergence of $D\ell_n$)**.** *Let $f, f_1, f_2 \in \mathcal{B}(\mathcal{X})$, and for $a \in [0,1]$ define $\tilde{f}_a := af_1 + (1-a)f_2$. Then there is a universal constant $C > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,*

$$\sup_{a\in[0,1]}\big|D\ell_n(\tilde{f}_a)(f) - D\ell_\star(\tilde{f}_a)(f)\big| \le \frac{Ce^{4\|f_0\|_\infty}e^{6(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)}}{q_1^2}\Big(\frac{s\|f\|_{L_2}}{\sqrt{n}} + \frac{s^2\|f\|_\infty}{n}\Big).$$

Proof of Lemma 32. By Lemmas 14 and 16, we have

$$\begin{aligned}
D\ell_n(\tilde{f}_a)(f) - D\ell_\star(\tilde{f}_a)(f) &= \int_0^1 \frac{DS_n(\tilde{f}_a,t)(f)}{S_n(\tilde{f}_a,t)}\,\mathrm{d}N(t) - \frac{1}{n}\sum_{i=1}^n f(X_i)N_i(1) \\
&\quad - \int_0^1 \frac{DS(\tilde{f}_a,t)(f)}{S(\tilde{f}_a,t)} S(f_0,t)\lambda_0(t)\,\mathrm{d}t + \int_0^1 DS(f_0,t)(f)\lambda_0(t)\,\mathrm{d}t \\
&= \int_0^1 \frac{DS_n(\tilde{f}_a,t)(f) - DS(\tilde{f}_a,t)(f)}{S_n(\tilde{f}_a,t)}\,\mathrm{d}N(t) \qquad (63) \\
&\quad + \int_0^1 \frac{DS(\tilde{f}_a,t)(f)}{S(\tilde{f}_a,t)} \frac{S(\tilde{f}_a,t) - S_n(\tilde{f}_a,t)}{S_n(\tilde{f}_a,t)}\,\mathrm{d}N(t) \qquad (64) \\
&\quad + \int_0^1 \frac{DS(\tilde{f}_a,t)(f)}{S(\tilde{f}_a,t)} \big(\mathrm{d}N(t) - S(f_0,t)\lambda_0(t)\,\mathrm{d}t\big) \qquad (65) \\
&\quad + \int_0^1 DS(f_0,t)(f)\lambda_0(t)\,\mathrm{d}t - \frac{1}{n}\sum_{i=1}^n f(X_i)N_i(1). \qquad (66)
\end{aligned}$$

We begin by bounding the term in (63). By Lemmas 30 and 31, and since $S(\tilde{f}_a,t) = \int_{\mathcal{X}} q(x,t)e^{\tilde{f}_a(x)}\,\mathrm{d}P_X(x) \geq e^{-(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)}q_1$, there exists a universal constant $C_1 > 0$ such that whenever $1 \leq s \leq e^{-2\|f_0\|_\infty}e^{-3(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)}q_1\sqrt{n}/C_1$, with probability at least $1 - e^{-s^2}$,

$$\begin{aligned}
&\sup_{a\in[0,1]}\left|\int_0^1 \frac{DS_n(\tilde{f}_a,t)(f) - DS(\tilde{f}_a,t)(f)}{S_n(\tilde{f}_a,t)}\,\mathrm{d}N(t)\right| \\
&\quad\leq \frac{\sup_{a\in[0,1]}\sup_{t\in[0,1]}\big|DS_n(\tilde{f}_a,t)(f) - DS(\tilde{f}_a,t)(f)\big|}{\inf_{a\in[0,1]}\inf_{t\in[0,1]} S(\tilde{f}_a,t) - \sup_{a\in[0,1]}\sup_{t\in[0,1]}|S_n(\tilde{f}_a,t) - S(\tilde{f}_a,t)|} \\
&\quad\leq \frac{(C_1/2)e^{2\|f_0\|_\infty}e^{2(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)}}{e^{-(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)}q_1 - (C_1/2)se^{2\|f_0\|_\infty}e^{2(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)}/\sqrt{n}} \\
&\qquad\times\left(\frac{s\|f\|_{L_2}}{\sqrt{n}} + \frac{s^2\|f\|_\infty}{n}\right) \\
&\quad\leq C_1 e^{2\|f_0\|_\infty}e^{3(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)}\left(\frac{s\|f\|_{L_2}}{\sqrt{n}q_1} + \frac{s^2\|f\|_\infty}{nq_1}\right).
\end{aligned}$$

Next we consider (64). By Lemma 15 and Cauchy–Schwarz, we have

$$\big|DS(\tilde{f}_a,t)(f)\big| \leq \int_{\mathcal{X}} q(x,t)e^{\tilde{f}_a(x)}|f(x)|\,\mathrm{d}P_X(x) \leq e^{\|f_0\|_\infty}e^{\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty}\|f\|_{L_2}.$$

Thus, by Lemmas 30 and 31, there exists a universal constant $C_2 > 0$ such that whenever

$1 \le s \le e^{-2\|f_0\|_\infty} e^{-3(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)} q_1\sqrt{n}/C_2$, with probability at least $1-e^{-s^2}$,

$$\begin{aligned}
&\sup_{a\in[0,1]}\left|\int_0^1 \frac{DS(\tilde f_a,t)(f)}{S(\tilde f_a,t)}\frac{S(\tilde f_a,t)-S_n(\tilde f_a,t)}{S_n(\tilde f_a,t)}\,\mathrm{d}N(t)\right| \\
&\quad\le \frac{\sup_{a\in[0,1]}\sup_{t\in[0,1]}|DS(\tilde f_a,t)(f)|}{\inf_{a\in[0,1]}\inf_{t\in[0,1]}S(\tilde f_a,t)} \\
&\qquad\times \frac{\sup_{a\in[0,1]}\sup_{t\in[0,1]}\big|S_n(\tilde f_a,t)-S(\tilde f_a,t)\big|}{\inf_{a\in[0,1]}\inf_{t\in[0,1]}S(\tilde f_a,t)-\sup_{a\in[0,1]}\sup_{t\in[0,1]}|S_n(\tilde f_a,t)-S(\tilde f_a,t)|} \\
&\quad\le C_2 e^{3\|f_0\|_\infty} e^{5(\|f_1-f_0\|_\infty\vee\|f_2-f_0\|_\infty)}\frac{s\|f\|_{L_2}}{\sqrt{n}q_1}.
\end{aligned}$$

We now turn to (65). Define the function class

$$\mathcal{G}=\left\{(X_i,T_i,I_i)\mapsto\int_0^1\frac{DS(\tilde f_a,t)(f)}{S(\tilde f_a,t)}\big(\mathrm{d}N_i(t)-S(f_0,t)\lambda_0(t)\,\mathrm{d}t\big):a\in[0,1]\right\},$$

with associated envelope function $F := 8e^{2\|f_0\|_\infty}e^{5(\|f_1-f_0\|_\infty\vee\|f_2-f_0\|_\infty)}\|f\|_{L_2}/q_1^2$. For $a,a'\in[0,1]$, by Lemma 15 and Cauchy–Schwarz,

$$\begin{aligned}
\sup_{t\in[0,1]}\big|DS(\tilde f_a,t)(f)-DS(\tilde f_{a'},t)(f)\big| &= \sup_{t\in[0,1]}\left|\int_{\mathcal{X}} q(x,t)\big(e^{\tilde f_a(x)}-e^{\tilde f_{a'}(x)}\big)f(x)\,\mathrm{d}P_X(x)\right| \\
&\le \sup_{x\in\mathcal{X}}\big|e^{\tilde f_a(x)}-e^{\tilde f_{a'}(x)}\big|\|f\|_{L_2} \\
&\le e^{\|f_0\|_\infty}e^{\|f_1-f_0\|_\infty\vee\|f_2-f_0\|_\infty}\big\|\tilde f_a-\tilde f_{a'}\big\|_\infty\|f\|_{L_2} \\
&\le e^{\|f_0\|_\infty}e^{\|f_1-f_0\|_\infty\vee\|f_2-f_0\|_\infty}|a-a'|\|f_1-f_2\|_\infty\|f\|_{L_2} \\
&\le 2e^{\|f_0\|_\infty}e^{2(\|f_1-f_0\|_\infty\vee\|f_2-f_0\|_\infty)}|a-a'|\|f\|_{L_2}. \qquad (67)
\end{aligned}$$

By a similar argument, we have

$$\sup_{t\in[0,1]}\big|S(\tilde f_a,t)-S(\tilde f_{a'},t)\big|\le 2e^{\|f_0\|_\infty}e^{2(\|f_1-f_0\|_\infty\vee\|f_2-f_0\|_\infty)}|a-a'|. \qquad (68)$$

Hence, by (67) and (68), since $N_i(1)\le 1$ for each $i\in[n]$ and $\int_0^1 S(f_0,t)\lambda_0(t)\,\mathrm{d}t\le 1$,

$$\begin{aligned}
&\left|\int_0^1\left(\frac{DS(\tilde f_a,t)(f)}{S(\tilde f_a,t)}-\frac{DS(\tilde f_{a'},t)(f)}{S(\tilde f_{a'},t)}\right)\big(\mathrm{d}N_i(t)-S(f_0,t)\lambda_0(t)\,\mathrm{d}t\big)\right| \\
&\quad\le \frac{2\sup_{t\in[0,1]}\big|DS(\tilde f_a,t)(f)-DS(\tilde f_{a'},t)(f)\big|}{\inf_{a\in[0,1]}\inf_{t\in[0,1]}S(\tilde f_a,t)} \\
&\qquad+\frac{2\sup_{a\in[0,1]}\sup_{t\in[0,1]}|DS(\tilde f_a,t)(f)|\sup_{t\in[0,1]}\big|S(\tilde f_a,t)-S(\tilde f_{a'},t)\big|}{\inf_{a\in[0,1]}\inf_{t\in[0,1]}S(\tilde f_a,t)^2} \\
&\quad\le \frac{8}{q_1^2}e^{2\|f_0\|_\infty}e^{5(\|f_1-f_0\|_\infty\vee\|f_2-f_0\|_\infty)}|a-a'|\|f\|_{L_2}=F|a-a'|.
\end{aligned}$$

Thus by setting $\mathcal{A} := \{0, \varepsilon, 2\varepsilon, \ldots, (\lceil 1/\varepsilon \rceil - 1)\varepsilon, 1\}$, given any $a \in [0,1]$, there exists $a' \in \mathcal{A}$ such that $|a' - a| \le \varepsilon$. It follows that $N\bigl(\varepsilon F, \mathcal{G}, \|\cdot\|_\infty\bigr) \le 2/\varepsilon$, so as $\int_0^1 \sqrt{\log(1/\varepsilon)}\,\mathrm{d}\varepsilon = \sqrt{\pi}/2$,

$$J := \int_0^1 \sqrt{1 + \log N\bigl(\varepsilon F, \mathcal{G}, \|\cdot\|_\infty\bigr)}\,\mathrm{d}\varepsilon \le \int_0^1 \sqrt{1 + \log(2/\varepsilon)}\,\mathrm{d}\varepsilon \le \sqrt{1+\log 2} + \frac{\sqrt{\pi}}{2} \le 3.$$

Therefore, by Chernozhukov et al. (2014, Theorem 5.2), there exists a universal constant $C_3 > 0$ such that

$$\mathbb{E}\biggl\{\sup_{a\in[0,1]}\biggl|\int_0^1 \frac{DS(\tilde{f}_a,t)(f)}{S(\tilde{f}_a,t)}\bigl(\mathrm{d}N(t) - S(f_0,t)\lambda_0(t)\,\mathrm{d}t\bigr)\biggr|\biggr\} \le C_3 e^{2\|f_0\|_\infty} e^{5(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)} \frac{\|f\|_{L_2}}{\sqrt{n}q_1^2}.$$

It follows by the bounded differences inequality (Boucheron et al., 2013, Theorem 6.2) that there exists a universal constant $C_4 > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,

$$\sup_{a\in[0,1]}\biggl|\int_0^1 \frac{DS(\tilde{f}_a,t)(f)}{S(\tilde{f}_a,t)}\bigl(\mathrm{d}N(t) - S(f_0,t)\lambda_0(t)\,\mathrm{d}t\bigr)\biggr| \le C_4 e^{2\|f_0\|_\infty} e^{5(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)} \frac{s\|f\|_{L_2}}{\sqrt{n}q_1^2}.$$

Finally, we consider (66). By Fubini's theorem and Lemma 15, for each $i \in [n]$,

$$\begin{aligned}\mathbb{E}\bigl(f(X_i)N_i(1)\bigr) &= \mathbb{E}\biggl(f(X_i)\int_0^1 R_i(t)e^{f_0(X_i)}\lambda_0(t)\,\mathrm{d}t\biggr)\\ &= \int_0^1\int_{\mathcal{X}} q(x,t)e^{f_0(x)}f(x)\,\mathrm{d}P_X(x)\lambda_0(t)\,\mathrm{d}t = \int_0^1 DS(f_0,t)(f)\lambda_0(t)\,\mathrm{d}t.\end{aligned}$$

Also, $|f(X_i)N_i(1)| \le \|f\|_\infty$ and $\mathbb{E}\bigl(f(X_i)^2N_i(1)^2\bigr) \le \|f\|_{L_2}^2$. Hence, by Bernstein's inequality,

$$\mathbb{P}\biggl(\biggl|\int_0^1 DS(f_0,t)(f)\lambda_0(t)\,\mathrm{d}t - \frac{1}{n}\sum_{i=1}^n f(X_i)N_i(1)\biggr| > s'\biggr) \le 2\exp\biggl(-\frac{ns'^2}{2\|f\|_{L_2}^2 + \frac{2}{3}\|f\|_\infty s'}\biggr)$$

for $s' > 0$. With $s' := 2s\|f\|_{L_2}/\sqrt{n} + 2s^2\|f\|_\infty/n$ for $s \ge 1$ and a universal constant $C_5 > 0$,

$$\biggl|\int_0^1 DS(f_0,t)(f)\lambda_0(t)\,\mathrm{d}t - \frac{1}{n}\sum_{i=1}^n f(X_i)N_i(1)\biggr| \le C_5\biggl(\frac{s\|f\|_{L_2}}{\sqrt{n}} + \frac{s^2\|f\|_\infty}{n}\biggr),$$

with probability at least $1 - e^{-s^2}$.

We deduce that there exists a universal constant $C_6 > 0$ such that for all s with $1 \le s \le e^{-2\|f_0\|_\infty}e^{-3(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)}q_1\sqrt{n}/C_2$, with probability at least $1 - e^{-s^2}$,

$$\sup_{a\in[0,1]}\bigl|D\ell_n(\tilde{f}_a)(f) - D\ell_\star(\tilde{f}_a)(f)\bigr| \le C_6 e^{4\|f_0\|_\infty} e^{6(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)}\biggl(\frac{s\|f\|_{L_2}}{\sqrt{n}q_1^2} + \frac{s^2\|f\|_\infty}{nq_1^2}\biggr). \tag{69}$$

Finally, we remove the restriction on s. By Lemma 33,

$$\sup_{a\in[0,1]} \left|D\ell_n(\tilde{f}_a)(f) - D\ell_\star(\tilde{f}_a)(f)\right| \le 4\|f\|_\infty,$$

so if $s > e^{-2\|f_0\|_\infty} e^{-3(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)} q_1\sqrt{n}/C_2$, then (69) holds by increasing the universal constant C_6 if necessary. □

Lemma 33 (Lipschitz property of ℓ_n and $\ell_\star$)**.** *Let $f_1, f_2 \in \mathcal{B}(\mathcal{X})$. Then*

$$|\ell_n(f_1) - \ell_n(f_2)| \le 2\|f_1 - f_2\|_\infty \quad \textit{and} \quad |\ell_\star(f_1) - \ell_\star(f_2)| \le 2\|f_1 - f_2\|_\infty.$$

Proof of Lemma 33. By Taylor's theorem and Lemmas 14 and 13, there exists $\tilde{f}$ on the line segment between f_1 and f_2 such that

$$\begin{aligned}
\left|\ell_n(f_1) - \ell_n(f_2)\right| &= \left|D\ell_n(\tilde{f})(f_1 - f_2)\right| \\
&= \left|\frac{1}{n}\sum_{i=1}^n \frac{DS_n(\tilde{f}, T_i)(f_1 - f_2)}{S_n(\tilde{f}, T_i)} N_i(1) - \frac{1}{n}\sum_{i=1}^n \big(f_1(X_i) - f_2(X_i)\big) N_i(1)\right| \\
&\le \frac{1}{n}\sum_{i=1}^n \frac{|DS_n(\tilde{f}, T_i)(f_1 - f_2)|}{S_n(\tilde{f}, T_i)} + \|f_1 - f_2\|_\infty \\
&\le \frac{1}{n}\sum_{i=1}^n \frac{\left|\sum_{j=1}^n R_j(T_i) e^{\tilde{f}(X_j)} \big(f_1(X_j) - f_2(X_j)\big)\right|}{\sum_{j=1}^n R_j(T_i) e^{\tilde{f}(X_j)}} + \|f_1 - f_2\|_\infty \\
&\le 2\|f_1 - f_2\|_\infty.
\end{aligned}$$

Moreover, again by Taylor's theorem and applying Lemmas 16 and 15, there exists $\check{f}$ on the line segment between f_1 and f_2 such that

$$\begin{aligned}
\left|\ell_\star(f_1) - \ell_\star(f_2)\right| &= \left|D\ell_\star(\check{f})(f_1 - f_2)\right| \\
&= \left|\int_0^1 DS(\check{f}, t)(f_1 - f_2)\frac{S(f_0, t)}{S(\check{f}, t)}\lambda_0(t)\,\mathrm{d}t - \int_0^1 DS(f_0, t)(f_1 - f_2)\lambda_0(t)\,\mathrm{d}t\right| \\
&\le \left|\int_0^1 \frac{\int_{\mathcal{X}} q(x,t) e^{\check{f}(x)} \big(f_1(x) - f_2(x)\big)\,\mathrm{d}P_X(x)}{\int_{\mathcal{X}} q(x,t) e^{\check{f}(x)}\,\mathrm{d}P_X(x)} S(f_0, t)\lambda_0(t)\,\mathrm{d}t\right| \\
&\qquad + \left|\int_0^1 \int_{\mathcal{X}} q(x,t) e^{f_0(x)} \big(f_1(x) - f_2(x)\big)\,\mathrm{d}P_X(x)\lambda_0(t)\,\mathrm{d}t\right| \\
&\le 2\|f_1 - f_2\|_\infty \int_0^1 S(f_0, t)\lambda_0(t)\,\mathrm{d}t \le 2\|f_1 - f_2\|_\infty \mathbb{E}\big(N(1)\big) \le 2\|f_1 - f_2\|_\infty,
\end{aligned}$$

as required. □

Lemma 34 (Convergence of ℓ_n)**.** *Let $f_1, f_2 \in \mathcal{B}(\mathcal{X})$. There exists a universal constant $C > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,*

$$\begin{aligned}
&\left|\ell_n(f_1) - \ell_\star(f_1) - \ell_n(f_2) + \ell_\star(f_2)\right| \\
&\qquad \le \frac{C e^{4\|f_0\|_\infty} e^{6(\|f_1-f_0\|_\infty \vee \|f_2-f_0\|_\infty)}}{q_1^2} \left(\frac{s\|f_1 - f_2\|_{L_2}}{\sqrt{n}} + \frac{s^2\|f_1 - f_2\|_\infty}{n}\right).
\end{aligned}$$

Proof of Lemma 34. By the mean value theorem applied to $\ell_n - \ell_\star$, there exists $\tilde{a}$ taking values in $[0,1]$ such that, defining $\tilde{f}_a := af_1 + (1-a)f_2$,

$$\begin{aligned}\big|\ell_n(f_1) - \ell_\star(f_1) - \ell_n(f_2) + \ell_\star(f_2)\big| &= \big|D\ell_n(\tilde{f}_{\tilde{a}})(f_1 - f_2) - D\ell_\star(\tilde{f}_{\tilde{a}})(f_1 - f_2)\big| \\ &\le \sup_{a\in[0,1]} \big|D\ell_n(\tilde{f}_a)(f_1 - f_2) - D\ell_\star(\tilde{f}_a)(f_1 - f_2)\big|.\end{aligned}$$

The result therefore follows by Lemma 32. □

E.3.12 Proof of Theorem 9

We use the result on local convergence of ℓ_n from Lemma 34 to derive a general-purpose cross-validation bound in Lemma 35. Next, Lemma 36 shows that $\gamma \mapsto \hat{f}_{n,\gamma}$ is Lipschitz with respect to the supremum norm. We then use these results to prove Theorem 9.

Lemma 35 (Cross-validation on finite sets)**.** *Let $\mathcal{A}$ be a non-empty finite set and suppose that for each $a \in \mathcal{A}$, $f_a \in \mathcal{B}(\mathcal{X})$ with $P_X(f_a) = 0$ and $\|f_a - f_0\|_\infty \le M$. Define $\hat{a} := \operatorname{argmin}_{a\in\mathcal{A}} \ell_n(f_a)$. Then there exists a universal constant $C > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,*

$$\|f_{\hat{a}} - f_0\|_{L_2} \le \frac{Ce^{5\|f_0\|_\infty} e^{8M}(1 + 1/\Lambda)}{q_1^3}\left(\min_{a\in\mathcal{A}} \|f_a - f_0\|_{L_2} + \frac{s + \sqrt{\log|\mathcal{A}|}}{\sqrt{n}}\right).$$

Proof of Lemma 35. Choose $a^\star \in \operatorname{argmin}_{a\in\mathcal{A}} \|f_a - f_0\|_{L_2}$. By definition of $\hat{a}$, we have

$$\begin{aligned}0 &\ge \ell_n(f_{\hat{a}}) - \ell_n(f_{a^\star}) \\ &= \big(\ell_n(f_{\hat{a}}) - \ell_\star(f_{\hat{a}}) - \ell_n(f_0) + \ell_\star(f_0)\big) - \big(\ell_n(f_{a^\star}) - \ell_\star(f_{a^\star}) - \ell_n(f_0) + \ell_\star(f_0)\big) \\ &\quad + \big(\ell_\star(f_{\hat{a}}) - \ell_\star(f_0)\big) - \big(\ell_\star(f_{a^\star}) - \ell_\star(f_0)\big). \end{aligned} \tag{70}$$

By Lemma 34 together with a union bound over $a \in \mathcal{A}$, there exists a universal constant $C_1 > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,

$$\begin{aligned}&\big|\ell_n(f_{\hat{a}}) - \ell_\star(f_{\hat{a}}) - \ell_n(f_0) + \ell_\star(f_0)\big| + \big|\ell_n(f_{a^\star}) - \ell_\star(f_{a^\star}) - \ell_n(f_0) + \ell_\star(f_0)\big| \\ &\qquad \le \frac{C_1 e^{4\|f_0\|_\infty} e^{6M}}{q_1^2}\left(\frac{s + \sqrt{\log|\mathcal{A}|}}{\sqrt{n}}\|f_{\hat{a}} - f_0\|_{L_2} + \frac{\big(s^2 + \log|\mathcal{A}|\big)M}{n}\right).\end{aligned}$$

For (70), we apply Taylor's theorem and Lemma 17 twice to obtain

$$\begin{aligned}&\big(\ell_\star(f_{\hat{a}}) - \ell_\star(f_0)\big) - \big(\ell_\star(f_{a^\star}) - \ell_\star(f_0)\big) \\ &\qquad \ge e^{-\|f_0\|_\infty} e^{-2M}\Lambda q_1 \|f_{\hat{a}} - f_0\|_{L_2}^2 - e^{\|f_0\|_\infty} e^{2M}\Lambda \|f_{a^\star} - f_0\|_{L_2}^2.\end{aligned}$$

Therefore, simplifying, we have that with probability at least $1 - e^{-s^2}$,

$$\begin{aligned}\|f_{\hat{a}} - f_0\|_{L_2}^2 \le{}& \frac{C_1 e^{5\|f_0\|_\infty} e^{8M}}{q_1^3\Lambda}\frac{s + \sqrt{\log|\mathcal{A}|}}{\sqrt{n}}\|f_{\hat{a}} - f_0\|_{L_2} \\ &+ \frac{e^{2\|f_0\|_\infty} e^{4M}}{q_1}\|f_{a^\star} - f_0\|_{L_2}^2 + \frac{C_1 e^{5\|f_0\|_\infty} e^{8M}}{q_1^3\Lambda}\frac{\big(s^2 + \log|\mathcal{A}|\big)M}{n}.\end{aligned}$$

We now again use the fact that if $x^2 \le ax + c$ for some $x, a, c \ge 0$, then $x \le a + \sqrt{c}$. Thus, there exists a universal constant $C_2 > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,

$$\|f_{\hat{a}} - f_0\|_{L_2} \le \frac{C_2 e^{5\|f_0\|_\infty} e^{8M}(1 + 1/\Lambda)}{q_1^3}\left(\|f_{a^\star} - f_0\|_{L_2} + \frac{s + \sqrt{\log|\mathcal{A}|}}{\sqrt{n}}\right). \qquad \square$$

Lemma 36 (Lipschitz property of $\hat{f}_{n,\gamma}$)**.** *Let $\gamma' \ge \gamma > 0$. Then*

$$\big\|\hat{f}_{n,\gamma'} - \hat{f}_{n,\gamma}\big\|_\infty \le \frac{2\sqrt{K}(\gamma' - \gamma)\sqrt{\log n}}{\gamma^{3/2}}.$$

Proof of Lemma 36. Since the result is clear if $\gamma' = \gamma$ or if $\hat{f}_{n,\gamma'} = \hat{f}_{n,\gamma}$, we may assume that $\gamma' > \gamma$ and $\|\hat{f}_{n,\gamma'} - \hat{f}_{n,\gamma}\|_{\mathcal{H}} > 0$. By definition of $\hat{f}_{n,\gamma'}$ and Taylor's theorem, there exists $\tilde{f}$ on the line segment between $\hat{f}_{n,\gamma}$ and $\hat{f}_{n,\gamma'}$ such that

$$\begin{aligned}
0 &\ge \ell_{n,\gamma'}(\hat{f}_{n,\gamma'}) - \ell_{n,\gamma'}(\hat{f}_{n,\gamma}) \\
&= \ell_{n,\gamma}(\hat{f}_{n,\gamma'}) - \ell_{n,\gamma}(\hat{f}_{n,\gamma}) + (\gamma' - \gamma)\big(\|\hat{f}_{n,\gamma'}\|_{\mathcal{H}}^2 - \|\hat{f}_{n,\gamma}\|_{\mathcal{H}}^2\big) \\
&= D\ell_{n,\gamma}(\hat{f}_{n,\gamma})(\hat{f}_{n,\gamma'} - \hat{f}_{n,\gamma}) + \frac{1}{2}D^2\ell_{n,\gamma}(\tilde{f})(\hat{f}_{n,\gamma'} - \hat{f}_{n,\gamma})^{\otimes 2} + (\gamma' - \gamma)\big(\|\hat{f}_{n,\gamma'}\|_{\mathcal{H}}^2 - \|\hat{f}_{n,\gamma}\|_{\mathcal{H}}^2\big).
\end{aligned}$$

Now $D\ell_{n,\gamma}(\hat{f}_{n,\gamma}) = 0$ and $D^2\ell_{n,\gamma}(\tilde{f})(f_1, f_1) \ge 2\gamma\|f_1\|_{\mathcal{H}}^2$ for $f_1 \in \mathcal{H}$. Further, $\ell_{n,\gamma}(\hat{f}_{n,\gamma}) \le \ell_{n,\gamma}(\hat{f}_{n,\gamma'})$ and $\ell_{n,\gamma'}(\hat{f}_{n,\gamma'}) \le \ell_{n,\gamma'}(\hat{f}_{n,\gamma})$, so $(\gamma' - \gamma)\|\hat{f}_{n,\gamma'}\|_{\mathcal{H}}^2 \le (\gamma' - \gamma)\|\hat{f}_{n,\gamma}\|_{\mathcal{H}}^2$ and hence $\|\hat{f}_{n,\gamma'}\|_{\mathcal{H}} \le \|\hat{f}_{n,\gamma}\|_{\mathcal{H}}$. Therefore,

$$\begin{aligned}
\gamma\|\hat{f}_{n,\gamma'} - \hat{f}_{n,\gamma}\|_{\mathcal{H}}^2 &\le (\gamma' - \gamma)\big(\|\hat{f}_{n,\gamma}\|_{\mathcal{H}}^2 - \|\hat{f}_{n,\gamma'}\|_{\mathcal{H}}^2\big) \\
&\le (\gamma' - \gamma)\|\hat{f}_{n,\gamma'} - \hat{f}_{n,\gamma}\|_{\mathcal{H}}\big(\|\hat{f}_{n,\gamma}\|_{\mathcal{H}} + \|\hat{f}_{n,\gamma'}\|_{\mathcal{H}}\big) \\
&\le 2(\gamma' - \gamma)\|\hat{f}_{n,\gamma'} - \hat{f}_{n,\gamma}\|_{\mathcal{H}}\|\hat{f}_{n,\gamma}\|_{\mathcal{H}},
\end{aligned}$$

so

$$\|\hat{f}_{n,\gamma'} - \hat{f}_{n,\gamma}\|_{\mathcal{H}} \le 2\frac{\gamma' - \gamma}{\gamma}\|\hat{f}_{n,\gamma}\|_{\mathcal{H}}.$$

Next, $S_n(f, T_i) \ge e^{f(X_i)}/n$ for $i \in [n]$ and $f \in \mathcal{H}$, so

$$\ell_n(f) = \frac{1}{n}\sum_{i=1}^n \log\big(S_n(f, T_i)\big)N_i(1) - \frac{1}{n}\sum_{i=1}^n f(X_i)N_i(1) \ge -\log n.$$

Similarly, $S_n(0_{\mathcal{X}}, T_i) \le 1$ so $\ell_n(0_{\mathcal{X}}) \le 0$. By definition of $\hat{f}_{n,\gamma}$, we have

$$-\log n + \gamma\|\hat{f}_{n,\gamma}\|_{\mathcal{H}}^2 \le \ell_n(\hat{f}_{n,\gamma}) + \gamma\|\hat{f}_{n,\gamma}\|_{\mathcal{H}}^2 = \ell_{n,\gamma}(\hat{f}_{n,\gamma}) \le \ell_{n,\gamma}(0_{\mathcal{X}}) = \ell_n(0_{\mathcal{X}}) \le 0,$$

so $\|\hat{f}_{n,\gamma}\|_{\mathcal{H}}^2 \le \gamma^{-1}\log n$. Thus, by Lemma 2,

$$\|\hat{f}_{n,\gamma'} - \hat{f}_{n,\gamma}\|_\infty \le \sqrt{K}\|\hat{f}_{n,\gamma'} - \hat{f}_{n,\gamma}\|_{\mathcal{H}} \le \frac{2\sqrt{K}(\gamma' - \gamma)\sqrt{\log n}}{\gamma^{3/2}}. \qquad \square$$

Proof of Theorem 9. We write $C_1^0, C_2^0, \dots$ for positive quantities depending only on $\|f_0\|_{\mathcal{H}}$, Λ, q_1, $\|1_{\mathcal{X}}\|_{\mathcal{H}}$, K and ξ. Let Γ' be an $n^{-3\xi}$-cover of $(\Gamma, |\cdot|)$ of cardinality at most $|\Gamma| \wedge n^{4\xi}$. For each $\gamma \in \Gamma$, by Lemma 36 and since $\gamma^- \geq n^{-\xi}$, there exists $\gamma' \in \Gamma'$ with

$$\big\|\hat{f}_{n,\gamma'} - \hat{f}_{n,\gamma}\big\|_\infty \leq \frac{2\sqrt{K} n^{-3\xi}\sqrt{\log n}}{n^{-3\xi/2}} \leq \frac{2\sqrt{K}}{n}. \tag{71}$$

Recall that $\gamma \mapsto H_{\infty,\gamma}$ is decreasing on $(0,\infty)$ and $\gamma \mapsto \gamma H_{\infty,\gamma}$ is increasing on $(0,\infty)$. Therefore, by Lemma 2, Theorem 4, and a union bound over $\gamma \in \Gamma'$, there exists $C_1^0 > 0$ such that if $\gamma^+ H_{\infty,\gamma^+} \leq 1/C_1^0$ and $1 \leq s \leq \sqrt{n}/(C_1^0 H_{\infty,\gamma^-})$, then with probability at least $1 - e^{-s^2}$,

$$\begin{aligned}
\max_{\gamma\in\Gamma'} \big\|\hat{f}_{n,\gamma} - f_0\big\|_\infty &\leq \max_{\gamma\in\Gamma'} \Big\{ \sqrt{H_{\infty,\gamma}} \big\|\hat{f}_{n,\gamma} - f_0\big\|_{\mathcal{H}_\gamma} \Big\} \\
&\leq C_1^0 \max_{\gamma\in\Gamma'} \left\{ \big(s + \sqrt{\log|\Gamma'|}\big) \frac{H_{\infty,\gamma}}{\sqrt{n}} + \sqrt{\gamma H_{\infty,\gamma}} \right\} \\
&\leq C_1^0 \left\{ \big(s + \sqrt{\log|\Gamma'|}\big) \frac{H_{\infty,\gamma^-}}{\sqrt{n}} + \sqrt{\gamma^+ H_{\infty,\gamma^+}} \right\}.
\end{aligned}$$

We deduce that there exists $C_2^0 > 0$ such that if $\gamma^+ H_{\infty,\gamma^+} \leq 1/C_2^0$, $H_{\infty,\gamma^-}\sqrt{\log|\Gamma'|}/\sqrt{n} \leq 1/C_2^0$ and $1 \leq s \leq \sqrt{n}/(C_2^0 H_{\infty,\gamma^-})$, then with probability at least $1 - e^{-s^2}$,

$$\max_{\gamma\in\Gamma'} \big\|\hat{f}_{n,\gamma} - f_0\big\|_{\mathcal{H}_\gamma} \leq \frac{1}{\sqrt{H_{\infty,\gamma^-}}} \quad \text{and} \quad \max_{\gamma\in\Gamma'} \big\|\hat{f}_{n,\gamma} - f_0\big\|_\infty \leq 1. \tag{72}$$

Moreover, for every $\gamma \in \Gamma'$, we have

$$\big\|P_X(\hat{f}_{n,\gamma}) 1_{\mathcal{X}}\big\|_\infty = \big|P_X(\hat{f}_{n,\gamma})\big| = \big|P_X(\hat{f}_{n,\gamma} - f_0)\big| \leq \big\|\hat{f}_{n,\gamma} - f_0\big\|_\infty.$$

Applying Lemma 35 to $\tilde{\ell}_n$ and $\big\{\hat{f}_{n,\gamma} - P_X(\hat{f}_{n,\gamma}) 1_{\mathcal{X}} : \gamma \in \Gamma'\big\}$ conditionally on the training data, on the event where (72) holds, there exists $C_3^0 > 0$ such that with probability at least $1 - e^{-s^2}$,

$$\big\|\hat{f}_{n,\hat{\gamma}} - P_X(\hat{f}_{n,\hat{\gamma}}) 1_{\mathcal{X}} - f_0\big\|_{L_2} \leq C_3^0 \left(\min_{\gamma\in\Gamma'} \big\|\hat{f}_{n,\gamma} - P_X(\hat{f}_{n,\gamma}) 1_{\mathcal{X}} - f_0\big\|_{L_2} + \frac{s + \sqrt{\log|\Gamma'|}}{\sqrt{n}} \right). \tag{73}$$

By Lemmas 29 and 28, (72) and a union bound, there exists $C_4^0 > 0$ such that, for $s \geq 1$, with probability at least $1 - e^{-s^2}$,

$$\max_{\gamma\in\Gamma'} \big|P_X\big(\hat{f}_{n,\gamma}\big)\big| \leq \max_{\gamma\in\Gamma'} \big|(P_n - P_X)\big(\hat{f}_{n,\gamma} - f_0\big)\big| + \big|(P_n - P_X)(f_0)\big| \leq \frac{C_4^0 s}{\sqrt{n}}. \tag{74}$$

Finally, by (71),

$$\min_{\gamma\in\Gamma'} \big\|\hat{f}_{n,\gamma} - f_0\big\|_{L_2} \leq \min_{\gamma\in\Gamma} \big\|\hat{f}_{n,\gamma} - f_0\big\|_{L_2} + \frac{2\sqrt{K}}{n}. \tag{75}$$

The conclusion follows from (73), (74) and (75). □

E.3.13 Proof of Theorem 10

In Lemma 37 we give a bound on the ℓ_1-covering number of the standard simplex. We then present a Bernstein-type tail bound for stochastic process suprema in Lemma 38 and use it to establish a result on cross-validation over the simplex in Lemma 39. These results are then used to prove Theorem 10.

Lemma 37 (Covering numbers of the standard simplex)**.** *Let $d \in \mathbb{N}$ and define the simplex $\Delta_d := \{x \in [0,1]^d : \|x\|_1 \le 1\}$. Then for $\varepsilon > 0$, we have $N(\varepsilon, \Delta_d, \|\cdot\|_1) \le (1 + 1/\varepsilon)^d$.*

Proof of Lemma 37. The volume (d-dimensional Lebesgue measure) of Δ_d is $\mathrm{Vol}_d(\Delta_d) = 1/d!$, while the volume of $B := \{x \in \mathbb{R}^d : \|x\|_1 \le 1\}$ is $\mathrm{Vol}_d(B) = 2^d/d!$. Writing N_{pack} for a packing number, we have

$$\frac{\varepsilon^d}{d!} N_{\mathrm{pack}}(\varepsilon, \Delta_d, \|\cdot\|_1) \le \mathrm{Vol}_d\big(\Delta_d + \varepsilon B/2\big).$$

If $u \in \Delta_d + \varepsilon B/2$, then $u + \varepsilon 1_d/2 \ge 0_d$ and $\|u + \varepsilon 1_d/2\|_1 \le 1 + \varepsilon$, so $\mathrm{Vol}_d(\Delta_d + \varepsilon B/2) \le (1+\varepsilon)^d \mathrm{Vol}_d(\Delta_d)$. As covering numbers are bounded by their corresponding packing numbers,

$$N(\varepsilon, \Delta_d, \|\cdot\|_1) \le N_{\mathrm{pack}}(\varepsilon, \Delta_d, \|\cdot\|_1) \le \left(\frac{1}{\varepsilon} + 1\right)^d. \qquad \square$$

Lemma 38 (Stochastic process suprema with Bernstein tails)**.** *Let $\mathcal{T}$ be a non-empty set equipped with two pseudo-metrics d_1 and d_2. Suppose that $\mathcal{T}$ has finite diameter D_1 under d_1 and D_2 under d_2, respectively. Let $(X_t)_{t\in\mathcal{T}}$ be a stochastic process that is separable in the sense of Lemma 47. Suppose that for all $t, t' \in \mathcal{T}$ and $s \ge 0$, we have*

$$\mathbb{P}\big(|X_t - X_{t'}| > s\big) \le 2\exp\left(-\frac{s}{d_1(t,t')}\right) + 2\exp\left(-\frac{s^2}{d_2(t,t')^2}\right).$$

Then there exists a universal constant $C > 0$ such that for any $t_0 \in \mathcal{T}$ and all $s \ge 1$, we have

$$\sup_{t\in\mathcal{T}} \big|X_t - X_{t_0}\big| \le C\int_0^\infty \sqrt{\log N(\varepsilon, \mathcal{T}, d_2)}\,\mathrm{d}\varepsilon + C\int_0^\infty \log N(\varepsilon, \mathcal{T}, d_1)\,\mathrm{d}\varepsilon + CsD_2 + Cs^2 D_1,$$

with probability at least $1 - e^{-s^2}$.

Proof of Lemma 38. This follows by Dirksen (2015, Theorem 3.5), using (2.3) therein to bound the Talagrand γ_1 and γ_2 functionals by the respective entropy integrals. $\square$

Lemma 39 (Cross-validation on the simplex)**.** *Let $f_0, \dots, f_d \in \mathcal{B}(\mathcal{X})$ with $P_X(f_j) = 0$ and $\|f_j - f_0\|_\infty \le M$ for each j. For $\theta \in \Delta_d$, write $\theta_0 := 1 - \|\theta\|_1$ and let $f_\theta := \sum_{j=0}^d \theta_j f_j$. Define $\hat\theta := \mathrm{argmin}_{\theta\in\Delta_d} \ell_n(f_\theta)$. Then there exists a universal constant $C > 0$ such that for all $s \ge 1$, with probability at least $1 - e^{-s^2}$,*

$$\|f_{\hat\theta} - f_0\|_{L_2} \le \frac{Ce^{5\|f_0\|_\infty} e^{8M}(1 + 1/\Lambda)}{q_1^3}\left(\min_{\theta\in\Delta_d} \|f_\theta - f_0\|_{L_2} + \frac{s + \sqrt{d}}{\sqrt{n}}\right).$$

Proof of Lemma 39. Let $\theta_\star \in \operatorname{argmin}_{\theta\in\Delta_d} \ell_\star(f_\theta)$ and write $\delta := \|f_{\theta_\star} - f_0\|_{L_2}$. For $\varepsilon > 0$, define $\Theta_\varepsilon := \{\theta \in \Delta_d : \varepsilon < \|f_\theta - f_{\theta_\star}\|_{L_2} \le 2\varepsilon\}$. Write $R_n(\theta) := \ell_n(f_\theta) - \ell_n(f_0) - \ell_\star(f_\theta) + \ell_\star(f_0)$ and observe that if $\hat\theta \in \Theta_\varepsilon$, then

$$\begin{aligned} 0 &\ge \inf_{\theta\in\Theta_\varepsilon} \{\ell_n(f_\theta) - \ell_n(f_{\theta_\star})\} = \inf_{\theta\in\Theta_\varepsilon} \{\ell_\star(f_\theta) - \ell_\star(f_{\theta_\star}) + R_n(\theta) - R_n(\theta_\star)\} \\ &\ge \inf_{\theta\in\Theta_\varepsilon} \{\ell_\star(f_\theta) - \ell_\star(f_{\theta_\star})\} - \sup_{\theta\in\Theta_\varepsilon} |R_n(\theta)| - |R_n(\theta_\star)|. \end{aligned} \tag{76}$$

For $\theta \in \Theta_\varepsilon$, by Taylor's theorem, the definition of $f_{\theta_\star}$ and Lemma 17, there exists $\tilde\theta$ on the line segment between θ and $\theta_\star$ with

$$\begin{aligned} \ell_\star(f_\theta) - \ell_\star(f_{\theta_\star}) &= D\ell_\star(f_{\theta_\star})(f_\theta - f_{\theta_\star}) + \frac{1}{2} D^2\ell_\star(f_{\tilde\theta})(f_\theta - f_{\theta_\star})^{\otimes 2} \\ &\ge \frac{1}{2} D^2\ell_\star(f_{\tilde\theta})(f_\theta - f_{\theta_\star})^{\otimes 2} \ge \frac{e^{-2M} e^{-\|f_0\|_\infty} \Lambda q_1 \varepsilon^2}{2}. \end{aligned} \tag{77}$$

By Lemma 34, there exists a universal constant $C > 0$ such that for all $s \ge 1$,

$$|R_n(\theta_\star)| \le \frac{C e^{4\|f_0\|_\infty} e^{6M}}{q_1^2} \left(\frac{s\delta}{\sqrt{n}} + \frac{s^2 M}{n} \right), \tag{78}$$

with probability at least $1 - e^{-s^2}$. Next, for $\theta, \theta' \in \Delta_d$, define $d_1(\theta, \theta') := \|f_\theta - f_{\theta'}\|_\infty / n$ so that $N(\xi, \Theta_\varepsilon, d_1) \le N(\xi, \Delta_d, d_1) \le N(n\xi/M, \Delta_d, \|\cdot\|_1) \le \big(1 + M/(n\xi)\big)^d$. Since $d_1(\theta, \theta') \le M\|\theta - \theta'\|_1/n$, we have $N(\xi, \Theta_\varepsilon, d_1) = 1$ for $\xi \ge M/n$, so it follows that

$$\begin{aligned} \int_0^\infty \log N(\xi, \Theta_\varepsilon, d_1)\, \mathrm{d}\xi &\le d \int_0^{M/n} \log\big(1 + M/(n\xi)\big)\, \mathrm{d}\xi = \frac{dM}{n} \int_0^1 \log(1 + 1/u)\, \mathrm{d}u \\ &= \frac{dM \log 4}{n}. \end{aligned}$$

Similarly, define the metric $d_2(\theta, \theta') := \|f_\theta - f_{\theta'}\|_{L_2}/\sqrt{n}$. Writing A for the matrix with entries $A_{jk} := \langle f_j, f_k \rangle_{L_2}$ for $j, k \in \{0, \dots, d\}$, we have that $d_2(\theta, \theta')^2 = (\theta - \theta')^\mathsf{T} A(\theta - \theta')/n$ and $\Theta_\varepsilon \subseteq \{\theta = (\theta_0, \dots, \theta_d) : (\theta - \theta_\star)^\mathsf{T} A(\theta - \theta_\star) \le 4\varepsilon^2\}$. Thus, by considering $\psi := A^{1/2}(\theta - \theta_\star)/(2\varepsilon)$, with B_{d+1} the $\|\cdot\|_2$-unit ball in $\mathbb{R}^{d+1}$, we have that $N(\xi, \Theta_\varepsilon, d_2) \le N(\sqrt{n}\xi/(2\varepsilon), B_{d+1}, \|\cdot\|_2) \le \big(1 + 4\varepsilon/(\sqrt{n}\xi)\big)^{d+1}$ by, e.g., Talagrand (2021, Exercise 2.5.9). Since $d_2(\theta, \theta') \le 4\varepsilon/\sqrt{n}$ for all $\theta, \theta' \in \Theta_\varepsilon$, we have $N(\xi, \Theta_\varepsilon, d_2) = 1$ whenever $\xi \ge 4\varepsilon/\sqrt{n}$, so by Jensen's inequality,

$$\begin{aligned} \int_0^\infty \sqrt{\log N(\xi, \Theta_\varepsilon, d_2)}\, \mathrm{d}\xi &\le \sqrt{d+1} \int_0^{4\varepsilon/\sqrt{n}} \sqrt{\log\big(1 + 4\varepsilon/(\sqrt{n}\xi)\big)}\, \mathrm{d}\xi \\ &\le 4\varepsilon \sqrt{\frac{d+1}{n}} \int_0^1 \sqrt{\log(2/u)}\, \mathrm{d}u \le 6\varepsilon \sqrt{\frac{d+1}{n}}. \end{aligned}$$

By Lemma 34, for each $\theta, \theta' \in \Theta_\varepsilon$, with probability at least $1 - e^{-s^2}$,

$$\begin{aligned} |R_n(\theta) - R_n(\theta')| &\le \frac{C e^{4\|f_0\|_\infty} e^{6M}}{q_1^2} \left(\frac{s\|f_\theta - f_{\theta'}\|_{L_2}}{\sqrt{n}} + \frac{s^2 \|f_\theta - f_{\theta'}\|_\infty}{n} \right) \\ &= \frac{C e^{4\|f_0\|_\infty} e^{6M}}{q_1^2} \big(s d_2(\theta, \theta') + s^2 d_1(\theta, \theta') \big). \end{aligned}$$

Therefore, by Lemma 38, there exists a universal constant $C > 0$ such that for all $s \geq 1$, with $\theta' \in \Theta_\varepsilon$ fixed and with probability at least $1 - e^{-s^2}$,

$$\sup_{\theta \in \Theta_\varepsilon} |R_n(\theta) - R_n(\theta')| \leq \frac{Ce^{4\|f_0\|_\infty} e^{6M}}{q_1^2} \left(\frac{\varepsilon(s + \sqrt{d})}{\sqrt{n}} + \frac{M(s^2 + d)}{n} \right).$$

Applying Lemma 34 again to $R_n(\theta')$, using (78) and by increasing C if necessary, we have

$$\sup_{\theta \in \Theta_\varepsilon} |R_n(\theta)| + |R_n(\theta_\star)| \leq \frac{Ce^{4\|f_0\|_\infty} e^{6M}}{q_1^2} \left(\frac{\varepsilon(s + \sqrt{d}) + s\delta}{\sqrt{n}} + \frac{M(s^2 + d)}{n} \right),$$

with probability at least $1 - e^{-s^2}$. By (76) and (77), writing $C_0 := e^{8M} e^{5\|f_0\|_\infty} q_1^{-3}(1 + 1/\Lambda)$ so that $C_0 \geq 1 \vee M$ and taking $\delta_n := CC_0\sqrt{d/n}$ and $s := (\delta + \varepsilon)\sqrt{n}/(CC_0)$, by increasing C if necessary,

$$\begin{aligned}
\mathbb{P}\big(\hat{\theta} \in \Theta_{\delta + \delta_n + \varepsilon}\big) &\leq \mathbb{P}\left(\sup_{\theta \in \Theta_{\delta + \delta_n + \varepsilon}} |R_n(\theta)| + |R_n(\theta_\star)| \geq \frac{e^{-2M} e^{-\|f_0\|_\infty} \Lambda q_1}{2} \big(\delta + \delta_n + \varepsilon\big)^2 \right) \\
&\leq \mathbb{P}\left(\sup_{\theta \in \Theta_{\delta + \delta_n + \varepsilon}} |R_n(\theta)| + |R_n(\theta_\star)| \geq \frac{Ce^{4\|f_0\|_\infty} e^{6M}}{q_1^2} \frac{(\delta + \delta_n + \varepsilon)^2}{2CC_0} \right) \\
&\leq \exp\left(-\frac{(\delta + \varepsilon)^2 n}{C^2 C_0^2} \right) \leq \exp\left(-\frac{\varepsilon^2 n}{C^2 C_0^2} \right).
\end{aligned}$$

Thus, by the geometric series formula and as $\{x/(1-x)\} \wedge 1 \leq 2x$ for all $x \in [0, 1)$,

$$\begin{aligned}
\mathbb{P}\big(\|f_{\hat{\theta}} - f_{\theta_\star}\|_{L_2} > \delta + \delta_n + \varepsilon\big) &= \mathbb{P}\left(\hat{\theta} \in \bigcup_{j=0}^{\infty} \Theta_{2^j(\delta + \delta_n + \varepsilon)} \right) = \sum_{j=0}^{\infty} \mathbb{P}\big(\hat{\theta} \in \Theta_{2^j(\delta + \delta_n + \varepsilon)}\big) \\
&= \sum_{j=0}^{\infty} \mathbb{P}\big(\hat{\theta} \in \Theta_{\delta + \delta_n + (2^j - 1)(\delta + \delta_n) + 2^j \varepsilon}\big) \leq 1 \wedge \sum_{j=0}^{\infty} \exp\left(-\frac{2^{2j} \varepsilon^2 n}{C^2 C_0^2} \right) \\
&\leq 1 \wedge \sum_{j=1}^{\infty} \exp\left(-\frac{j\varepsilon^2 n}{C^2 C_0^2} \right) \leq 2 \exp\left(-\frac{\varepsilon^2 n}{C^2 C_0^2} \right).
\end{aligned}$$

Taking $\varepsilon := C^2 C_0 s/\sqrt{n}$ and then increasing C yields

$$\mathbb{P}\left(\|f_{\hat{\theta}} - f_{\theta_\star}\|_{L_2} > \delta + \frac{C^2 \big(s + \sqrt{d}\big) e^{8M} e^{5\|f_0\|_\infty} (1 + 1/\Lambda)}{q_1^3 \sqrt{n}} \right) \leq 2e^{-C^2 s^2} \leq e^{-s^2}, \tag{79}$$

for all $s \geq 1$. Writing $\check{\theta} \in \operatorname{argmin}_{\theta \in \Delta_d} \|f_\theta - f_0\|_{L_2}$, by definition of $\theta_\star$, Taylor's theorem and Lemma 17, for some $\tilde{f}$ on the line segment between f_0 and $f_{\theta_\star}$,

$$\ell_\star(f_{\hat{\theta}}) - \ell_\star(f_0) \geq \ell_\star(f_{\theta_\star}) - \ell_\star(f_0) = \frac{1}{2} D^2 \ell_\star(\tilde{f})(f_{\theta_\star} - f_0)^{\otimes 2} \geq \frac{1}{2} e^{-2M} e^{-\|f_0\|_\infty} \Lambda q_1 \delta^2.$$

On the other hand, by Taylor's theorem and Lemma 17 again, for some $\check{f}$ on the line segment between f_0 and $f_{\check{\theta}}$,

$$\ell_\star(f_{\check{\theta}}) - \ell_\star(f_0) = \frac{1}{2} D^2 \ell_\star(\check{f})(f_{\check{\theta}} - f_0)^{\otimes 2} \leq \frac{1}{2} e^{2M} e^{\|f_0\|_\infty} \Lambda \min_{\theta \in \Delta_d} \|f_\theta - f_0\|_{L_2}^2.$$

Combining the last two displays yields

$$\delta \le \frac{e^{2M} e^{\|f_0\|_\infty}}{\sqrt{q_1}} \min_{\theta \in \Delta_d} \|f_\theta - f_0\|_{L_2}. \tag{80}$$

After increasing the universal constant $C > 0$ if necessary, we conclude by (79) and (80) that for all $s \ge 1$,

$$\|f_{\hat\theta} - f_0\|_{L_2} \le \frac{C e^{5\|f_0\|_\infty} e^{8M}(1 + 1/\Lambda)}{q_1^3} \bigg(\min_{\theta \in \Delta_d} \|f_\theta - f_0\|_{L_2} + \frac{s + \sqrt{d}}{\sqrt{n}} \bigg),$$

with probability at least $1 - e^{-s^2}$, as required. □

Proof of Theorem 10. We write $C_1^0, C_2^0, \ldots$ for positive quantities depending only on $\|f_0\|_{\mathcal{H}}$, Λ, q_1, $\|1_{\mathcal{X}}\|_{\mathcal{H}}$, M, K and ξ, recalling from Lemma 2 that $\|f_0\|_\infty^2 \le K \|f_0\|_{\mathcal{H}}^2$. For $\gamma \in \Gamma$, let $\check\theta(\gamma) \coloneqq \operatorname{sargmin}_{\theta \in \Theta} \tilde\ell_n(\check f_{n,\gamma,\theta})$. Recall that $\max_{m \in \mathcal{M}} \|\tilde f_m\|_\infty \le M$ almost surely and $\gamma^- \ge n^{-\xi}$. Further, by the proof of Theorem 9, there exists $C_1^0 > 0$ such that if $\gamma^+ H_{\infty,\gamma^+} \le 1/C_1^0$, $H_{\infty,\gamma^-} \sqrt{(\log|\Gamma|) \wedge \log n}/\sqrt{n} \le 1/C_1^0$ and $1 \le s \le \sqrt{n}/(C_1^0 H_{\infty,\gamma^-})$, then $\max_{\gamma \in \Gamma} \|\hat f_{n,\gamma} - f_0\|_\infty \le 1$ on an event $\mathcal{A}_n$ with probability at least $1 - e^{-s^2}$. Applying Lemma 39 on $\mathcal{A}_n$ to $\tilde\ell_n$ and $\{\check f_{n,\gamma,\theta} - P_X(\check f_{n,\gamma,\theta}) : \theta \in \Theta\}$, conditionally on the training data and on $(\hat f_m : m \in \mathcal{M})$, there exists $C_2^0 > 0$ such that for each $\gamma \in \Gamma$, with probability at least $1 - e^{-s^2}$,

$$\big\| \check f_{n,\gamma,\check\theta(\gamma)} - P_X(\check f_{n,\gamma,\check\theta(\gamma)}) - f_0 \big\|_{L_2} \le C_2^0 \bigg(\min_{\theta \in \Theta} \big\| \check f_{n,\gamma,\theta} - P_X(\check f_{n,\gamma,\theta}) - f_0 \big\|_{L_2} + \frac{s + \sqrt{|\mathcal{M}|}}{\sqrt{n}} \bigg).$$

Let $\Gamma' \subseteq \Gamma$ be as defined in the proof of Theorem 9. By Lemma 36, for any $\gamma \in \Gamma$, there exists $\gamma' \in \Gamma'$ with

$$\sup_{\theta \in \Theta} \big\| \check f_{n,\gamma',\theta} - \check f_{n,\gamma,\theta} \big\|_\infty = \big\| \hat f_{n,\gamma'} - \hat f_{n,\gamma} \big\|_\infty \le \frac{2\sqrt{K}(\gamma' - \gamma)\sqrt{\log n}}{\gamma^{3/2}} \le \frac{2\sqrt{K \log n}}{n^{3\xi/2}} \le \frac{2\sqrt{K}}{n}.$$

By Lemma 35, as $|\Gamma'| \le |\Gamma| \wedge n^{4\xi}$, with probability at least $1 - e^{-s^2}$,

$$\begin{aligned} &\big\| \check f_{n,\check\gamma,\check\theta} - P_X(\check f_{n,\check\gamma,\check\theta}) - f_0 \big\|_{L_2} \\ &\quad \le C_3^0 \bigg(\min_{(\gamma,\theta) \in \Gamma' \times \Theta} \big\| \check f_{n,\gamma,\theta} - P_X(\check f_{n,\gamma,\theta}) - f_0 \big\|_{L_2} + \frac{s + \sqrt{|\mathcal{M}| + \log n}}{\sqrt{n}} \bigg). \end{aligned} \tag{81}$$

Observe that

$$P_X(\tilde f_m) = P_X\big(\hat f_m - P_n(\hat f_m) 1_{\mathcal{X}}\big) = (P_X - P_n)(\hat f_m)$$

and $\|\hat f_m\|_\infty \le M$ almost surely for each $m \in \mathcal{M}$. Therefore, applying (74), Hoeffding's inequality and a union bound, there exists $C_4^0 > 0$ such that with probability at least $1 - e^{-s^2}$,

$$\max_{(\gamma,\theta) \in \Gamma' \times \Theta} \big| P_X\big(\check f_{n,\gamma,\theta}\big) \big| \le \max_{\gamma \in \Gamma'} \big| P_X\big(\hat f_{n,\gamma}\big) \big| + \max_{m \in \mathcal{M}} \big| P_X\big(\tilde f_m\big) \big| \le C_4^0 \frac{s + \sqrt{\log|\mathcal{M}|}}{\sqrt{n}}. \tag{82}$$

The result follows from (81) and (82). □

E.4 Proofs for Section 5

For the reader's convenience, we state a useful result for computing the Hilbert space norm of a constant function in Lemma 40.

Lemma 40 (Conditions for $1_{\mathcal{X}} \in \mathcal{H}$)**.** *Let $\mathcal{H}$ be an RKHS on $\mathcal{X}$ with kernel k. Then $1_{\mathcal{X}} \in \mathcal{H}$ if and only if there exists $s > 0$ such that $(x, y) \mapsto k(x, y) - s$ is a kernel on $\mathcal{X}$, in which case $1/\|1_{\mathcal{X}}\|_{\mathcal{H}}^2 = \sup\{s > 0 : k(\cdot, \cdot) - s \text{ is a kernel on } \mathcal{X}\}$.*

Proof of Lemma 40. See Paulsen and Raghupathi (2016, Theorem 3.11), setting $f = 1_{\mathcal{X}}$ and $s = 1/c^2$ in their notation. □

Proof of Proposition 11. We begin by showing equivalence of the two expressions for $\kappa(\boldsymbol{A})$. Let $v_1, \dots, v_n \in \mathbb{R}^n$ form an orthonormal basis of eigenvectors of $\boldsymbol{A}$, with associated eigenvalues $\lambda_1, \dots, \lambda_n \geq 0$. For $\delta > 0$, the matrix $\boldsymbol{A} + \delta \boldsymbol{I}_n$ has the same eigenvectors, but now with eigenvalues $\lambda_1 + \delta, \dots, \lambda_n + \delta > 0$, so we may write $(\boldsymbol{A} + \delta \boldsymbol{I}_n)^{-1} = \sum_{i=1}^n \frac{1}{\lambda_i + \delta} v_i v_i^\mathsf{T}$. Therefore $1_n^\mathsf{T}(\boldsymbol{A} + \delta \boldsymbol{I}_n)^{-1} 1_n = \sum_{i=1}^n \frac{1}{\lambda_i + \delta}(1_n^\mathsf{T} v_i)^2$. We now proceed by cases. Suppose first that there exists $v \in \mathbb{R}^n$ with $\boldsymbol{A}v = 0$ and $1_n^\mathsf{T} v = 1$. Then we can find $i_0 \in [n]$ with $\lambda_{i_0} = 0$ and $1_n^\mathsf{T} v_{i_0} \neq 0$. Thus $\frac{1}{\lambda_{i_0} + \delta}(1_n^\mathsf{T} v_{i_0})^2 \to \infty$ as $\delta \searrow 0$, so $1_n^\mathsf{T}(\boldsymbol{A} + \delta \boldsymbol{I}_n)^{-1} 1_n \to \infty$ as $\delta \searrow 0$. Otherwise, for all $i \in [n]$, either $\lambda_i > 0$ or $1_n^\mathsf{T} v_i = 0$. In this case, $\sum_{i=1}^n \frac{1}{\lambda_i + \delta}(1_n^\mathsf{T} v_i)^2 \to \sum_{i=1}^n \lambda_i^+ (1_n^\mathsf{T} v_i)^2 = 1_n^\mathsf{T} \sum_{i=1}^n \lambda_i^+ v_i v_i^\mathsf{T} 1_n^\mathsf{T} = 1_n^\mathsf{T} \boldsymbol{A}^+ 1_n$ as $\delta \searrow 0$, where $\lambda_i^+ := 1/\lambda_i$ if $\lambda_i > 0$ and $\lambda_i^+ := 0$ if $\lambda_i = 0$. This verifies the desired equivalence of the two expressions for $\kappa(\boldsymbol{A})$.

Next, we show that

$$\kappa(\boldsymbol{A}) = \sup\big\{s \geq 0 : \boldsymbol{A} - s 1_n 1_n^\mathsf{T} \succeq 0\big\}. \tag{83}$$

Suppose for now that $\boldsymbol{A}$ is positive definite, so invertible. Then for $s > 0$, we have $\boldsymbol{A} - s 1_n 1_n^\mathsf{T} \succeq 0$ if and only if $v^\mathsf{T} 1_n 1_n^\mathsf{T} v / v^\mathsf{T} \boldsymbol{A} v \leq 1/s$ for all $v \in \mathbb{R}^n \setminus \{0\}$. Equivalently, defining $w := \boldsymbol{A}^{1/2} v$, this holds precisely when

$$\frac{1}{s} \geq \sup_{w \in \mathbb{R}^n \setminus \{0\}} \frac{w^\mathsf{T} \boldsymbol{A}^{-1/2} 1_n 1_n^\mathsf{T} \boldsymbol{A}^{-1/2} w}{\|w\|_2^2} = 1_n^\mathsf{T} \boldsymbol{A}^{-1} 1_n.$$

It follows that the largest such s is given by $1/1_n^\mathsf{T} \boldsymbol{A}^{-1} 1_n$. We now consider the case where $\boldsymbol{A}$ is not invertible. Define $g : [0, \infty)^2 \to \mathbb{R}$ by $g(s, \delta) := \lambda_{\min}(\boldsymbol{A} + \delta \boldsymbol{I}_n - s 1_n 1_n^\mathsf{T})$, which is continuous in (s, δ) by Weyl's inequality (e.g., Samworth and Shah, 2026, Theorem A.10.17). Further, $s \mapsto g(s, \delta)$ is decreasing for each $\delta \geq 0$ and $\delta \mapsto g(s, \delta)$ is increasing for each $s \geq 0$. For $\delta \geq 0$, define $s^\star(\delta) := \sup\{s \geq 0 : g(s, \delta) \geq 0\}$, noting that $s^\star$ is an increasing function on $[0, \infty)$, that $s^\star(\delta) = 1/1_n^\mathsf{T}(\boldsymbol{A} + \delta \boldsymbol{I}_n)^{-1} 1_n$ for $\delta > 0$, and $g\big(s^\star(\delta), \delta\big) \geq 0$ by continuity. Now take $\varepsilon > 0$, so that $g\big(s^\star(0) + \varepsilon, 0\big) < 0$. By continuity, there exists $\delta > 0$ such that $g\big(s^\star(0) + \varepsilon, \delta\big) < 0$, so $s^\star(\delta) < s^\star(0) + \varepsilon$. But since $\varepsilon > 0$ was arbitrary, we deduce that

$$s^\star(0) = \lim_{\delta \searrow 0} s^\star(\delta) = \lim_{\delta \searrow 0} \frac{1}{1_n^\mathsf{T}(\boldsymbol{A} + \delta \boldsymbol{I}_n)^{-1} 1_n} = \kappa(\boldsymbol{A}).$$

This establishes (83).

Now $k(x, y) - s$ is a kernel if and only if $\boldsymbol{K}(x_1, \dots, x_n) - s 1_n 1_n^\mathsf{T} \succeq 0$ for all $n \in \mathbb{N}$ and $x_1, \dots, x_n \in \mathcal{X}$, or equivalently if $s \leq \inf_{n \in \mathbb{N}} \inf_{x_1, \dots, x_n \in \mathcal{X}} \kappa\big(\boldsymbol{K}(x_1, \dots, x_n)\big)$. The result follows by Lemma 40. □

E.5 Proofs for Appendix C

Proof of Proposition 12. We write $C_1^0, C_2^0, \ldots$ for quantities depending only on $\max_{k\in\mathcal{K}} \|f_0\|_{\mathcal{H}_k}$, Λ, q_1, $\max_{k\in\mathcal{K}} \|1_{\mathcal{X}}\|_{\mathcal{H}_k}$, $\max_{k\in\mathcal{K}} K_k$ and ξ. Let Γ' be as in the proof of Theorem 9. For each $\gamma \in \Gamma$ and $k \in \mathcal{K}$, by (71), there exists $\gamma' \in \Gamma'$ with

$$\big\|\hat{f}_{n,\gamma',k} - \hat{f}_{n,\gamma,k}\big\|_\infty \le \frac{2\sqrt{K}}{n}.$$

By Lemma 2, Theorem 4 and a union bound over $\gamma \in \Gamma'$ and $k \in \mathcal{K}$, there exists $C_1^0 > 0$ such that if $\gamma^+ H_{\infty,\gamma^+,\mathcal{K}} \le 1/C_1^0$ and $1 \le s \le \sqrt{n}/(C_1^0 H_{\infty,\gamma^-,\mathcal{K}})$, then with probability at least $1 - e^{-s^2}$,

$$\begin{aligned}
\max_{\gamma\in\Gamma',\,k\in\mathcal{K}} \big\|\hat{f}_{n,\gamma,k} - f_0\big\|_\infty &\le \max_{\gamma\in\Gamma',\,k\in\mathcal{K}} \Big\{\sqrt{H_{\infty,\gamma,k}}\big\|\hat{f}_{n,\gamma,k} - f_0\big\|_{\mathcal{H}_{k,\gamma}}\Big\} \\
&\le C_1^0 \max_{\gamma\in\Gamma',\,k\in\mathcal{K}} \bigg\{\big(s + \sqrt{\log|\mathcal{K}| + \log|\Gamma'|}\big)\frac{H_{\infty,\gamma,k}}{\sqrt{n}} + \sqrt{\gamma H_{\infty,\gamma,k}}\bigg\} \\
&\le C_1^0 \bigg\{\big(s + \sqrt{\log|\mathcal{K}| + \log|\Gamma'|}\big)\frac{H_{\infty,\gamma^-,\mathcal{K}}}{\sqrt{n}} + \sqrt{\gamma^+ H_{\infty,\gamma^+,\mathcal{K}}}\bigg\}.
\end{aligned}$$

Thus, there exists $C_2^0 > 0$ such that if $\gamma^+ H_{\infty,\gamma^+,\mathcal{K}} \le 1/C_2^0$, $H_{\infty,\gamma^-,\mathcal{K}}\sqrt{\log|\mathcal{K}| + \log|\Gamma'|}/\sqrt{n} \le 1/C_2^0$ and $1 \le s \le \sqrt{n}/(C_2^0 H_{\infty,\gamma^-,\mathcal{K}})$, then with probability at least $1 - e^{-s^2}$,

$$\max_{\gamma\in\Gamma',\,k\in\mathcal{K}} \big\|\hat{f}_{n,\gamma,k} - f_0\big\|_\infty \le 1. \tag{84}$$

Moreover, for every $\gamma \in \Gamma'$ and $k \in \mathcal{K}$, we have

$$\big\|P_X(\hat{f}_{n,\gamma,k})1_{\mathcal{X}}\big\|_\infty = \big|P_X(\hat{f}_{n,\gamma,k})\big| = \big|P_X(\hat{f}_{n,\gamma,k} - f_0)\big| \le \big\|\hat{f}_{n,\gamma,k} - f_0\big\|_\infty.$$

Applying Lemma 35 to $\tilde{\ell}_n$ and $\big\{\hat{f}_{n,\gamma,k} - P_X(\hat{f}_{n,\gamma,k})1_{\mathcal{X}} : \gamma \in \Gamma', k \in \mathcal{K}\big\}$ conditionally on the training data, on the event where (84) holds, there exists $C_3^0 > 0$ such that with probability at least $1 - e^{-s^2}$,

$$\begin{aligned}
&\big\|\hat{f}_{n,\hat{\gamma}_{\mathcal{K}},\hat{k}} - P_X(\hat{f}_{n,\hat{\gamma}_{\mathcal{K}},\hat{k}})1_{\mathcal{X}} - f_0\big\|_{L_2} \\
&\quad\le C_3^0\bigg(\min_{\gamma\in\Gamma',\,k\in\mathcal{K}} \big\|\hat{f}_{n,\gamma,k} - P_X(\hat{f}_{n,\gamma,k})1_{\mathcal{X}} - f_0\big\|_{L_2} + \frac{s + \sqrt{\log|\mathcal{K}| + \log|\Gamma'|}}{\sqrt{n}}\bigg).
\end{aligned}$$

The result now follows by arguing as in the proof of Theorem 9. □

E.6 Inequalities for U-processes

Let $(\mathcal{X}, \mathcal{A})$ be a measurable space and $X_1, \ldots, X_n$ be independent and identically distributed random variables taking values in a measurable space $\mathcal{X}$ where $n \ge 2$. Let $\mathcal{F}$ be a non-empty class of bounded, symmetric, Borel measurable functions $f : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ that is separable with respect to the topology of pointwise convergence. In other words, there exists a countable subset $\mathcal{F}' \subseteq \mathcal{F}$ such that for every $f \in \mathcal{F}$, there exists a sequence (f_n) in $\mathcal{F}'$ with $f_n(x) \to f(x)$

as $n \to \infty$ for every $x \in \mathcal{X}$. Assume that $\mathbb{E}\{f(X_1, x)\} = 0$ for all $x \in \mathcal{X}$ and $f \in \mathcal{F}$. Suppose that there exists a function $F : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ with $f(x, y) \leq F(x, y)$ for all $x, y \in \mathcal{X}$ and $f \in \mathcal{F}$, and further $\mathbb{E}\{F(X_1, X_2)^2\} < \infty$. Define $\mathcal{I} := \{(i, j) : i, j \in [n], i < j\}$ and for $f \in \mathcal{F}$, define the degenerate second-order U-process and its decoupled version by

$$U_n(f) := \frac{2}{n(n-1)} \sum_{(i,j)\in\mathcal{I}} f(X_i, X_j), \qquad V_n(f) := \frac{2}{n(n-1)} \sum_{(i,j)\in\mathcal{I}} f(X_i, X'_j),$$

where $X'_1, \ldots, X'_n$ are independent copies of $X_1, \ldots, X_n$. Now let $\varepsilon_1, \ldots, \varepsilon_n, \varepsilon'_1, \ldots, \varepsilon'_n$ denote independent and identically distributed Rademacher variables that are independent of $X_1, \ldots, X_n, X'_1, \ldots, X'_n$. Define randomised versions of V_n with and without decoupling by

$$W_n(f) := \frac{2}{n(n-1)} \sum_{(i,j)\in\mathcal{I}} \varepsilon_i \varepsilon_j f(X_i, X_j), \qquad Y_n(f) := \frac{2}{n(n-1)} \sum_{(i,j)\in\mathcal{I}} \varepsilon_i \varepsilon'_j f(X_i, X'_j).$$

For a stochastic process $\big(Z(f)\big)_{f\in\mathcal{F}}$, we write $\|Z\|_{\mathcal{F}} := \sup_{f\in\mathcal{F}} |Z(f)|$.

Lemma 41 (Decoupling)**.** *Let $\phi : [0, \infty) \to [0, \infty)$ be convex and increasing. Then*

$$\mathbb{E}\left\{\phi\left(\frac{1}{4}\|V_n\|_{\mathcal{F}}\right)\right\} \leq \mathbb{E}\big\{\phi\big(\|U_n\|_{\mathcal{F}}\big)\big\} \leq \mathbb{E}\big\{\phi\big(8\|V_n\|_{\mathcal{F}}\big)\big\}.$$

Proof of Lemma 41. See the proof of de la Peña and Giné (1999, Theorem 3.1.1). □

Lemma 42 (Randomisation)**.** *Let $\phi : [0, \infty) \to [0, \infty)$ be convex and increasing. Then*

$$\mathbb{E}\big\{\phi\big(\|V_n\|_{\mathcal{F}}\big)\big\} \leq \mathbb{E}\big\{\phi\big(4\|Y_n\|_{\mathcal{F}}\big)\big\}.$$

Proof of Lemma 42. We apply twice the randomisation lemma for independent processes from van der Vaart and Wellner (1996, Lemma 2.3.6), noting that the outer expectations may be replaced with expectations by the separability condition on $\mathcal{F}$; this also means that we do not need to insist that the random variables are coordinate projections on a product space. Hence

$$\begin{aligned}
\mathbb{E}\big\{\phi\big(\|V_n\|_{\mathcal{F}}\big)\big\} &= \mathbb{E}\left[\mathbb{E}\left\{\phi\left(\sup_{f\in\mathcal{F}}\left|\frac{2}{n(n-1)} \sum_{(i,j)\in\mathcal{I}} f(X_i, X'_j)\right|\right) \,\middle|\, X'_1, \ldots, X'_n\right\}\right] \\
&\leq \mathbb{E}\left\{\phi\left(2\sup_{f\in\mathcal{F}}\left|\frac{2}{n(n-1)} \sum_{(i,j)\in\mathcal{I}} \varepsilon_i f(X_i, X'_j)\right|\right)\right\} \\
&= \mathbb{E}\left[\mathbb{E}\left\{\phi\left(2\sup_{f\in\mathcal{F}}\left|\frac{2}{n(n-1)} \sum_{(i,j)\in\mathcal{I}} \varepsilon_i f(X_i, X'_j)\right|\right) \,\middle|\, \varepsilon_1, \ldots, \varepsilon_n, X_1, \ldots, X_n\right\}\right] \\
&\leq \mathbb{E}\big\{\phi\big(4\|Y_n\|_{\mathcal{F}}\big)\big\}.
\end{aligned}$$

□

Corollary 43 (Decoupling and randomisation)**.** *We have*

$$\mathbb{E}\big\{\phi\big(\|U_n\|_{\mathcal{F}}\big)\big\} \leq \mathbb{E}\big\{\phi\big(128\|W_n\|_{\mathcal{F}}\big)\big\}.$$

Proof of Corollary 43. Let

$$\mathcal{G} := \big\{ g : \big(\{0,1\} \times \mathcal{X}\big) \times \big(\{0,1\} \times \mathcal{X}\big) \to \mathbb{R} : g\big((\varepsilon, x), (\varepsilon', x')\big) = 128\varepsilon\varepsilon' f(x, x') \text{ with } f \in \mathcal{F} \big\}.$$

Then by Lemmas 41 and 42,

$$\begin{aligned} \mathbb{E}\big\{\phi\big(\|U_n\|_{\mathcal{F}}\big)\big\} &\le \mathbb{E}\big\{\phi\big(8\|V_n\|_{\mathcal{F}}\big)\big\} \le \mathbb{E}\big\{\phi\big(32\|Y_n\|_{\mathcal{F}}\big)\big\} = \mathbb{E}\bigg\{\phi\bigg(\frac{1}{4}\|V_n\|_{\mathcal{G}}\bigg)\bigg\} \\ &\le \mathbb{E}\big\{\phi\big(\|U_n\|_{\mathcal{G}}\big)\big\} \le \mathbb{E}\big\{\phi\big(128\|W_n\|_{\mathcal{F}}\big)\big\}. \end{aligned}$$

□

Lemma 44 (Rademacher chaos moment equivalence)**.** *Let* $\mathbf{X} = (X_1, \ldots, X_n)$*. For* $k \in [1, \infty)$ *and* $f \in \mathcal{F}$*, define* $\|W_n(f)\|_{k|\mathbf{X}} := \mathbb{E}\big\{|W_n(f)|^k \mid \mathbf{X}\big\}^{1/k}$*. Then*

$$\|W_n(f)\|_{k|\mathbf{X}} \le \max(1, k-1) \cdot \|W_n(f)\|_{2|\mathbf{X}}.$$

Proof of Lemma 44. For $k \in [1, 2]$, the result follows from Hölder's inequality. For $k > 2$, we can apply de la Peña and Giné (1999, Theorem 3.2.2) conditionally on $\mathbf{X}$ with $d = 2$, $p = 2$ and $q = k$ to obtain the result. □

Define $\psi : [0, \infty) \to [0, \infty)$ by $\psi(x) := e^x - 1$, and for a random variable A, the Orlicz norms $\|A\|_\psi := \inf\big\{c > 0 : \mathbb{E}\big(\psi(|A|/c)\big) \le 1\big\}$ and $\|A\|_{\psi|\mathbf{X}} := \operatorname{ess\,inf}\big\{c(\mathbf{X}) \text{ measurable} : \mathbb{E}\big[\psi\big(\frac{|A|}{c(\mathbf{X})}\big) \mid \mathbf{X}\big] \le 1 \text{ almost surely}\big\}$, where ess inf denotes the essential infimum.

Lemma 45 (Exponential bound for Rademacher chaoses)**.** *For every* $f \in \mathcal{F}$*,*

$$\big\|W_n(f)\big\|_{\psi|\mathbf{X}} \le 3\|W_n(f)\|_{2|\mathbf{X}}.$$

Proof of Lemma 45. By Stirling's formula, $k! > \sqrt{2\pi k}(k/e)^k$ for $k \in \mathbb{N}$, so by the monotone convergence theorem and Lemma 44, with $a := 3$ and noting that $(1 - 1/k)^k \le 1/e$,

$$\begin{aligned} \mathbb{E}\bigg\{\psi\bigg(\frac{|W_n(f)|}{a\|W_n(f)\|_{2|\mathbf{X}}}\bigg) \,\bigg|\, \mathbf{X}\bigg\} &= \sum_{k=1}^{\infty} \frac{1}{k!} \frac{\|W_n(f)\|_{k|\mathbf{X}}^k}{a^k \|W_n(f)\|_{2|\mathbf{X}}^k} \\ &\le \frac{1}{a} + \frac{1}{2a^2} + \frac{4}{3a^3} + \sum_{k=4}^{\infty} \frac{(k-1)^k}{a^k k!} \le \frac{1}{a} + \frac{1}{2a^2} + \frac{4}{3a^3} + \sum_{k=4}^{\infty} \frac{1}{\sqrt{2\pi k}} \bigg(\frac{e}{a}\bigg)^k (1 - 1/k)^k \\ &\le \frac{1}{a} + \frac{1}{2a^2} + \frac{4}{3a^3} + \frac{1}{e\sqrt{8\pi}} \sum_{k=4}^{\infty} \bigg(\frac{e}{a}\bigg)^k \le \frac{1}{a} + \frac{1}{2a^2} + \frac{4}{3a^3} + \frac{1}{e\sqrt{8\pi}} \frac{e^4}{a^3(a-e)} \le 1. \end{aligned}$$

□

Definition 46 (Covering numbers)**.** *Let* $(\mathcal{T}, d)$ *be a non-empty pseudometric space. For* $\varepsilon > 0$*, we say that a non-empty finite set* $\mathcal{C} \subseteq \mathcal{T}$ *is an* ε-cover *of* $(\mathcal{T}, d)$ *if* $\sup_{t \in \mathcal{T}} \min_{t' \in \mathcal{C}} d(t, t') \le \varepsilon$*. We write* $N(\varepsilon, \mathcal{T}, d)$ *for the minimal cardinality of such a cover, and set* $N(\varepsilon, \mathcal{T}, d) = \infty$ *if no such cover exists. We say that* $(\mathcal{T}, d)$ *is* totally bounded *if* $N(\varepsilon, \mathcal{T}, d) < \infty$ *for each* $\varepsilon > 0$*.*

Lemma 47 (Chaining with Orlicz norms)**.** *Let* $(\mathcal{T}, d)$ *be a non-empty, totally bounded pseudometric space with diameter* $D \in (0, \infty)$ *and let* $(X_t)_{t \in \mathcal{T}}$ *be a stochastic process. Suppose that* $(X_t)_{t \in \mathcal{T}}$ *is separable in the sense that there exists a countable set* $\mathcal{T}' \subseteq \mathcal{T}$ *such that for all*

$t \in \mathcal{T}$, there exists a sequence (t_n) in $\mathcal{T}'$ with $t_n \to t$ and $X_{t_n} \to X_t$ almost surely. Assume that $\|X_s - X_t\|_\psi \le d(s,t)$ for all $s,t \in \mathcal{T}$. Then for any $t_0 \in \mathcal{T}$,

$$\Big\| \sup_{t\in\mathcal{T}} |X_t| \Big\|_\psi \le \|X_{t_0}\|_\psi + 8e \int_0^{D/4} \log\big(1 + N(\varepsilon, \mathcal{T}, d)\big)\, \mathrm{d}\varepsilon,$$

where $N(\varepsilon, \mathcal{T}, d)$ denotes the ε-covering number of $\mathcal{T}$ with respect to d.

Proof of Lemma 47. Let $N \ge 2$. Following the proof of van der Vaart and Wellner (1996, Lemma 2.2.2) with $c = 1$ and $y \ge 1$, for random variables $Z_1, \dots, Z_N$, we have

$$\mathbb{E}\left\{ \psi\left(\frac{\max_{j\in[N]} |Z_j|}{y \max_{j\in[N]} \|Z_j\|_\psi} \right) \right\} \le \frac{N}{\psi(y)} + \psi(1).$$

Now if Z is a non-negative random variable, $a > 0$ and $b \ge 1$, then $\psi\big(Z/(ab)\big) \le \psi(Z/a)/b$ by convexity of ψ, so $\|Z\|_\psi \le \inf_{a>0} a \max\big\{\mathbb{E}\big(\psi(Z/a)\big), 1\big\}$. Hence, with $y = \psi^{-1}(N) = \log(1+N) \ge 1$,

$$\begin{aligned} \Big\| \max_{j\in[N]} |Z_j| \Big\|_\psi &\le y \max_{j\in[N]} \|Z_j\|_\psi \max\left[\mathbb{E}\left\{ \psi\left(\frac{\max_{j\in[N]} |Z_j|}{y \max_{j\in[N]} \|Z_j\|_\psi} \right) \right\}, 1 \right] \\ &\le e \log(1+N) \max_{j\in[N]} \|Z_j\|_\psi. \end{aligned} \tag{85}$$

Let $\delta > 0$ and suppose that $t_1, \dots, t_N, s_1, \dots, s_N \in \mathcal{T}$ satisfy $\max_{j\in[N]} d(s_j, t_j) \le \delta$. By (85),

$$\Big\| \max_{j\in[N]} \big| X_{t_j} - X_{s_j} \big| \Big\|_\psi \le e\delta \log(1+N).$$

Let $\tilde{\mathcal{T}}$ be a finite subset of $\mathcal{T}$ that has strictly positive diameter and contains t_0. Set $\tilde{\mathcal{T}}_0 = \{t_0\}$, which is a D-cover of $(\tilde{\mathcal{T}}, d)$, and for $k \in \mathbb{N}$ let $\tilde{\mathcal{T}}_k$ be a $(2^{-k}D)$-cover of $(\tilde{\mathcal{T}}, d)$ with cardinality $N_k := N\big(2^{-k}D, \tilde{\mathcal{T}}, d\big)$. Now, for $k \in \mathbb{N}_0$, define $\pi_k : \tilde{\mathcal{T}} \to \tilde{\mathcal{T}}_k$ by[2] $\pi_k(t) := \operatorname{argmin}_{s\in\tilde{\mathcal{T}}_k} d(s,t)$, so that $d\big(t, \pi_k(t)\big) \le 2^{-k}D$ for all $t \in \tilde{\mathcal{T}}$. Choose $K \in \mathbb{N}$ sufficiently large that $2^{-K}D < \min\{d(s,t) : s,t \in \tilde{\mathcal{T}}, d(s,t) > 0\}$. Observing that if $d(s,t) = 0$ then $\|X_s - X_t\|_\psi = 0$, so $X_s = X_t$ almost surely, and from the fact that π_K is surjective, we have

$$\max_{t\in\tilde{\mathcal{T}}} |X_t - X_{t_0}| = \max_{t\in\tilde{\mathcal{T}}_K} |X_t - X_{t_0}| = \max_{t\in\tilde{\mathcal{T}}} |X_{\pi_K(t)} - X_{t_0}|$$

almost surely. Noting that $d\big(\pi_{k+1}(t), \pi_k \circ \pi_{k+1}(t)\big) \le 2^{-k}D$, we have

$$\Big\| \sup_{t\in\tilde{\mathcal{T}}} \big| X_{\pi_{k+1}(t)} - X_{\pi_k \circ \pi_{k+1}(t)} \big| \Big\|_\psi \le e \cdot 2^{-k} D \log(1 + N_{k+1}).$$

[2] Formally, we order $\tilde{\mathcal{T}}_k$ and in the definition of π_k we take the smallest element of the argmin set.

Since $N(\varepsilon, \tilde{\mathcal{T}}, d) \le N(\varepsilon/2, \mathcal{T}, d)$,

$$\begin{aligned}
\Big\|\max_{t\in\tilde{\mathcal{T}}} |X_t - X_{t_0}|\Big\|_\psi &\le \sum_{k=0}^{K-1} \Big\|\max_{t\in\tilde{\mathcal{T}}} |X_{\pi_{k+1}\circ\cdots\circ\pi_K(t)} - X_{\pi_k\circ\pi_{k+1}\circ\cdots\circ\pi_K(t)}|\Big\|_\psi \\
&\le e\sum_{k=0}^{K-1} 2^{-k} D \log(1+N_{k+1}) \le 4e\sum_{k=0}^{\infty}\int_{2^{-(k+2)}D}^{2^{-(k+1)}D} \log\big(1+N(\varepsilon,\tilde{\mathcal{T}},d)\big)\,\mathrm{d}\varepsilon \\
&\le 4e\int_0^{D/2} \log\big(1+N(\varepsilon,\tilde{\mathcal{T}},d)\big)\,\mathrm{d}\varepsilon \le 4e\int_0^{D/2} \log\big(1+N(\varepsilon/2,\mathcal{T},d)\big)\,\mathrm{d}\varepsilon \\
&= 8e\int_0^{D/4} \log\big(1+N(\varepsilon,\mathcal{T},d)\big)\,\mathrm{d}\varepsilon.
\end{aligned}$$

Taking an increasing sequence of finite sets whose union is the countable set $\mathcal{T}'$, and since the above bound does not depend on the set $\tilde{\mathcal{T}}$, the monotone convergence theorem yields

$$\Big\|\sup_{t\in\mathcal{T}'}|X_t|\Big\|_\psi \le \big\|X_{t_0}\big\|_\psi + \Big\|\sup_{t\in\mathcal{T}'}|X_t - X_{t_0}|\Big\|_\psi \le \big\|X_{t_0}\big\|_\psi + 8e\int_0^{D/4}\log\big(1+N(\varepsilon,\mathcal{T},d)\big)\,\mathrm{d}\varepsilon.$$

Finally the result follows by separability, since $\sup_{t\in\mathcal{T}}|X_t| = \sup_{t\in\mathcal{T}'}|X_t|$ almost surely. □

Lemma 48 (Orlicz norm supremum bound for symmetrised U-statistics)**.** *Let $\mathcal{Q}$ denote the set of finitely supported probability measures on $\mathcal{X}\times\mathcal{X}$, and for a measurable $f : \mathcal{X}\times\mathcal{X}\to\mathbb{R}$ and $Q\in\mathcal{Q}$, define $\|f\|_{Q,2}^2 := \mathbb{E}_{(Y_1,Y_2)\sim Q}\{f(Y_1,Y_2)^2\}$ and $\|f\|_2^2 := \mathbb{E}\{f(X_1,X_2)^2\}$. Let $\|F\|_\infty := \sup_{x_1,x_2\in\mathcal{X}}|F(x_1,x_2)|$ and*

$$J := \sup_{Q\in\mathcal{Q}}\int_0^1 \log\big(1+N(\varepsilon\|F\|_{Q,2},\mathcal{F},\|\cdot\|_{Q,2})\big)\,\mathrm{d}\varepsilon,$$

with the covering number taken with respect to the pseudo-metric induced by $\|\cdot\|_{Q,2}$. Then

$$\big\|\|nW_n\|_{\mathcal{F}}\big\|_\psi \le 140\|F\|_\infty J.$$

Proof of Lemma 48. The result is clear if $\|F\|_\infty = 0$, so assume that $\|F\|_\infty > 0$. Note that W_n is a Rademacher chaos of degree two conditional on $\mathbf{X}$, and for $f\in\mathcal{F}$ define $\|f\|_{n,2}^2 := \frac{2}{n(n-1)}\sum_{(i,j)\in\mathcal{I}} f(X_i,X_j)^2$. If $(i,j),(i',j')\in\mathcal{I}$, then $\mathbb{E}(\varepsilon_i\varepsilon_j\varepsilon_{i'}\varepsilon_{j'}) = 0$ unless $(i,j)=(i',j')$. Hence, by Lemma 45,

$$\begin{aligned}
\big\|W_n(f)\big\|_{\psi|\mathbf{X}}^2 \le 9\|W_n(f)\|_{2|\mathbf{X}}^2 &= \frac{36}{n^2(n-1)^2}\,\mathbb{E}\bigg\{\bigg(\sum_{(i,j)\in\mathcal{I}}\varepsilon_i\varepsilon_j f(X_i,X_j)\bigg)^2\,\bigg|\,\mathbf{X}\bigg\} \\
&= \frac{36}{n^2(n-1)^2}\sum_{(i,j)\in\mathcal{I}}\sum_{(i',j')\in\mathcal{I}} f(X_i,X_j)f(X_{i'},X_{j'})\mathbb{E}(\varepsilon_i\varepsilon_j\varepsilon_{i'}\varepsilon_{j'}) \\
&= \frac{36}{n^2(n-1)^2}\sum_{(i,j)\in\mathcal{I}} f(X_i,X_j)^2 = \frac{18}{n(n-1)}\|f\|_{n,2}^2 \le \frac{36}{n^2}\|f\|_{n,2}^2. \qquad (86)
\end{aligned}$$

For $f, f' \in \mathcal{F}$, by linearity of W_n and (86), we have $\big\|nW_n(f) - nW_n(f')\big\|_{\psi|\mathbf{X}}^2 \le 36\|f - f'\|_{n,2}^2$. The diameter of $\mathcal{F}$ under $\|\cdot\|_{n,2}$ is at most $2\|F\|_{n,2}$, so by Lemma 47 with $d(f, f') = 6\|f - f'\|_{n,2}$,

$$\begin{aligned}
\big\|\|nW_n\|_{\mathcal{F}}\big\|_{\psi|\mathbf{X}} &\le \|nW_n(f_0)\|_{\psi|\mathbf{X}} + 8e\int_0^{3\|F\|_{n,2}} \log\big(1 + N(\varepsilon, \mathcal{F}, 6\|\cdot\|_{n,2})\big)\,\mathrm{d}\varepsilon \\
&\le 6\|f_0\|_{n,2} + 48e\|F\|_{n,2}\int_0^{1/2} \log\big(1 + N(\varepsilon\|F\|_{n,2}, \mathcal{F}, \|\cdot\|_{n,2})\big)\,\mathrm{d}\varepsilon \\
&\le 6\|F\|_{n,2} + 48e\|F\|_{n,2}J \le \|F\|_{n,2}J\Big(\frac{6}{J} + 48e\Big) \le 140\|F\|_\infty J,
\end{aligned}$$

where the last inequality follows as $J \ge \log 2$ and $6/\log 2 + 48e \le 140$. Therefore,

$$\mathbb{E}\bigg\{\psi\bigg(\frac{\|nW_n\|_{\mathcal{F}}}{140\|F\|_\infty J}\bigg)\bigg\} = \mathbb{E}\bigg[\mathbb{E}\bigg\{\psi\bigg(\frac{\|nW_n\|_{\mathcal{F}}}{140\|F\|_\infty J}\bigg)\,\bigg|\,\mathbf{X}\bigg\}\bigg] \le 1,$$

as required. □

Lemma 49 (Orlicz norm supremum bound for U-statistics)**.** *We have*

$$\big\|\|nU_n\|_{\mathcal{F}}\big\|_\psi \le 17920\|F\|_\infty J.$$

Therefore, for $s \ge 1$,

$$\mathbb{P}\Big(\|nU_n\|_{\mathcal{F}} \ge 33104s\|F\|_\infty J\Big) \le e^{-s}.$$

Proof of Lemma 49. By Corollary 43, we have

$$\mathbb{E}\bigg\{\psi\bigg(\frac{\|U_n\|_{\mathcal{F}}}{128\big\|\|W_n\|_{\mathcal{F}}\big\|_\psi}\bigg)\bigg\} \le \mathbb{E}\bigg\{\psi\bigg(\frac{\|W_n\|_{\mathcal{F}}}{\big\|\|W_n\|_{\mathcal{F}}\big\|_\psi}\bigg)\bigg\} \le 1,$$

so that by Lemma 48,

$$\big\|\|nU_n\|_{\mathcal{F}}\big\|_\psi \le 128\big\|\|nW_n\|_{\mathcal{F}}\big\|_\psi \le 128 \times 140\|F\|_\infty J = 17920\|F\|_\infty J.$$

Now, for $s \ge 1$ and a random variable Z, we have by Markov's inequality that

$$\mathbb{P}\big(|Z| \ge s\|Z\|_\psi\big) = \mathbb{P}\bigg\{\psi\bigg(\frac{|Z|}{\|Z\|_\psi}\bigg) \ge \psi(s)\bigg\} \le \frac{1}{\psi(s)} = \frac{1}{e^s - 1} \le \frac{1}{(e-1)^s} = e^{-s\log(e-1)},$$

so

$$\begin{aligned}
\mathbb{P}\Big(\|nU_n\|_{\mathcal{F}} \ge 33104s\|F\|_\infty J\Big) &\le \mathbb{P}\bigg(\|nU_n\|_{\mathcal{F}} \ge \frac{17920s}{\log(e-1)}\|F\|_\infty J\bigg) \\
&\le \mathbb{P}\bigg(\|nU_n\|_{\mathcal{F}} \ge \frac{s}{\log(e-1)}\big\|\|nU_n\|_{\mathcal{F}}\big\|_\psi\bigg) \le e^{-s}.
\end{aligned}$$

□

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